SUCA 106

UNDERGRADUATE COURSE

BCA - COMPUTER APPLICATIONS

FIRST YEAR

SECOND SEMESTER

ALLIED PAPER - II

MATHEMATICS - II

INSTITUTE OF DISTANCE EDUCATION

UNIVERSITY OF MADRAS

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WELCOME

Warm Greetings.

It is with a great pleasure to welcome you as a student of Institute of Distance Education, University of Madras. It is a proud moment for the Institute of Distance education as you are entering into a cafeteria system of learning process as envisaged by the University Grants Commission. Yes, we have framed and introduced Choice Based Credit System(CBCS) in Semester pattern from the academic year 2018-19. You are free to choose courses, as per the Regulations, to attain the target of total number of credits set for each course and also each degree programme. What is a credit? To earn one credit in a semester you have to spend 30 hours of learning process. Each course has a weightage in terms of credits. Credits are assigned by taking into account of its level of subject content. For instance, if one particular course or paper has 4 credits then you have to spend 120 hours of self-learning in a semester. You are advised to plan the strategy to devote hours of self-study in the learning process. You will be assessed periodically by means of tests, assignments and quizzes either in class room or laboratory or field work. In the case of PG (UG), Continuous Internal Assessment for 20(25) percentage and End Semester University Examination for 80 (75) percentage of the maximum score for a course / paper. The theory paper in the end semester examination will bring out your various skills: namely basic knowledge about subject, memory recall, application, analysis, comprehension and descriptive writing. We will always have in mind while training you in conducting experiments, analyzing the performance during laboratory work, and observing the outcomes to bring out the truth from the experiment, and we measure these skills in the end semester examination. You will be guided by well experienced faculty.

I invite you to join the CBCS in Semester System to gain rich knowledge leisurely at your will and wish. Choose the right courses at right times so as to erect your flag of success. We always encourage and enlighten to excel and empower. We are the cross bearers to make you a torch bearer to have a bright future.

With best wishes from mind and heart,

DIRECTOR

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BACELLOR OF COMPUTER APPLICATIONS

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SYLLABUS

OBJECTIVE OF THE COURSE

This course gives an exposure to fundamental mathematics for logic implementation

Unit-I INTEGRAL CALCULUS:

Bernoulli's formula. Reduction formulae - $\int_0^{\frac{\pi}{2}} \sin^n x dx$, $\int_o^{\frac{\pi}{2}} sin^n x dx$, $\int_0^{\frac{\pi}{2}} \cos^n x dx$, $\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x \, dx$ (m, n being positive integers), Fourier series for functions in (a, a+2δ), Half range sine and cosine series

Unit-II DIFFERENTIAL EQUATIONS

Ordinary Differential Equations: Second order non- homogeneous differential equations with constant coefficients of the form $ay'' + by' + cy = X$ where X is of the form $e^{ax} \cos bx$ and $e^{ax} \sin bx$.

Partial Differential Equations: Formation, complete integrals and general integrals, four standard types and solving lagrange's linear equation Pp +Qq= R

Unit-III LAPLACE TRANSFORMS:

Laplace transformations of standard functions and simple properities, inverse Laplace transforms, Application to solution of linear differential equations up to 2nd order- simple problems.

Unit – IV VECTOR DIFFERENTIATION

Introduction, Scalar point functions, Vector point functions, Vector differential operator ∇ , Gradient, Divergence, Curl, Solenoidal, irrotational, identities.

Unit – V VECTOR INTEGRATION

Line, surface and volume integrals, Gauss, Stoke's and Green's theorems (without proofs). Simple problems on these.

Book for Reference:

1. S. Narayanan and T.K. Manickavasagam Pillai – Ancillary Mathematics, S. Viswanathan Printers, 1986, Chennai.
2. P. Duraipandian and S.UdayaBaskaran, Allied Mathematics, Vol. I & II Muhil Publications Chennai.
3. P.R. Vittal - Margham Publication, 2001, Chennai.

BACELLOR OF COMPUTER APPLICATIONS

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SCHEME OF LESSONS

LESSON - 1

INTEGRAL CALCULUS

BERNOULLIS AND REDUCTION FORMULAE

Learning Objectives

After reading this lesson you will be able to

- ❖ understand the concept of Bernoulli's formulae
- ❖ undersand the concept of Reduction formulae
- ❖ solve the problems of Bernoulli's and Reduction formulae

Structure

1.1 Introduction

A reduction formula is a linear relationship which connects the given integrals with other integrals which is of the same type but a lower index, then the given integral. the original integral or the given integral can be evaluated by successive applications of this reduction formula. This definition of reduction formula can be easiliy understand from the following book works and sovled problems.

1.2 Bernoulli's Formulae

Bernoullis formula is an extension of the formula of integration by parts

$$\int udv = uv - u^1v_1 + u^{11} v_2 - u^{111} v_3 + \ldots \text{ (Bernoullis formulae)}$$

Where $u' = \frac{du}{dx}, u'' = \frac{d^2u}{dx^2} \ldots.$

$$v' = \int vdx, v_2 = \int v_1dx, v_3 = \int v_2dx \ldots$$

If u', u'', u'''... denotes first, second, third defrivatives of the function u and v_1, v_2, v_3... are the successive integrals of functions of the function v.

Example 1

Evaluate $\int e^{ax}.x^3dx$

Solution

Here $u = x^3$

$u' = 3x^2$

$u'' = 6x$

$u''' = 6$

$$dv = e^{ax} dx$$

$$\int dv = \int e^{ax}dx$$

$$v = \frac{e^{ax}}{a}$$

$$v_1 = \frac{e^{ax}}{a^2}$$

$$v_2 = \frac{e^{ax}}{a^3}$$

$$v_3 = \frac{e^{ax}}{a^4} \text{ .... and so on.}$$

Bernoullis formula is

$$\int udv = uv - u'\,v_1 + u''\,v_2 - u'''\,v_3 + ...$$

$$\int e^{ax}.x^3dx = x^3\left(\frac{e^{ax}}{a}\right) - 3x^2\left(\frac{e^{ax}}{a^2}\right) + 6x\left(\frac{e^{ax}}{a^3}\right) - 6\left(\frac{e^{ax}}{a^4}\right) + c$$

$$= e^{ax}\left[\frac{x^3}{a} - \frac{3x^2}{a^2} + \frac{6x}{a^3} - \frac{6}{a^4}\right] + c$$

$$\int e^{ax}x^3dx = \frac{e^{ax}}{a^4}\,[a^3x^3 - 3a^2\,x^2 + 6ax - 6] + c$$

Example 2

Evaluate $\int x^2e^{-2x}dx$

Solution

Bernoullis formula is given by

$$\int udv = uv - u'v_1 + u''\,v_2 - u'''v_3 + ...$$

$$\int x^e e^{-2x}dx \quad = \quad x^2\left[\frac{e^{-2x}}{-2}\right] - 2x\left[\frac{e^{-2x}}{4}\right] + 2\left[\frac{e^{-2x}}{-8}\right] + c$$

By takng $u = x^3$, $dv = e^{-2x}dx$

$$= \quad e^{-2x}\left[-\frac{x^2}{2}-\frac{x}{2}-\frac{1}{4}\right]+c$$

$$= \quad \frac{-e^{-ex}}{4}\left[2x^2+2x+1\right]+c$$

Example 3

Evaluate $\int x^3 \cos 3x dx$

Solution

Here $u = x^3$

$u' = 3x^2$

$u'' = 6x$

$u''' = 6$

$dv = \cos 3x dx$

$$v = \frac{\sin 3x}{3}$$

$$v_1 = \frac{-\cos 3x}{9}$$

$$v_2 = \frac{-\sin 3x}{27}$$

$$v_3 = \frac{\cos 3x}{81}$$

$$\therefore \int x^3 \cos 3x dx = \frac{x^3 \sin 3x}{3} - \frac{3x^2}{9} (-\cos 3x) + \frac{6x}{27} (-\sin 3x) - \frac{6}{81} (\cos 3x) + c$$

$$= \frac{1}{3} \left[x^3 \sin 3x + x^2 \cos 3x - \frac{2}{3} x \sin 3x - \frac{2}{9} \cos 3x \right] + C$$

Example 4

Evaluate $\int x^5 \cos \frac{x}{2} dx$

Solution

Here $u = x^5$

$u' = 5x^4$

$u'' = 20x^3$

$u''' = 60x^2$

$u^{iv} = 120x$

$u^{v} = 120$

$$dv = \cos \frac{x}{2}$$

$$v = 2 \sin \frac{x}{2}$$

$$v_1 = -4 \cos \frac{x}{2}$$

$$v_2 = -8 \sin \frac{x}{2}$$

$$v_3 = 16\cos\frac{x}{2}$$

$$v_4 = 32\sin\frac{x}{2}$$

$$v_5 = -64\cos\frac{x}{2}$$

$$\therefore \int x^5\cos\frac{x}{2}dx = 2x^5\sin\frac{x}{2} - 5x^4\left(-4\cos\frac{x}{2}\right) + 20x^3\left(-8\sin\frac{x}{2}\right)$$

$$-60x^2\left(16\cos\frac{x}{2}\right) + 120\,x\left(32\sin\frac{x}{2}\right) - 120\left(-64\cos\frac{x}{2}\right)$$

$$= 2\left[x^5\sin\frac{x}{2} + 10x^4\cos\frac{x}{2} - 80x^3\sin\frac{x}{2} - 480x^2\cos\frac{x}{2}\right.$$

$$\left. + 1920x\sin\frac{x}{2} + 3840\cos\frac{x}{2}\right] + C$$

Check Your Progress- I

1. Evaluate $\int x^2e^x dx$

2. Evaluate $\int x^3\cos 2x dx$

3. Evaluate $\int x^4\sin\frac{x}{2}dx$

4. Evaluate $\int \left(x^3 + 2x^2 + 3\right)\sin x\, dx$

5. Evaluate $\int x^3e^{-3x}dx$

Check Your Answers - I

1. $e^x (x^2-2x+2)+c$

2. $\frac{1}{2}\left[x^3 \sin 2x + \frac{3x^2 \cos 2x}{2} - \frac{3x \sin 2x}{2} - \frac{3 \cos 2x}{4}\right] + c$

3. $-2x^4 \cos \frac{x}{2} + 16x^3 \sin \frac{x}{2} + 96\, x^2 \cos \frac{x}{2} - 384\, x \sin \frac{x}{2} - 768 \cos \frac{x}{2} + c$

1.3 Reduction Formula

1. Obtain a reduction formula for $\int x^n e^{ax} dx$

Solution

Given $I_n = \int x^n e^{ax} dx$

Let $u = x^n$; $dv = e^{ax}\, dx$

$du = nx^{n-1}dx$; $v = \frac{e^{ax}}{a}$ ($\because$ using Integral by parts)

$$I_n = \frac{x^n e^{ax}}{a} - \int \frac{e^{ax}}{a}.nx^{n-1}dx$$

$$= \frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} dx$$

$$\therefore I_n = \frac{x^n e^{ax}}{a} - \frac{n}{a}\, I_{n-1}$$

2. Obtain a reduction formula for $\int x^m (\log x)^n dx$

Solution

Let $I_{m,n} = \int x^m (\log x)^n dx$

put $u = (\log x)^n$ and $dv = x^m dx$

$du = n \frac{(\log x)^{n-1}}{x} . dx$ and $v = \frac{x^{m+1}}{m+1}$

Then, applying by parts rule,

$$I_{m,n} = \left[(\log x)^n \frac{x^{m+1}}{m+1} \right] - \int \frac{x^{m+1}}{m+1} .n \frac{(\log x)^{n-1}}{x} dx$$

$$= \frac{x^{m+1}}{m+1} (\log x)^n - \frac{n}{m+1} \int x^m (\log x)^{n-1} dx$$

i.e., $$I_{m,n} = \frac{x^{m+1}}{m+1} (\log x)^n - \frac{n}{m+1} I_{m, n-1}$$

3. Obtain the Reduction formula for $\int \sin^n x dx$

Solution

Let $I_n = \int \sin^n x dx$

$$= \int \sin^{n-1} x . \sin x dx$$

put $u = \sin^{n-1} x$ $dv = \sin x$

$du = (n-1) \sin^{n-2} x . \cos x dx$; $v = -\cos x$

Then, applying by parts rule, $\int udv = uv - \int vdu$

$$I_n = \sin^{n-1}x\,(-\cos x) - \int (n-1)\sin^{n-2}x.\ \cos x.\ (-\cos x)\ dx$$

$$= -\sin^{n-1}x\,(-\cos x) + (n-1)\int \sin^{n-2}x.\ \cos^2 x dx$$

$$= -\sin^{n-1}x\ .\cos x + (n-1)\int \sin^{n-2}x\,(1-\sin^2 x)dx$$

$$= -\sin^{n-1}x\ .\cos x + (n-1)\int \sin^{n-2}x dx - (n-1)\int \sin^n x dx$$

$$= -\sin^{n-1}x\ .\cos x + (n-1)\ I_{n-2} - (n-1)\ I_n$$

$$= -\sin^{n-1}x\ .\cos x + (n-1)\ I_{n-2}$$

$$\therefore I_n = \frac{\sin^{n-1}x.\cos x}{n} + \frac{n-1}{n}.I_{n-1}$$

4. Obtain the Reduction formula for $\int \cos^n x dx$

Solution

$$\text{Let } I_n = \int \cos^n x dx$$

$$= \int \cos^{n-1}x.\ \cos x\ dx$$

$$= \cos^{n-1}x.\sin x - \int (n-1).\ \cos^{n-2}x.\ (-\sin x).\ \sin x dx \text{ (use Integral by parts)}$$

$$= \cos^{n-1}x.\sin x + (n-1)\int \cos^{n-2}x\,(1-\cos^2 x)\ dx$$

$$= \cos^{n-1}x.\sin x + (n-1)\ .I_{n-2} - (n-1)\ I_n$$

$$\therefore I_n = \frac{\cos^{n-1}x.\sin x}{n} + \frac{n-1}{n}.I_{n-2}$$

4. Reduction formula for $\int_0^{\frac{\pi}{2}} \sin^n x dx$

Solution

Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$

$$= \int_0^{\frac{\pi}{2}} \sin^{n-1} . \sin x dx$$

put u = $\sin^{n-1} x$; dv = $\sin x \, dx$

du = $(n-1) \sin^{n-2} x . \cos x dx$; v = $-\cos x$

using Integrate by parts $\int udv = uv - \int vdu$

$$I_n = \left[-\cos x \sin^{n-1} x\right]_0^{\pi/2} + \int_0^{\frac{\pi}{2}} \cos x (n-1) . \sin^{n-2} x . \cos x dx$$

$$= 0 + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x . \cos^2 x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} (1 - \sin^2 x) dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \sin^n x dx$$

$$I_n = (n-1).I_{n-2}-(n-1)\ I_n$$

$$\therefore (1+n-1)\ I_n = (n-1).\ I_{n-2}$$

$$\therefore I_n = \frac{(n-1)}{n}.I_{n-2}$$

Evaluate of I_n;

Case (i) : Let n be an even initeger

$$I_n = \frac{n-1}{n}\ .\ I_{n-2}$$

$$= \frac{n-1}{n}.\frac{n-3.}{n-2}\ .\ I_{n-4}$$

Proceedings in this way,

$$I_n = \frac{n-1}{n}.\frac{n-3.}{n-2}.\frac{n-5}{n-4}....\frac{1}{2}I_0$$

But $I_0 = \int_0^{\pi/2} dx = \frac{\pi}{2}$

$$\Rightarrow\ I_n = \frac{n-1}{n}.\frac{n-3.}{n-2}.\frac{n-5}{n-4}....\frac{1}{2}.\frac{\pi}{2}$$

Case (2) : Let n be an Odd Integer

Then $I_n = \frac{n-1}{n}.\frac{n-3.}{n-2}.....\frac{2}{3}.I_1$

But $I_1 = \int_0^{\pi/2} \sin x dx = [-\cos x]_0^{\pi/2} = 1$

which implies, $I_n = \frac{n-1}{n}.\frac{n-3.}{n-2}.\frac{n-5}{n-4}....\frac{2}{3}.1$

5. Reduction for $\int_0^{\pi/2} \cos^n x dx$

Solution

Let $I_n = \int_0^{\pi/2} \cos^n x dx$

$= \int_0^{\pi/2} \cos^{n-1} . \cos x dx$

put $u = \cos^{n-1} x$; $dv = \cos x dx$

$du = (n-1) \cos^{n-2} x.(-\sin x)dx$; $v = \sin x$

$\therefore I_n = \left[\cos^{n-1} . \sin x dx\right]_0^{\pi/2} + \int_0^{\pi/2} \sin x (n-1) \cos^{n-2} x . \sin x dx$

$= 0 + (n-1) \int_0^{\pi/2} \cos^{n-2} x . (1 - \cos^2 x) dx$

$= (n-1). I_{n-2} - (n-1).I_n.$

$I_n + (n-1).I_n = (n-1).I_{n-2}$

$(1+n-1).I_n = (n-1).I_{n-2}$

$\therefore I_n = \frac{n-1}{n} .I_{n-2}$

Proceedings as in the last problem

When n is even

$$I_n = \frac{n-1}{n}.\frac{n-3.}{n-2}.\frac{n-5}{n-4}....\frac{1}{2}.\frac{\pi}{2}$$

When n is odd,

$$I_n = \frac{n-1}{n}.\frac{n-3.}{n-2}.\frac{n-5}{n-4}....\frac{2}{3}$$

7. Obtaion Reduction formula for $\int \tan^n xdx$

Solution

Let $I_n = \int \tan^n xdx$

$= \int \tan^{n-2} x \tan^2 xdx$ $\quad$ ($\tan^2 x = \sec^2 x - 1$)

$= \int \tan^{n-2}.(\sec^2 x - 1)dx$

$= \int \tan^{n-2}.d(\tan x) - \int \tan^{n-2} xdx$

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

8. If $I_n = \int_0^{\pi/2} \tan^n xdx$, prove that $I_n + I_{n-2} = \frac{1}{n-1}$ and hence evaluate I_5.

Solution

$$I_n = \int_0^{\pi/4} \tan^n xdx$$

$$= \int_0^{\pi/4} \tan^{n-2} x(\sec^2 x - 1)dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \, d(\tan x) - \int_0^{\pi/4} \tan^{n-2} x \, dx$$

$$I_n = \left[\frac{\tan^{n-1} x}{n-1}\right]_0^{\frac{\pi}{4}} - I_{n-2}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

$$I_n = \frac{1}{n-1} - I_{n-2}$$

Put n =5, $I_5 = \frac{1}{4} - I_3$

$$I_3 = \frac{1}{3} - I_1$$

$$\therefore I_5 = \frac{1}{4} - \left(\frac{1}{2} - I_1\right)$$

$$= \frac{1}{4} - \frac{1}{2} + I_1$$

$$= -\frac{1}{4} + \int_0^{\pi/4} \tan x \, dx = \left[\frac{-1}{4} + \log(\sec x)\right]_0^{\pi/4}$$

$$\therefore I_5 = \log\sqrt{2} - \frac{1}{4}$$

9. Reduction formula for $\int \sec^n x dx$

Solution

$$I_n = \int \sec^n x dx$$

$$= \int \sec^{n-2} x \sec^2 x dx$$

$$= \tan x.\sec^{n-2}x - \int \tan x (n-2).\sec^{n-3} x.\sec x \tan x dx$$

$$= \tan x.\sec^{n-2}x - (n-2) \int \sec^{n-2} x.\tan^2 x dx$$

$$= \tan x.\sec^{n-2}x - (n-2) \int \sec^{n-2} x(\sec^2 x - 1) dx$$

$$= \tan x.\sec^{n-2}x - (n-2).I_n + (n-1).I_{n-2}.$$

$$\therefore I_n = \frac{\tan x.\sec^{n-2} x}{n-1} + \frac{n-2}{n-1}.I_{n-2}.$$

Note 1:

Reduction formula for

$$I_n = \int \cot n x dx = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}$$

Note 2:

$$I_n = \int_{\pi/\Delta}^{\pi/2} \cot^n x dx = \frac{1}{n-1} - I_{n-2} \text{(or)}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1}$$

10. Obtain a reduction formula for $\int_0^{\pi/2} x^n \sin x dx$

Solution

$$\text{Let } I_n = \int_0^{\pi/2} x^n \sin x dx$$

$$= \left[-x^n \cos x\right]_0^{\pi/2} + \int_0^{\pi/2} n x^{n-1} \cos x dx$$

$$= 0 + n \int_0^{\pi/2} x^{n-1} \cos x dx$$

$$= n \left[x^{n-1} \sin x\right]_0^{\pi/2} - \int_0^{\pi/2} (n-1) x^{n-2} . \sin x dx$$

$$= n\left[\left(\frac{\pi}{2}\right)^{n-1} - (n-1).I_{n-2}\right]$$

$$\therefore \; I_n + n(n-1)I_{n-2} = n\left(\frac{\pi}{2}\right)^{n-1}$$

11. Reduction formula for $\int_0^{\pi/2} \sin^m x . \cos^n x dx$

Solution

$$\text{Let } I_{m,n} = \int_0^{\pi/2} \sin^{m-1} x . \sin x . \cos^n x dx$$

$$= \quad -\int_0^{\pi/2} \sin^{m-1} x.\cos^n .xd(\cos x) \text{, (Use Integral by parts)}$$

$$= \quad \left[-\sin^{m-1} x.\frac{\cos^{n+1} x}{n+1}\right]_0^{\pi/2} + \int_0^{\pi/2} \frac{\cos^{n+1} x}{n+1} \ (m-1)\ \sin^{m-2}x.\cos xdx$$

$$= \quad \frac{m-1}{n+1} \int_0^{\pi/2} \cos^{n+2} x \sin^{m-2} xdx$$

$$= \quad \frac{m-1}{n+1} \int_0^{\pi/2} \sin^{m-2} x \cos^n x(1-\sin^2 x)dx$$

$$= \quad \frac{m-1}{n+1} \int_0^{\pi/2} \sin^{m-2} x \cos^n xdx - \frac{m-1}{n+1} \int_0^{\pi/2} \sin^m x \cos^n xdx$$

$$\therefore\ I_{m,n} \quad = \quad \frac{m-1}{n+1}.I_{m-2,n} - \frac{m-1}{n+1}.I_{m,n.}$$

$$\left(1+\frac{m-1}{n+1}\right).I_{m,n} \quad = \quad \frac{m-1}{n+1}.I_{m-2,n}$$

$$\left(\frac{m-1}{n+1}\right) I_{m,n} \quad = \quad \frac{m-1}{n+1}.I_{m-2,n}$$

$$\therefore \quad I_{m,n} \quad = \quad \frac{m-1}{n+1}.I_{m-2,n}$$

This is a required reduction formula for $I_{m,n}$

evaluate $I_{m,n}$

Case (1) : Suppose m is even

$$I_{m,n} \quad = \quad \frac{m-1}{m+n}.I_{m-2,n}$$

$$I_{m-2,\,n} = \frac{m-3}{m+n-2} \cdot I_{m-4,n}$$

$$I_{m-4,\,n} = \frac{m-5}{m+n-4} \cdot I_{m-6,n}$$

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$$I_{4,\,n} = \frac{3}{n+4} \cdot I_{2,n}$$

$$I_{2,\,n} = \frac{1}{n+2} \cdot I_{0,n}$$

Multiplying all these results we get

$$I_{m,n} = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdot \frac{m-5}{m+n-4} \cdots \frac{3}{n+4} \cdot \frac{1}{n+2} \cdot I_{0,n}$$

$$\text{But } I_{0,n} = \int_0^{\pi/2} \cos^n x dx = \frac{n-1}{m} \cdot \frac{n-3}{n-2} \cdots\cdots \frac{1}{2} \cdot \frac{\pi}{2}, \text{ if n is even}$$

$$= \frac{n-1}{m} \cdot \frac{n-3}{n-2} \cdots\cdots \frac{2}{3}, \text{ if n is even}$$

If m is Even and n is Even.

$$\text{Then, } I_{m,n} = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdot \frac{m-5}{m+n-4} \cdots \frac{3}{n+4} \cdot \frac{1}{n+2} \cdot \frac{n-1}{n} \cdot \frac{n-3}{n-1} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \qquad (1)$$

If m is Even and n is ODD,

$$\text{Then, } I_{m,n} = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdot \frac{m-5}{m+n-4} \cdots \frac{3}{n+4} \cdot \frac{1}{n+2} \cdot \frac{n-1}{n} \cdot \frac{n-3}{n-1} \cdots \frac{2}{3} \qquad (2)$$

Case (2) Suppose m is ODD

Then, $I_{m,n} = \frac{m-1}{m+n} . I_{m-2,n}$

$$I_{m-2,n} = \frac{m-1}{m+n-2} . I_{m-4,n}$$

$$. . . . .$$

$$. . . . .$$

$$. . . . .$$

$$I_{5,n} = \frac{4}{n+5} . I_{3,n}$$

$$I_{3,n} = \frac{2}{n+3} . I_{1,n}$$

Multiplying all these results,

$$I_{m,n} = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdots \frac{4}{n+5} \cdot \frac{2}{n+3} I_{1,n}$$

But $I_{m,n} = \int_0^{\pi/2} \sin x . \cos^n x dx$

$$= \int_0^{\pi/2} -\cos^n (\cos x)$$

$$= \left[-\frac{\cos^{n+1} x}{n+1} \right]_0^{\pi/2}$$

$$= \frac{1}{n+1}$$

$$=> I_{m,n} = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdots \frac{4}{n+5} \cdot \frac{2}{n+3} \cdot \frac{1}{n+1}$$

1.4 Solved Problems

Problem1

Evaluate $\int_0^{\pi/2} \sin^7 x dx$

Solution

Let $I_n = \int_0^{\pi/2} \sin^n x dx$

using the formula,

$$I_n = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \cdot \cdot \frac{2}{3}$$, when n is odd, we get

put n =7

$$I_7 = \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3}$$

$$= \frac{48}{105}$$

$$I_7 = \frac{16}{35}$$

Problem 2

Evaluate $\int_0^{\pi/2} \cos^8 x dx$

Solution

Using the formula

$$I_n = \int_0^{\pi/2} \cos^n x dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{1}{2} \cdot \frac{\pi}{2}$$

when n is even, we get put n = 8

$$I_8 = \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$I_7 = \frac{35\pi}{256}$$

Problem 3

Evaluate $\int_0^{\pi/2} \sin^6 x . \cos^4 x dx$

Solution

$$\int_0^{\pi/2} \sin^6 x . \cos^4 x dx = \int_0^{\pi/2} \sin^6 x . \cos^4 x dx$$

$$= \int_0^{\pi/2} \sin^6 x (\cos^2 x)^2 dx$$

$$= \int_0^{\pi/2} \sin^6 x (1 - \sin^2 x)^2 dx$$

$$= \int_0^{\pi/2} \sin^6 x (1 + \sin^4 x - 2\sin^2 x) dx$$

$$= \int_0^{\pi/2} \sin^6 x dx + \int_0^{\pi/2} \sin^{10} x dx - 2\int_0^{\pi/2} \sin^8 x dx$$

$$= \frac{5}{6}.\frac{3}{4}.\frac{1}{2}.\frac{\pi}{2}+\frac{9}{10}.\frac{7}{8}.\frac{5}{6}.\frac{3}{4}.\frac{1}{2}.\frac{\pi}{2}-2\left[\frac{7}{8}.\frac{5}{6}.\frac{3}{4}.\frac{1}{2}.\frac{\pi}{2}\right]$$

$$= \frac{15\pi}{96}+\frac{63\pi}{512}-\frac{355}{128}$$

$$= \frac{3\pi}{512}$$

Problem 4

Evaluate $\int_0^{\pi/2} \sin^6 x.\cos^5 x dx$

Solution

$$\int_0^{\pi/2} \sin^6 x.\cos^5 x dx = \frac{4}{\pi}.\frac{2}{9}.\frac{1}{7}=\frac{8}{693}$$

(using reduction formula)

Problem 5

Evaluate $\int_0^{\pi/2} \sin^7 x.\cos^4 x\ dx$

Solution

$$\int_0^{\pi/2} \sin^7 x.\cos^4 x = \frac{3}{11}.\frac{1}{9}.\frac{6}{7}.\frac{4}{5}.\frac{2}{3}$$

$$= \frac{16}{1155}$$

(using reduction formula)

Problem 6

Evaluate $\int_0^{\pi/2} \sin^8 x.\cos^6 x\ dx$

Solution

Here m=8, n =6 (both even)

$$\int_0^{\pi/2} \sin^m x.\cos^n x dx = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdots \frac{1}{n+2} \cdot \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{1}{2} \cdot \frac{\pi}{2}$$

$$\int_0^{\pi/2} \sin^8 x.\cos^6 x dx = \frac{5}{14} \cdot \frac{3}{12} \cdot \frac{1}{10} \cdot \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

Problem 7

Evaluate $\int_0^1 x^2(1-x^2)^{3/2} dx$

Solution

Put x = sin θ => dx =cosθ dθ

when x = 0, => θ= 0

x = 1, => $\theta = \frac{\pi}{2}$

$$\therefore \int_0^1 x^2(1-x^2)^{3/2} dx = \int_0^{\pi/2} \sin^2\theta(1-\sin^2\theta)^{3/2}.\cos\theta d\theta$$

$$= \int_0^{\pi/2} \sin^2\theta.\cos^3\theta.\cos\theta d\theta$$

$$= \int_0^{\pi/2} \sin^2\theta - \cos^4\theta d\theta$$

(m=2,n=4 both even)

$$\int_0^{\pi/2} \sin^m \cos^n \theta d\theta = \frac{n-1}{m+n}.\frac{n-3}{m+n-2}\cdots\frac{1}{m+2}.\frac{m-1}{m}.\frac{m-3}{m-2}\cdots\frac{1}{2}.\frac{\pi}{2}$$

$$= \frac{4-1}{2+4}.\frac{4-3}{2+4-2}.\frac{2-1}{2}.\frac{\pi}{2}$$

$$= \frac{3}{6}.\frac{1}{4}.\frac{1}{2}.\frac{\pi}{2}$$

$$= \frac{\pi}{32}$$

Problem 8

Evaluate $\int x^4(\log x)^3 dx$ using reduction formula

Solution

We know that

$$I_{m,n} = \int x^m(\log x)^n dx = \frac{x^{m+1}(\log x)^n}{m+1} \cdot \frac{n}{m+1} I_{m,\,n-1}$$

$$\therefore \quad I_{4,3} = \int x^4(\log x)^3 dx$$

$$\therefore \quad I_{4,3} = \frac{x^5(\log x)^3}{5} - \frac{3}{5} I_{4,2}$$

$$= \frac{x^5(\log x)^3}{5} - \frac{3}{5}\left\{\frac{x^5(\log x)^2}{5} - \frac{2}{5}.I_{4,1}\right\}$$

$$= \frac{x^5(\log x)^3}{5} - \frac{3}{25}\ x^5(\log x)^2 + \frac{6}{25}\left[\frac{x^5(\log x)'}{5} - \frac{2}{5}.I_{4,0}\right]$$

$$= \frac{x^5(\log x)^3}{5} - \frac{3}{25}\ x^5(\log x)^2 + \frac{6}{125}\ x^5 \log x - \frac{6}{125}\left[\int x^4 dx\right]$$

$$= \frac{x^5(\log x)^3}{5} - \frac{3}{25}\ x^5(\log x)^2 + \frac{6}{125}\ x^5 \log x - \frac{6}{125} \cdot \frac{x^5}{5}$$

$$= x^5\left[\frac{(\log x)^3}{5} - \frac{3}{25}(\log x)^2 + \frac{6}{125}\log x - \frac{6}{625}\right]$$

Check Your Progress - II

1. The value of the integral $\int_0^{\pi/2} \sin^8 x.\cos^6 x$ is ________________

2. Reduction formula for $\int \tan^6 x dx$ is ______________________

3. The value of the integral $\int \sec^5 x dx$ is ______________

Check Your Answers - II

1. $\frac{5\pi}{4096}$

2. $\frac{\tan^5 x}{5} - \frac{\tan^3 x}{3} + \tan x - x$

3. $\frac{\sec^2 x \tan x}{4} + \frac{3}{8}\sec x \tan x + \frac{3}{8}\ \log(\sec x + \tan x)$

1.5 Recap

1. Bernouills formula

$$\int udv = uv = u'xv_1 + u''v_2 - u'''v_3 + ...$$

2. Reduction formula

(i) $\int_0^{\pi/2} \sin^n x dx = \frac{n-1}{1} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{1}{2} \cdot \frac{\pi}{2}$ (if n is even)

$= \frac{n-1}{1} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{2}{3}$ (if n is odd)

(ii) $\int_0^{\pi/2} \cos^n x dx = \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{1}{2} \cdot \frac{\pi}{2}$ (if n is even)

$= \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdot \frac{n-5}{n-4} \cdots \frac{2}{3}$ (if n is odd)

(iii) $\int_0^{\pi/2} \sin^m x \cos^n x dx = \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdot \frac{m-5}{m+n-4} \cdots \frac{3}{n+4} \cdot \frac{1}{n+2} \cdot$

$\frac{n-1}{n} \frac{n-3}{n-1} \cdots \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$ (if m is even & n is even)

$= \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdots \frac{3}{n+4} \cdot \frac{1}{n+2} \cdot \frac{n-3}{n} \cdot \frac{n-3}{n-2} \cdots \frac{2}{3}$

(if m is even & n is odd)

$= \frac{m-1}{m+n} \cdot \frac{m-3}{m+n-2} \cdots \frac{4}{n+5} \cdot \frac{2}{n+3} \cdot \frac{1}{n+1}$

(if m is odd)

1.7 Exercise

I. Evaluate

(1) $\int \sec^3 x dx$

(2) $\int_0^{\pi/2} \sin^6 x dx$

(3) $\int_0^{\pi/2} \cos^8 x dx$

(4) $\int_0^{\pi/2} \sin^9 x dx$

(5) $\int_0^{\pi/2} \sin^5 x . \cos^6 x dx$

(6) $\int_0^{\pi/2} \cos^5 x . \sin^4 x dx$

(7) $\int_0^{\pi/2} \sin^7 x . \cos^3 x dx$

(8) obtain a reduction formula for $I_n = \int x^n \cos x dx$

(9) $\int_0^{\pi/2} \sin^5 x dx$

(10) $\int_0^{\pi/2} \sin^6 x . \cos^4 x dx$

1.7 Check Your Answers

1. $\frac{\sec x \tan x}{2} + \frac{1}{2} \log (\sec x + \tan x)$

2. $\frac{5\pi}{32}$

3. $\frac{35\pi}{512}$

4. $\frac{384}{945}$

5. $\frac{8}{693}$

6. $\frac{4}{15}$

7. $\frac{1}{40}$

8. $I_n = x^n \sin x + n x^{n-1} x \cos x - n(n-1).\ I_{n-2}.$

9. $\frac{8}{15}$

10. $\frac{3\pi}{512}$

LESSON - 2
FOURIER SERIES OR FOURIER EXPANSION

Learning Objectives

After reading this lesson you will be able to

- ❖ understand the concept and find the problem of Fourier expansion
- ❖ know the fourier series for odd and even function.
- ❖ solve the problems of Fourier series

Structure

2.1 Introduction

Fourier Series

It has been shown by Fourier that a function f(x) which has only a finite number of discontinuities can be expressed as a trigonometric series in a given range of x in the form

$$f(x) = \frac{a_0}{2} + (a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + \ldots\ldots + a_n \cos nx + \ldots\ldots.\infty)$$

$$+ (b_1 \sin x + b_2 \sin 2x + b_3 \sin 3x + \ldots\ldots + b_n \sin nx + \ldots\ldots.\infty)$$

Fourier has shown that the expansion of f(x) in the above form is possible only if it satisfies Dirichlet conditions.

Dirichlet conditions

These conditions called *Dirichlet* conditions are stated below.

Let $f(x)$ be defined in the interval $c < x < c + 2\pi$ with period 2π and satisfy the following conditions.

i) $f(x)$ is single valued.

ii) It has a finite number of discontinuities in a period of 2π.

iii) It has a finite number of maxima and minima in a given period.

iv) $\int_0^{c+2\pi} |f(x)|dx$ is convergent.

Properties of Definite Integrals

We state the following properties of definite integral which are used in the evaluation a_0, a_n & b_n.

(i) $$\int_0^{c+2\pi} \sin x dx = \left[\frac{-\cos nx}{n}\right]_c^{c+2\pi} = 0$$

(ii) $$\int_0^{c+2\pi} \cos nx dx = \left[\frac{\sin nx}{n}\right]_c^{c+2\pi} = 0$$

2.2 Definition of Fourier Series

If f (x) satisfies the above condition,then it is possible to express f(x) as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \text{ .... (1)}$$

Here the coefficients a_0 a_n, b_n are called fourier coefficients

Where

$$a_0 = \frac{1}{\pi}\int_{c}^{c+2\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_{c}^{c+2\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{c}^{c+2\pi} f(x)\sin nx\,dx$$

Note 1

1. If c =0, then the interval of definition for f(x) is $(0, 2\pi)$ and in this case,

$$a_0 = \frac{1}{\pi}\int_{0}^{2\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_{0}^{2\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{0}^{2\pi} f(x)\sin nx\,dx$$

2. If $c = -\pi$, then the interval of expansion for f(x) is $(-\pi, \pi)$ and in this interval

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$$

Value of function f(x) at a point

Note 1

The fourier series coverges to f(x) at all points, where f(x) is continuous.

(i.e.,) If $x = x_0$ is a point of continuity, the sum of the fourier series at $x = x_0$ is $f(x_0)$

Note 2

If $x = x_0$ is a point of discontinuity, then the value of f(x) at $x = x_0$ is

$$\frac{f(x_0+0)+f(x_0-0)}{2}$$

Solved Problems

Problem 1

Obtain a Fourier expansion for the function $f(x) = \frac{1}{2}(\pi - x),\ 0 < x < 2\pi$.

Solution

Let the Fourier series for the function f(x) be

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \qquad (1)$$

Then

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} \frac{1}{2}(\pi - x)\,dx$$

$$= \frac{1}{2\pi}\left[\pi x - \frac{x^2}{2}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}[2\pi^2 - 2\pi^2] = 0$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\ dx$$

$$= \frac{1}{\pi}\int_0^{2\pi} \frac{1}{2}(\pi - x)\cos nx\ dx$$

$$= \frac{1}{2\pi}\left[(\pi - x)\frac{\sin nx}{n} - (-1)\left(\frac{-\cos nx}{n^2}\right)\right]_0^{2\pi}$$

(using Bernoulis formula)

$$= \frac{1}{2\pi}\left[(\pi - x)\frac{\sin nx}{n} - \frac{\cos nx}{n^2}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left[-\frac{1}{n^2} + \frac{1}{n^2}\right] = 0$$

$$\boxed{a_n = 0}$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\ dx$$

$$= \frac{1}{2\pi}\int_0^{2\pi} (\pi - x)\sin nx\ dx$$

$$= \frac{1}{2\pi}\left[(\pi - x)\left(\frac{-\cos nx}{n}\right) - (-1)\left(\frac{-\sin nx}{n^2}\right)\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left[\pi.\frac{1}{n} + \frac{\pi}{n}\right]$$

$$b_n = \frac{1}{n}$$

(i) becomes, $f(x) = \dfrac{\pi - x}{2} = \displaystyle\sum_{n=1}^{\infty} \frac{1}{n} \sin nx$

$$\therefore f(x) = \sin x + \frac{\sin 2x}{2} + \frac{\sin 3x}{2} + \dots$$

Problem 2

If $f(x) = \left(\dfrac{\pi - x}{2}\right)^2$ in $(0, 2\pi)$ show that $f(x) = \dfrac{\pi^2}{12} + \displaystyle\sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$ and deduce that $\dfrac{1}{1^2} + \dfrac{1}{3^2} + \dfrac{1}{5^2} + \dots \infty = \dfrac{\pi^2}{8}$

Proof

Given $f(x) = \left(\dfrac{\pi - x}{2}\right)^2$

The fourier series is given by

$$f(x) = a_0 \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

where

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x)\, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} \left[\frac{\pi - x}{2}\right]^2 dx$$

$$= \frac{1}{4\pi}\left[-\frac{(\pi-x)^3}{3}\right]_0^{2\pi}$$

$$= \frac{1}{4\pi}\left[\frac{\pi^3}{3}+\frac{\pi^3}{3}\right]$$

$$= \frac{1}{4\pi}\left[\frac{2\pi^3}{3}\right]$$

$$\boxed{a_0 = \frac{\pi^2}{6}}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos nx\,dx$$

$$= \frac{1}{\pi}\int_0^{2\pi}\left(\frac{\pi-x}{2}\right)^2 \cos nx\,dx = \frac{1}{4\pi}\int_0^{2\pi}(\pi-x)^2.\cos nx dx$$

$$= \frac{1}{4\pi}\left[(\pi-x)^2\left(\frac{\sin nx}{n}\right)-2(\pi-x)(-1)\left(\frac{-\cos nx}{n^2}\right)+2(-1)^2\left(\frac{-\sin nx}{n^3}\right)\right]_0^{2\pi}$$

$$= \frac{1}{4\pi}\left[(\pi-x)^2\frac{\sin nx}{n}-\frac{2(\pi-x)\cos nx}{n^2}-\frac{2\sin nx}{n^3}\right]_0^{2\pi}$$

$$= \frac{1}{4\pi}\left[\frac{2\pi}{n^2}+\frac{2\pi}{n^2}\right]$$

$$\boxed{a_n = \frac{1}{n^2}}$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} f(x)\sin nx\,dx$$

$$= \frac{1}{4\pi}\int_0^{2\pi}(\pi - x)^2 \sin nx \, dx$$

$$= \frac{1}{4\pi}\left[(\pi - x)^2\left(\frac{-\cos nx}{n}\right) - 2(\pi - x)(-1)\left(\frac{-\sin nx}{n^2}\right) + 2\left(\frac{\cos nx}{n^2}\right)\right]_0^{2\pi}$$

$$= \frac{1}{4\pi}\left[-\frac{\pi^2}{n} + \frac{2}{n^3} + \frac{\pi^2}{n} - \frac{2}{n^3}\right] = 0$$

$\boxed{b_n = 0}$

(i) becomes,

$$\left(\frac{\pi - x}{2}\right)^2 = \frac{\pi^2}{12} + \sum_1^{\infty}\frac{1}{n^2}\cos nx \qquad (2)$$

Hence proved

put x =0, in (2), $\frac{\pi^2}{4} = \frac{\pi^2}{12} + \sum\frac{1}{n^2}$

$$\therefore \quad \sum\frac{1}{n^2} = \frac{\pi^2}{4} - \frac{\pi^2}{12} = \frac{\pi^2}{6}$$

$$\text{i.e.,} \quad \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\infty = \frac{\pi^2}{6} \qquad (3)$$

put $x = \pi$

$$0 = \frac{\pi^2}{12} + \sum\left(\frac{1}{n^2}(-1)^n\right)$$

$$0 = \frac{\pi^2}{12} - \frac{1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \ldots\infty$$

Therefore

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \ldots = \frac{\pi^2}{12} \qquad (4)$$

Adding (3) and (4)we get

$$2\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\infty\right) = \frac{\pi^2}{6}+\frac{\pi^2}{12}=\frac{3\pi^2}{12}$$

$$\therefore \quad \frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\infty = \frac{\pi^2}{8}$$

Hence proved

Problem 4

Obtain a fourier series for the function f (x) = x sin x, $0 < x < 2\pi$.

Solution

f (x) = x sin x, $0 < x < 2\pi$.

The fourier series is given by

$$f(x) = \frac{a_0}{2}+\sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \qquad (1)$$

where

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} x \sin x dx$$

$$= \frac{1}{\pi}\left[x(-\cos x)-1(-\sin x)\right]_0^{2\pi}$$

$$= \frac{1}{\pi}\left[-2\pi\right]$$

$$\boxed{a_0 = -2}$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} x \sin x \cos nx dx$$

$$= \frac{1}{2\pi}\int_0^{2\pi} x\ [\sin(n+1)x-\sin(n-1)x]\ dx$$

$$= \quad \frac{1}{2\pi}\left\{x\left[\frac{-\cos(n+1)x}{n+1}+\frac{\cos(n-1)x}{n-1}\right]-\right.$$

$$\left.\left[-\frac{\sin(n+1)x}{(n+1)^2}+\frac{\sin(n-1)}{(n-1)^2}\right]\right\}_0^{2\pi} \text{ for n>1}$$

$$= \quad \frac{1}{2n}\left[2\pi\left(-\frac{1}{n+1}+\frac{1}{n-1}\right)\right] \text{ for n>1}$$

$$= \quad \frac{(n+1)-(n-1)}{n^2-1}$$

$$\boxed{a_n = \frac{2}{n^2-1}}$$

when n=1,

$$a_1 \quad = \quad \frac{1}{\pi}\int_0^{2\pi} x\sin x\cos x dx$$

$$= \quad \frac{1}{2\pi}\int_0^{2\pi} x \sin 2x dx$$

$$= \quad \frac{1}{2\pi}\left[x\left(\frac{-\cos 2x}{2}\right)-1\left(\frac{-\sin 2x}{2}\right)\right]_0^{\pi}$$

$$= \quad \frac{1}{2\pi}\left[2\pi\left(-\frac{1}{2}\right)\right]$$

$$\boxed{a_1 = -\frac{1}{2}}$$

$$b_n \quad = \quad \frac{1}{\pi}\int_0^{2\pi} x\sin x\sin nx dx$$

$$= \quad \frac{1}{2\pi}\int_0^{2\pi} x\ [\cos(n-1)x-\cos(n+1)x]dx$$

$$= \frac{1}{2\pi}\left\{x\left[\frac{\sin(n-1)x}{(n-1)} - \frac{\sin(n+1)x}{(n+1)}\right]\right.$$

$$\left.-\left[-\frac{\cos(n-1)x}{(n-1)^2} + \frac{\cos(n+1)x}{n+1}\right]\right\}_0^{2\pi} \text{ for } n>1\backslash$$

$$= \frac{1}{2\pi}\left[-\left(\frac{-1}{(n-1)^2} + \frac{1}{(n+1)^2}\right) + \left(-\frac{1}{(n-1)^2} - \frac{1}{(n+1)^2}\right)\right]$$

$\boxed{b_n = 0}$

when n =1,

$$b_1 = \frac{1}{\pi}\int_0^{2\pi} x\sin^2 x\,dx$$

$$= \frac{1}{2\pi}\int_0^{2\pi} x\ (1-\cos 2x)dx$$

$$= \frac{1}{2\pi}\left[x\left(x - \frac{\sin 2x}{2}\right) - \left(\frac{x^2}{2} + \frac{\cos 2x}{4}\right)\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left[\frac{x^2}{2} - \frac{x\sin 2x}{2} - \frac{\cos 2x}{4}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left[2\pi^2 - \frac{1}{4} + \frac{1}{4}\right]$$

$\boxed{b_1 = \pi}$

$$(1) \Rightarrow f(x) = \frac{a_0}{2} + \sum_1^{\infty} a_n \cos nx + \sum_1^{\infty} b_n \sin nx$$

$$= -1-\frac{1}{2}\cos x + \sum_{n=2}^{\infty}\frac{2}{n^2-1}\cos nx + \pi\sin x$$

$$\therefore f(x) = -1+\pi\sin x - \frac{\cos x}{2} + \sum_{n=2}^{\infty}\frac{2\cos nx}{n^2-1}$$

Problem 5

Obtain a Fourier expansion for f(x) $\begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$

Hence show that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + ... = \frac{\pi^2}{8}$

Solution

Given f(x) $\begin{cases} -\pi, & \text{when } -\pi < x < 0 \\ x, & \text{when } 0 < x < \pi \end{cases}$

The fourier series is,

$$f(x) = \frac{a_0}{2} + \sum_1^\infty a_n \cos nx + \sum_1^\infty b_n \sin nx$$

where,

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f(x)dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{0} -\pi dx + \frac{1}{\pi}\int_0^{\pi} e^x dx$$

$$= \frac{1}{\pi}\left[(-\pi x)_{-\pi}^{0} + \left(\frac{x^2}{2}\right)_0^{\pi}\right]$$

$$= \frac{1}{\pi}\left[-\pi^2 + \frac{\pi^2}{2}\right] - \frac{1}{\pi}\left(-\frac{\pi^2}{2}\right)$$

$$a_0 = \frac{-\pi}{2}$$

$$a_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} -\pi\cos nx dx + \int_{0}^{\pi} x\cos nx dx\right]$$

$$= \frac{1}{\pi}\left[\left(\frac{-\pi\sin nx}{n}\right)_{-\pi}^{0} + \left(\frac{x\sin nx}{n} - \left(-\frac{\cos nx}{n^2}\right)\right)_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[\frac{\cos n\pi}{n^2} - \frac{1}{n^2}\right]$$

$$\therefore a_n = \frac{1}{\pi n^2}(\cos n\pi - 1)$$

$$b_n = \frac{1}{\pi}\left[\int_{-\pi}^{0} -\pi\sin nx dx + \int_{0}^{\pi} x\sin nx dx\right]$$

$$= \frac{1}{\pi}\left[\left(\frac{\pi\cos nx}{n}\right)_{-\pi}^{0} + \left(\frac{-\cos nx}{n} - 1\left(\frac{-\sin nx}{n^2}\right)\right)_{0}^{\pi}\right]$$

$$= \frac{1}{\pi}\left[\frac{\pi}{n} - \frac{\pi\cos n\pi}{n} - \frac{\pi\cos\pi}{n}\right]$$

$$b_n = \frac{1}{\pi}(1 - 2\cos n\pi)$$

$$(1) \Rightarrow f(x) = \frac{-\pi}{4} + \sum_{1}^{\infty}\frac{1}{\pi n^2}(\cos n\pi - 1)\cos nx + \sum_{1}^{\infty}\frac{1-2\cos n\pi}{n}.\sin nx$$

$$= \quad \frac{-\pi}{4}+\frac{1}{\pi}\left[-\frac{2\cos x}{1^2}-\frac{2}{3^2}\cos 3x-\frac{2}{5^2}\cos 5x...\right]+$$

$$\left[\frac{3\sin x}{1}-\frac{\sin 2x}{2}+\frac{3\sin 3x}{3}-\frac{\sin 4x}{4}+...\right]$$

$$= \quad \frac{-\pi}{4}-\frac{2}{\pi}\left[\frac{\cos x}{1^2}+\frac{\cos 3x}{3^2}+\frac{\cos 5x}{5^2}+...\right]+$$

$$\left[\frac{3\sin x}{1}-\frac{\sin 2x}{2}+\frac{3\sin 3x}{3}-...\right]$$

put x =0,

$$\text{RHS} \quad = \quad \frac{-\pi}{4}+\frac{2}{\pi}\left[\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+...\right]$$

since x = 0 is a point of discontinuity

$$f(0) \quad = \quad \frac{f(0-0)+f(0+0)}{2}=\frac{-\pi+0}{2} = \frac{-\pi}{2}$$

$$\therefore \frac{-\pi}{2} \quad = \quad \frac{-\pi}{4}-\frac{2}{\pi}\left[\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+...\right]$$

$$\therefore \frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+...\infty=\frac{\pi^2}{8}$$

Hence proved

2.3 Fourier Series ODD and Even Functions

ODD Function

(i) A function f(x) is said to be odd,if f(-x) =-f(x) and we know that for such a function $\int_{-a}^{a} f(x)dx = 0$

If f(x) in $-\pi < x < \pi$ is an odd function of x,

Then,

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx dx = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} f(x)\sin nx dx$$

ஃ If f(x) is an odd function in the interval $-\pi<x<\pi$, then the fourier expansion for f(x) is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \text{ where } b_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\sin nx dx$$

Even Function

(ii) A function f(x) is said to be an even function if $f(-x) = f(x)$.

If f(x) is an even function of x, then $\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$

∴ If f(x) in even function in $-\pi < x < \pi$

Then,

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)dx$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = 0$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where $a_0 = \frac{2}{\pi}\int_{0}^{\pi} f(x)dx$; $a_n = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx$

Simple Problems

Problem 1

Obtain the Fourier series for the function $f(x) = |x|$ in $-\pi < x < \pi$ and deduce $\frac{1}{1^2} + \frac{1}{3^2} + ... = \frac{\pi^2}{8}$

Solution

Given f (x) = |x|

f (-x) = |-x| = |x| = f (x)

(i.e.,) f (x) is even function

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \qquad \text{........... (1)}$$

where

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x\ dx$$

$$= \frac{2}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi} = \pi$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\ dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x\cos nx\ dx$$

$$= \frac{2}{\pi}\left[x\left(\frac{\sin nx}{n}\right) - 1\ \left(\frac{-\cos nx}{n^2}\right)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{\cos n\pi}{n^2} - \frac{1}{n^2}\right]$$

$\therefore a_n \quad = \dfrac{-2}{\pi n^2}\left[1-(-1)^n\right]$

(i) becomes,

$$f(x) = \frac{\pi}{2} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1-(-1)^n}{n^2} \cos nx$$

$$\Rightarrow |x| = \frac{\pi}{2} - \frac{4}{\pi}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\infty\right]$$

put x = 0.

$$0 = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots\infty\right)$$

therefore, $\dfrac{1}{1^2} + \dfrac{1}{3^2} + \dfrac{1}{5^2} + \ldots\ldots = \dfrac{\pi^2}{8}$

Hence Proved

Problem 2

Obtain the Fourier series for the function f(x) = x^2 in the internal $-\pi < x < \pi$ and hence show that

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\ldots\infty = \frac{\pi^2}{6} \text{ and}$$

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \ldots\ldots\infty = \frac{\pi^2}{12}$$

Solution

Given f (x) = x^2

f (-x) = $(-x)^2 = x^2$ = f (x)

$\therefore$ f (x) is an even function

$$\therefore \quad f(x) = \frac{a_0}{2} + \sum a_n \cos nx \qquad \text{............ (1)}$$

where

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x^2\ dx$$

$$= \frac{2}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi}$$

$$a_0 = \frac{2\pi^2}{3}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\ dx$$

$$= \frac{2}{\pi}\int_0^{\pi} x^2 \cos nx\ dx$$

$$= \frac{2}{\pi}\left[\frac{x^2 \sin x}{n} - \frac{2x(-\cos nx)}{n^2} + \frac{2(-\sin nx)}{n^3}\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{2\pi \cos n\pi}{n^2}\right]$$

$$a_n = \frac{4}{n^2}\ (-1)^n$$

(i) becomes,

$$f(x) = \frac{a_0}{2} + \sum_{1}^{\infty} a_n \cos nx$$

$$= \frac{\pi^2}{3} + \sum_{1}^{\infty} \frac{4}{n^2}(-1)^n.\cos nx$$

$$x^2 = \frac{\pi^2}{3} - 4\left[\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} \ldots\right] \qquad (2)$$

put x = π in (2)

$$(2) \Rightarrow \pi^2 = \frac{\pi^2}{3} - 4\left[-\frac{1}{1^2} - \frac{1}{2^2} - \frac{1}{3^2} - \ldots\right]$$

$$\frac{2\pi^2}{3} = 4\left[\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots..\infty\right]$$

$$\therefore \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\infty = \frac{\pi^2}{6}$$

put x = $-\pi$

$$(2) => \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\infty = \frac{\pi^2}{6}$$

At x=π, the value of the function is $\frac{1}{2}$ [f(π-0)+(π+0)]

put x = 0,in(2)

$$(2) \Rightarrow 0 = \frac{\pi^2}{3} - 4\left[-\frac{1}{1^2} - \frac{1}{2^2} - \frac{1}{3^2} + \ldots\right]$$

$\therefore \quad \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \ldots\infty = \frac{\pi^2}{12}$

Hence Proved

Problem 3

Find the fourier series of the function

$$f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$$

Solution

Given $\quad f(x) \quad = \quad \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$

$f(-x) \quad = \quad \begin{cases} -k, & -\pi < -x < 0 \quad \text{(i.e.,) } 0 < x < \pi \\ k, & 0 < -x < \pi \quad \text{(i.e.,) } -\pi < x < 0 \end{cases}$

$\therefore \quad f(x) = -f(x)$

$\therefore \quad f(x)$ is an odd function

$$f(x) = \sum b_n \sin nx \qquad \ldots\ldots\ldots\ldots (1)$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} b_n \sin nx dx$$

$$= \frac{2}{\pi}\int_0^{\pi} k \sin nx dx$$

$$= \frac{2k}{\pi}\left[-\frac{\cos nx}{n}\right]_0^{\pi}$$

$\therefore b_n \quad = \quad \frac{2k}{\pi n}[1-\cos nx] = \frac{-2k}{\pi}\left(\frac{1-(-1)^n}{n}\right)$

(1) becomes

$$(1) \Rightarrow f(x) = \frac{2k}{\pi}\sum_{n \text{ is odd}}\left(\frac{1-(-1)^n}{n}\right)\sin nx$$

$$= \frac{4k}{\pi}\sum_{n \text{ is odd}} \frac{\sin nx}{n}$$

$$\therefore f(x) = \frac{4k}{\pi}\sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1}$$

Problem 4

Show that for all values of x on $(-\pi,\pi)$ $\frac{x}{2} = \sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \frac{\sin 4x}{4} + \ldots\infty$

Proof

Given $f(x) = \frac{x}{2}$; (f(x) = - f(x) is on odd function)

$\therefore$ f(x) is an odd function,

$$f(x) = \sum_{n=0}^{0} b_n \sin nx$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

$$= \frac{2}{\pi}\int_0^{\pi} \frac{x}{2}\sin nx\,dx$$

$$= \frac{1}{\pi}\left[\frac{x(\cos nx)}{n} - 1\left(\frac{-\sin nx}{n^2}\right)\right]_0^{\pi}$$

$$= \frac{1}{\pi}\left[\frac{-\pi\cos n\pi}{n}\right]$$

$$\therefore b_n = \frac{-(-1)^n}{n}$$

$$(1) \Rightarrow f(x) = -\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin nx$$

$$\text{(i.e.,)}\ \frac{x}{2} = \sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} \ldots\infty$$

Hence proved.

Problem 5

Obtain the fourier series for the functionf(x)givenby

$$f(x) = \begin{cases} -x+1, & -\pi < x < 0 \\ x+1, & 0 < x < \pi \end{cases} \text{ Hence prove that}$$

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$$

Solution

$$\text{Given } f(x) = \begin{cases} -x+1, & -\pi < x < 0 \\ x+1, & 0 < x < \pi \end{cases}$$

$$f(-x) = \begin{cases} -x+1, & 0 < -x = \pi \\ -x+1, & -\pi < x < 0 \end{cases}$$

$$\therefore \quad f(x) = \frac{a_0}{2} + \sum a_n \cos nx \qquad \ldots\ldots\ldots\ldots (1)$$

where

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx$$

$$= \frac{2}{\pi}\int_{-\pi}^{\pi} (x+1)dx$$

$$= \frac{2}{\pi}\left[\frac{x^2}{2}+x\right]_0^{\pi} = \pi+2$$

$$a_n = \frac{2}{\pi}\int_{-\pi}^{\pi} (x+1)\cos nx dx$$

$$= \frac{2}{\pi}\left[(x+1)\frac{\sin nx}{n}+\frac{\cos nx}{n^2}\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[+\frac{(-1)^n-1}{n^2}\right]$$

$$a_n = \left[\frac{2[(-1)^n-1]}{n^2.\pi}\right]$$

$$(1) \Rightarrow f(x) = \frac{\pi+2}{2}+\frac{2}{\pi}\sum\left(\frac{(-1)^n-1}{n^2}\right)\cos nx$$

$$= \frac{\pi+2}{2}-\frac{4}{\pi}\left[\frac{\cos x}{1^2}+\frac{\cos 3x}{3^2}+\frac{\cos 5x}{5^2}+\ldots\right]$$

put x=0

$$\Rightarrow f(0) = \frac{f(0-0)+(0+0)}{2} = \frac{1+1}{2} = 1$$

$$1 = \frac{\pi}{2} + 1 - \frac{4}{\pi}\left[\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\right]$$

$$\frac{\pi}{2} = \frac{4}{\pi}\left[\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\right]$$

$$\therefore \qquad \frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\infty=\frac{\pi^2}{8}$$

Hence proved

2.4 Recap

1. Fourier Series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

where

$$a_0 = \frac{1}{\pi}\int_0^{e+2\pi} f(x)dx$$

$$a_n = \frac{1}{\pi}\int_0^{e+2\pi} f(x)\cos nx dx$$

$$b_n = \frac{1}{\pi}\int_0^{e+2\pi} f(x)\sin nx dx$$

2. If $x=x_0$ is a point at discontinuity, then the value of f(x) at $x = x_0$ is $\frac{f(x_0+0)+(x_0-0)}{2}$.

3. A Fouriers Series is an odd function in the interval $-\pi<x<\pi$, i if $f(-x) = -f(x)$ then, $f(x) \sum_{n=1}^{\infty} b_n \sin nx$.

Where $b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx dx$.

4. A Fourier Series is said to be an even function if f(-x)= f(x) then f(x) = $\frac{a_0}{2}+\sum_{n=1}^{\infty} a_n \cos nx$

where $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx$, $a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx dx$.

2.5 Exercises

1. Find the fourier series for f(x) = $\frac{1}{2}(\pi - x)$ in $-\pi<x<\pi$.

2. Obtain the fourier series for f(x) = $x\sin x$ in $0<x<2\pi$.

3. Find the Fourier series for the function

$$f(x) = \begin{cases} 0 & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$$

4. Find a Fourier series for the function f(x) = $x+x^2$ in the interval $-\pi<x<\pi$.

5. Obtain a fourier series for f(x) = e^x in $-\pi<x<\pi$.

6. Determine fourier expansion for f(x) = $\begin{cases} -\pi & 0 < x < \pi \\ x-\pi & \pi < x < 0 \end{cases}$ and show that $\sum_{n=1}^{\infty} \frac{1}{(2r+1)^2} = \frac{\pi^2}{8}$.

7. Obtain the fourier series for the function f(x) = x^2, in $-\pi \le x \le \pi$ and hence derive the results.

(i) $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\infty=\frac{\pi^2}{6}$

(ii) $\frac{1}{1^2}-\frac{1}{2^2}+\frac{1}{3^2}+\ldots\infty=\frac{\pi^2}{12}$

(iii) $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots\infty=\frac{\pi^2}{8}$

8. If f(x) = $\frac{1}{\pi}$ x in $-\pi < x < \pi$ show that

$$f(x) = \frac{2}{\pi}\left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} \ldots \infty\right)$$

9. Show that the fourier series for $f(x) = x^2$, in $-\pi < x < \pi$ is given by

$$f(x) = \pi^2 - \left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} \ldots \infty\right)$$

10. Show that the fourier series for f(x) =x in $-\pi < x < \pi$ is given by

$$f(x) = 2\left(\sin x - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} \ldots \infty\right)$$

LESSON - 3
HALF RANGE FOURIER SERIES

Learning Objectives

After reading this lesson you will be able to

- ❖ understand the concept of Half range fourier series
- ❖ Find the problem for fourier cosine and sine series
- ❖ solve the problems for half range fourier series

Structure

3.1 Introduction

3.2 Objectives

3.3 Half range fourier of cosine and sine series

3.4 Solved Problems

3.5 Recap

3.6 Exercise

3.1 Introduction

If a function is defined over the range 0 to π, instead of 0 to 2π or $-\pi$ to π, it may be expanded in a series of sine terms only or of cosine terms only. This is particularly useful in the solution of some partial differential equations whose boundary conditions may restrict us to a series which contains sine terms only or cosine terms only. The series produced is called a half range Fourier series. In such cases, we may have to construct a function in the full range 0 to 2π, or $-\pi$ to π in such a way that the Fourier expansion for this series is valid in half range.

3.3 Half range fourier of Cosine and Sine series

Suppose we are given a function f(x) defined in the interval $(0, \pi)$. Here we extend the function in the interval $(-\pi, 0)$ and then find the Fourier coefficients in the interval $(-\pi, \pi)$. Such an extended function should converge to given function f(x) between 0 to π also. For such an extension here we consider two types of function.

I. Suppose f(x) is defined in the interval $(0, \pi)$. We define a extension of f(x) as follows and call the new function as F(x).

Let $F(x) = f(x), \quad 0 < x < \pi$

and $F(x) = f(-x) \quad -\pi < x < \pi$

Clearly F(x) is an even function in the interval $-\pi < x < \pi$ and it is identified with f(x) in $0 < x < \pi$.

The Fourier series for F(x) is

$$F(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x) \cos nx dx$$

Since $F(x) = f(x)$ in $0 < x < \pi$, the function f(x) in $0 < x < \pi$ is given by the expansion.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

II. Suppose f(x) is defined in the interval $0 < x < \pi$. Then we define an extension of f(x) as follows :

$F(x) = f(x)$ in $0 < x < \pi$.

$= -F(-x)$ in $-\pi < x < \pi$

Clearly F(x) is an odd function in the interval $-\pi < x < \pi$ and it is identical with f(x) in $0 < x < \pi$.

The Fourier series for the odd function F(x) is

$$F(x) = \sum_{1}^{\infty} b_n \sin nx$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

Since F(x) = f(x), in $0 < x < \pi$, the Fourier series for f(x) in $0 < x < \pi$ is

$$f(x) = \sum_{1}^{\infty} b_n \sin nx$$

where $b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin x\,dx$

III) A Fourier cosine series of a function f(x) in $0 < x < \pi$ is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx \text{ and } a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

iv) A Fourier sine series of f(x) in $0 < x < \pi$ is given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx.$$

3.4 Solved Problems

Problem 1

Obtain the Fourier series for the function f(x) = x in $0 < x < \pi$ and deduce the sum of the series

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\infty$$

Solution

Given f (x) = x, in $0 < x < \pi$

we note that, the given series for f(x) is a cosine series for f(x) in the half range $(0,\pi)$

∴ Let us now find the half range cosine series for f(x)

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x\,dx$$

$$= \frac{2}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi} = \pi$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x\cos nx\, dx$$

$$= \frac{2}{\pi}\left[\frac{x\sin nx}{n} - \left(\frac{-\cos nx}{n^2}\right)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{\cos n\pi - 1}{n^2}\right]$$

$$= \frac{-2}{\pi}\left[\frac{1-(-1)^n}{n^2}\right]$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$$= \frac{\pi}{2} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^2}\cos nx$$

$$f(x) = \frac{\pi}{2} - \frac{2}{\pi}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\infty\right]$$

Since 0 is a point of discontinuity

$$f(x) = \frac{f(0-0)+f(0+0)}{2} - \frac{(0+0)}{2} = 0 \qquad [\because f(x)=x^2]$$

$$0 = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\ldots\right)$$

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots.$$

Hence Proved

Problem 2

Obtain a fourier sinie series for unity in $0 < x < \pi$.

Solution

Given $f(x) = 1$; $0 < x < \pi$.

The half range sine series for f(x) is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} 1.\sin nx\,dx$$

$$= \frac{2}{\pi}\left[-\frac{\cos nx}{n}\right]_0^{\pi}$$

$$b_n = \frac{2}{\pi n}[1-(-1)^n]$$

$$f(x) = \frac{2}{\pi}\sum_{n=1}^{\infty} [1-(-1)^n]\,\frac{\text{siinnx}}{n}$$

Therefore, $f(x) = \frac{4}{\pi}\left[\frac{\sin x}{1}+\frac{\sin 3x}{3}+\frac{\sin 5x}{5}+\ldots\right]$

Problem 3

If f(x) = $\begin{cases} kx, & 0 \le x \le \frac{\pi}{2} \\ k(\pi - x), & \frac{\pi}{2} \le x \le \pi \end{cases}$ show that

f(x) = $\frac{4k}{\pi}\left[\frac{\sin x}{1^2} - \frac{\sin 3x}{3^2} + \frac{\sin 5x}{5^2} - \ldots\right]$ and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi}{8}$.

Solution

We have to the half range sine series for f(x),

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

Where

$$b_n = \frac{2}{\pi}\int_0^{\pi/2} f(x) \text{ sininx } dx + \frac{2}{\pi}\int_0^{\pi/2} k(\pi\text{-}x).\text{ sininx } dx$$

$$= \frac{2k}{\pi}\left[\frac{x(-\cos nx)}{x} - \frac{1(-\sin nx)}{n^2}\right]_0^{\pi/2} +$$

$$\frac{2k}{\pi}\left[(\pi - x)\left(\frac{-\cos nx}{n}\right) - (-1)\left(\frac{-\sin nx}{n^2}\right)\right]_{\pi/2}^{\pi}$$

$$= \frac{2k}{\pi}\left[\frac{\sin n\pi/2}{n^2} + \frac{\sin n\pi/2}{n^2}\right]$$

$$= \frac{2k}{\pi}\left[\frac{2\sin n\pi/2}{n^2}\right]$$

$$\therefore b_n = \frac{4k}{\pi}\left[\frac{\sin n\pi/2}{n^2}\right]$$

$$b_1 = \frac{4k}{\pi}\cdot\frac{1}{1^2} \quad ; \quad b_2 = 0$$

$$b_2 = \frac{4k}{\pi}\cdot\left(\frac{-1}{3^2}\right) \quad ; \quad b_4 = 0$$

$$b_5 = \frac{4k}{\pi}\cdot\frac{1}{5^2} \quad ; \quad b_6 = 0 \quad \text{etc}$$

$$\therefore f(x) = \frac{4k}{\pi}\left[\frac{\sin x}{1^2} - \frac{\sin 3x}{3^2} + \frac{\text{siin}5x}{5^2} - \ldots\right]$$

put $x = \frac{\pi}{2}$,

$$\Rightarrow \quad f\left(\frac{\pi}{2}\right) = k\frac{\pi}{2}$$

$$\Rightarrow \quad \frac{k\pi}{2} = \frac{4k}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\infty\right]$$

$$\therefore \quad \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$$

Hence proved

Problem 4

Find the fourier cosine series for $f(x) = x(\pi - x)$, $0 \leq x \leq \pi$.

Solution

$f(x) = x(\pi - x)$, $0 \leq x \leq \pi$.

we have to find a half range cosine series of f(x)

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx \qquad (1)$$

Where

$$a_0 = \frac{2}{\pi}\int_0^{\pi} x(\pi\text{-}x)dx = \frac{2}{\pi}\int_0^{\pi} (\pi x\text{-}x^2dx$$

$$= \frac{2}{\pi}\left[\frac{\pi x^2}{2} - \frac{x^3}{3}\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{\pi^3}{2} - \frac{x^3}{3}\right]$$

$$= \frac{2}{\pi}\frac{\pi^3}{6}$$

$$\boxed{a_0 = \frac{\pi^2}{3}}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} (\pi x\text{-}x^2)\cos nx dx$$

$$= \frac{2}{\pi}\left[(\pi x - x^2).\frac{\sin nx}{n} - (\pi - 2x)\left(\frac{-\cos nx}{n^2}\right) + (-2)\left(\frac{-\sin nx}{n^3}\right)\right]_0^{\pi}$$

$$= \frac{2}{\pi}\left[\frac{-\pi\cos n\pi}{n^2} - \frac{\pi}{n^2}\right]$$

$$\therefore a_n = -2\left[\frac{1+\cos n\pi}{n^2}\right]$$

putting n = 0,1,2...

$a_1 = 0 \quad ; a_2 = \frac{-4}{2^2}$

$a_3 = 0 \quad ; a_4 = \frac{-4}{4^2}$

$a_5 = 0 \quad ; a_6 = \frac{-4}{6^2}$ etc

$\therefore f(x) = \frac{\pi^2}{6} - 4\left[\frac{\cos 2x}{2^2} + \frac{\cos 4x}{4^2} + \frac{\cos 6x}{6^2} + ...\infty\right]$

$$\therefore \pi x - x^2 = \frac{\pi^2}{6} - 4\left[\frac{\cos 2x}{2^2} + \frac{\cos 4x}{4^2} + \frac{\cos 6x}{6^2} + ...\infty\right] \qquad (2)$$

Note (1) :

put $x = \frac{\pi}{2}$ in (2)

$$\frac{\pi}{2} \cdot \frac{\pi^2}{4} = \frac{\pi^2}{6} - 4\left(-\frac{1}{2^2} + \frac{1}{4^2} - \frac{1}{6^2} + ...\right)$$

$$\frac{\pi}{2} \cdot \frac{\pi^2}{4} - \frac{\pi^2}{6} = 4\left(\frac{1}{2^2.1^2} - \frac{1}{2^2.2^2} + \frac{1}{2^2.3^2} + ...\infty\right)$$

(i.e.,) $\quad \frac{\pi^2}{1^2} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + ...\infty$

Problem 5

If f(x) $= x$ for $0 < x < \frac{\pi}{2}$

$= \pi - x$ for $\frac{\pi}{2} < x < \pi$

show that $f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left(\frac{\cos 2x}{1^2} + \frac{\cos 6x}{2^2} + \frac{\cos 10x}{5^2} + ...\right)$

Solution

We note that the given series for f(x) is a cosine series for f(x) in the half range $(0,\pi)$.

Let us now find the half range cosine series for f(x)

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx$$

$$= \frac{2}{\pi}\int_0^{\pi} xdx + \frac{2}{\pi}\int_{\pi/2}^{\pi} (\pi-x)dx$$

$$= \frac{2}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi/2} + \frac{2}{\pi}\left[\pi x - \frac{x^2}{2}\right]_{\pi/2}^{\pi}$$

$$= \frac{2}{\pi}\cdot\frac{\pi^2}{8} + \frac{2}{\pi}\left[\left(\pi^2 - \frac{\pi^2}{2}\right) - \left(\frac{\pi^2}{2} - \frac{\pi^2}{8}\right)\right]$$

$$= \frac{\pi}{4} + \pi - \frac{3\pi}{4}$$

$$a_0 = \frac{\pi}{2}$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x\cos nx dx + \frac{2}{\pi}\int_{\pi/2}^{\pi} (\pi-x)\cos nx dx$$

$$= \frac{2}{\pi}\left[\frac{x\sin nx}{n} - \frac{1(-\cos nx)}{n^2}\right]_0^{\pi/2} +$$

$$\frac{2}{\pi}\left[(\pi - x)\frac{\sin nx}{n} - (-1)\left(\frac{-\cos nx}{n^2}\right)\right]$$

$$\therefore a_n = \frac{2}{\pi}\left[\frac{2\cos n\pi/2}{n^2} - \frac{1+\cos n\pi}{n^2}\right]$$

Also, a_1 =0, a_3 = 0, a_5 = 0 ...

$$a_3 = \frac{2}{\pi}\left[-\frac{4}{2^2}\right] = \frac{2}{\pi}\left(\frac{-1}{1^2}\right)$$

$$a_4 = \frac{2}{4^2} - \frac{2}{4^2} = 0$$

$$a_6 = \frac{2}{\pi}\left[-\frac{2}{6^2} - \frac{2}{6^2}\right] = \frac{2}{\pi}\left(-\frac{4}{2^2.3^2}\right) = \frac{-2}{\pi}.\frac{1}{3^2}$$

$$a_{10} = \frac{2}{\pi}\left[-\frac{2}{10^2} - \frac{2}{10^2}\right] = \frac{-2}{\pi}\left(\frac{1}{5^2}\right)$$

$$\therefore f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left[\frac{\cos 2x}{1^2} + \frac{\cos 6x}{3^2} + \frac{\cos 10x}{5^2} + \ldots\infty\right]$$

Check Your Progress

1. Find the half range cosine series for the function f (x) = x^2 is the range $0 \leq x \leq \pi$. and Lorce find $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots$

....

2. Obtain the cosine series of f (x) = $\begin{cases} \frac{\pi}{2}x & \text{in } 0 \leq x < \frac{\pi}{2} \\ \frac{\pi}{2}(\pi - x) & \text{in } \frac{\pi}{2} \leq x < \pi \end{cases}$

....

3. Obtain the cosine series of f (x) = x sin x, $0 < x < \pi$

....

3.5 Recap

1. Fourier cosine series

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where

$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)dx$$

$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx dx$$

2. Fourier sine series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx dx$$

3.6 Exercise

1. Find the fourier sine series of $\frac{\pi x}{8}(\pi - x)$ is $(0,\pi)$.
2. Fiind the fourier sine series of $f(x) = x$ in$(0,\pi)$.
3. Find the fourier cosine series of $f(x) = x^2, 0 < x < \pi$.
4. Find the fourier cosine series of $f(x) = \sin x,, 0 < x < \pi$.
5. Obtain a cosine series for $f(x)=e^x$,in $0 < x < \pi$.
6. Obtain a sine series for $f(x) = \sin x,\ 0 \le x \le \frac{\pi}{4}$.

   $= \cos x,\ \frac{\pi}{4} \le x \le \frac{\pi}{2}$
7. Obtain a half range sine series for $f(x) = \pi x - x^2$ in the range $(0,\pi)$
8. Find the fourier sine series for $f(x) = \cos x,\ 0 < x < \pi$.
9. Obtain the cosine series for $f(x) = \begin{cases} \cos x & 0 < x < \pi/2 \\ 0 & \pi/2 < x < \pi \end{cases}$

LESSON - 4

DIFFERENTIAL EQUATIONS - ORDINARY DIFERENTIAL EQUATIONS

Learning Objectives

After reading this lesson you will be able to

- ❖ know the complementary function
- ❖ know the special type of particular integral
- ❖ known how to solve the simple problems of Second order Differential Equation

Structure

4.1 Introduction

Students are already familiar with formation and solution of some types of linear differential equations of second and first order with constant co-efficients. Before we take up the study of remaining types of such equations, we shall recall the various notations and working rules resulted to the solutions of such equation. This will be of use to purusue the study of not only the remaining types of IDE with constant Coefficients, but also linear differential equations with variable co-efficiants and simultaneous differential equations.

4.2 Ordinary Differential Equtaion

Definition

Differential equations which involve only one independent variable are called ordinary Differential Equation.

For Example

1. $5\frac{d^2y}{dx}+6\frac{dy}{dx^2}+3y=0$

2. $x^3\frac{d^3y}{dx^3}+2x^2\frac{d^2y}{dx^2}+3x\frac{dy}{dx}-3y=x^2+X$

Order

The order of the hightest Differential Coefficient that occur in a differential equation is called the order of the D.E.

Degree

The degree of an equtaion is the degree of the hightest differential coefficient which occurs in it after the equtaion has been made rational and integral.

For Example

$\left(\frac{d^3y}{dx^2}\right)^5+\left(\frac{d^3y}{dx^3}\right)+\frac{dy}{dx}-4y=x^2\sin x$ Order is 3 Degree is 2.

4.2.1 Second order Differential Equations with Constant Coefficients

A second order linear differential equation is an equation that can be written in the form

$$a_2(x)\frac{d^2y}{dx^2}+a_1(x)\frac{dy}{dx}+a_0(x)\,y=b(x) \qquad (1)$$

Here $a_2(x)$, $a_1(x)$, $a_0(x)$ and $b(x)$ are continuous functions of x on an interval D. When a_0, a_1, a_2 are constants, we say that the equation has constant co-efficients; otherwise it has variable co-efficients.

Now we are interested in the linear equation, where $a_2(x) \neq 0$.

In that case, we can write the above equation in the form

$$\frac{d^2y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = g(x) \qquad (2)$$

where $p(x) = \dfrac{a_1(x)}{a_2(x)}$ and $q(x) = \dfrac{a_0(x)}{a_2(x)}$

Associated with each equation (2) is of the form,

$$\frac{d^2y}{dx^2} + p(x) \frac{dy}{dx} + q(x)y = 0 \qquad (3)$$

where is obtained from (2) by replacing g(x) by zero. We say that (2) is a non-homogeneous equation and (3) is the corresponding homogeneous equation.

The general form of linear differential equation of second order with constant coefficient is

$$a_2 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = (x),$$

where a_0, a_1,a_2 are constant and f(x) is a function of x (4)

is called a linear differential equaiton of order two. If we use the differential operator D for $\dfrac{d}{dx}$

So that $Dy = \dfrac{dy}{dx}$

The above equation can be written as

$\therefore \phi(D)(y) = F(x)$

For example $\dfrac{d^2y}{dx^2} - 3\dfrac{dy}{dx} + 2y = \sin x$

This is equation of the form $\phi(D)y = F(x)$

where $\phi(D) = D^2 - 3D + 2$ and $F(x) = \sin x$

Theorem 4.3.1

If y_1 and y_2 are solutions of $\phi(D) = 0$ then $c_1y_1+c_2y_2$ is also a solution.

Proof

Since y_1 and y_2 are solutions of $\phi(D)y = 0$

$$\phi(D)y_1 = 0 \qquad (1)$$

$$\phi(D)y_2 = 0 \qquad (2)$$

Multiplying (1) by c_1 and (2) by c_2 we get

$\phi(D)\,(c_1y_1) = 0$

$\phi(D)\,(c_2y_2) = 0$ since $\phi(D)$ is a linear operator.

Also $\phi(D)\,(c_1y_1+c_2y_2) = 0$

$\therefore y = c_1y_1+c_2y_2$ is also a solution of $\phi(D)y = 0$

This result can be extended to n solutions $y_1, y_2 \ldots y_n$

i.e. If $y_1, y_2 \ldots y_n$ are the solutions of $\phi(D)y = 0$

Then

$y = c_1y_1 + c_2y_2 + \ldots + c_ny_n$ is also solution of $\phi(D)y = 0$

Note

This equation has n arbitrary constants and hence the equation $(D-m_1), (D-m_2) \ldots (D-m_n)\, y = 0$ has solution

$$y = c_1e^{m_1x} + c_2e^{m_2x} + \ldots + c_ne^{m_nx}$$

4.3 Solution of Second Order non-homogenous equation with constant coefficients

Let us now discuss the solutions depending upon the nature of the roots of the second order linear differential equation.

Step (1)

Consider the linear differential equation

$$a_2 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_0y = F(x) \qquad (1)$$

Step (2)

Write the given equation (1) in terms of the operator $D = \frac{d}{dx}$

$$(a_2D^2 + a_1D + a_0)\, y = F(x)$$

Step (3)

Find the complementary function and particular integral.

Step (4)

Whenever $F(x) = 0$, the solution of the D.E. (1) is y = C.F and whenever F(x) is the solution of the D.E. of (1) is y = C.F. +P.I.

Case i

To find complementary function If the auxillary equation has real and distinct root, say m_1 and m_2.

Then the solution of equation (1) is

$y = Ae^{m_1 x} + Be^{m_2 x}$

Case ii

Suppose the roots are real and equal say $m_1 = m_2 = m$

The solution of equation (1) is $y = (Ax+B)e^{mx}$

Case iii

Suppose the roots are complex say, $m = \alpha \pm i\beta$

(i e) $m_1 = \alpha+i\beta$ and $m_2 = \alpha-i\beta$

The solution of equation (1) is

$y = e^{\alpha x}(A\cos\beta x + B\sin\beta x)$

Theorem 4.4.1

If $y = u(x)$ is the solution of $\phi(D)y = F(x)$ and $y = v(x)$ is the solution of $\phi(D)y = 0$.

Then the complete solution of $\phi(D)y = F(x)$ is $y=u(x) + v(x)$

Proof

Since $y=u(x)$ is solution of $\phi(D)y = F(x)$

we have $\phi(D)u(x) = F(x)$ (1)

∴ y=v(x) is the solution of $\phi(D)y = 0$

$$\Rightarrow \phi(D)\, v(x) = 0 \qquad (2)$$

Adding (1) and (2),

$$\phi(D)u(x) + \phi(D)v(x) = F(x)$$

$$\phi(D)\,[u+v] = F(x)$$

Therefore $y=u(x) + v(x)$ is a solution of $\phi(D)y = F(x)$.

To find Particular Integral

If $y = u(x)$ has no arbitrary constant, it is called a particular solution of $\phi(D)y = F(x)$.

Then $y=u(x) + v(x)$ will also contain 'n' arbitrary constants and is called the complete solution or general solution of $\phi(D)(y) = F(x)$. $u(x)$ is called complementary function of the given Differential Equtaion and $v(x)$ is called the particular integral of the given DE.

4.3.1 Some for special types of Particular Integral

Let us look into some short cut methods of evaluating particular integrals where the function F(x) on the RHS is of special type. We will discuss the following types of special functions which appear on the RHS of the linear DE.

Let us now discuss these cases in evaluation P.I :

Type 1

Suppose $F(x) = e^{ax}$, where a is constant

We have

$$\text{P.I} = \frac{1}{\phi(D)}.e^{ax} \text{ (Replace D by a)}$$

$$= \frac{1}{\phi(a)} \cdot e^{ax},\ \phi(a) \neq 0$$

Type 2

Suppose F(x) = sinax or cos ax, where a is constant

In this case, the following rules are used.

$$P.I = \frac{1}{\phi(D^2)} \text{ sinax or cos ax}$$

(Replacing D^2 by $-a^2$)

(Here D = a)

$$= \frac{1}{\phi(-a^2)} \text{. sinax or cos ax} \quad (\text{if } \varphi(-a^2) \neq 0)$$

Note :

When finding P.I the above rules are to be applied in parts, as f(D) will not be in general of the form $\phi(D)^2$. This means that, we have to first replace D^2 by a^2. D^3 by a^2D, D^4 by a^4, etc.,

Type 3

Suppose $f(x) = x^m$, where m is a positive integral

$$\text{Here} \quad P.I = \frac{1}{\phi(D)}.x^m$$

$$= [\phi(D)]^{-1} x^m$$

We can expand $[\varphi(D)]^{-1}$ in ascending powers of D as for as the term containing D^m and operate on x^m term by term.

Here it will be useful to remember the following binomial Theorem.

$(1-D)^{-1} = (1+D+D^2+ \dots \infty)$

$(1+D)^{-1} = (1-D+D^2-D^3 + \dots \infty)$

$(1-D)^{-2} = (1+2D + 3D^2 + \dots \infty)$

$(1+D)^{-2} = (1-2D+3D^2 - \dots \infty)$

Type 4

Suppose $F(x) = e^{ax}$ sinax or e^{ax} cos ax

Here we have

$$P.I = \frac{1}{\phi(D)} e^{ax} \sin ax \text{ or } e^{ax} \cos ax$$

$$= e^{ax} \frac{1}{\phi(D+a)} \quad \sin ax \text{ or } \cos ax$$

RHS can be evaluated using the type -2 we get the required P.I

Note : This rule is referred to as exponential shift rule.

Type 5

Suppose F(x) = xv, where v is any function of x. (Here u(x) = sinax or cosax)

we have,

$$P.I = \left[x - \frac{\phi'(D)}{\phi(D)}\right] \cdot \frac{v}{\phi(D)}$$

(v is any function of x)

Example 1

Solve $(D^2–4D+3)\ y = 0$

Solution

The auxiliary equation is

$m^2 – 4m + 3 = 0$

i.e. $(m–1)\ (m–3)\ y = 0$

$\therefore$ the roots are 1,3,

$\therefore$ the general solution is $y = Ae^x + Be^{3x}$

Example 2

Solve $(D^2–6D+9)\ y= 0$

Solution

The auxiliary equation is $m^2-6m+9=0$

$(m-3)\ (m-3) = 0$

$\therefore$ The roots are $m = 3,3$ (∴ roots are equals)

$\therefore$ The general solution is $y = e^{3x}\ (Ax+B)$

Example 3

Solve $(D^2+D+1)y+0$

Solution

The auxiliary equation is $m^2 + m + 1= 0$

the roots are m $= \frac{-1 \pm i\sqrt{3}}{2}$, Here $\alpha = \frac{-1}{2}$, $\beta = \frac{\sqrt{3}}{2}$

$\therefore$ the solution is y = $e^{-\frac{1}{2}x}\left[A\cos\frac{\sqrt{3}}{2}x + B\cos\frac{\sqrt{3}}{2}x\right]$

4.5 Simple Problems

Problem 1

Solve $(D^2-3D+2)\,y = e^{5x}$

Solution

The auxiliary equation is $m^2-3m+2=0$

$(m-1)(m-2)=0$

$m = 1,2.$

The roots are distinct $(m_1 \neq m_2)$, $m_1 = 1$, $m_2 = 2$.

The complementary function is $y = Ae^{m_1x} + Be^{m_2x}$

$y = Ae^{x} + Be^{2x}$

Particular Integral :

$$PI = \frac{1}{D^2-3D+2}e^{5x} \text{ (Replace D by a, Here a = 5)}$$

$$= \frac{1}{25-15+2}e^{5x}$$

$$= \frac{e^{5x}}{12}$$

∴ The General Solution is $y = C.I + P.I$

$$y = Ae^{x} + Be^{2x} + \frac{e^{5x}}{2}$$

Problem 2

Solve $(D^2 – D – 2)\, y = e^{2x} + e^{x}$

Solution

The auxiliary equation is $m^2 – m – 2 = 0$

$$(m+2)\,(m+1) = 0$$

$$m = 2, –1$$

∴ The complementary function is $y = Ae^{m_1 x} + Be^{m_2 x}$

$$y = Ae^{2x} + Be^{-x}$$

$$PI = \frac{1}{D^2 - D - 2}\, e^{2x} + \frac{1}{D^2 - D - 2} e^{x}$$

$$= \frac{1}{4 - 2 - 2} e^{2x} + \frac{1}{1 - 1 - 2} e^{x}$$

$$= \frac{x}{2D - 1} e^{2x} + \frac{1}{-2} e^{x}$$

$$PI = \frac{x}{2(2) - 1}\, e^{2x} - \frac{1}{2} e^{x}$$

∴ the complete solution is $y = C.I + P.I$

$$y = Ae^{2x} + Be^{x} + \frac{1}{3}\, xe^{2x} - \frac{1}{2} e^{x}$$

Problem 3

Solve $(D^2-6D+9)\,y=e^{3x}$

Solution

The auxiliary equation is $m^2 - 6m + 9 = 0$

$(m-3)^2 = 0$

$m = 3,3.$

The complementary function is

$y = (Ax+B)\, e^{mx}$

$y = (Ax+B)\, e^{3x}$

Particular Integral

$$P.I = \frac{1}{(D^2-6D+9)}\, e^{3x}, \text{ (failure case)}$$

$$= \frac{1}{(D-3)^2}\, e^{3x}$$

$$= \frac{x}{2D-6}\, e^{3x} \text{ (Again failure case Diffy. } \phi(D)\text{)}$$

$$= \frac{x^2}{2}\, e^{3x}$$

$$P.I = \frac{x^2}{2}\, e^{3x}$$

$\therefore$ the complete solution is

$$y = (Ax+B)\, e^{3x} + \frac{x^2}{2}\, e^{3x}$$

Problem 4

Solve $(D^2+D-2)y = \sin 2x$

Solution

the auxiliary equation is $m^2+m-2 = 0$

$(m+2)(m-1) = 0$

$m = -2, 1$ (The roots are distinct)

the complementary function is

$y = Ae^x + Be^{-2x}$

$$\text{PI} = \frac{1}{D^2+D-2}\sin 2x$$

$$= \frac{1}{-4+D-2}\sin 2x$$

$$= \frac{1}{D-6}\sin 2x$$

$$= \frac{D+6}{(D-6)(D+6)}\sin 2x$$

$$= \frac{(D+6)\sin 2x}{D^2-6^2}$$

$$= \frac{(D+6)}{D^2-36}\sin 2x$$

$$= \frac{(D+6)\sin 2x}{2^2-36}$$

$$= \frac{-1}{40}(D\sin 2x + 6\sin 2x)$$

$$= \frac{-1}{40}(2\cos2x + 6\sin2x)$$

$$\text{P.I} = \frac{-1}{20}(\cos2x + 3\sin2x)$$

$\therefore$ the general solution is

$$y = Ae^{x} + Be^{-2x} - \frac{1}{20}(\cos2x + 3\sin2x)$$

Problem 5

Solve $(D^2–4D–12)\, y = \sin x \sin2x$

Solution

$$(D^2–4D–12)y = \frac{1}{2}[\cos x - \cos3x]$$

we know that,

$[\cos(A+B) - \cos(A-B) = -2\sin A \sin B]$

The auxiliary equation is $m^2–4m–12 = 0$

$(m–6)(m+2) = 0$

$m = 6, –2$

The complementary function is $y = Ae^{6x} + Be^{-2x}$

$$\text{PI} = \frac{1}{D^2 - 4D - 12} \cdot \frac{1}{2}(\cos3x\text{-}\cos3x)$$

$$= \frac{1}{(D^2 - 4D - 12)} \cdot \frac{1}{2}\cos x - \frac{1}{D^2 - 4D - 12} \cdot \frac{1}{2}\cos3x$$

$$= \frac{1}{-(4D+11)} \cdot \frac{1}{2}\cos x + \frac{4D-21}{16D^2-441} \cdot \frac{1}{2}\cos x$$

$$= \frac{-(4D-11)}{\left(16D^2-121\right)} \cdot \frac{1}{2}\ \cos x + \frac{4D-21}{16D^2-441} \cdot \frac{1}{2}\cos 3x$$

$$= \frac{1}{137}\ (4D\text{-}11)\ \frac{1}{2}\ \cos x - \frac{(4D-21)}{585} \cdot \frac{1}{5}\cos 3x$$

$$= \frac{1}{274}\ (-4\sin x - 11\ \cos x) - \frac{(-12\sin 3x - 21\cos 3x)}{1170}$$

$$= -\frac{4\sin x + 11\cos x}{274} + \frac{(4\sin 3x + 7\cos 3x)}{390}$$

$\therefore$ The general solution is

$$y = Ae^{6x} + Be^{-2x} - \frac{4\sin x + 11\cos x}{274} + \frac{4\sin 3x + 7\cos 3x}{390}$$

Problem 6

Solve $(D^2+a^2)y = \sin ax + a\cos ax$

Solution

The auxiliary equation is $m^2+a^2 = 0$

$\therefore m = \pm a$

the complementary function is $y = A\cos ax + B\sin ax$

Particular Integral

$$PI = \frac{1}{D^2+a^2}\ \sin ax + \frac{1}{D^2+a^2}\ a\cos ax$$

$$= \text{IP of } \frac{1}{(D+ai)(D-ai)} e^{iax} + \text{RP of } \frac{1}{(D+ai)(D-ai)} ae^{iax}$$

$$= \text{IP of } \frac{1}{2ai} e^{iax} + \frac{1}{2ai} xae^{iax}$$

$$= \text{IP of } \frac{-ix}{2a} (\cos ax + i\sin ax) + \text{RP of } \frac{xa}{2ai} (\cos ax + i\sin ax)$$

$$\text{P.I} = \frac{x\cos ax}{2a} + \frac{x\sin ax}{2}$$

The general solution is

$$y = A\cos ax + B\sin ax - \frac{x\cos ax}{2a} + \frac{x\sin ax}{2}$$

Problem 7

Solve $(D^2+D+1)\, y = x$

Solution

The auxiliary equation is $m^2+m+1 = 0$

$$m = \frac{-1 \pm \sqrt{3}}{2}$$

The complementary function is

$$y = e^{\frac{-x}{2}}\, x \left[A \cos \frac{\sqrt{3}}{2} + B \sin \frac{\sqrt{3}}{2}\right]$$

$$\text{PI} = \frac{1}{D^2+D+1} x$$

$$= [1+D+D^2)]^{-1}\, x$$

$$= \left[1-(D+D^2) + (D+D^2)^2 + ...\right]x$$

$$= [1-D]\,x$$

$$P.I = x-1$$

$\therefore$ The complete solution is

$$y = A\cos\frac{\sqrt{3}}{2}x + B\sin\frac{\sqrt{3}}{2}x + x-1$$

Problem 8

Solve $(D^2+4)y = x^2$

Solution

The auxiliary equation is $m^2 + 4 = 0$

$$m = \pm 2i$$

The complementary function is

$$y = A\cos 2x + B\sin 2x$$

$$PI = \frac{1}{D^2+4}\,.\,x^2$$

$$= \frac{1}{4\left(1+\frac{D^2}{4}\right)}.x^2$$

$$= \frac{1}{4}\left[1+\frac{D^2}{4}\right]^{-1}.x^2$$

$$= \frac{1}{4}\left[1-\frac{D^2}{4}+...\right].x^2$$

$$= \frac{1}{4}\left[x^2 - \frac{1}{2}\right]$$

$\therefore$ The complete solution is

$$y = A\cos 2x + B\sin 2x + \frac{1}{4}\left(x^2 - \frac{1}{2}\right)$$

Problem 9

Solve $(D^2 – 2D+1)y = x^2 + 1 + \sin 2x$

Solution

The auxiliary equation is $m^2 - 2m + 1 = 0$

$(m-1)^2 = 0$

$m = 1, 1$ (roots are equal)

The complementary function is $y = e^x (Ax + B)$

$$PI_1 = \frac{1}{D^2 - 2D + 1}.(x^2 + 1) + \frac{1}{D^2 - 2D + 1}\sin 2x$$

$$= (1-D)^{-2}(x^2+1) + \frac{1}{-4 - 2D + 1}.\sin 2x$$

$$= (1+2D+3D^2)(x^2+1) + \frac{1}{-(2D+3)}.\sin 2x$$

$$= x^2 + 1 + 4x + 6 + \frac{(2D-3)}{4D^2 - 9}\sin 2x, \text{ [Replace } D^2 \text{by} -a^2\text{, Here } a = 2]$$

$$P.I = x^2 + 4x + 7 + \frac{1}{25}(4\cos 2x - 3\sin 2x)$$

$\therefore$ the complete solution is

$$y = e^x (Ax+B) + x^2 + 4x + 7 + \frac{1}{27}(4 \cos 2x - 3\sin 2x)$$

Problem 10

Solve $(D^2-4D+3)\, y = x^3 e^{2x}$

Solution

The auxiliary equation is $m^2-4m+3=0$

$$(m-1)(m-3) = 0$$

$$m = 1,3$$

The complementary function is $y = Ae^x + Be^{3x}$

$$PI = \frac{1}{D^2-4D+3} \cdot x^3 e^{2x}$$

$$= e^{2x} \frac{1}{(D+2)^2 - 4(D+2)+3} x^3$$

$$= e^{2x} \frac{1}{D^2+4D+4-4D-8+3} x^3$$

$$= e^{2x} \frac{1}{D^2-1} x^3$$

$$= -e^{2x} \frac{1}{1-D^2} x^3$$

$$= -e^{2x} (1-D^2)^{-1} x^3$$

$$= -e^{2x} (1+D^2)x^3$$

$$= -e^{2x} (x^3+6x)$$

$\therefore$ The complete solution is $y = Ae^x + Be^{3x} - e^{2x} (x^3 + 6x)$

Problem 11

Solve $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^x\cos2x$

Solution

The auxiliary equation is $m^2 – 5m + 6 = 0$

$(m–2)(m–3) = 0$

$m = 2,3$

The complementary function is $y = Ae^{2x} + Be^{3x}$

$$P.I = \frac{1}{D^2 - 5D + 6} \cdot e^x \cos2x$$

$$= e^x \frac{1}{(D+1)^2 - 5(D+1) + 6} \cos2x$$

$$= e^x \frac{1}{D^2 - 3D + 2} \cos2x$$

$$= e^x \frac{1}{-4 - 3D + 2} \cos2x$$

$$= -e^x \frac{1}{3D + 2} \cos2x$$

$$= -e^x \frac{3D - 2}{9D^2 - 4} \cos2x$$

$$= -e^x \frac{(3D - 2)\cos2x}{-40}$$

$$\therefore P.I = \frac{e^x}{40}(-6 \sin2x - 2\cos2x)$$

$$= \frac{-e^x}{20}(3\mathrm{sini}2x + \cos2x)$$

$\therefore$ The complete solution is y = C.I + P.I

$$y = Ae^{2x} + Be^{3x} - \frac{e^x}{20}(3\sin 2x + \cos 2x)$$

Problem 12

Solve $\frac{d^2y}{dx^2} + 4y = x \sin x$

Solution

$(D^2+4)\, y = x \sin x$

the auxiliary equation is $m^2 + 4 = 0$

i e $m = \underline{+}\, 2i$

the complementary function is $y = A \cos 2x + B \sin 2x$

$$PI = \frac{1}{D^2+4}\, x \sin x$$

$$= \left[x - \frac{2D}{D^2+4}\right] \frac{1}{D^2+4} \sin x$$

$$= \left[x - \frac{2D}{D^2+4}\right] \frac{1}{-1+4} \sin x \quad \text{(Replacing } D^2 \text{by} -a^2 \text{ Here a =1)}$$

$$= \frac{1}{3}\left[x - \frac{2D}{D^2+4}\right] \sin x$$

$$= \frac{1}{3}\left[x \sin x - \frac{2\cos x}{3}\right]$$

$$P.I = \frac{x \sin x}{3} - \frac{2\cos x}{9}$$

$\therefore$ The complete solution is

$$y = A\cos 2x + B\sin 2x + \frac{x\sin x}{3} - \frac{2\cos x}{9}$$

Problem 13

Solve $(D^2-2D+1)\,y = xe^x \sin x$

Solution

The auxiliary equation is $m^2-2m+1 = 0$

$$m = 1, 1$$

$\therefore$ The complementary function is $y = e^x (Ax+B)$

$$PI = \frac{1}{D^2-2D+1}\, xe^x \sin x$$

$$= e^x \frac{1}{(D+1)^2-2(D+1)+1}\, x\sin x$$

$$= e^x \frac{1}{D^2+1+2D-2D-2+1}\, x\sin x$$

$$= e^x \frac{1}{D^2}\, x\sin x$$

$$= e^x \left[x - \frac{2D}{D^2}\right] . \frac{1}{D^2} \sin x$$

$$= e^x \left[x - \frac{2}{D}\right] . (-\sin x)$$

$$P.I = -e^x (x\sin x + 2\cos x)$$

The General solution is

$$y = e^{x}(Ax+B) - e^{x}\,(x \sin x + 2\cos x)$$

Problem 13

Solve $(D^2+3D+2)\,y = e^{2x} + x^2 + \sin x$

Solution

Given $(D^2+3D+2)\,y = 0$

The auxiliary equation is $m^2+3m+2 = 0$

$$(m+1)(m+2) = 0$$

$$m = -1, -2.$$

The complementary function is

$y = Ae^{-x} + Be^{-2x}$

$$PI_1 = \frac{1}{D^2+3D+2}\,e^{2x}$$

$$= \frac{1}{4+6+2}\,e^{2x}$$

$$= \frac{1}{12}\,e^{2x}$$

$$PI_2 = \frac{1}{D^2+3D+2}\,x^2$$

$$= \frac{1}{2\left(1+\frac{3D+D^2}{2}\right)}\,x^2$$

$$= \frac{1}{2}\left[1+\left(\frac{3D+D^2}{2}\right)\right]^{-1} x^2$$

$$= \frac{1}{2}\left[1-\frac{(3D+D^2)}{2}+\frac{(3D+D^2)^2}{4}+...\right]x^2$$

$$= \frac{1}{2}\left[1-\frac{3D}{2}-\frac{D^2}{2}+\frac{9D^2}{4}\right]x^2$$

$$= \frac{1}{2}\left[1-\frac{3D}{2}+\frac{7D^2}{4}\right]x^2$$

$$P.I_2 = \frac{1}{2}\left[x^2-3x+\frac{7}{2}\right]$$

$$P.I_3 = \frac{1}{D^2+3D+2}.\sin x$$

$$= \frac{1}{-1+3D+2}\sin x, \text{ (Replace } D^2 \text{ by } -a^2)$$

$$= \frac{1}{3D+1}\sin x$$

$$= \frac{3D-1}{9D^2-1}\sin x$$

$$= \frac{-1}{10}(3D-1)\sin x, \left(D=\frac{d}{dx}\right)$$

$$PI_3 = \frac{-1}{10}(3\cos x - \sin x)$$

$$\therefore PI = P.I_1 + PI_2 + PI_3$$

$$= \frac{1}{12}e^{2x} + \frac{1}{2}\left[x^2-3x+\frac{7}{2}\right] - \frac{1}{10}[3\cos x - \sin x]$$

$\therefore$ The general solution is y = C.F + P.I

$$y = Ae^{-x} + Be^{-2x} + \frac{1}{12}e^{2x} + \frac{1}{2}\left[x^2 - 3x + \frac{7}{2}\right] - \frac{1}{10}[3\cos x - \sin x]$$

4.6 Recap

1. Solution of second order non-homogenous equation

   (i) If Roots are distincts, say $m^1 \neq m^2$

   Then the solution is y = $Ae^{m_1x} + Be^{m_2x}$

   (ii) If Roots are equal say $m_1 = m_2$

   Then the solution is y = e^{mx} (Ax+B)

   (iii) If the roots are imaginary, say $\alpha + i\beta$

   the solution is y = $e^{\alpha x}$ [A$\cos\beta x$ + B $\sin\beta x$]

2. Method of obtaining the particular integral

   1. f(x) = e^{ax}
   2. f(x) = sinax of cosax
   3. f(x) = x^m
   4. f(x) = e^{ax} sinax or e^{ax} cosax (i.e., e^{ax}.v)
   5. f(x) = xv, where v is any function of x

4.7 Exercise

I. Solve the following differential equations :

1. $(D^2 - 6D + 13)\, y = 0$

2. $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 8y = 0$

3. $\frac{d^2y}{dx^2}+4\frac{dy}{dx}+3y = e^{-3x}$

4. $D^2y = e^x$

5. $\frac{d^2y}{dx^2}-2\frac{dy}{dx}-3y = 8\cos 2x$

6. $(D^2-4D+3)y = \sin 3x.\cos 2x$

7. $(D^2+D+2)y = e^x+\cos x$

8. $(D^2-4D+4)y = \sin^2 x$

9. $(D^2+4)y=7+\cos 2x$

10. $(D^2+4D-5)y = e^{3x}+4\cos 4x$

11. $(D^2+1)\,y = e^x+\sin x$

12. $\frac{d^2y}{dx^2}-4y = (1+e^x)^2 + 3$

II. Solve the following differential equations

1. $(D^2+3D+2)y = x^2$

2. $(D^2-2D+1)y = x^3$

3. $(D^2-D+1)\,y = x^3 - 3x^2 + 1$

4. $(D^2-6D+9)\,y = x^2 + e^x$

5. $(D^2-4D)\,y = x^2 - 1$

6. $(D^2+9)\,y = x^2$

7. $(D^2+D-2)\,y = x+\cos x$

8. $(D^2-1)y = 2e^x+3x$

9. $(D^2-2D+1)\ y = x+1$

10. $(D^2+D-2)\ y = x^2-2x+3$

III. Solve the following differential equations

1. $(D^2+4D+4)y = e^{-x}\sin 2x$

2. $(D^2-2D+4)y = e^x\sin x$

3. $(D^2-2D+5)y = e^{2x}\sin 2x$

4. $(D^2-7D+18)y = x^2.e^{2x}$

5. $(D^2+9)y = (x^2+1)\ e^{3x}$

6. $(D^2-3D+2)y = 2e^{3x}\sin 2x$

7. $(D^2-5D-6)y = e^{2x}(1+x)$

8. $(D^2-4D+3)y = 2xe^{3x}+3e^x\cos 2x$

9. $(D^2+4)y = x^2\sin^2 x$

10. $(D^2-D)y = 3xe^x$

11. $(D^2-2D+1)\ y = x^2e^x$

12. $(D^2-D+2)y = 58e^x\cos 3x$

LESSON - 5

PARTIAL DIFFERENTIAL EQUATIONS - FORMATION AND TYPES OF PDE

Learningn Objectives

After studying this lesson you should be in a position to

- form PDF by elimination of arbitrary constants
- form PDF by eliminating arbitrary function
- Different solution of PDE - Standard types

Structure

5.1 Introduction

In many situations we come across a quantity whose value deponds on the values of more then one variable, For example,

(i) the volume of a right circular cylinder is a function of the base radius and height.

(ii) the volume of a cuboid depends on its length, breadth and height

If the value of the variable quantity 'u' depends on the values of several others, variable quantitied x,y,z... we say that it is a function of x,y,z....and it is denoted as u = f(x,y,z); x,y,z are called independent variable and 'u' is called the dependent variable.

5.2 Partial Differential Equation

A partial differential equation is an equation which contains one or more partial derivatives.The order of the PDE is that of the derivative of highest order in the equation.

For example, $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial x} = z$ (1)

$x^2 = \frac{\partial^2 z}{\partial x^2} + y^2 \frac{\partial^2 z}{\partial y^2} + xy \frac{\partial^2 z}{\partial x \partial y} = 0$ (2) is called

the PDE of first order and second orders respectively.

Let Z = f(x,y) be a function of two independent variables x and y

Then, $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$ are the first order partial derivatives and

$\frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}$ are the second order partial derivatives.

For convenience we use the follwing conventional notations for partial derivatives,

$$p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}, r = \frac{\partial^2 z}{\partial x^2}, s = \frac{\partial^2 z}{\partial x \partial y}, \ t = \frac{\partial^2 z}{\partial y^2}$$

Then, equations (1) and (2) can be written as

$xp + yq = z$ (3)

$x^2r+y^2t+xys = 0$ (4)

In ordinary differential equation we have seen that the differential equations are formed by eliminating arbitary constants. In the same way PDE are also formed by eliminating the arbitary constants from a given relation between te variables.

Also partial differential equations may be formed by the eliminating of arbitary function of the variables. Hence PDE may be formed by ;

(i) Eliminating arbitary constants

(ii) Eliminating arbitary functions

5.2.1 Formation of PDE by Eliminating the Arbitrary Constants

Consider z to be a dependent variable and x and y to be independent variables.

Let $f(x,y,z, a, b) = 0$ ... (1)

Here there are arbitary constants a and b.

Eliminating a and b, we need three equations.

Differentiating (1) partially w.r. to x and y, we get.

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial x} = 0 \quad \text{(i.e.,)} \quad \frac{\partial f}{\partial x} + p\frac{\partial f}{\partial z} = 0 \qquad \ldots \qquad (2)$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} \cdot \frac{\partial f}{\partial y} = 0 \quad \text{(i.e.,)} \quad \frac{\partial f}{\partial x} + q\frac{\partial f}{\partial z} = 0 \qquad \ldots \qquad (3)$$

In general, we can eliminate the arbitary constants from (1), (2) and (3) and the resulting equation will be of the form $\phi(x, y, z, p, q) = 0$ (4)

Example 1

Form the partial differential equating by eliminating the arbitrary constants from $z = (x^2 +a) (y^2 +b)$

Solution

Given $z = (x^2 +a) (y^2 +b)$ (1)

Differentiating (1) w.r. x and y we get,

$p = 2x (y^2 +b)$ (2)

$q = 2y (x^2 + a)$ (3)

Multiplying (1) and (2) we get,

$pq = 4xy(x^2+a) (y^2+b)$

Therefore,

$pq = 4xyz$ ($\because$ using (1)

This is the required PDE.

Example 2

Find the partial differential equation by eliminating the arbitary constants a and b from log log (az-1) = x+ay+b

Solution

Given $\log(az-1) = x+ay+b$ (1)

Diffy, partially with respect to x and y we get,

$$\frac{1}{az-1}\, a.\frac{\partial z}{\partial x} = 1$$

$$\Rightarrow \quad \frac{1}{az-1}\, ap = 1 \qquad (1)$$

and

$$\frac{1}{az-1}\ a.\frac{\partial z}{\partial y} = a$$

=> $$\frac{1}{az-1}\ aq = a \qquad (2)$$

Dividing (2) by (1) we get

$$\frac{q}{p} = a \qquad (3)$$

From equation (1),

$$ap = az-1$$

$$a(z-p) = 1 \text{ (or) } \frac{q}{p}\ (z-p) = 1 \qquad (ஃ \text{ by } (3))$$

ஃ $q(z-p)$ -p

This is the required P.D.E

Example 3

Form the partial differential equation by eliminating a and b from $z = ax + by + a^2+b^2$

Solution

Given $z = ax + by + a^2+b^2$ (1)

Differentiating 'p' w.r. to x and y, we get

$$p = a\ ;\ q=b \qquad (2) \text{ Here } \left(ஃ\ \frac{\partial z}{\partial x} = p,\ \frac{\partial z}{\partial y} = q\right)$$

Subsuiting these in (1) we get

$$z = px + qy + p^2+q^2$$

This is the required PDE

Example 4

Form the PDE by eliminating the arbitary constants in $z = (x-a)^2 + (y-b)^2$.

Solution

Given : $z = (x-a)^2 + (y-b)^2$ .... (1)

Here we have two arbitary constans a and b. To eliminate these two arbitarry constant, we need two or more equations connecting a and b.

Diff, (1) 'p' w.r. to x and y we get,

$$\frac{\partial z}{\partial x} = p = 2\,(x-a) \qquad \text{and} \qquad (2)$$

$$\frac{\partial z}{\partial y} = q = 2(y-b) \qquad (3)$$

From equation (2) we get.

$$x-a = \frac{p}{2} \qquad (4)$$

$$(3) \Rightarrow \quad y-b = \frac{q}{2} \qquad (5)$$

Substituting (4) and (5) in (1) we get

$$z = \left(\frac{p}{2}\right)^2 + \left(\frac{q}{2}\right)^2$$

$$\therefore 4z = p^2+q^2$$

Example 5

Find the equation of all spheres with centres on the z axis

Solution

Let the centre equation of the sphere be

$x^2+y^2+(z-1)^2 = r^2$

Diffy, 'p' w.r. to x and y we get,

$2x + 2p (z-c) = 0 \Rightarrow (z-c) p = -x$ (1)

$2y + 2q (z-c) = 0 \Rightarrow (z-c) q = -y$ (2)

Dividing (1) by (2) we get,

$\frac{p}{q} = \frac{x}{y}$ (or) py = xq

Problem 3

Form the partial D.E. by eleminating a, b, c from $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Solution

Given $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ . ... (1)

Differentiating 'p' w.r. to x and y we get

$$\frac{2x}{a^2} + \frac{2z}{c^2} p = 0 \quad (2)$$

$$\frac{2y}{b^2} + \frac{2z}{c^2} q = 0 \quad (3)$$

Again Diff. (2) 'p' .respect to y

$$0+\frac{2}{c^2}(zs+qp)=0$$

Therefore $\boxed{zs + qp = 0}$

Note

More than one PDE is possible in this problem

These PDE are $xzr + xp - zp = 0$

$yzt + yq^2 - zq = 0$

Check Your Problems - I

1. From the partial differential equation by eliminating the arbitary constants.

(i) $z = ax + by + \sqrt{a^2+b^2}$

(ii) $z = (x+a^n)(y+b^m)$

(iii) $2z = (ax+y)^2+b$

(iv) $z = (x+a)(y+b)$

(v) $z = ax+by+ab$

(vi) $z = ax^3+by^3$

(vi) $a(x^2+y^2) + bz^2=1$

Check Your Answers - I

(i) $z = ax + by + \sqrt{a^2+b^2}$

(ii) $z = pq$

(iii) $px+qy = q^2$

(iv) $pq = z$

(v) $z = px+qy+pq$

(vi) $px + qy = 3z$

(vii) $px = qx$.

5.2.2 Formulation of PDE by Elimination of arbitrary function

Here we consider a few example to show now PDE can be obtained by eliminating the arbitary funtions.

Example 1

From the PDE by eliminating the arbitary function from $z = f(x^2+y^2)$

Solution

Given $z = f(x^2+y^2)$ (1)

Diffy, (1) 'p' w.r. to x and y we get,

$$\frac{\partial z}{\partial x} = f^1 (x^2+y^2).\ 2x \Rightarrow \frac{p}{2x} = f' (x^2+y^2) \quad (2)$$

$$\frac{\partial z}{\partial y} = f^1 (x^2+y^2).\ 2y \Rightarrow \frac{q}{2y} = f' (x^2+y^2) \quad (3)$$

Eliminating from (2) and (3) we get,

$$\frac{p}{2x} = \frac{q}{2y}$$

Here => $\boxed{py = qx}$

Example 2

Form the PDE by eliminating arbitary function from $xyz = \phi\,(x^2+y^{2-}z^2)$.

Solution

Given $xyz = \phi\,(x^2+y^{2-}-z^2)$ (1)

Diffy, (1) 'p' w.r.t. to x and y we get,

$xyp + yz = \phi^1\,(x^2+y^2-z^2)\,(2x-2zp)$ (2)

$xyq + xz = \phi^1\,(x^2+y^2-z^2)\,(2y-2zq)$ (3)

Dividingn (2) by (3) we get,

$$\frac{yz+xyp}{xz+xyq} = \frac{x-zp}{y-zp}$$

Cross multiplying,

$(yz + xyp)\,(y–zq) = (x–zp)\,(xz+xyq)$

$zy^2+xy^2p–yz^2q - xyzpq = x^2z+x^2yq-xz^2p–xyzpq$

(or)

$px\,(y^2+z^2) - qy\,(z^2+x^2) = z\,(x^2-y^2)$

This is the required PDE

Example 3

Form the PDE by eliminating the arbitary function from $lx+my+nz = f(x^2+y^2+z^2)$

Solution

Given $lx+my+nz = f(x^2+y^2+z^2)$ (1)

Diffy, (1) 'p' w.r.t. to x we get

$l + np = f^1\,(x^2+y^2+z^2)\,(2x-2zp)$ (2)

Diffy, (1) 'p' w.r. to y we get

$m + nq = f^1 (x^2+y^2+z^2) (2y+2zq)$ (3)

Dividing (2) by (3) we get,

$$\frac{l+np}{m+nq} = \frac{x+zp}{y+zq}$$

$(l+np) (y+zq) = (x+zp) (m+nq)$

$ly + npy + lzq = mx + mzp + nxq$

$(mz-ny) p + (nx-lz) q = ly - mx$

This is required PDE

Example 4

Eliminating the aritary function from $z = xy + f(x^2+y^2)$

Solution

Given $z = xy + f(x^2+y^2)$ (1)

Dify (1) 'p' w.r. to x, we get,

$p = y+f'(x^2+y^2).2x$ (2)

Dify (1) 'p' w.r. to y, we get,

$q = x+f'(x^2+y^2).2y$ (3)

Eliminating (2) and (3) we get,

$(p-y)y = (q-x).x$

$py - y^2 = qx - x^2$

$py - qx = y^2-x^2$ which is a required PDE.

Example 5

Eliminate the arbitary function in $z = f_1(y+2x) + f_2(y-3x)$

Solution

Given $z = f_1(y+2x) + f_2(y-3x)$ (1)

Diffy (1) 'p' w.r. to x and y we get,

$$\frac{\partial z}{\partial x} = 2f_1'(y+2x) - 3f_2'(y-3x) \quad (2)$$

and

$$\frac{\partial z}{\partial y} = f_1'(y+2x) + f_2'(y-3x) \quad (3)$$

$$(2) \Rightarrow \frac{\partial^2 z}{\partial x^2} = 4f_1''(y+2x) + 9f_2''(y-3x) \quad (4)$$

$$(3) \Rightarrow \frac{\partial^2 z}{\partial y^2} = f_1''(y+2x) + f_2''(y-3x) \quad (5)$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2f_1''(y+2x) - 3f_2''(y-3x) \quad (6)$$

Eliminating the arbitary function, we get

$$\therefore \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = 0$$

Example 6

From the PDE by eliminating the arbitary functin in $xy + z^2 = f(x+y+z)$

Solution

Given $xy + z^2 = f(x+y+z)$ (1)

The equation can be written as $f(x+y+z, xy + z^2) = 0$

(i.e.,) $f(u,v) = 0$, where $u = x+y+z$, $v= xy+z^2$

Diffy (1) 'p' w.r. to x and y we get,

$$\frac{\partial f}{\partial u}(1+p) + \frac{\partial f}{\partial v}(y+2zp) = 0$$

$$\frac{\partial f}{\partial v}(1+q) + \frac{\partial f}{\partial v}(x+2xq) = 0$$

Eliminating $\frac{\partial f}{\partial u}, \frac{\partial f}{\partial v}$ we get

$$\begin{vmatrix} 1+p & y+2zp \\ 1+q & x+2zq \end{vmatrix} = 0$$

$(1+p)(x+2zq) - (1+q)(y-2zp) = 0$

$x+2zq + xp + 2zqp - y–2zp - qy - 2zpq = 0$

$p(x-2z) – q(y-2z) = y-x$

This is the required PDE

Check Your Progress - II

1. Form the PDE by eliminating arbitory function from the following

   (i) $z = f(xy)$

   (ii) $z = f(x-y)$

(iii) $z = f\left(\frac{y}{x}\right)$

(iv) $z = f(x^2-y^2)$

(v) $z = x+y+f(xy)$

(vi) $z = (x+y)\, f(x^2-y^2)$

(vii) $z = f(x+z) + g\,(x+y)$

(viii) $ax+by+cz = f(x^2+y^2+z^2)$

(ix) $x+y+z = f(xy+z^2)$

(x) $z = f(x+4y) + g(x-4y)$

(xi) $z = f(2x+3y) + yg\,(2x+3y)$

II. Form the PDE by eliminating the arbitory function from the following

(i) $z = \phi\,(x+ct) + \phi\,(x-ct)$

(ii) $z = \phi\,(x) + \phi\,(y)$

(iii) $\phi(x+y+z,\, xy+z^2) = 0$

(iv) $\phi(x^2+y^2+z^2,\, xyz) = 0$

(v) $z = f(lx-my)$

(vi) $z = e^y f(x+y)$

(vii) $\phi(x^3-y^3,\, x^2-z^2) = 0$

(viii) $\phi(xyz,\, x^2+y^2+z^2) = 0$

Check Your Answers - II

I. (i) $z = pq$

(ii) $p+q = 0$

(iii) $px+qy = 0$

(iv) $qx+py = 0$

(v) $px-qy = x-y$

(vi) $yp+xq=z$

(vii) $qr\ (1+p+q)s + (1+p)t = 0$

(viii) $(bz-cy)\ p + (cx-az)q = ay - bx$

(ix) $(x+2z)\ p - (y+2z)q = y-x$

(x) $16r-t = 0$

(xi) $t + \frac{q}{4} r = 3s$

ii. (i) $c\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial t^2}$

(ii) $\frac{\partial^2 z}{\partial x \partial y} = 0$

(iii) $p(x-2z) + q(2z-y) = y-x$

(iv) $px\ (z^2-y^2) + qy\ (x^2-z^2) = z(y^2-x^2)$

(v) $pm - lq = 0$

(vi) $q = p+z$

(vii) $y^2zp+x^2zq = xy^2$

(viii) $px\ (z^2-y^2) +qy\ (x^2-z^2) = z\ (y^2-x^2)$

5.4 Different Solutions of Partial Differential Equations

Non - Linear Differential Equation of First Order - Standard Types

A partial differential equation invoving the first order partial derivations $\frac{\partial z}{\partial x}$ (=p) and $\frac{\partial z}{\partial y}$ (=q) is called a linear partial differential equation. Partial differential equations which involve the higher powers of p,q, pq are called non-linear partial differential equations. Linear PDE are obtained by eliminating arbitrary constants or arbitary functions.

Complete Solution

A solution in which the number of arbitrary constants is equal to the number of independent variables is called complete integral or complete solution. The complete solution is also referred to as fomplete integral.

For example, the partial differential equations,

$$f(x,y,z,p,q) = 0 \qquad (1)$$

may be obtained from the equation, here p=a, q=b

$$g(x,y,z,a,b) = 0 \qquad (2)$$

In this case (2) is called the complete solution of (1). There are two independent variable. In the solution (2) there are two arbitary constants.

Singular Solution

The complete solution represents two parameter family of surfaces which may or may not have an envelope.

The envelope, if it exits, is obtained by eliminating 'a' and 'b'.

If g(x,y,z,a,b) = 0 is the complete integral of a PDE

Then, the eliminating of a and b, from the equation, Diff, (2) w.r. to 'a' and 'b' we get,

$\frac{\partial g}{\partial a} = 0$; $\frac{\partial g}{\partial b} = 0$. If the eliminent ϕ (x,y,z) = 0 exists.

It is called the singular solution or singular integral.

General solution

In the non-linear PDe f(x,y,z,a,b) = 0.

In the If f(x,y,z,a,b) = 0 the complete solution is g(x,y,z,a,b) = 0. It may be possible for the existence of a relationship between the arbitrary constant a and b.

Let b = ϕ (a)

Then the complete solution takes the form

$$g(x,y,z,a, \phi(a)) = 0 \qquad (1)$$

Diffy, 'p' w.r. to 'a', we get

$$\frac{\partial}{\partial a} g(x,y,z, a, \phi(b)) = 0 \qquad (2)$$

If this eliminate if it exists it called the general solution of equation (1). This solution is also referred to as general integral.

Example 1

Consider the partial differential equation

$z = px+qy - (p^2+q^2)$

Solution

$$\text{Given } z = px + gy - (p^2+q^2) \qquad (1)$$

Let p = a, q = b

then $z = ax+by - (a^2+b^2)$ is the complete solution of (1)

To find singular solution :

Let $g(x,y,z,a,\phi(b)) = z - ax - by + a^2 + b^2 = 0$

Diffy, (2) 'p' w.r. to a and b, we get

$$\frac{\partial}{\partial a} g(x,y,z,a,b) = 0 \Rightarrow -x+2a = 0$$

$$\frac{\partial}{\partial b} (x,y,z,a,b) = 0 \Rightarrow -y+2b=0$$

Eliminating a and b from the above equations, we get

$\therefore x^2+y^2= 4z$

This solution is called the singular solution.

To find General solution :

Let b = a is complete solution for (2)

Then,`

$$(2) \Rightarrow z - a(x+y) + 2a^2 = 0 \qquad (3)$$

Diffy (3) 'p' w.r to 'a' we get

$$\frac{\partial g}{\partial a} = 0 \Rightarrow -(x+y)+4a = 0 \qquad (4)$$

Eliminating 'a' from (3) and (4) we get

$\therefore 8z=(x+y)^2$ is called the general solution of equation (1)

5.3.1 Standard Types of PDE

Let us consider four different types of non linear PDE and the procedure for obtaining their complete solution.

Standard type I

Equation of the form $F(p,q) = 0$ (1)

Here the PDE involves only p and q and the variable x, y,z are absent.

Consider the equtaion $z = ax+by+c$ (2)

where a and b are connected by the relation $f(a,b) = 0$

Differentiating the above equation partially with respect to x and y we get here $p = a$, $q = b$.

Substituting, $f(a,b) = 0$,

We can find 'b' interms of 'a', say $b = g(a)$, substituting this (2) we get,

(2) => $z = ax + g(a)\, y + c$ (3)

We see that, (3) is the complete solution of (1)

to find the singular solution, we have to determine the eliminate of 'a' and c from (3)

(3) => $z = ax+g(a)\, y +c$

Diffy, (2) 'p' w.r. to 'a' & 'c' we get

$0 = x + g'(a)\, y$

As the above equation is inconsistent no singular solutions will exists for the PDE of the type $F(p,q) = 0$

To obtain the genreal solution,

Take $c = \phi(a)$

Then, the complete solution becomes one parameter family.

(3) => $z = ax+g(a)y+\phi(a)$ (4)

Diffy, (4) 'p' w.r. to 'a' we get,

$0 = a+g'(a) + \phi'(a)$ (5)

Eliminating 'a' from the above two equations we get the general solution.

Example 1

Solve $p^2+q^2 = 1$

Solution

Given $p^2+q^2 = 1$

This equation of the type $f(p,q) = 0$

The complete solution of this PDE is the form

$z = ax+by+c$ (1)

Diffy, (1) 'p' to x and y we get

Here $p = a$, $q = b$

Substituting in the given PDE, we have

$a^2+b^2 = 1 \Rightarrow b = \sqrt{1-a^2}$

(1) => $z = ax + \left(\sqrt{1-a^2}\right) y + c$ (2)

This is the required complete solution of (1). Where a and c are entritrary constants

There is no singular integral to the PDE of this type.

To obtain the general solution;

Take c = ϕ(a)

(2) => z = ax + $\left(\sqrt{1-a^2}\right)$ y + ϕ(a)

Diffy, (3) 'p' w.r.t. to 'a' we get,

$$0 = x - \left(\frac{a}{\sqrt{1-a^2}}\right) y + \phi'(a) \qquad (3)$$

The Eliminating of 'a' between from (2) and (3) gives the general solution.

Example 2

Solve $\sqrt{p}+\sqrt{q}=1$

Solution

Given $\sqrt{p}+\sqrt{q}=1$

The equation is of the type f (p,q) = 0

The complete solution is of the form

$$z = ax + by + c \qquad (1)$$

Diffy, (1) 'p' w.r.t. to x and y, we get

p = a and q = b

Therefore the given equation (1) becomes $\sqrt{a}+\sqrt{b}=1$

The complete solution is

$$z = ax + \left(1-\sqrt{a}\right)^2 y + c \qquad (2)$$

There is no singular integral of (2)

General solution :

Take c = ϕ(a)

(2) => z = ax + $\left(1-\sqrt{a}\right)^2$ + ϕ(a)

Diffy, 'p' w.r. to 'a' we get,

0 = x - $\left(1-\sqrt{a}\right)\frac{1}{\sqrt{a}}$ y + ϕ'(a) (3)

Eliminating of 'a' between equation (2) and (3) gives the genreal solution.

Example 3

Solve p + q = pq

Solution

Given p + q = pq

This equation is of the type f(p,q) = 0

The complete solution is of the form

z = ax + by + c (1)

Diffy, (1) 'p' w.r. to x and y we get

p = a and

q = b

The given equation becomes

(a) => a + b = ab

a = b (a-1) = > b = $\frac{a}{a-1}$

Therefore the complete solution is

$$z = ax + \frac{a}{a-1}y + c \qquad (2)$$

This type of equation as no singular solution

General Solution :

Let c = $\phi(a)$

(2) => $z = ax + \frac{a}{a-1}y + \phi'(a)$

Diffy, (3) 'p' w.r.t. to 'a', we get,

$$0 = x + \left(\frac{(a-1).1-a}{(a-1)^2}\right) y + \phi'(a)$$

$$0 = \frac{-1}{(a-1)^2} y + \phi'(a) \qquad (3)$$

Eliminating of 'a' between (2) and (3) gives the general solution

Example 4

Solve pq = 1

Solution

Solve pq = 1

This equation is of the type f(p,q) = 0

The complete solution is of the form

$$z = ax + by + c \qquad (1)$$

Diffy, (1) 'p' w.r.t. to x and y we get

p = a

q = b

The given equation becomes

(1) => a b = 1 = > b = $\frac{1}{a}$

The complete solution is

z = $ax + \frac{1}{a}y + c$ (2)

where a and c are arbitrary constants

There is no singular solution

General Solution :

Take c = $\phi(a)$

(2) => z = $ax + \frac{1}{a}y + \varphi(a)$

Diffy, 'p' w.r. to 'a' we get

$$0 = x - \frac{1}{a^2}y + \phi'(a) \qquad (3)$$

the Eliminating of 'a' between (2) and (3) gives the general solution

Standard Type II (Clairaul's form)

Equation of the form z = px +qy + f(p,q)

Now consider the equation z = ax+by +f(a,b) (1)

Where a and b are aritrarory constants

Diffy, 'p' w.r. to x and y we get,

p = $\frac{\partial z}{\partial x}$ = a (2) and

$$q = \frac{\partial z}{\partial y} = b \qquad (3)$$

Therefore by eliminating the arbitrary constants from (1), (2) and (3) we get,

$z = px+qy+f(p,q)$ (4)

Therefore (1), can be considered as the complete solution of the given PDE

This type of PDE is knwon as extended Clairout's form. The complete solution consists of a two parameter family of planes.

The singular solution if it exists is a surface having the complete solution as tangent planes.

Example 1

Solve $z = px+qy+ab$

Solution

Solve $z = px+qy+ab$

This equation is of the 'Clairaut's form

To find complete solution :

The complete solution is obtained by replacing p by a and q by b. Where a and b are arbitrary constants.

(i.e.,) The complete solution is $z = px+qy+ab$

Diffy (q) 'p' w.r. to 'a' and 'b' we get

$0 = x+b \Rightarrow b=-x$ (2)

$0=y+a \Rightarrow a=-y$ (3)

Eliminating a & b from (1) (2) and (3) we get,

(1) => $z = -xy-xy+xy$

(i.e.,) $z+xy=0$

This gives the singular solution of the given PDE and to get the general solution.

Take $b = \phi(a)$

(1) => $z = ax + \phi(a)\, y + a\phi(a)$ (4)

Diffy, (4) 'p' w.r. to 'a' we get

$0 = x + \phi'(a)\, y + a\phi'(a) + \phi(a)$ (5)

Eliminating of 'a' from (4) and (5) we get the general solution

Example 2

Find the complex and singular solution of

$z = xp + yq + p^2 - q^2$

Solution

$z = xp + yq + p^2 - q^2$

This is the Clairaut's form. there p = a, q = b

Therefore the complete solution is $z = xp + yq + a^2 - b^2$ (1)

Where a & b are arbitrary constants.

Diffy (1) 'p' w.r. to 'a' and 'b' we get

$0 = x+2a \quad => a = \frac{-x}{2}$

$0=y-2b \qquad \Rightarrow b=\frac{y}{2}$

Substituting in (1) we get

$$z = \frac{-x^2}{2}+\frac{y^2}{2}+\frac{x^2-y^2}{4}$$

(i.e.,) $4z = y^2-x^2$ which is the singular solution

Example 3

Solve $z = px + qy + p^2 q^2$

Solution

Given $z = px + qy + p^2 q^2$

This equation is a PDE of Clairaut's form

Therefore the complete solution is $z = ax + by + a^2b^2$

Where a and b are arbitrary constants.

Diffy (1) 'p' w.r. to 'a' and 'b' we get

$0 = x+2ab^2 \qquad (2)$

$0=y+2a^2b \qquad (3)$

$(2) \Rightarrow x = -2ab^2$

$$= \frac{-2ay^2}{4a^4} \qquad (\therefore \text{by}(3)\ (\sim b = \frac{-y}{2a^2})$$

$$\therefore x = \frac{-y^2}{2a^3}$$

$$z = x.\sqrt[3]{\frac{y^2}{2x}} - y\sqrt[3]{\frac{x^2}{2y}}$$

$$z = -2^{\frac{2}{3}}x^{\frac{2}{3}}y^{\frac{2}{3}}$$

This is the singular solution, where b = f(a)

The complete solution is $z = ax + f(a)\,y + a^2(f(a))^2$ (4)

Diffy, 'p' w.r. to 'a' we get

$0 = a + f'(a)y + 2a^2f(a)f'(a) + 2a(f(a))^2$ (5)

Eliminate of 'a' from (4) and (5) gives the general solution.

Example 4

Solve $z = px + qy + \sqrt{1+p^2+q^2}$

Solution

Given $z = px + qy + \sqrt{1+p^2+q^2}$ (1)

This is equation of the form Clairaut's type

Here p = a, q = b

The complete iintegral is z = = ax + by + $\sqrt{1+a^2+b^2}$ (2)

To find Singular Integral

Diffy (2) 'p' w.r. to 'a' and 'b' and then equating zero, we get,

$$\frac{\partial z}{\partial a} = x + \frac{a}{\sqrt{1+a^2+b^2}} = 0 \qquad (3)$$

$$\frac{\partial z}{\partial b} = y + \frac{b}{\sqrt{1+a^2+b^2}} = 0 \qquad (4)$$

From (3)

$$x^2 = \frac{a^2}{1+a^2+b^2} \qquad (5)$$

From (4)

$$y^2 = \frac{b^2}{1+a^2+b^2} \qquad (6)$$

From (5) and (6) we get,

$$x^2+y^2 = \frac{a^2+b^2}{1+a^2+b^2}$$

$$1-(x^2+y^2) = 1-\frac{a^2+b^2}{1+a^2+b^2} = \frac{1}{1+a^2+b^2}$$

$$\therefore \sqrt{1+a^2+b^2} = \frac{1}{\sqrt{1-x^2-y^2}} \qquad (7)$$

Sustutitng (7) in (3) and (4) we get

$$a = \frac{-x}{\sqrt{1-x^2-y^2}}$$

$$b = \frac{-y}{\sqrt{1-x^2-y^2}} \qquad (8)$$

Substituting (7) and (8) in (2) we get,

$$z = \frac{-x^2}{\sqrt{1-x^2-y^2}} - \frac{y^2}{\sqrt{1-x^2-y^2}} + \frac{1}{\sqrt{1-x^2-y^2}}$$

$$= \frac{1-x^2-y^2}{\sqrt{1-x^2-y^2}}$$

$\therefore$ z $= \sqrt{1-x^2-y^2}$ or $z^2 = 1-x^2 - y^2$

$\therefore$ $x^2 + y^2 + z^2 = 1$

which is the singular integral

Standard Type III

Equation of the form F(z,p,q) = 0

Procedure for solving PDE of the type F (Z,p,q) = 0

Step 1

We assume z = F(x+ay), where u x+ay

(i.e.,) z = f(u)

Step 2

Replace $p = \frac{\partial z}{\partial x} = \frac{dz}{du} \cdot \frac{\partial u}{\partial x} = \frac{dz}{du}$

$q = \frac{\partial z}{\partial y} = \frac{dz}{du} \cdot \frac{\partial u}{\partial y} = a\frac{dz}{du}$ respectively in the given PDE

Step 3

Solve the resulting ODE of first order

(i.e.,) $f\left(z, \frac{dz}{du}, a\frac{dz}{du}\right) = 0$

which is an ODE of first order.

Example 1

Solve $z = p^2+q^2$ (i)

Solution

Solve $z = p^2+q^2$

Assume $z = f(x+ay) = f(u)$ is a solution of (i) (ii)

put $x+ay = u$ in (ii)

then $z = f(u)$ (iii)

Diff 'p' w.r. to x and y, we get

$$p = \frac{dz}{du}; \ q = a\frac{dz}{du} \quad \text{(iv)}$$

The given equation becomes,

$$z = \left(\frac{dz}{du}\right)^2 + a^2\left(\frac{dz}{du}\right)^2$$

$$z = \left(\frac{dz}{du}\right)^2 (1+a^2)$$

$$\frac{dz}{du} = \frac{\sqrt{z}}{\sqrt{1+a^2}}$$

$$\text{(i.e.,)} \ \frac{dz}{\sqrt{z}} = \frac{du}{\sqrt{1+a^2}}$$

Integrating, (v) we get

$$\int \frac{dz}{\sqrt{z}} = \frac{1}{\sqrt{1+a^2}} \int du$$

$$=> 2\sqrt{z} = \frac{u}{\sqrt{1+a^2}} + b$$

$$=> 2\sqrt{z} = \frac{x+ay}{\sqrt{1+a^2}} + b \qquad (u= x +ay)$$

which gives the complete integral.

(or)

$4(1+a^2)\, z = (x+ay+b^2)$ (2)

This is the complete solution

Diffy, (2) 'p' w.r. to a and b we get

$8az = 2\,(x+ay+b)y$ (3)

$0 = 2(x+ay+b)1$ (4)

Substuting (4) in (3) we get

$\therefore$ z = 0, This is the Singular solution

Example 2

Find the complete solution of $p(1+q) = qz$

Solution

Given $p(1+q) = qz$

Assume that $z = f(x+ay)$, put $x + ay = u$

then $p = \frac{dz}{du}$; $q = a\,\frac{dz}{du}$

The given equation becomes,

$$\frac{dz}{du}\left(1+a\frac{dz}{du}\right) = a\frac{dz}{du}.z$$

i.e., $1+\frac{adz}{du} = az$

$$\frac{dz}{du} = \frac{az=1}{a}$$

$$\frac{adz}{az-1} = du$$

On Integrating, we get

$$\int\frac{a}{az-1}dz = \int du$$

log (az-1) = u+C

Therefore, log (az-1) = x+ay+c

This is the complete solution

Example 3

Solve $p^3 + q^3 = 8z$

Solution

Given $p^3 + q^3 = 8z$

Assume that z = f(x+ay) = f(u), put u = x+ay

then, $p = \frac{dz}{du}$; $q = a\frac{dz}{du}$

The given equation becomes,

$$\left(\frac{dz}{du}\right)^3 (1+a^3) = 8z$$

$$\frac{dz}{du} = \frac{2(z)^{\frac{1}{3}}}{(1+a^3)^{\frac{1}{3}}}$$

$$\frac{dz}{z^{\frac{1}{3}}} = \frac{2du}{(1+a^3)^{\frac{1}{3}}}$$

Integrating,

$$\frac{3z^{2/3}}{2} = \frac{2u}{(1+a^3)^{\frac{1}{3}}} + c$$

$$\therefore 3\ (1+a^3)^{\frac{1}{3}} \cdot z^{2/3} = 4(x+ay) + c \qquad (2)$$

This is the complete solution

Standard type - IV

Equation of the form $f_1(x,p) = f_2(y,q)$

In this type of equation z is absent. Also the terms containing p and x can be separated from those containing q and y

let $\quad f_1(x_1p) = f_2(x,q) = k \qquad (1)$

(i.e.,) $\quad f_1(x,p) = k$, solving for p we get $p = f_1(x)$

$f_2(y,q) = k$, solving for q we get $q = f_2(y)$

Also $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial x} dy.$

$\therefore$ $dz = pdx + qdy$

$= f_2(x)dx+f_2(y)dy$

Integrating, $\int dz = \int f_1(x)\, dx + \int f_2(y)\, dy$

$\therefore$ $z = \int f_1(x)\, dx + \int f_2(y)\, dy + c$

This is the complete solution

Example 1

Solve $p + q = x+y$

Solution

Solve $p + q = x+y$ (1)

This equation of the form $f_1(x,p) = f_2(y,q)$

We can write $p - x = y - q$

let $p-x = k \Rightarrow p = x+k$

$y-q = k \Rightarrow q = y-k$

Then

$dz = pdx + qdy$

$dz = (x+k)\, dx + (y-k)\, dy$

Integrating,

$$z = \frac{(x+k)^2}{2} + \frac{(y-k)^2}{2} + c$$

This is the required complete solution of (1).

There is no singular and general integral is found as usual.

Example 2

Solve p + q = sin x+siny

Solution

Given p + q = sin x+siny

=> p -sinx = siny-q

This is the equation of the form $f_1(x,p) = f_2(y,q)$

let p-sinx = k => p = k+sinx

siny-q = k => q = siny-k

Then

dz = pdx + qdy

dz = (k+sinx) dx + (siny-k) dy

Integrating,

z = (kx-cosx) - (ky+cosy) + c

z = k(x-y) - (cosx+cosy) + c

∴ which is the complete solution.

Example 3

Solve $p^2 + q^2 = x+y$

Solution

Given $p^2 + q^2 = x+y$

$p^2 - x = y - q^2$

Equation of the form $f_1(x,p) = f_2(y,q)$

let $p^2 - x = k \Rightarrow p = \pm\sqrt{x+k}$

$y - q^2 = k \Rightarrow q = \pm\sqrt{y+k}$

Then

$$dz = pdx + qdy$$

$$= \pm\left(\sqrt{x+k}\right)dx \quad \pm\left(\sqrt{y+k}\right)dy$$

Integrating,

$$z = \pm\frac{2}{3}(x+k)^{3/2} \pm \frac{2}{3}(y+k)^{3/2} + c$$

$$= \pm\frac{2}{3}\left[(x+k)^{3/2} + (y+k)^{3/2}\right] + c$$

Example 4

Solve $p - x^2 = q+y^2$

Solution

Given $p - x^2 = q+y^2$

Equation of the form $f_1(x,p) = f_2(y,q)$

let $p^2 - x = q + y^2 = k$

$p = k + x^2$; $q = k - y^2$

Then

$$dz = pdx + qdy$$

$$= (k+x^2)\, dx + (k-y^2)dy$$

Integrating,

$$z = kx+\frac{x^3}{3}+ky-\frac{y^3}{3}+b$$

$$\therefore \quad z = kx+ky+\frac{x^3}{3}-\frac{y^3}{3}+b$$

which is a complete integral

Check Your Progress - III

1. A solution obtained by given particular values to the arbitary constant in a complete integral is called a ________________

2. The solution of $x+y\frac{\partial z}{\partial x} = 0$ is ____________

Check Your Answers - III

1. Particular Integral

2. $z = \frac{x^2}{\partial y}+f(x)$

5.5 Recap

1. Partial Differential equation

Notation of PDE : $p = \frac{\partial z}{\partial x}$; $q = \frac{\partial z}{\partial y}$; $r = \frac{\partial^2 z}{\partial x^2}$; $S = \frac{\partial^2 z}{\partial x \partial y}$; $t = \frac{\partial^2 z}{\partial y^2}$

2. PDE can be obtained by the elimination of

   (i) Arbitrary constant from a relation between x,y,z

   (ii) Arbitrary function of these variables.

3. Different solution of PDE - Standard types

   type - I, $F(p,q) = 0$

   type - II $z=px+qy+f(p,q)$ (Clairaut's form)

   type - III $F(z,p,q) = 0$

   type - IV $f_1(x,p) = f_2(y,q)$

5.6 Exercise

1. Obtain the Complete solution and singular solution of the following PDE

   (i) Solve $p^2+q^2=4$

   (ii) Solve $p^2+q^2=npq$

   (iii) Solve $pq+p+q=0$

   (iv) Solve $z = px+qy+p^2+pq+q^2$

   (v) Solve $z = px+qy+2\sqrt{pq}$

   (vi) Solve $z = px+qy+\log pq$

   (vii) Solve $z = px+qy+p^2+q^2$

   (viii) Solve for complete soluton for $z^2 (p^2z^2+q^2) = 1$

   (ix) Solve $p^3+q^3=8z$

   (x) Solve $pq=z$

   (xi) Solve $pz=1+q^2$

   (xii) Solve $p-q = x^2+y^2$

(xiii) Solve p+q = px+qy

(xiv) Solve $\frac{x}{p}+\frac{y}{q}=1$

(xv) Solve $xp-y^2q^2=1$

5.7 Check Your Answer

I. (i) $z = ax \pm y\sqrt{4-a^2}+c$

(ii) $z = ax + \frac{ay}{2}\left[\sqrt{n}+\sqrt{n^2-4}\right]+1$

(iii) $z = -ax - \frac{a}{a+1}y + c$

(iv) $z = \left(\frac{2x-y}{3}\right)x + \left(\frac{2x-x}{3}\right)y + \left(\frac{2x-y}{3}\right)^2 + \frac{(2x-y)(2y-x)}{9} + \left(\frac{2x-x}{3}\right)^2$

(v) $z = ax+by+2\sqrt{ab}$; $xy = 1$

(vi) z = ax + by + log ab ; z+2 + logxy= 0

(vii) $z = ax^2+by^2+\sqrt{ab}$; xy = 1 (put $x^2=x$; $y^2=y$)

(ix) $3(1+a^3)^{1/3}.z^{2/3} = 4(x+ay)+b$

(x) $4az = (x+ay+b)^2$

(xi) $z \pm \left[z\sqrt{z^2-4a^2}-4a^2\log\left(z+\sqrt{z^2-4a^2}\right)\right] = 4(x+ay) + b$

(xii) $z = \frac{-x^2}{4} \pm \frac{1}{2}\left[\frac{k^2}{2}\sinh^{-1}\frac{x}{k} + \frac{x\sqrt{x^2+k^2}}{2}\right] + \frac{k^2 y}{4} + b$

(xiii) $z = a \log\left(\frac{y-1}{x-1}\right) + b$

(xiv) $2z = \frac{x^2}{a} + \frac{y^2}{1-a} + b$

(xv) $z = a \log x \pm \sqrt{a-1}\,\log y + b$

Lesson - 6

LAGRANGE'S METHOD

Learning Objectives

Our aim is to study about

- Definition of Lagranges' equation
- Lagranges' auxiliary equation
- Lagrange's partial differential equations and its solutions.

Structure

6.1 Introduction

6.2 Lagrange's Linear Partial Differential Equations

6.3 Methods of Solution

6.4 Recap

6.5 Exercises

6.1 Introduction

In this lesson, we shall study partial differential equation of the first order and of the first degree in the derivatives. Equations containing only three variables will be studied. It will be noted that the method of solution presented here can be applied to equations involving any number of variables.

6.2 Lagrange's Linear Partial Differential Equations

The partial differential equation of the form $P_p+Q_q = R$ where P, Q, R are function of x,y,z and $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$ is called Lagrange's Linear differential equation.

Procedure to solving $P_p+Q_q = R$

Step - 1 : form the auxiliary equation $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$.

Step - 2 : Find two independent solutions of the auxiliary equations, say $u = c_1$ and $v = c_2$

Step - 3 : Then the solution of the given PDE is $\phi(u,v) = 0$ (or) $u = \phi(v)$.

Example 1

Solve $x^2p + y^2q = z^2$

Solution

Given $x^2p + y^2q = z^2$

This is a Lagrange's linear equation of the form $P_p+Q_q = R$, Here $P = x^2$, $Q = y^2$, $R = z^2$,

The auxiliary equation are $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$

$$\Rightarrow \frac{dx}{x^2}=\frac{dy}{y^2}=\frac{dz}{z^2}$$

Consider,

$$\frac{dx}{x^2}=\frac{dy}{y^2}$$

Integrating

$$\frac{1}{x}-\frac{1}{y}=c$$

Consider

$$\frac{dy}{y^2}=\frac{dz}{z^2}$$

Integrating

$$\frac{1}{y}-\frac{1}{z}=c^2$$

The general solution is $\phi\left(\frac{1}{x}-\frac{1}{y},\frac{1}{y}-\frac{1}{z}\right) = 0$

Example 2

Solve (y+z)p + (z+x) q = x+y

Solution

Given (y+z)p + (z+x) q = x+y

This is a Lagrange's linear equation of the form $P_p+Q_q = R$,

Here p = y + z, Q = z+x, R = x+y

The auxiliary equation are $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$

$$= \frac{dx}{y+z}=\frac{dy}{z+x}=\frac{dz}{x+y}$$

(i.e.,) $\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z} = \frac{dz-dx}{z-x} = \frac{dx+dy+xz}{2(x+y+z)}$

Consider first two members and Integrating we get,

$$\int\frac{d(x-y)}{x-y} = \int\frac{d(y-z)}{y-z}$$

$\log (x-y) = \log(y-z) + \log c_1$

$\log\left(\frac{x-y}{y-z}\right) = \log c_1$

$$\Rightarrow \frac{x-y}{y-z} = c_1 \qquad (1)$$

Consider first and last members and integrating we get,

$$\int \frac{d(x-y)}{x-y} = \int \frac{d(x+y+z)}{2(x+y+z)}$$

$$\log(x-y) = \frac{1}{2}\log(x+y+z) + \log c_2$$

$$\therefore \log \frac{(x-y)^2}{x+y+z} = \log C_2$$

$$= \frac{(x-y)^2}{x+y+z} = C_2$$

$$\therefore \text{ The general solution is } \phi\left[\frac{x-y}{y-z}, \frac{(x-y)^2}{x+y+z}\right] = 0$$

Example 3

Solve x(y-z)p+y(z-x)q = z(x-y)

Solution

Given x(y-z)p+y(z-x)q = z(x-y)

This is a Lagrange's equation of the form $P_p+Q_q = R$,

Here p = x(y - z) ; Q = y(z-x) ; R = z(x-y)

The auxiliary equation are $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{xy-xz} = \frac{dy}{yz-yx} = \frac{dz}{zx-zy} = \frac{dx+dy+dz}{0}$$

$= > dx + dy + dz = 0$

On Integrating we get, $x+y+z = C_1$ (1)

Consider,

$$\frac{\frac{dx}{x}}{y-z} = \frac{\frac{dy}{y}}{z-x} = \frac{\frac{dz}{z}}{x-y} = \frac{\frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z}}{0}$$

$$= > \frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z} = 0$$

Integrating, $\log x + \log y + \log z = \log c_2$

$\therefore xyz = c_2$

The solution is $\phi\,[x+y+z,\ xyz] = 0$

Example 4

Solve $(mz-ny)p - (nx-lz) = ly - mx$

Solution

Given $(mz-ny)p - (nx-lz) = ly - mx$

This is a Lagrange's equation of the form $P_p+Q_q = R$,

Here $p = mz-ny$; $Q = -(nx-lz)$; $R = ly - mx$

The auxiliary equation are $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$= \frac{dx}{mz-ny} = \frac{dy}{lz-nx} = \frac{dz}{ly-mx}$$

using multiplier's x, y, z we get each ratio

$$\frac{xdx+ydy+zdz}{x(mz-ny)+y(lz-nx)+z(ly-mx)} = \frac{xdx+ydy+zdz}{0}$$

$\therefore$ $x^2+y^2+z^2 = C_1$ is one solution

Also, using multiplier's l,m,n we get, each ratio,

$$\frac{ldx + mdy + ndz}{l(mz - ny) + m(lz - nx) + n(ly - mx)} = \frac{ldx + mdy + ndz}{0}$$

$\therefore$ $lx+y+nz = C_2$ is another solution

$\therefore$ The genreal solution is $\phi\ [x^2+y^2+z^2,\ lx+y+nz) = 0$

Example 5

Solve $(x^2-yz)\ p + (y^2-zx)q = z^2-xy$

Solution

This is a Lagrange's equation of the form $P_p+Q_q = R$,

Here $p = x^2 - yz$; $Q = y^2-zx$; $R = z^2-xy$

The auxiliary equation are $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$=> \frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy}$$

This each ratio is equal to

$$\frac{dx - dy}{(x + y + z)(x - y)} = \frac{dy - dz}{(x + y + z)(y - z)} = \frac{dz - dx}{(x + y + z)(z - x)}$$

$$\frac{d(x - y)}{x - y} = \frac{d(y - z)}{y - z} = \frac{d(z - x)}{z - x}$$

$$\therefore \qquad \frac{d(x - y)}{x - y} = \frac{d(y - z)}{y - z}$$

Integrating,

$$\frac{x-y}{y-z} = C_1$$

Also using x,y,z each of the subsidiary equation

$$\frac{xdx+ydy+zdz}{x^3+y^3+z^3-3xyz} = \frac{xdx+ydy+zdz}{(x+y+z)(x^2+y^2+z^2-xy-yz-zx)}$$

and is also equal to, $\dfrac{xdx+ydy+zdz}{x^2+y^2+z^2-yz-zx-xy}$

$$\therefore \frac{xdx+ydy+zdz}{x+y+z} = \frac{dx+dy+dz}{1}$$

xdx + ydy + zdz = (x+y+z) d (x+y+z)

Integrating

$$x^2+y^2+z^2 = (x+y+z)^2$$

$$\therefore xy + yz + zx = C_2$$

The Solution is

$$\phi\left[\frac{x-y}{y-z}, xy+yz+zx\right] = 0$$

Example 6

Solve $x(y^2+z)\, p - y\, (x^2+z) = z\, (x^2-y^2)$.

Solution

This is a Lagrange's equation of the form $P_p+Q_q = R$,

Here $P = x(y^2+z)$; $Q = -y\,(x^2+z)$; $R= z\,(x^2-y^2)$.

The auxiliary equation are $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$=> \frac{dx}{x(y^2+z)} = \frac{dy}{-y(x^2+z)} = \frac{dz}{z(x^2-y^2)}$$

$$\frac{\frac{dx}{x}}{y^2+z} = \frac{\frac{dy}{y}}{-(x^2+z)} = \frac{\frac{dz}{z}}{x^2-y^2} = \frac{\frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z}}{0}$$

$\therefore \frac{dx}{x}+\frac{dy}{y}+\frac{dz}{z} = 0$

Integrating

$xyz = C_1$

Also multiplying, x,y, -1 each ratio is equal to

$$\frac{xdx+ydy-dz}{x^2(y^2+z)-y^2(x^2+z)-z(x^2-y^2)}$$

$$= > \frac{xdx+ydy+zdz}{0} = 0$$

$\therefore$ xdx + ydy - dz = 0

Integrating

$x^2 + y^2 - 2z = C_2$

The general solution is $\phi\left[xyz,\ x^2+y^2-2x\right] = 0$

6.3 Methods of solutions of $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

6.3.1 Method of grouping

Consider the auxiliary equaiton $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

Take any two function say first two fractions or last two fractions or first and last fraction Suppose that one of the variables is either absent or cancels out from any two fractions of the given equaitons. Then an integral can be obtained by using the usual methods. The same procedure can be repeated with another set of two fractions of given equations.

6.3.2 Method of Multipliers

Choose the multipliers l, m and n such that lP + mQ + nR = 0, l.m and n are constants or functions of x, y and z

We have $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{ldx + mdy + ndz}{lP + mQ + nR}$

when lP + mQ + nR = 0, ldx + mdy + ndz = 0. Here ldx + mdy + ndz = 0 is an exact differential equation on integration we get u (x,y,z) = C_1 Similarly by selecting another set of multipliers we get v (x,y,z) = C_2

Problem 1

Solve px + qy = x

Solution

This is of Lagrange's type of partial differential equation where

P =x, Q =y, R = x

The auxiliary equaitons are

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{x}$$

Taking the first two fractions

$$\frac{dx}{x} = \frac{dy}{y}$$

Integrating, log x = log y + log c_1

i.e., $\frac{x}{y} = \log C_1$

i.e., $\frac{x}{y} = C_1$

Taking first and last fraction, $\frac{dx}{x} = \frac{dz}{x}$

i.e., dx = dz

Integrating, x = z + C_2

x - z = C_2

Hence the general solution of the equation is

$$f\left(\frac{x}{y}, x - z\right) = 0$$

Problem 2

Solve p tanx + q tan y = tanz

Solution

This is a Lagrange's type of P.D.E where

P = tax, Q = tany, R = tanz

The auxiliary equations are

$$\frac{dx}{\tan x} = \frac{dy}{\tan y} = \frac{dz}{\tan z}$$

Taking the first two equations, we get

$$\frac{dx}{\tan x} = \frac{dy}{\tan y}$$

i.e., cot x dx = cot y dy

Integrating, log sinx = log siny + log c

$$\log \frac{\sin x}{\sin y} = \log c_1$$

$$\therefore \frac{\sin x}{\sin y} = c_1$$

Taking the last two equations, we get

$$\frac{dy}{\tan y} = \frac{dz}{\tan z}$$

i.e., cot y dy = cot z dz

Integrating, log sin y = log sinz + log C_2

$$\log \frac{\sin y}{\sin z} = \log c_2$$

$$\frac{\sin y}{\sin z} = c_2$$

$\therefore$ The solution of the given equation is $f\left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z}\right) = 0$

Problem 3

Solve $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$

Solution

Here $P = \sqrt{x}, Q = \sqrt{y}, R = \sqrt{z}$

The Lagrange'sauxiliary equations are

$$\frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$$

consider $\frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}}$

i.e. $\frac{dx}{2\sqrt{x}} = \frac{dy}{2\sqrt{y}}$

Integrating, $\sqrt{x} = \sqrt{y} + c_1$

i.e. $\sqrt{x} - \sqrt{y} = c_1$

Taking $\frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$

$$\frac{dy}{2\sqrt{y}} = \frac{dz}{2\sqrt{z}}$$

Integrating, $\sqrt{y} = \sqrt{z} + c_2$

$$\sqrt{y} - \sqrt{z} = c_2$$

$\therefore$ The solution of the given PDE is

$$f\left(\sqrt{x} - \sqrt{y}, \sqrt{y} - \sqrt{z}\right) = 0$$

Problem 4

Solve p - q = log (x +y)

Solution

Here P =1, Q = -1, R = log (x+y)

The Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{-1} = \frac{dz}{\log(x+y)}$$

Consider the first two equations

$$\frac{dx}{1} = \frac{dy}{-1}$$

i.e., $dx = -dy$

Integrating, $x = -y + C_1$

$$x + y = C_1 \qquad \text{.............} (1)$$

Taking last two fractions

$$\frac{dy}{-1} = \frac{dz}{\log(x+y)}$$

$$-dy = \frac{dz}{\log(c_1)} \qquad \text{by (1)}$$

Integrating, $-y = \dfrac{z}{\log c_1} + c_2$

$$-y - \frac{z}{\log c_1} = c_2$$

i.e. $-y - \dfrac{z}{\log(x+y)} = c_2$

$\therefore$ The solution of the given equation is

$$f\left(x+y, -y - \frac{z}{\log(x+y)}\right) = 0$$

Problem 5

Solve p + q = x + y + z

Solution

Lagrange's auxiliary equations are

$$\frac{dx}{1} = \frac{dy}{1} = \frac{dz}{x+y+z}$$

Taking the first two fractions, we get

$$\frac{dx}{1} = \frac{dy}{1}$$

$\Rightarrow$ dx = dy

Integrating $x = y + c_1$ .... (1)

Taking the last two fractions

$$dy = \frac{dz}{x+y+z}$$

i.e., $dy = \frac{dz}{c_1+y+y+z}$ by (1)

$$= \frac{dz}{c_1+2y+z}$$

i.e., $\frac{dz}{dy} = c_1 + 2y + z$

i.e., $\frac{dz}{dy} - z = c_1 + 2y$, which is linear in z.

Its Integrating Factor, $e^{\int pdy} = e^{-y}$

and its solution is

$$ze^{-y} = \int (c_1 + 2y)e^{-y}dy + c_2$$

$$= (c_1 + 2y)(-e^{-y}) - \int -e^{-y}2dy + c_2$$

$$= -e^{-y}(c_1 + 2y) + 2(-e^{-y}) + c_2$$

$$= -e^{-y}(c_1 + 2y + 2) + c_2$$

$= -e^{-y}(x - y + 2y + 2) + c_2$ by (1)

$$= -e^{-y}(x + y + 2) + c_2$$

$\therefore$ $(x+y+z+2)e^{-y} = c_2$

Hence the required solution is

$$f\left[x-y,\ (x + y + z + 2)e^{-y}\right] = 0$$

Problem 6

Solve (y+z) p + (z+x) q = x + y

Solution

This is a Langrange's type of partial differential equation where P = y + z

Q = z +x, R = x + y

The Lagrange's auxiliary equations are

$$\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y}$$

Also each ratio = $\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z} = \frac{dz-dx}{-(z-x)}$ = $\frac{dx+dy+dz}{2(x+y+z)}$

Choosing first two terms, we have

$$\frac{dx-dy}{x-y} = \frac{dy-dz}{y-z}$$

Integrating, log (x-y) = log (y-z) + log k

i.e., $\frac{x-y}{y-z} = c_1$

Consider the third and fourth ratio, we have

$$\frac{dz-dx}{-(z-x)} = \frac{dx+dy+dz}{2(x+y+z)}$$

Integrating, -log (z-x) = $\frac{1}{2}\log(x+y+z) + \log m$

i.e., $(z-x)^2 (x + y + z) = c_2$

Hence the required solution is

$$f\left(\frac{x-y}{y-z},\ (z-x)^2(x+y+z)\right) = 0$$

Problem 7

Solve $(x^2 - yz)\, p + (y^2 - zx)\, q = (z^2 - xy)$

Solution

The Lagrange's auxiliary equations are

$$\frac{dx}{x^2 - yz} = \frac{dy}{y^2 - zx} = \frac{dz}{z^2 - xy} \qquad \text{.............} \ (1)$$

Each of the ratios are equal to

$$\frac{dx - dy}{(x^2 - yz) - (y^2 - zx)} = \frac{dy - dz}{(y^2 - zx) - (z^2 - xy)} = \frac{dz - dx}{(z^2 - xy) - (x^2 - yz)}$$

i.e., $$.\ \frac{d(x - y)}{(x - y)(x + y + z)} = \frac{d(y - z)}{(y - z)(x + y + z)} = \frac{d(z - x)}{(z - x)(x + y + z)}$$

i.e., $$\frac{d(x - y)}{x - y} = \frac{d(y - z)}{y - z} = \frac{d(z - x)}{z - x}$$

From these ratios, we get $\dfrac{d(x - y)}{x - y} = \dfrac{d(y - z)}{(y - z)};\ \ \dfrac{d(y - z)}{y - z} = \dfrac{d(z - x)}{z - x}$

which on integration gives,

$$\log (x-y) = \log (y-z) + \log c_1$$

and $$\log (y-z) = \log (z-x) + \log c_2$$

i.e., $$\frac{x - y}{y - z} = c_1 \text{ and } \frac{y - z}{z - x} = c_2$$

These two solutions are not independent in nature. Because

$$\frac{x - y}{y - z} + 1 = \frac{x - z}{y - z} = \left(\frac{y - z}{z - x}\right)^{-1}$$

$$\Rightarrow \frac{y - z}{z - x} \text{ is not independent of } \frac{x - y}{y - z}$$

Therefore, we are to get another solution

Each ratio of (1)

$$= \frac{xdx + ydy + zdz}{x(x^2 - yz) + y(y^2 - zx) + z(z^2 - xy)} = \frac{dx + dy + dz}{x^2 - yz + y^2 - zx + z^2 - xy}$$

i.e., $$\frac{xdx + ydy + zdz}{x^3 + y^3 + z^3 - 3xyz} = \frac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx}$$

i.e., $$\frac{xdx + ydy + zdz}{(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)} = \frac{dx + dy + dz}{x^2 + y^2 + z^2 - xy - yz - zx}$$

$$\frac{xdx + ydy + zdz}{x + y + z} = \frac{dx + dy + dz}{1}$$

i.e., $xdx + ydy + zdz = (x+y+z)\,(dx + dy + dz)$

Integrating, $\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = \frac{(x + y + z)^2}{2} + c$

i.e., $x^2 + y^2 + z^2 = (x + y + z)^2 + c$

i.e., $x^2 + y^2 + z^2 - (x + y + z)^2 = c$

i.e., $-2xy - 2yz - 2zx = c_2$

i.e., $xy + yz + zx = c_2$

Hence the required solution is

$$f\left(\frac{x - y}{y - z}, xy + yz + zx\right) = 0$$

Problem 8

Solve $(x^2 - y^2 - z^2)\, p + 2xyq - 2zx = 0$

Solution

This is a Lagrange's type of partial differentiual equation

where $P = x^2 - y^2 - z^2$, $Q = 2xy$, $R = -2zx$

The Lagrange's auxiliary equaitons are

$$\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2zx}$$

Consider the last two fractions

$$\frac{dy}{2xy} = \frac{dz}{2xz}$$

i.e. $$\frac{dy}{y} = \frac{dz}{z}$$

Integrating, log y = log z + log c_1

i.e., $\log \frac{y}{z} = \log c_1$

i.e., $\frac{y}{z} = c_1$

Choose the multipliers x, y, z

$$\frac{dx}{x^2 - y^2 - z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} = \frac{xdx + ydy + zdz}{z\left(x^2 - y^2 - z^2\right) + 2xy^2 + 2xz^2}$$

Consider $\frac{dy}{2xy} = \frac{xdx + ydy + zdz}{x^3 + xy^2 + xz^z}$

$$\frac{dy}{2xy} = \frac{xdx + ydy + zdz}{x\left(x^2 + y^2 + z^2\right)}$$

$$\frac{dy}{y} = \frac{2xdx + 2ydy + 2zdz}{x^2 + y^2 + z^2}$$

Integrating, log y = log (x^2 +y^2+ z^2) + log c_2

Therefore, $\frac{y}{x^2 + y^2 + z^2} = c_2$

Hence the required solution is

$$f\left(\frac{y}{z}, \frac{y}{x^2 + y^2 + z^2}\right) = 0$$

Check Your Progress

1. The Lagranges' equation is

   ....

2. The Lagrange's auxiliary equations are $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$. True or False.

   ....

3. Solution of px + qy = x is

   ....

4. The Lagrange's auxiliary equation of p + q = x + y + z are

   ....

6.5 Recap

1. Lagrange's partial Differential Equation : Pp + Qq = R

   Its auxiliary equations are $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$

2. Use $\frac{a}{u}=\frac{b}{v}=\frac{c}{w}=\frac{la+mb+nc}{lu+mv+nw}$

3. Lagrange's Linear PDE : P_p+Q_q = R

   It's Auxiliary eqution = $\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$

Check Your Answers

I. 1. Pp + Qq = R

2. True

3. $f\left(\frac{x}{y}, x-z\right)=0$

4. $\frac{dx}{1}=\frac{dy}{1}=\frac{dz}{x+y+z}$

6.6 Exercises

I. Solve the following equations

1. $p \cot x + q \cot y = \cot z$

2. $y^2 p - xyq = x(z - 2y)$

3. $xp + yq = z$

4. $p + q = 1$

5. $x^2 p + y^2 q = z^2$

6. $y^2 zp + zx^2 q = xy^2$

7. $xzp + yzq = xy$

8. $\frac{y-z}{yz} p + \frac{z-x}{zx} q = \frac{x-y}{xy}$ (or) $x(y-z)p + y(z-x)q = z(x-y)$

9. $(3z - 4y)p + (4x - 2z)q = 2y - 3x$

II. Solve the following differential equations

(i) $(3z-4y)p + (4x-2z)q = 2y-3x$

(ii) $y^2 zp + x^2 zq = y^2 x$

(iii) $(y+z)p + (z+x)q = x+y$

(iv) $(x^2-yz)p + (y^2-zx)q = z^2-xy$

(v) $x^2 p + y^2 q = 4xz$

(vi) $z(x+y)p + z(x-y)q = x^2+y^2$

(vii) $xp + yq = 2z$

(viii) $x(z^2-y^2)p + y(x^2-z^2)q = z(y^2-x^2)$

Check Your Answers

(i) $\phi(2x+3y+4z,\ x^2+y^2+z^2) = 0$

(ii) $\phi(x^3-y^3,\ x^2-z^2) = 0$

(iii) $\phi(x+y+z)\ (y-x)^2 = 0$

(iv) $\phi\left[\frac{1}{x}+\frac{1}{y}+\frac{1}{z}, xyz\right] = 0$

(v) $\phi\left[\frac{1}{y}-\frac{1}{x}, \frac{x^4}{z}\right] = 0$

(vi) $\phi(x^2-y^2-z^2,\ 2xy-z^2) = 0$

(vii) $\phi\left(\frac{x}{y}, \frac{y^2}{z}\right) = 0$

(viii) $\phi[x^2+y^2+z^2,\ xyz] = 0$

LESSON - 7
LAPLACE TRANSFORM

Learning Objectives

After studying this lesson you should be able to

- Define Laplace Transform
- State basic properties of Laplace Transform and prove them
- Find Laplace Transform of standard functions.
- Find Laplace transforms of given function by using properties and Laplace Transforms Table
- Find Laplace Transforms of given function by using the general theorems

Structure

7.1 Introduction

The Laplace Transform was first introduced by Pierre Laplace in 1779 in his research on probability. G. Doetsch devloped the use of Laplace Transform to solve differential equation. His work in 1930s served to justify the operational calculus used by Oliver Heaviside the importance of Laplace Transform lies in converting a diferential equation into an algebraic equation.

The Laplace transform is a powerful mathematical technque method useful to the engineers and scientists, as it enabled them to solve linear differential equation with given initial conditions by using algebraic method. The Laplace Transform technique can also be used to solve systems of differential eautations, PDE and Integral equations.

Starting with the definition of Laplace transform we shall diserves below the properties of Laplace Transform and derive theh transforms of some functions which usually occur in the solution of linear differential equation.

7.2 Definition of Laplace Transform

Let f (t) be a real valued function of the variable t defined for t > 0. Then the Laplace Transform of f (t) defined by

$$L\,[f\,(t)] = \int_0^\infty e^{-st} f(t)\,dt$$

provided the integral exists.

Clearly the integral is a function of the parameter's and f(t) is denoted by the symbol $\bar{f}$ (s) (or) F(s) unless we have to deal with the Laplace transforms of more then one function, we shall denote the laplace transform of f(t) as $\phi(s)$.

$$\text{i.e., } \bar{f}\,(s) = \int_0^\infty e^{-st} f(t)\,dt = \phi(s)$$

The operation of multiplying f(t) by e^{-st} and integrating te product with respect to 't' between 0 and ∞ is called Laplace transformation.

Note:

1. The parameter s used in the definition of Laplace transform is a real or complex number, but we shall assume it to be a real positive number sufficiently large to ensure the existence of the integral that defines th eLaplace transform.

2. Laplace transforms of all function do not exists. For example, L (tant) and L $L\left(e^{t^2}\right)$ do not exists.

   We give below th esufficient conditions (without proof) for the existence of Laplace transform of a function f(t)

Conditions for the existence of Laplace transform. If the function f(t) defined for $t \geq 0$ is

(i) Piecewise continious in every finite interval in the range $t \geq 0$ and

(ii) of the exponential order, then L [f(t)] exists.

Note : 2

1. A function f(t) is said to be piecewise continious in the finite interval $a \leq t \leq b$, if the interval can be divided into a finite number of sub intervals such that

   (i) f(t) is continious at every point inside each of the sub-intervals and

   (ii) f(t) has finite limits as t approaches the and points of each sub-intervals from the interior of the sub intervals.

2. A function f(t) is said to be the exponential order, if $|f(t)|$ = < $Me^{\alpha t}$, for all $t \geq 0$ and some constant M and α or equivalently, if

   $\lim_{t \to \infty} \{e^{-\alpha t} f(t)\} = a$ finite quantity.

Most the function that represent physical quantitites and that we encounter in differential equations satisfy the condition stated above and hence may be assumed to have Laplace transform.

7.3 Laplace transform of elementary functions

1. Laplase transform of 1

$$L[1] = \int_0^\infty e^{-st}, dt$$

$$= \left[\frac{e^{-st}}{-s}\right]_0^\infty$$

$$\therefore L[1] = \frac{1}{s}, \text{ if } s > 0$$

2. Laplase transform of t

$$L[t^2] = \int_0^\infty e^{-st}\, t\, dt$$

$$= \left[\frac{e^{-st}}{-s}.t\right]_0^\infty - \int_0^\infty \frac{e^{-st}}{-s}\, dt$$

$$= 0 + \left[\frac{e^{-st}}{-s}\right]_0^\infty$$

$$\therefore L(t) = \frac{1}{s^2}, \text{if } s > 0$$

3. Laplase transform of t^2

$$L[t^2] = \int_0^\infty e^{-st}\, t^2\, dt$$

$$= \left[\frac{e^{-st}}{-s}.t^2\right]_0^\infty - \int_0^\infty \frac{e^{-st}}{-s}.\, 2t\, dt$$

$$= 0 + \frac{2}{s}\int_0^\infty e^{-st}.tdt, \text{if } s > 0$$

$$= \frac{2}{s}.\frac{1}{s^2} = \frac{\lfloor 2}{s^3}$$

$$L[t^2] = \frac{\lfloor 2}{s^3}$$

Similarly,

$$L[t^3] = \int_0^\infty e^{-st}\, t^3\, dt$$

$$= \left[\frac{e^{-st}}{-s}t^3\right]_0^\infty - \int_0^\infty \frac{e^{-st}}{-s} 3t^2 dt$$

$$= \quad 0 + \frac{3}{s}\int_0^\infty e^{-st}t^2 dt \quad s > 0$$

$$= \quad \frac{3}{s}\cdot\frac{2}{s^3} = \frac{\lfloor 3}{s^4} \text{ and so on.}$$

$$L(t^n) \quad = \quad \frac{\lfloor n}{S^4}$$

Similarly, $L(t^n) = \dfrac{\lfloor n}{S^{n+1}}$

4. Laplace transform of e^{-at}

$$L[e^{-at}] \quad = \int_0^\infty e^{-st}e^{-at}dt$$

$$= \int_0^\infty e^{-(s+a)t}dt$$

$$= \left[\frac{e^{-(s+a)t}}{-(s+a)}\right]_0^\infty$$

$$= \frac{1}{s+a}, \text{if } s > -a$$

Note

Also the Laplace Transform of e^{at}

$L(e^{at}) = \dfrac{1}{s-a}$,if $s > a$

5. Laplace transform of sinat

$$L[\text{sinat}] \quad = \quad \int_0^\infty e^{-st}\sin at dt$$

We know that

$$\int_0^\infty e^{-st}\sin bx dt \quad = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos x) + k$$

$$\therefore L\,(\text{sinat}) = \left[\frac{e^{-st}}{s^2+a^2}(-s\,\text{sinat - a cosat}\right]_0^\infty$$

$$= \frac{a}{s^2+a^2}\text{ , if s > 0}$$

5. Laplace transform of cosat

$$L\,[\text{cosat}] = \int_0^\infty e^{-st}\cos atdt$$

$$= \left[\frac{e^{-st}}{s^2+a^2}(-s\ \text{cosat +a sinat})\right]_0^\infty$$

We know that

$$\int_0^\infty e^{-st}\cos bxdx = \frac{e^{ax}}{a^2+b^2}\ \text{(acosbx - bsinbx) +k}$$

$$\therefore L(\text{cosat}) = \frac{S}{s^2+a^2}\text{ , if s > 0}$$

7. Laplase transform of sinhat

$$L\,[\text{sinhat}] = \int_0^\infty e^{-st}\sinh atdt$$

$$= \int_0^\infty e^{-st}\frac{e^{at}-e^{-at}}{2}dt$$

$$= \frac{1}{2}\int_0^\infty e^{-(s-a)t}dt - \frac{1}{2}\int_0^\infty e^{-(s+a)t}dt$$

$$= \frac{1}{2(s-a)} - \frac{1}{2(s+a)} \text{ ; if } s > a$$

$$= \frac{s+a-s+a}{2(s^2-a^2)}$$

$$\therefore L(\sinh at) = \frac{a}{s^2-a^2}$$

8. Laplace transform of coshat

$$[\cosh at] = \int_0^\infty e^{-st} \cosh at\,dt$$

$$= \int_0^\infty e^{-st}\left(\frac{e^{at}+e^{-at}}{2}\right)dt$$

$$= \frac{1}{2}\int_0^\infty e^{-(s-a)t}dt + \frac{1}{2}\int_0^\infty e^{-(s+a)t}dt$$

$$= \frac{1}{2}.\frac{1}{s-a} + \frac{1}{2(s+a)}$$

$$\therefore L(\cosh at) = \frac{s}{s^2-a^2}, \text{if } s > |a|$$

7.4 Some Important Properties of Laplace Transforms

1. Linearity Property

If $f_1(t)$ and $f_2(t)$ are two functions of t defined for positive values of t, and if c is a constant, then

$$L[f_1(t) + f_2(t)] = L[f_1(t)] + L[f_2(t)] \text{ and}$$

$$L[c(t)] = cL[f(t)]$$

Proof

$$L[f_1(t) + f_2(t)] = \int_0^\infty e^{-st}\left[f_1(t) + f_2(t)\right]dt$$

$$= \int_0^\infty e^{-st} f_1(t)dt + \int_0^\infty e^{-st} f_2(t)dt$$

$$= L[f_1(t)] + L[f_2(t)]$$

and

$$L[cf(t)] = \int_0^\infty e^{-st} c.f(t)dt$$

$$= c\int_0^\infty e^{-st} f(t)dt$$

$$= cL[f(t)]$$

2. Shifting Property

If $L[f(t)] = \bar{f}(s)$ then $L[e^{-at} f(t)] = \bar{f}(s+a)$

Proof

We know that,

$$L[f(t)] = \int_0^\infty e^{-st} f(t)dt = \bar{f}(s)$$

$$L[e^{-st} f(t)] = \int_0^\infty e^{-st} e^{-at} f(t)dt$$

$$= \int_0^\infty e^{-(s+a)t} f(t)\,dt$$

$$= \bar{f}(s+a)$$

Note:

If $L[f(t)] = \bar{f}(s)$

Then, $L[e^{at} f(t)] = \bar{f}(s-a)$

3. Change of Scale Property

If $L[f(t)] = \bar{f}(s)$, then $L[f(at)] = \frac{1}{a}\bar{f}\left(\frac{s}{a}\right)$

Proof

We know that,

$$L[f(t)] = \int_0^\infty e^{-st} f(t)\, dt,$$

$$L[f(at)] = \int_0^\infty e^{-st} f(at)\, dt$$

Put $at = t'$

$adt = dt'$ or $dt = \frac{dt'}{a}$

$$\therefore L[f(at)] = \int_0^\infty e^{-st} f(at)\, dt$$

$$= \int_0^\infty e^{-\frac{s}{a}.t'} f(t') \frac{dt'}{a}$$

$$= \frac{1}{a}\int_0^\infty e^{-\frac{s}{a}t'} f(t')dt'$$

$$= \frac{1}{a}\int_0^\infty e^{-\frac{s}{a}t} f(t)dt$$

$$\therefore L(f(at)) = \frac{1}{a}\bar{f}\left(\frac{s}{a}\right).$$

4. Laplace Transform on derivatives

If L [f(t)] = $\bar{f}$ (s) then

(i) L [f' (t)] = s $\bar{f}$ (s) - f (0)

(ii) L [f'' (t)] = s² $\bar{f}$ (s) -s f(0) - f' (0)

Proof

(i) $L[f(t)] = \int_0^\infty e^{-st} f(t)\,dt$

$$L[f'(t)] = \int_0^\infty e^{-st} f'(t)\,dt$$

$$= \left[e^{-st} f(t)\right]_0^\infty - \int_0^\infty e^{-st}(-s) f(t)\,dt \qquad (\because \text{by parts rule})$$

$$= -f(0) + s\int_0^\infty e^{-st} f(t)\,dt$$

$$\therefore L[f'(t)] = -f(0) + s.\bar{f}(s) \qquad (1)$$

(ii) $L[f''(t)] = \int_0^\infty e^{-st} f''(t)\,dt$

$$= \left[e^{-st} f'(t)\right]_0^\infty - \int_0^\infty e^{-st}(-s) f'(t)\,dt$$

$$= 0 - f'(0) + s\int_0^\infty e^{-st} f'(t)\,dt$$

$= \quad -f'(0) + s[s.\bar{f}(s) - f(0)] \qquad$ by (1)

$\therefore L[f''(t)] \quad = \quad s^2\bar{f}(s) - sf(0) - f'(0)$

Note

In general

$L[f^n(t) = s^n\,\bar{f}(s) - s^{n-1} f(0) - s^{n-2} f'(0) \ldots\ldots sf^{n-2}(0) - f^{n-1}(0)$

5. Laplace Transform of Integrals

If $L[f(t)] = \bar{f}(s)$ then $L\left[\int_0^t f(u)\,du\right] = \frac{\bar{f}(s)}{s}$

Proof

$L[f(t)] \quad = \quad \int_0^\infty e^{-st} f(t)\,dt$ and

Let $g(t) \quad = \quad \int_0^t f(u)\,du$

Then $g(0) = 0$, Also $g'(t) = f(t)$

Taking Laplace Transform on both sides

$L[g'(t)] = L[f(t)]$

(i.e.,) $\quad s\bar{g}(s) - g(0) = \bar{f}(s)$

$s\bar{g}(s) = \bar{f}(s)$

$\therefore \bar{g}(s) = \frac{\bar{f}(s)}{s}$

Therefore $= L\left[\int_0^t f(u)\,du\right] = \frac{\bar{f}(s)}{s}$

7. Multiplication by t

If L [f (t)] = $\bar{f}$ (s) then L [tf (t)] = $-\frac{d}{ds}\bar{f}(s)$

Proof

$$L[f(t)] = \int_0^{\infty} e^{-st} f(t)dt$$

$$\text{i.e., } \bar{f}(s) = \int_0^{\infty} e^{-st} f(t)dt$$

$$\therefore \frac{d}{ds}\bar{f}(s) = \frac{d}{ds}\int_0^{\infty} e^{-st} f(t)dt$$

$$= \int_0^{\infty} \frac{\partial}{\partial s} e^{-st} f(t)\, dt$$

$$= \int_0^{\infty} e^{-st}(-t)\, f(t)\, dt$$

$$= -\int_0^{\infty} e^{-st} t.f(t)\, dt$$

$$= \int_0^{\infty} e^{-st} f(t)dt \qquad (1)$$

$$= -L[t\,f(t)]$$

Diffy, (1) again w.r.t. 's'

$$\frac{d^2}{ds^2}\bar{f}(s) = \frac{-d}{ds}\int_0^\infty e^{-st}(t)f(t)dt$$

$$= \int_0^\infty \frac{\partial}{\partial s} e^{-st} f(t)dt$$

$$= -\int_0^\infty e^{-st}(-t)[tf(t)]dt$$

$$= \int_0^\infty e^{-st}t^2 f(t)dt$$

$$\therefore \quad L\left[t^2 f(t)\right] = (-1)^2 \frac{d^2}{ds^2}\bar{f}(s)$$

Similarly

$$L[t^2 f(t)] = (-1)^3 \frac{d^3}{ds^3}\bar{f}(s)$$

In general,

$$L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}\bar{f}(s)$$

7.5 Simple Problems

Problem 1

Find the Laplace transform of the following functions

(i) t^3 (ii) e^{-4t} (iii) $\cos\frac{t}{2}$ (iv) $\sin 4t$

Solution

(i) $L(t^3) = \frac{3!}{s^4}$

(ii) $L(e^{-4t}) = \frac{1}{s+4}$

(iii) $L\left(\cos\frac{t}{2}\right) = \frac{3}{s^2+\frac{1}{4}} = \frac{4s}{4s^2+1}$

(iv) $L(\sin 4t) = \frac{4}{s^2+16}$

Problem 2

Find $L\left[(7e^{3t} + 6t^4 - 5\sin 6t + 2\cos 3t)\right]$

Solution

$L[7e^{3t} + 6t^4 - 5\sin 6t + 2\cos 3t)$

$$= 7L\left(e^{3t}\right) + 6L(t^4) - 5L(\sin 6t) + L(\cos 3t)\backslash$$

$$= 7.\frac{1}{s-3} + 6.\frac{4!}{s^5} - 5\left(\frac{6}{s^2+36}\right) + 2\left(\frac{s}{s^2+9}\right)$$

$$= \frac{7}{s-3} + \frac{144}{s^5} - \frac{30}{s^2+36} + \frac{25}{3^2+9}$$

Problem 3

Find the Laplace transform of the following functions

(i) $\sin^2 3t$ (ii) $\cos^3 t$ (iii) $\sin^3 2t$ (iv) $\cos^4 t$

Solution

(i) $\sin^2 t = \dfrac{1-\cos 6t}{2}$ $\quad\left(\sin 2A = \dfrac{1-\cos 2A}{2}\right)$

$$L[\sin^2 3t] = \left(\frac{1}{2}-\frac{1}{2}\cos 6t\right)$$

$$= \frac{1}{2}L(1) - \frac{1}{2}L(\cos 6t)$$

$$= \frac{1}{2s} - \frac{3}{2(s^2+36)}$$

(ii) we know that,

$$\cos^3 t = 4\cos^3 t - 3\cos t$$

$$4\cos^3 t = \cos 3t + 3\cos t$$

$$\cos^3 t = \frac{1}{4}\cos 3t + \frac{3}{4}\cos t$$

$$L(\cos^3 t) = \frac{1}{4}L(\cos 3t) + \frac{3}{4}L(\cos t)$$

$$= \frac{1}{4}\frac{s}{s^2+9} + \frac{3s}{4(s^2+1)}$$

$$= \frac{s[(s^2+1)+3(s^2+9)]}{4(s^2+9)(s^2+1)}$$

$$= \frac{s(4s^2+28)}{4(s^2+1)(s^2+9)}$$

$$\therefore L(\cos^3 t) = \frac{s(s^2+7)}{(s^2+1)(s^2+9)}$$

(iii) $\sin 3A = 3\sin A - 4\sin^3 A$

$4\sin^3 A = 3\sin A - \sin 3A$

$\sin^3 A = \frac{3}{4}\sin A - \frac{1}{4}\sin 3A$

$\sin^3 2t = \frac{3}{4}\sin 2t - \frac{1}{4}\sin 6t$

$L(\sin^3 2t) = \frac{3}{4}L(\sin 2t) - \frac{1}{4}L(\sin 6t)$

$= \frac{3}{4}\ \frac{s}{s^2+4} - \frac{1}{4}\ \frac{6}{s^2+36}$

$= \frac{6[(s^2+36)-(s^2+4)]}{4(s^2+4)\,(s^2+36)}$

$\therefore L(\sin^3 2t) = \frac{48}{(s^2+4)(s^2+36)}$

(iv) $\cos^4 t = (\cos 2t)^2$

$= \left(\frac{1+\cos 2t}{2}\right)^2$

$= \frac{1}{4}\ \left[1+2\cos 2t+\cos^2 2t\right]$

$= \frac{1}{4}\ \left[2+4\cos 2t+\frac{1+\cos 4t}{2}\right]$

$= \frac{1}{8}\ \left[3+4\cos 2t+\cos 4t\right]$

$$= \quad \frac{1}{8}\left[3 + 4\cos 2t + \cos 4t\right]$$

$$L(\cos^4 t) \quad = \quad \frac{1}{8}\left[3L(1) + 4L(\cos 2t) + L(\cos 4t)\right]$$

$$= \quad \frac{1}{8}\left[\frac{3}{s} + 4\frac{s}{s^2+4} + \frac{s}{s^2+16}\right]$$

Problem 4

Find L [$\cos^2 5t$]

Solution

$$L(\cos^2 5t) \quad = \quad L\left[\frac{1+\cos 10t}{2}\right]$$

$$= \quad \frac{1}{2}\left[L(1) + L(\cos 10t)\right] \qquad \left(\therefore \cos^2 A = \frac{1+\cos 2t}{2}\right)$$

$$= \quad \frac{1}{2}\left[\frac{1}{s} + \frac{s}{s^2+100}\right]$$

$$= \quad \frac{1}{2}\left[\frac{s^2+100+s^2}{s(s^2+100)}\right]$$

$$= \quad \frac{s^2+50}{s(s^2+100)}$$

Problem 5

Find L [sin5tcos3t]

Solution

$$L\left[\sin 5t \ \cos 3t\right] = \quad L\left(\frac{2\sin 5t\cos 3t}{2}\right)$$

$$= \frac{1}{2} L(\sin 8t + \sin 2t)$$

$$= \frac{1}{2}\left[\frac{8}{s^2+64}+\frac{2}{s^2+4}\right]$$

$$= \frac{5s^2+80}{(s^2+64)\,(s^2+4)}$$

Problem 6

Find L [sin6t sin2t]

Solution

$$L\,[\sin 6t \sin 2t] = L\left(\frac{2\sin 6t.\sin 2t}{2}\right)$$

$$= L\left(\frac{\cos 5t - \cos 8t}{2}\right)$$

$$= \frac{1}{2}\left[\frac{s}{s^2+25}-\frac{s}{s^2+64}\right]$$

$$= \frac{1}{2}\left[\frac{39s}{(s^2+25)(s^2+64)}\right]$$

Problem 7

Find L [$\sin^2 t \cos^3 t$]

Solution

$$\sin^2 t\ .\ \cos^3 t = \frac{1-\cos 2t}{2}\,.\,\frac{\cos 3t + 3\cos t}{4}$$

$$= \frac{1}{8}\ [\cos 3t - \cos 3t - \cos 2t + 3\cos t - 3\cos 2t \cos t]$$

$$= \frac{1}{8}\ [\cos 3t - \frac{1}{2}\ \cos 5t - \frac{1}{2}\cos t + 3\cos t - \frac{3}{2}\cos 3t - \frac{3}{2}\cos t]$$

$$= \frac{1}{8}\ [-\frac{1}{2}\ \cos 5t - \frac{1}{2}\cos 3t + \cos t]$$

$$= \frac{1}{16}\ [2\cos t - \cos 3t - \cos 5t]$$

$$L(\sin^2 t\cos^3 t) = \frac{1}{16}\ [2L(\cos t) - L(\cos 3t) - L(\cos 5t)]$$

$$= \frac{1}{16}\ \frac{2s}{s^2+1} - \frac{s}{s^2+9} - \frac{s}{s^2+25}$$

Problem 8

Find $L\,[t^4 - t^2 - t + \sin\sqrt{2t}\,]$

Solution

$$L\,[t^4 - t^2 - t + \text{siin}\sqrt{2t}\,] = L[t^4] - L[t^2] - L[t] + L[\sin\sqrt{2t}\,]$$

$$= \frac{4!}{s^5} - \frac{2!}{s^3} - \frac{1}{s^2} + \frac{\sqrt{2}}{s^2+2}$$

Problem 9

Find $L\,[f(t)]$ if $f(t) = \begin{cases} 5, & 0 < t < 3 \\ 0, & t > 3 \end{cases}$

Solution

We know that,

$$L\,[f(t)] = \int_0^\infty e^{-st} f(t)\,dt$$

$$= \int_0^3 e^{-st}(5)\,dt + \int_3^\infty e^{-st}(0)\,dt$$

$$= \quad 5\int_0^3 e^{-st}dt + 0$$

$$= \quad 5\left[\frac{e^{-st}}{-s}\right]_0^3$$

$$= \quad \frac{5(1-e^{-3s})}{s}$$

Problem 10

Find L [f(t)] when f(t) = $\begin{cases} \sin t, & \text{when } 0 < t < 3 \\ 0, & \text{when } t > \pi \end{cases}$

Solution

We know that,

$$L\,[f(t)] \quad = \quad \int_0^\infty e^{-st} f(t)dt$$

$$= \quad \int_0^\pi e^{-st}\sin t\,dt$$

we know that, $\int_0^\pi e^{ax}\sin bx dx = \frac{e^{ax}}{a^2+b^2}$ (asinbx - bcosbx) + k

$$= \quad \left[\frac{e^{-st}}{s^2+1}\left(-s\sin t + \cos t\right)\right]_0^\pi$$

$$= \quad \frac{e^{-3\pi}}{s^2+1}(-1) - \frac{1}{s^2+1}$$

$$= \quad \frac{-(e^{-s\pi}+1}{s^2+1}$$

Problem 11

Find the Laplase Transform of the following functions.

$$f(t) = \begin{cases} e^{2t}, & 0 < t < 3 \\ 1, & t > 3 \end{cases}$$

Solution

$$L\,[f(t)] \quad = \quad \int_0^{\infty} e^{-st} f(t)dt$$

$$= \quad \int_0^{3} e^{-st}e^{2t}dt + \int_3^{\infty} e^{-st}.dt$$

$$= \quad \left[\frac{e^{-(s-2)t}}{-(s-2)}\right]_0^3 + \left[\frac{e^{-st}}{-s}\right]_3^{\infty}$$

$$= \quad \frac{e^{-(s-2)}}{-(s-2)} + \frac{1}{s-2} + \frac{1}{s}e^{-3s}$$

Problem 12

Find the Laplase Transform of the following functions.

(i) $e^{7t}\sin^2 t$ (ii) e-3tcos2t (iii) $te^{2t}\cos 5t$

(iv) $t^2\cos 4t$ (v) $(1+e^{-2t})^2$ (vi) $2t^{2-t}$e-t+cos4t

Solution

(i) $L[e^{7t}\sin^2 t] \quad = \quad L\left[e^{7t}\,\frac{(1-\cos 2t)}{2}\right]$

$$= \quad \frac{1}{2}\,L(e^{7t}) - \frac{1}{2}\,L(e^{7t}\cos 2t)$$

$$= \frac{1}{2}\frac{1}{s-7} - \frac{1}{2}\cdot\frac{s-7}{(s-7)^2+4}$$

(ii) $L[\cos 2t] = \frac{s}{s^2+4}$ (by shifting property, $L\left(e^{-at}\right) = \vec{f}(s+a)$)

$L[e^{-3t}\cos 2t] = \frac{s+3}{(s+3)^2+4}$

(iii) $L(\cos 5t) = \frac{s}{s^2+25}$

$L(t\cos 5t) = \frac{-d}{ds}\left(\frac{s}{s^2+25}\right)$

$$= \frac{-\left[(s^2+25)-s(2s)\right]}{(s^2+25)^2}$$

$$= \frac{s^2-25}{(s^2+25)^2}$$

$L(te^{2t}\cos 2t) = \frac{(s-2)^2-25}{\left[(s-2)^2+25\right]^2}$ (∴ by shifting property)

$$= \frac{s^2-4s-21}{(s^2-4s+29)^2}$$

(iv) $L(\cos 4t) = \frac{s}{s^2+4}$

$L(t^2\cos 4t) = \frac{d^2}{ds^2}\left(\frac{s}{s^2+4}\right)$

$$= \frac{d}{ds}\left[\frac{(s^2+4)1-s2s}{(s^2+4)^2}\right]$$

$$= \frac{d}{ds}\left[\frac{4-s^2}{(s^2+4)^2}\right]$$

$$= \frac{(s^2+4)^2(-2s)-(4-s^2).2(s^2+4)2s}{(s^2+4)}$$

$$= \frac{2s^3-24s}{(s^2+4)^3}$$

(v) $(1+e^{-2t})^2 = 1+2e^{-2t}+e^{-4t}$

$L[(1+e^{-2t})^2] = L[1+2e^{-2t}+e^{-4t}]$

$= L(1) + 2L(e^{-2t}) + L(e^{-4t})$

$$= \frac{2}{s}+\frac{2}{s+2}+\frac{1}{s+4}$$

(vi) $L(\cos 4t) = \frac{s}{s^2+16}$

$L(t^2) = \frac{2!}{s^2}$

$\therefore L(e^{-t}.t^2) = \frac{2}{(s+1)^3}$

$\therefore L(2t^2e^{-t}-t + \cos 4t] = 2L[t^2e^{-t}]-L(t) + L(\cos 4t)$

$$= 2\frac{2}{(s+1)^3} - \frac{1}{s^2}+\frac{s}{s^2+16}$$

$$= \qquad \frac{4}{(s+1)^3} - \frac{1}{s^2} + \frac{s}{s^2+16}$$

Example 12

Find the Laplase Transform for the following functions.

(i) t^2 coshat (ii) $\frac{\sin^2 t}{t}$ (iii) $\frac{\cos 3t - \cos 2t}{t}$

(iv) $\frac{e^{3t} - e^{-2t}}{t}$

Solution

(i) $L[t^2 \text{coshat}] = L\left[\frac{t^2(e^{at} + e^{-at})}{2}\right]$

$$= \frac{1}{2} L[t^2 e^{at}] + \frac{1}{2}\ [t^2 e^{-at}]$$

$$= \frac{1}{2} \cdot \frac{2}{(s-a)^3} + \frac{1}{2}\frac{2}{(s+a)^3}$$

$$= \frac{(s+a)^3 + (s-a)^3}{(s^2-a)^3}$$

$$= \frac{2s^3 + 6a^2 s}{(s^2-a^2)^3}$$

(iii) $L[\sin^2 t] = L\left[\frac{1-\cos 2t}{2}\right]$

$$= \frac{1}{2} L(1\text{-}\cos 2t)$$

$$= \quad \frac{1}{2}\left(\frac{1}{s} - \frac{s}{s^2+4}\right)$$

$$\therefore L\left(\frac{\sin^2 t}{t}\right) = \quad \frac{1}{2}\int_s^\infty \left(\frac{1}{s} - \frac{s}{s^2+4}\right) ds$$

$$= \quad \frac{1}{2}\left[\log s - \frac{1}{2}\log(s^2+4)\right]_s^\infty$$

$$= \quad \left[\frac{1}{2}\log - \frac{s}{\sqrt{s^2+4}}\right]_s^\infty$$

$$= \quad \left[-\frac{1}{2}\log\frac{\sqrt{s^2+4}}{s}\right]_s^\infty$$

$$= \quad \left[\frac{-1}{2}\log\sqrt{1+\frac{4}{s^2}}\right]_s^\infty$$

$$= \quad \frac{1}{2}\log\sqrt{1+\frac{4}{s^2}}$$

$$= \quad \frac{1}{2}\log\frac{\sqrt{s^2+4}}{s}$$

(iii) $L(\cos 3t - \cos 2t) = \quad \dfrac{s}{s^2+9} - \dfrac{s}{s^2+4}$

$$\left(\frac{\cos 3t - \cos 2t}{t}\right) = \quad \int_s^\infty \left(\frac{s}{s^2+9} - \frac{s}{s^2+4}\right) ds$$

$$= \quad \frac{1}{2}\left[\log(s^2+9)-\frac{1}{2}\log(s^2+4)\right]_s^\infty$$

$$= \quad \frac{1}{2}\left[\log(s^2+9)-\frac{1}{2}\log(s^2+4)\right]_s^\infty$$

$$= \quad \frac{1}{2}\left[\log\frac{s^2+9}{s^2+4}\right]_s^\infty$$

$$= \quad \frac{1}{2}\log\left(\frac{1+\frac{9}{s^2}}{1+\frac{4}{s^2}}\right)_s^\infty$$

$$= \quad \frac{1}{2}\left[0-\log\left(\frac{1+\frac{9}{s^2}}{1+\frac{4}{s^2}}\right)\right]$$

$$= \quad \frac{1}{2}\log\left(\frac{s^2+4}{s^2+9}\right)$$

(iv) $L(e^{3t}-e^{-2t}) \quad = \quad \frac{1}{s-3}-\frac{1}{s+2}$

$$L\left[\frac{e^{3t}-e^{-2t}}{t}\right] \quad = \quad \int_s^\infty\left(\frac{1}{s-3}-\frac{1}{s+2}\right)ds = \left[\log\left(\frac{s-3}{s-2}\right)\right]_s^\infty$$

$$= \quad \left(\log\left(\frac{1-\frac{3}{s}}{1+\frac{2}{s}}\right)\right)_s^\infty$$

$$= \quad 0\text{-}\log\left(\frac{1-\frac{3}{s}}{1+\frac{2}{s}}\right)$$

$$= \quad \log\left(\frac{s+2}{s-3}\right)$$

7.6 Laplace transform of Periodic functions

Definition

A function f (t) is said to be periodic function, it there exists, a constant p (>0), such that f(t+P) - f(t) = f(t), for all values of t. Now f(t+2P) = f(t+P+P) = f(t+P) = f(t), for all t, In general, f(t+nP) = f(t), for all t, when 'n' is an integral (positive negative) P is called the period of the function.

Unlike other functions whose Laplace transform are expressed in terms of an integral over the semi infinite interval $0 \leq t \leq \infty$, the L.T of a periodic functionf(t) with period P can be expressed in terms of the integral of e^{-st} f(t) over th efinite interval (O,P) as established in the following theorem.

Theorem

If f (t) has period T and is piecewise continuous on [0, T]. then

$$L\,[f\,(t)] = \frac{1}{1-e^{-st}}\int_0^T e^{-st} f(t)dt$$

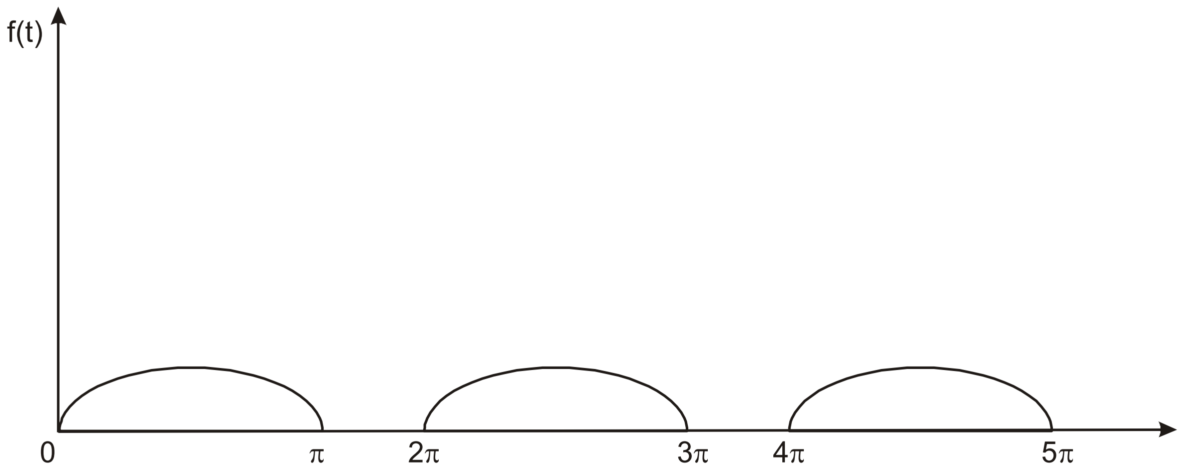
Example 1

Graph the function f (t) = $\begin{cases} \sin t, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$ extended periodically with period 2π and find its Laplace transform

$$f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$$

extending periodically with period 2π. and its Laplac transform

Solution



Here period is 2π.

$$\therefore L\,[f(t)] = \frac{1}{1-e^{-2\pi s}} \int_0^{2\pi} e^{-st} f(t)dt \qquad \text{(by the above)}$$

$$= \frac{1}{1-e^{-2\pi s}} \int_0^{\pi} e^{-st} \sin t dt$$

$$= \frac{1}{1-e^{-2\pi s}} \left[\frac{e^{-st}}{s^2+1} \left(-s \sin t - \cos t \right) \right]_0^{\pi}$$

$$= \frac{1}{1-e^{-2\pi s}} \frac{1+e^{-\pi s}}{s^2+1}$$

$$= \frac{1}{\left(1-e^{-\pi s}\right)\left(s^2+1\right)}$$

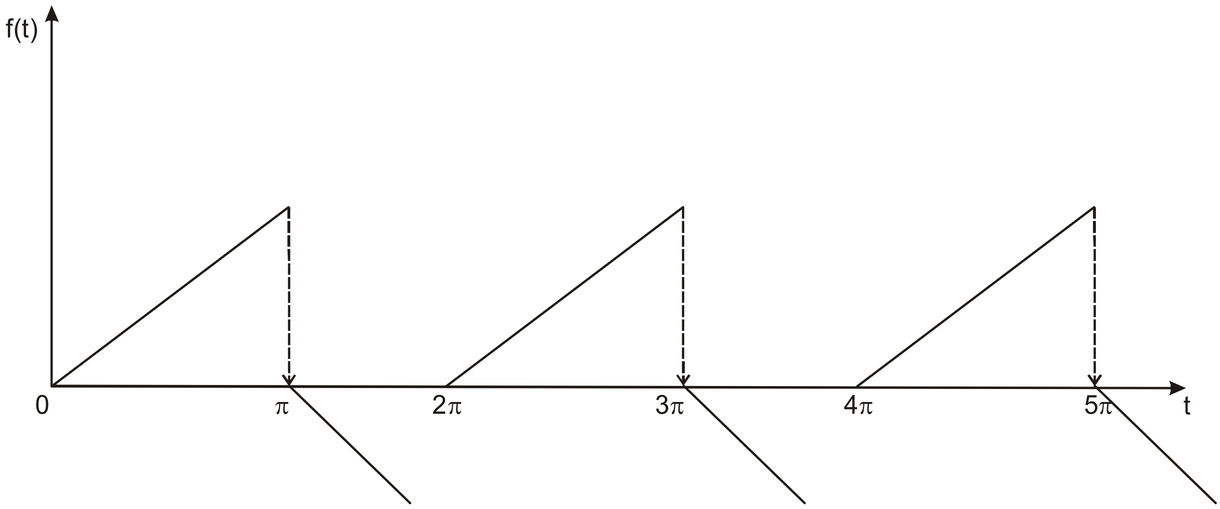
Example 2

Graph the following function its Laplace transform f (t) = $\begin{cases} t, & 0 < t < \pi \\ \pi\text{-}t & \pi < t < 2\pi \end{cases}$ where f(t) = f(t+2π)

Solution

The funtion f(t) is a periodic function of period 2π.

Since f(t) = f(t+2π). The graph of f(t) is given



by above theorem,

$$L[f(t)] = \frac{1}{1-e^{-2\pi s}} \int_0^{2\pi} e^{-st} f(t)dt$$

$$= \frac{1}{1-e^{-2\pi s}} \left[\int_0^{\pi} e^{-st} t dt + \int_0^{2\pi} e^{-st} (\pi - t) dt \right]$$

$$= \frac{1}{1-e^{-2\pi s}} \left[\left(\frac{e^{-st}}{-s} \right)_0^{\pi} - \int_0^{\pi} \frac{e^{-st}}{-s} dt + \left(\frac{e^{-st}}{-s} (\pi - t) \right)_{\pi}^{2\pi} + \int_{\pi}^{2\pi} \frac{e^{-st}}{-s} dt \right]$$

$$= \frac{1}{1-e^{-2\pi s}}\left[\frac{-\pi}{s}e^{-s\pi}-\frac{1}{s^2}e^{-s\pi}+\frac{1}{s^2}+\frac{\pi}{2}e^{-s\pi s}+\frac{1}{s^2}e^{-2\pi s}-\frac{1}{s^2}e^{-\pi s}\right]$$

$$= \frac{1}{1-e^{-2\pi s}}\left[-\frac{\pi}{s}e^{-\pi}(1-e^{-s\pi})-\frac{1}{s^2}e^{-s\pi}(1-e^{-s\pi})+\frac{1}{s^2}(1-e^{-s\pi})\right]$$

$$= \frac{1}{1+e^{-2\pi s}}\left[-\frac{\pi}{s}e^{-s\pi}-\frac{1}{s^2}e^{-s\pi}+\frac{1}{s^2}\right]$$

$$= \frac{1-(1+s\pi)e^{-s\pi}}{s^2(1+e^{-s\pi})}$$

Check Your Progress - I

1. FindL($5-e^{2t}+6t^2$) = ________________

2. Find $\left(e^{2t}.t^n\right)$ = _________________

7.7 Recap

1. Laplase transform of f(t) = $\int_0^\infty e^{-st}f(t)dt = \bar{f}(s)$

2. L(t^n) = $\frac{L^n}{s^{n+1}}$

3. L(e^{-at}) = $\frac{1}{s+a}$, s >-a

4. L(sinat) = $\frac{a}{s^2+a^2}$ if s > 0

5. $L(\cos at) = \frac{s}{s^2+a^2}$ if $s > 0$

6. Linearity Property : $L[f_1(t) + f_2(t)] = L[f_1(t)] + L[f_2(t)]$

   $L[cf(t)] = cL[f(t)]$

7. Shifting Property : If $L[f(t)] = \bar{f}(s)$, then $L[e^{at}f(t)] = \bar{f}(s-a)$

8. Change of Scale Property : $L[f(at)] = \frac{1}{a}\bar{f}\left(\frac{s}{a}\right)$

9. Laplase transform of Derivatives :

$$L(f'(t)] = s\bar{f}(s) - f(0)$$

$$L(f''(t)] = s^2 f(s) - sf(0) - f'(0)$$

10. Multiple by 't' $L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}\vec{f}(s)$

7.8 Exercise

1. Find the Laplace transform of the folloiwing functions

   (i) cosh5t

   (ii) t^5-4t^3+3

   (iii) $\sin^4 t$

   (iv) sin3tcos2t

   (v) $f(t) = \begin{cases} 0 & , \quad 0 < t < 2 \\ t & , \quad t > 2 \end{cases}$

   (vi) $L(t^2+1)^2$

   (vii) $L(2t^2-e^{-t})$

2. Find the Laplace transform of the following functions

   (i) te^{-3t}

   (ii) t^2e^{5t}

   (iii) $t\sin 3t$

   (iv) $t^2\cos 3t$

   (v) $e^{3t}\cos 6t - t^3 + e^t$

   (vi) $\dfrac{1-\cos t}{t}$

   (vii) $\dfrac{\cos t - \cos 2t}{t}$

   (viii) $\dfrac{1-\cos t}{t}$

   (ix) $te^{-2t}\sin 3t$

   (x) $t^3 - te^t + e^{4t}\cos t$

3. Find the Laplace transform of the following functions

   (i) $t^2 + e^t\cos t$

   (ii) $t\sin 2t.\sin 5t$

   (iii) $te^{2t}\cos 5t$

   (iv) $(t-1)^4$

   (v) $(1-e^{-t})^2$

   (vi) $\sin^2 t\cos t$

   (vii) $\sin^3 t\cos^2 t$

(viii) $t^3\cos t$

4. Evaluate the following Integrals using Laplase transform

(i) $\int_0^\infty \frac{e^{-3t} - e^{-4t}}{t} dt$

(ii) $\int_0^\infty \frac{e^{-t}\sin t}{t} dt$

(iii) $\int_0^\infty te^{-2t}\cos t \, dt$

(iv) $\int_0^\infty t^3e^{-t}\sin t \, dt$

5. For a square wave function of f(t) = $\begin{cases} -1 & \text{for} \quad 0 < t < \frac{\pi}{2} \\ -1 & \text{for} \quad \frac{\pi}{2} < 6 < \pi \end{cases}$

6. If f(t) is the staircase function such that

f(t) = $\begin{cases} b & \text{when} \quad 0 < t < a \\ 2b & \text{when} \quad a < t < 2a \end{cases}$ and so forth show that L[f(t)] = $\frac{t}{s(1-e^{-\pi s})}$

LESSON - 8

INVERSE LAPLACE TRANSFORM AND APPLICATION TO THE FIRST AND SECOND ORDER LINEAR DIFFERENTIAL EQUATION

Learning Objectives

- define inverse Laplace transform of f(t)
- to know important properties of Inverse Laplace Transforms
- find L^{-1} of standard fractions
- to know solving differential equations of Laplace Transform

Structure

8.1 Introduction

In this chapter, we study the diferent methods of obtaining the inverse Laplace Transform f(t) when we are given the transform function $\bar{f}$ (s).

That is we seek an inverse mapping for the laplace transform, Then, we use the properties of Laplace Transform to convert a Linear differential with constant coefficient into an algebric equation in the constant coefficient into an algebraic equation in the Laplace Transform variable equation in the Laplace transform variable. Using Laplace transform technique we seek the solution for the given DE with initial conditions, we also use LT techniques to solve Linear simultaneous differential equations with initial conditions.

8.2 Inverse Laplace Transforms

Let as find the inverse laplace transform function of S using the above properties. In many cases partial fractions method will enable us to find the inverse laplace transform easily. We also use the properties of inverse laplace transform to determine the inverse laplace transform in some other cases.

Definition

If L [f (t)] = $\bar{f}$ (s), then f (t) is called the inverse Laplace transform of $\bar{f}$ (s) and this we write symbolically

f(t) = L^{-1} [$\bar{f}$ (s)].

Example,

If $L(e^{-5t}) = \frac{1}{s+5}$, then $L^{-1}\left(\frac{1}{s+5}\right) = e^{-5t}$

8.2.1 Inverse Laplace Transform of some functions :

No.	$\bar{f}$ (s)	f(t)
1.	$\frac{1}{s}$	1
2.	$\frac{1}{s^{n+1}}$	$\frac{t^n}{n!}$
3.	$\frac{1}{s+a}$	e^{-at}
4.	$\frac{1}{s^2+a^2}$	$\frac{\sin at}{a}$

5.	$\frac{s}{s^2+a^2}$	cosat
6.	$\frac{1}{s^2-a^2}$	$\frac{\text{sinhat}}{a}$
8.	$\frac{s}{s^2-a^2}$	coshat
8.	$\frac{1}{(s+a)^2+b^2}$	$\frac{1}{b}e^{-at}$ sinbt
9.	$\frac{s+a}{(s+a)^2+b^2}$	e^{-at} cosbt
10.	$\frac{s^2-a^2}{(s^2+a^2)^2}$	tcosat
11.	$\frac{2as}{(s^2+a^2)^2}$	tsinat

8.2.2 Important Properties of Inverse Laplace Transform

1. Linearity Property :

If $L^{-1}[\bar{f}(s)] = f(t)$ then

$L^{-1}[c_1\bar{f}_1(s)+c_2\bar{f}_2(s)] = c_1f_1(t) + c_2f_2(t)$

This may be extended to more than two functions.

2. Shifting Property

If $L^{-1}[\bar{f}(s)] = f(t)$, then $L^{-1}[\bar{f}(s-a)] = e^{at} f(t)$

3. Change of Scale Property

If $L^{-1}[\bar{f}(s)] = f(t)$, then $L^{-1}[\bar{f}(as)] = \frac{1}{a} f\left(\frac{t}{a}\right)$

4. Inverse Laplace transform of Derivatives

If $L^{-1}[\bar{f}(s)] = f(t)$, then

$$L^{-1}[\bar{f}^{(n)}(s)] = L^{-1}\left[\frac{d^n}{ds^n}\bar{f}(s)\right] = (-1)^n t^n f(t)$$

5. Inverse Laplace transform of Integrals

If $L^{-1}[\bar{f}(s)] = f(t)$, then

$$L^{-1}\left[\int_s^{\infty} \bar{f}(u)du\right] = \frac{f(t)}{t}$$

6. Multiplication by s

If $L^{-1}[\bar{f}(s)] = f(t)$ and $f(0) = 0$

then,

$L^{-1}[s\bar{f}(s)] = f'(t)$ then

If $f(0) = 0$ then, $L^{-1}[s\bar{f}(s) - f(0)] = f'(t)$.

7. Division by s,

If $L^{-1}[\bar{f}(s)] = f(t)$, then,

$$L^{-1}\left[\frac{\bar{f}(s)}{s}\right] = \int_0^t f(u)du$$

Problem 1

(i) $L^{-1}\left(\frac{1}{s^2+16}\right)$ (ii) $L^{-1}\left(\frac{1}{s^5}\right)$ (iii) $L^{-1}\left(\frac{s}{s^2+3}\right)$

(iv) $L^{-1}\left(\frac{7s}{s^2-25}\right)$ (v) $L^{-1}\left(\frac{5}{s-3}\right)$

Solution

(i) $L^{-1}\left(\frac{1}{s^2+16}\right) = L^{-1}\left(\frac{1}{s^2+4^2}\right) = \frac{\sin 4t}{4}$

(ii) $L^{-1}\left(\frac{1}{s^5}\right) = \frac{t^4}{4!} = \frac{t^4}{24}$

(iii) $L^{-1}\left(\frac{s}{s^2+3}\right) = L^{-1}\left(\frac{s}{s^2+\left(\sqrt{3}\right)^2}\right) = \cos\sqrt{3}\,t$

(iv) $L^{-1}\left(\frac{7s}{s^2-25}\right) = 7L^{-1}\left(\frac{s}{s^2-5^2}\right) = 7\cosh 5t$

(v) $L^{-1}\left(\frac{5}{s-3}\right) = 5e^{3t}$

Problem 2

Find inverse of Laplace Transform $L^{-1}\left(\frac{8s-4}{s^2-4s+5}\right)$

Solution

$$L^{-1}\left(\frac{8s-4}{s^2-4s+5}\right) = L^{-1}\left(\frac{8s-4}{(s-2)^2+1}\right)$$

$$= L^{-1}\left(\frac{8(s-2)+12}{(s-2)^2+1}\right)$$

$$= 8L^{-1}\left[\frac{s-2}{(s-2)^2+1^2}\right] + 12L^{-1}\left[\frac{1}{(s-2)^2+1^2}\right]$$

$$= 8e^{2t}\cos t+12e^{2t}\sin t$$

Problem 3

Find $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right]$

Solution

We have $\frac{d}{ds}\left[\frac{1}{(s^2+a^2)}\right] = \frac{-2s}{(s^2+a^2)^2}$

Thus,

$$\frac{s}{(s^2+a^2)^2} = \frac{1}{2}\frac{d}{ds}\left(\frac{1}{s^2+a^2}\right)$$

$$L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right] = \frac{1}{2}L^{-1}\left[\frac{d}{ds}\left(\frac{1}{s^2+a^2}\right)\right]$$

$$= \frac{1}{2}.t\left(\frac{\sin at}{a}\right)$$

$$= \frac{t\sin at}{2a}$$

Problem 4

$$L^{-1}\left\{\log\left(1+\frac{1}{s^2}\right)\right\}$$

Solution

Let $\bar{f}(s) = \log\left(1+\frac{1}{s^2}\right) = L\,[f(t)$

Then,

$$f(s) = \frac{-2}{s(s^2+1)} = -2\left[\frac{1}{s}-\frac{s}{s^2+1}\right]$$

$$L^{-1}\,[\bar{f}(s)] = -2\left\{L^{-1}\left(\frac{1}{s}\right)-L^{-1}\left(\frac{s}{s^2+1}\right)\right\}$$

$$= -2\ (1-\cos t)$$

But $L[tf(t)] = -f'(s) = L^{-1}\,[f'(s)] = -tf(t)$

$$f(t) = \frac{2(1-\cos t)}{t}$$

Problem 5

Find $L^{-1}\left(\frac{1}{s^3(s^2+1)}\right)$

Solution

$$L^{-1}\left[\frac{1}{s^2+1}\right] = \text{sint}$$

$$L^{-1}\left[\frac{1}{s(s^2+1)}\right] = \int_0^t \sin u du = [-\cos u]_0^t$$

$$= \text{1-cost}$$

$$L^{-1}\left[\frac{1}{s^2(s^2+1)}\right] = \int_0^t \text{(1-cost)dt = t-sint}$$

$$L^{-1}\left[\frac{1}{s^3(s^2+1)}\right] = \int_0^t \text{(t-sint)dt}$$

$$= \frac{t^2}{2} + \cos t - 1$$

Problem 6

Find $L^{-1}\left[\frac{7s-1}{(s+1)(s+2))s+3)}\right]$

Solution

Let $\frac{7s-1}{(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}$ (1)

7s-1 = A (s+2))s+3) + B (s+1))s+3) + C (s+1) (s+2) (2) put s=-1, in (2) $-8 = A(1)(2)$ => $\boxed{A = -4}$

put s=-2, in (2) $-15 = B(-1)(1)$ => $\boxed{B = 15}$

put s = -3, in (2) $-22 = c(-2)(-1)$ => $\boxed{c = -11}$

(1) => ∴ $\frac{7s-1}{(s+1)(s+2)(s+3)} = \frac{-4}{s+1} + \frac{15}{s+2} - \frac{11}{s+3}$

Taking L^{-1} on both sides,

$$L^{-1}\left[\frac{7s-1}{(s+1)(s+2)(s+3)}\right] = -4L^{-1}\left(\frac{1}{s+1}\right) + 15L^{-1}\left(\frac{1}{s+2}\right) - 11L^{-1}\left(\frac{1}{s+3}\right)$$

$$= -4e^{-t} + 15e^{-2t} - 11e^{-3t}$$

Problem 7

Find $L^{-1}\left[\frac{s^2+9s+2}{(s-1)^2(s+2)}\right]$

Solution

Let $\frac{s^2+9s+2}{(s-1)^2(s+2)} = \frac{A}{s+2} + \frac{B}{s-1} + \frac{C}{(s-1)^2}$ (1)

∴ $s^2+9s+2 = A(s-1)^2 + B(s-1)(s+2) + c(s+2)$ (2)

put s = 1, in (2) $c = 4$

put s =-2, in (3) $4-18+2 = A(-3)^2$

$-12 = 9A$ $\boxed{A = -\frac{4}{3}}$

Equating the co-efficinent of s^2,

(2) => $A+B = 1$

=> $B = 1+\frac{4}{3}$

$$\boxed{B = \frac{7}{3}}$$

(1) => $\therefore \frac{s^2+9s+2}{(s-1)^2(s+2)} = \frac{-4}{3(s+2)} + \frac{7}{3(s-1)} + \frac{4}{(s-1)^2}$

Taking L^{-1} on bothsides,

$$L^{-1}\left[\frac{s^2+9s+2}{(s-1)^2(s+2)}\right] = \frac{-4}{3} L^{-1}\left(\frac{1}{s+2}\right) + \frac{7}{3} L^{-1}\left(\frac{1}{s-1}\right) + 4L^{-1}\left(\frac{1}{(s-1)^2}\right)$$

$$= \frac{-4}{3} e^{-2t} + \frac{7}{3} e^{t} + 4e^{t}.t$$

Problem 8

Find $L^{-1}\left[\frac{s^2}{(s^2+4)(s^2+9)}\right]$

Solution

Let $\frac{s^2}{(s^2+4)(s^2+9)} = \frac{A}{s^2+4} + \frac{B}{s^2+9}$ (1)

$\therefore s^2 = A(s^2+9) + B(s^2+4)$ (2)

put $s^2=-4$, $-4 = 5A$, => $\boxed{A = -\frac{4}{5}}$

put $s^2=-9$, $-9 = -5B$, => $\boxed{B = \frac{9}{5}}$

$$(2) \Rightarrow \quad \therefore \frac{s^2}{(s^2+4)(s^2+9)} = \frac{-4}{5(s^2+4)} + \frac{9}{5(s^2+4)}$$

Taking L^{-1} on both sides,

$$L^{-1}\left[\frac{s^2}{(s^2+4)(s^2+9)}\right] = \frac{-4}{5}L^{-1}\left(\frac{1}{s^2+4}\right) + \frac{9}{5}L^{-1}\left(\frac{1}{s^2+4}\right)$$

$$= \frac{-4}{5}\frac{\sin 2t}{2} + \frac{9}{5}\frac{\sin 3t}{3}$$

$$= \frac{-2}{5}\sin 2t + \frac{9}{15}\sin 3t$$

Problem 9

Find $L^{-1}\left[\frac{s+2}{(s-4)(s^2+1)}\right]$

Solution

$$\text{Let } \frac{s+2}{(s-4)(s^2+1)} = \frac{A}{s-4} + \frac{Bs+C}{s^2+1} \qquad (1)$$

$$\therefore s^2 = A(s^2+1) + (Bs+c)(s-4) \qquad (2)$$

put $s=4$, $6 = 17A$ $\Rightarrow A = \frac{6}{17}$

Equating the co-efficient of s^2, $A+B = 0$

$$\Rightarrow \boxed{B = \frac{-6}{17}}$$

Equating the constant term,

$$A - 4C = 2$$

$$=> \quad 4C = A - 2$$

$$4C = \frac{6}{17} - 2$$

$$C = \frac{-7}{17}$$

$$\therefore \frac{s+1}{(s-4)(s^2+1)} = \frac{6/17}{s-4} - \frac{\frac{6}{7}s + \frac{7}{17}}{s^2+1}$$

$$= \frac{6}{17(s-4)} - \frac{6s+7}{17(s^2+1)}$$

Taking Inverse Laplase Transform on both sides,

$$L^{-1}\left[\frac{s+1}{(s-4)(s^2+1)}\right] = \frac{6}{17} L^{-1}\left(\frac{1}{s-4}\right) - \frac{6}{17} L^{-1}\left(\frac{s}{s^2+1}\right) - \frac{7}{17} L^{-1}\left(\frac{1}{s^2+1}\right)$$

$$= \frac{6}{17} e^{4t} - \frac{6}{7} \cos t - \frac{7}{17} \sin t$$

Problem 10

Find $L^{-1}\left[\frac{2s^2+10s}{(s^2-2s+5)(s+1)}\right]$

Solution

$$L^{-1}\left[\frac{2s^2+10s}{(s^2-2s+5)(s+1)}\right] = L^{-1}\left[\frac{2s^2+10s}{[(s-1)^2+4][(s-1)+2]}\right]$$

$$= e^{t} L^{-1}\left[\frac{2(s+1)^2+10(s+1)}{(s^2+4)(s+2)}\right]$$

$$= \quad e^t L^{-1}\left[\frac{2(s+1)+(s+6)}{(s+2)(s^2+4)}\right]$$

Let $\frac{2(s+1)(s+6)}{(s+2)(s^2+4)} = \frac{A}{s+2} + \frac{Bs+C}{s^2+4}$

$\therefore$ 2(s+1) (s+6) = A (s^2+4) + (s+2) (Bs+C)

put s = -2, 2(-1) 4 = A(8) = > A=-1

Equating the co-efficient of s^2, A+B = 2

$\therefore$ B = 3

Equating the co effiient of constant term,

4A +2c = 12

=> $\boxed{C= 8}$

$$\therefore L^{-1}\left[\frac{2s^2+10s}{(s^2-2s+5)(s+1)}\right] = e^t\left[L^{-1}\left(\frac{-1}{s+2}+L^{-1}\left(\frac{3s+8}{s^2+4}\right)\right)\right]$$

$$= e^t\left[L^{-1}\left(\frac{-1}{s+2}\right)+3L^{-1}\left(\frac{s}{s^2+4}\right)+8L^{-1}\left(\frac{1}{s^2+4}\right)\right]$$

$$= e^t\,[-e^{-2t} + 3\cos 2t + 4\sin 2t]$$

$$= 3e^t\cos 2t + 4e^t\sin 2t - e^{-t}$$

Problem 11

Find the Laplase Inverse of $\frac{10}{(s+2)^6}$

Solution

$$L^{-1}\left[\frac{10}{(s+2)^6}\right] \quad = \quad 10L^{-1}\left[\frac{1}{(s+2)^6}\right]$$

$$= \quad 10e^{-2t}\,L^{-1}\left[\frac{1}{s^6}\right]$$

$$= \quad 10e^{-2t}\,\frac{t^5}{5!}$$

$$= \quad e^{-2t}\,\frac{t^5}{12}$$

Problem 12

Find the inverse Laplace Transform of $\frac{2(s+1)}{(s^2+2s+2)^2}$

Solution

$$\text{let } L^{-1}\left[\frac{2(s+1)}{(s^2+2s+2)^2}\right] \quad = L^{-1}\left[\frac{2(s+1)}{[(s+1)^2+1]^2}\right]$$

$$= \; e^{-t}L^{-1}\left[\frac{2s}{(s^2+1)^2}\right]$$

Let $\bar{f}(s) = \frac{1}{s^2+1}$; $f(t) = \sin t$

$$\frac{d}{ds}\,\bar{f}(s) = \frac{-1}{(s^2+1)^2}.2s$$

$$-\frac{d}{ds}\,\bar{f}(s) = \frac{2s}{(s^2+1)^2}$$

But $\quad L^{-1}\left[\frac{-d}{ds}\bar{f}(s)\right] \qquad = L^{-1}\left[\frac{2s}{(s^2+1)^2}\right]$

$$L^{-1}\left[\frac{2(s+1)}{(s^2+2s+2)^2}\right] = e^{-t}.t.f(t)$$

$$= t.e^{-t}.\sin t$$

Problem 13

Find the Inverse Laplase transform of $\log\left(\frac{1+s}{s}\right)$

Solution

Let $\quad \bar{f}(s) \;=\; \log\left(\frac{1+s}{s}\right)$

$$= \log(1+s) - \log s$$

$$\frac{d}{ds}\bar{f}(s) = \frac{1}{s+1} - \frac{1}{s}$$

$$L^{-1}\left[\frac{-d}{ds}\bar{f}(s)\right] = L^{-1}\left(\frac{1}{s}\right) - L^{-1}\left(\frac{1}{s+1}\right)$$

$$tf(t) = 1-e^{-t}$$

$$\therefore f(t) = \frac{1-e^{-t}}{t}$$

Problem 14

Find $L^{-1}\left[\frac{s+2}{(s-2)^7}\right]$

Solution

$$L^{-1}\left[\frac{s+2}{(s-2)^7}\right] \quad = L^{-1}\left[\frac{s+2+4}{(s-2)^7}\right]$$

$$= e^{2t}\, L^{-1}\left[\frac{s+4}{s^7}\right]$$

$$= e^{2t}\, L^{-1}\left[\frac{1}{s^6}+\frac{4}{s^7}\right]$$

$$= e^{2t}\left[\frac{t^5}{L^5}+4.\frac{t^6}{L^6}\right]$$

$$= \frac{e^{2t}}{720}\;\left[6t^5+4t^6\right]$$

Problem 15

Find $L^{-1}\left[\log\left(\frac{s^2+9}{s^2+1}\right)\right]$

Solution

Let $\bar{f}(s) = \log\left(\frac{s^2+9}{s^2+1}\right) = \log(s^2+9) - \log(s^2+1)$

$$\frac{d}{ds}\bar{f}(s) = \frac{2s}{s^2+9} - \frac{2s}{s^2+1}$$

$$L^{-1}\left[-\frac{d}{ds}\bar{f}(s)\right] = L^{-1}\left[\frac{2s}{s^2+9}-\frac{2s}{s^2+1}\right]$$

$$t.f(t) = 2\cos t - 2\cos 3t$$

$$f(t) = \frac{2(\cos t - \cos 3t)}{t}$$

8.3 Solving Differential Equations using Laplace Transform

Problem 1

Solve $\frac{d^2y}{dt^2}$ - $\frac{dy}{dt}$ - 2y = 0 given y(0) = -2, y' (0) = 5

Solution

$$\frac{d^2y}{dt^2} - \frac{dy}{dt} - 2y = 0$$

Taking Laplace Transform both sides,

$$L\left[\frac{d^2y}{dt^2}\right]-L\left[\frac{dy}{dt}\right] -2L(y) = 0$$

$s^2L(y) - sy(0) - y'(0) - [sL(y) - y(0)] - 2L(y) = 0$

$L(y)[s^2-s-2] - sy(0) - y'(0) + y(0) = 0$

Given y (0) = -2 y(0) = -5

=> $(s^2-s-2) L(y) +2s - 5 - 2 = 0$

$(s^2-s-2) L(y) = 7-2s$

$$L(y) = \frac{7-2s}{s^2-s-2}$$

$$\therefore y = L^{-1}\left[\frac{7-2s}{(s-2)(s+1)}\right] \qquad (1)$$

Let $\frac{7-2s}{(s-2)(s+1)} = \frac{A}{s-2}+\frac{B}{s+1}$

$7-2s = A(s+1) + B(s-2)$

put $s=-1$, $9 = -3B$, $\boxed{B = -3}$

$s=2$, $3 = 3A$, $\boxed{A = 1}$

$(1) \Rightarrow y = L^{-1}\left[\frac{1}{s-2}-\frac{3}{s+1}\right] = e^{2t}-3e^{-t}$

$y = e^{2t}-3e^{-t}$

Problem 2

Solve $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 5y = 4e^{3t}$ given $y(0) = 2$, $y'(0) = 7$

Solution

$$\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 5y = 4e^{3t}$$

Taking L.T on bothsides,

$$L\left[\frac{d^2y}{dt^2}\right] - 4L\left[\frac{dy}{dt}\right] + 5L(y) = 4L(e^{3t})$$

$$s^2L(y) - sy(0) - y'(0) - 4[L(y) - y(0)] + 5L\,(y) = \frac{4}{s-3}$$

$$(s^2-4s+5)L(y) = \frac{4}{s-3} + 2s - 1$$

$$L(y) = \frac{4+(2s-1)(s-3)}{(s^2-4s+5)(s-3)}$$

Let $\frac{4+(2s-1)(s-3)}{(s^2-4s+5)(s-3)} = \frac{A}{s-3} + \frac{Bs+c}{s^2-4s+5}$

$4+(2s-1)\,(s-3) = A\,(s^2-4s+5) + (s-3)\,(Bs+c)$

put s=3, $\quad 4 = A\,(9-12+5)$

$\Rightarrow \quad A = 2$

Equating the co-efficient of s^2, $\quad A + B = 2$

$\Rightarrow \quad B = 0$

Equating the constant term,

$7 = 5A-3C$

$C=1$

$$L(y) = \frac{2}{s-3} + \frac{1}{s^2-4s+5}$$

$$y = L^{-1}\left(\frac{2}{s-3}\right) + L^{-1}\left[\frac{1}{s^2-4s+5}\right]$$

$$= 2e^{3t} + L^{-1}\left(\frac{1}{(s-2)^2+1}\right)$$

$$y = 2e^{3t} + e^{2t}\sin t$$

Problem 3

Using Laplace transform, solve the equtaion $\frac{d^2y}{dt^2}$ + 4 $\frac{dy}{dt}$ +13y = 2e^{-t} given y(0) = 0, y'(0) = -1

Solution

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 13y = 2e^{-t}$$

Taking L.T on both sides,

$$L\left[\frac{d^2y}{dt^2}\right] + 4L\left[\frac{dy}{dt}\right] + 13L(y) = 2L(e^{-t})$$

$$s^2L(y) - sy(0) - y'(0) + 4[sL(y) - y(0)] + 13L(y) = \frac{2}{s+1}$$

$$(s^2+4s+13)L(y) - sy(0) - y'(0) - 4y(0) = \frac{2}{s+1}$$

$$(s^2+4s+13)L(y) = \frac{2}{s+1} - 1$$

$$= \frac{1-s}{1+s}$$

$$L(y) = \frac{1-s}{(1+s)(s^2+4s+13)} \quad (\because \text{ using partial fraction})$$

Let $$\frac{1-s}{(1+s)(s^2+4s+13)} = \frac{A}{1+s} + \frac{Bs+c}{s^2+4s+13} \qquad (1)$$

$$1-s = A(s^2-4s+13) + (1+s)(Bs+c) \qquad (2)$$

put s=-1, $2 = 10A \Rightarrow A = \frac{1}{5}$

Equating the co-efficient of s^2, $A + B = 0$

$$\Rightarrow \quad B = \frac{-1}{5}$$

Equating the coefficient of constant term,

$$13A + C = 1$$

$$c = \frac{-8}{5}$$

$$L(y) = \frac{\frac{1}{5}}{s+1} - \frac{\frac{1}{5}s + \frac{8}{5}}{s^2 + 4s + 13}$$

$$= \frac{1}{5} \cdot \frac{1}{s+1} - \frac{1}{5} - \frac{s}{s^2 + 4s + 13} - \frac{8}{5} \cdot \frac{4}{s^2 + 4s + 13}$$

$$y = \frac{1}{5} L^{-1}\left[\frac{1}{s+1}\right] - \frac{1}{5} L^{-1}\left[\frac{s}{(s+2)^2 + 9}\right] - \frac{8}{5} L^{-1}\left[\frac{1}{(s+2)^2 + 9}\right]$$

$$= \frac{1}{5} e^{-t} - \frac{1}{5} e^{-2t} \cos 3t - \frac{8}{5} e^{-2t} \frac{\sin 3t}{3}$$

$$y = \frac{3e^{-t} - 3e^{-2t} \cos 3t - 8e^{-2t} \sin 3t}{15}$$

Problem 4

Using Laplace transform, solve the equtaion $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = e^{-t}$ given $y = 0$, $\frac{dy}{dt} = 1$, when $t = 0$

Solution

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = e^{-t}$$

Taking L.T on both sides,

$$L\left[\frac{d^2y}{dt^2}\right] + 5L\left[\frac{dy}{dt}\right] + 6L(y) = L(e^{-t})$$

$$s^2L(y) - sy(0) - y'(0) + 5[sL(y) - y(0)] + 6L\,(y) = \frac{1}{s+1}$$

$$(s^2+5s+6)L(y)-1 = \frac{1}{s+1}$$

$$(s^2+5s+6)L(y) = \frac{s+2}{s+1}$$

$$L(y) = \frac{1}{(s+1)(s+3)}$$

Let $\frac{1}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$

$1 = A\,(s+3) + B(s+1)$

put $s=-1$, $1 = A(2) \Rightarrow A = \frac{1}{2}$

put $s=-3$, $-2B=1, \Rightarrow B = \frac{-1}{2}$

$$L(y) = \frac{\frac{1}{2}}{s+1} - \frac{\frac{1}{2}}{s+3}$$

$$y = \frac{1}{2} L^{-1}\left(\frac{1}{s+1}\right) - \frac{1}{2} L^{-1}\left(\frac{1}{s+3}\right)$$

$$\therefore y = \frac{1}{2} [e^{-t}-e^{-3t}]$$

Problem 5

Using Laplace transform, solve $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 5y = e^{-2t}$ given that $y = 0$, $\frac{dy}{dt} = 1$, when $t = 0$.

Solution

Given $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 5y = e^{-2t}$

Taking Laplase Transform,

$$L\left[\frac{d^2y}{dt^2}\right] + 6L\left[\frac{dy}{dt}\right] + 5L(y) = L(e^{-2t})$$

$$s^2L(y) - sy(0) - y'(0) + 6[sL(y) - y(0)] + 5L(y) = \frac{1}{s+2}$$

$$(s^2+6s+5) L(y)-1 = \frac{1}{s+2}$$

$$L(y) \frac{s+3}{(s+2)(s+1)(s+5)} = \frac{A}{s+2} + \frac{B}{s+1} + \frac{C}{s+5}$$

$$\therefore s+3 = A(s+1)(s+5) + B(s+2)(s+5) + C(s+1)(s+2)$$

put s=-1, B(1) (4) = 2 => $B = \frac{1}{2}$

put s=-2, A(-1) (3) = 1 => $A = \frac{-1}{3}$

put s=-5, C(-4) (-3) = -2 => $C = \frac{-1}{6}$

$$L(y) = \frac{-1}{3(s+1)} + \frac{1}{2(s+2)} - \frac{1}{6(s+5)}$$

$$y = \frac{-1}{3}L^{-1}\left(\frac{1}{s+1}\right) + \frac{1}{2}L^{-1}\left(\frac{1}{s+2}\right) - \frac{1}{6}L^{-1}\left(\frac{1}{s+5}\right)$$

$$\therefore \quad y = \frac{-1}{3}e^{-t} + \frac{1}{2}e^{-2t} - \frac{1}{6}e^{-5t}$$

8.3.1 Solving simultoneous Linear Differential Equations by using Laplace Transform :

Problem 21

Using the method of Laplace transform, solve $\frac{dx}{dt}$ +2x-3y=2t ; $\frac{dy}{dt}$ - 3x+2y = e^{2t}.given

x(0) = 0, y(0) = 0.

Solution

$$\frac{dx}{dt} + 2x - 3y = 2t$$

Taking L.T on both sides

$$L\left[\frac{dx}{dt}\right] + 2L(x) - 3L(y) = 2L(t)$$

$$sL(x) - x(0) + 2L(x) - 3L(y) = \frac{2}{s^2}$$

$$\therefore (s+2)\,L(x) - 3L(y) = \frac{2}{s^2}$$

$$\frac{dy}{dt} - 3x + 2y = e^{2t}$$

Taking L.T on both sides,

$$L\left[\frac{dy}{dt}\right] - 3L(x) + 2L(y) = L\,(e^{2t})$$

$$SL(y) - y(0) - 3L\,(x) + 2L(y) = \frac{1}{s-2}$$

$$-3L\,(x) + (s+2)\,L(y) = \frac{1}{s-2}$$

$$(s+2) = L(x) - 3L(y) = \frac{2}{s^2} \qquad (1)$$

$$-3L(x) + (s+2)\,L(y) = \frac{1}{s-2} \qquad (2)$$

$$(1) \times (s+2) => (s+2)^2\,L(x) - 3(s+2)\,L(y) = \frac{2(s+2)}{s^2}$$

$$(2) \times 3 => -9L\,(x) + 3\,(s+2)\,L(y) = \frac{3}{s-2}$$

Adding $(s^2+4s-5)\,)L(x) \quad = \quad \frac{2(s+2)}{s^2} + \frac{3}{s-2}$

$$= \frac{2(s^2-4)+3s^2}{s^2(s-2)}$$

$$L(x) = \frac{5s^2-8}{s^2(s-2)(s^2+4s-5)} = \frac{5s^2-8}{s^2(s-2)(s+5)(s-1)}$$

$$\Rightarrow L(x) = \frac{A}{s}+\frac{B}{s^2}+\frac{C}{s-2}+\frac{D}{s+5}+\frac{E}{s-1} \qquad (1)$$

$\therefore$ $5s^2-8$ = $As(s-2)(s+5)(s-1) + B(s-2)(s+5)(s-1) + cs^2(s+5)(s-1) + Ds^2(s-2)(s-1) + Es^2(s-2)(s+5)$

put $s=0$, $-8=10B$ $\Rightarrow B = \frac{-4}{5}$

$s=1$, $-3=-6E$, $\Rightarrow E = \frac{1}{2}$

$s=2$, $12=28c$, $\Rightarrow C = \frac{3}{7}$

$s=-5$ $117 = D(25)(-7)(-6)$, $D = \frac{39}{350}$

Equating the co-efficient of s^4

$A+C+D+E = 0$

$A = \frac{-3}{7} - \frac{39}{350} - \frac{1}{2}$

$A = \frac{-182}{175}$

$$(I) \Rightarrow x = \frac{-182}{175} - \frac{4}{5}t + \frac{3}{7}e^{2t} + \frac{39}{350}e^{-5t} + \frac{1}{2}e^{t}$$

$$3y = \frac{dx}{dt} + 2x\text{-}t$$

$$= \frac{-4}{5} + \frac{6}{7}e^{2t} - \frac{195}{350}\ e^{-5t} + \frac{1}{2}\ e^{t} - \frac{364}{175} - \frac{8}{5}t + \frac{6}{7}\ e^{2t} + \frac{78}{350}\ e^{-5t} + e^{t}\text{-}t$$

$$= \frac{-504}{175} - \frac{13}{5}t + \frac{12}{7}\ e^{2t} - \frac{117}{350}e^{-5t} + \frac{3}{2}e^{t}$$

$$\therefore y = \frac{-168}{175} - \frac{13}{5}t + \frac{12}{7}\ e^{2t} - \frac{391}{350}e^{-5t} + \frac{1}{2}e^{t}$$

Problem 7

Using LT, Solve $\frac{dx}{dt} = 2x - 3y$

$\frac{dy}{dt} = y - 2x$ Given that x(0) = 8, y(0) = 3

Solution

Given $\frac{dx}{dt} = 2x - 3y$

Taking L.T. on bothsides

$L\left[\frac{dx}{dt}\right]$ = 2L (x) - 3 L(y)

sL(x) - x(0) = L(x) - 3L (y)

(s-2) L(x) - 8 + 3L(y) = 0 (1)

and

$\frac{dy}{dt}$ = y - 2x

Taking L.T. on both sides

$L\left[\frac{dy}{dt}\right]$ = L(y) - 2L(x)

sL(x) - y(0) = L(y) - 2L(x)

sL(y) -3 - L(y) + 2L (x) = 0

2L (x) + (s-1) L(y) - 3 = 0 (2)

(1) => (s-2) L(x) + 3 L(y) -8 = 0

(2) = > 2L (x) + (s-1) L(y)-3 = 0

Solving, we get,

$$\frac{L(x)}{8(s-1)-9} = \frac{L(y)}{-16+3(s-2)} = \frac{1}{(s-2)(s-1)-6}$$

$$\therefore \quad L(x) = \frac{8s-17}{s^2-3s-4} \; ; \; L(y) = \frac{3s-22}{s^2-3s-4}$$

$$\frac{8s-17}{s^2-3s-4} = \frac{A}{s-4} + \frac{B}{s+1}$$

$\therefore$ 8s-17 = A(s+1) + B (s-4)

put s=4, 15=5A => A = 3

s=-1, -25=5B => B = 5

$$\therefore \quad x = L^{-1}\left(\frac{3}{s-4}\right) + L^{-1}\left(\frac{5}{s+1}\right)$$

$x = 3e^{4t} + 5e^{-t}$

Let $\dfrac{3s-22}{s^2-3s-4} = \dfrac{C}{s-4}+\dfrac{D}{s+1}$

$3s-22 = C(s+1) + D(s-4)$

put s=4, -10=5c C = -2

s=-1 -25=-5D D = 5

$\therefore \quad y = L^{-1}\left(\dfrac{-2}{s-4}\right) + L^{-1}\left(\dfrac{5}{s+1}\right)$

$y = -2e^{4t}+5e^{-t}$

Problem 8

Solve $\dfrac{dy}{dt}+ay = x;\ \dfrac{dx}{dt}+ax=y$, given that x=0, y = 1, when t = 0

Solution

Given $\dfrac{dy}{dt}+ay = x;\ \dfrac{dx}{dt}+ax=y$

Taking L.T on bothsides,

$L\left[\dfrac{dy}{dt}\right] + aL(y) = L(x) \quad ; \quad L\left[\dfrac{dx}{dt}\right]+9L(x) = L(y)$

$sL(y) - y(0) +aL(y) = L(x) \quad ; \quad sL(x) - x(0) + aL(x) = L(y)$

$sL(y) - 1 + aL(y) = L(x) \quad ; \quad sL(x) + aL(x) = L(y)$

$(s+a)\ L(y) - L(x) = 1$

$(s+a)\ L(x) - L(y) = 0$

Solving these equation

$(s+a)^2 L(x) - L(x) = 1$

$$\Rightarrow L(x) = \frac{1}{(s+a)^2 - 1}$$

$$\therefore x = L^{-1}\left\{\frac{1}{(s+a)^2 - 1}\right\}$$

$$\boxed{x = e^{-at}\sinh t}$$

$$\Rightarrow L(y) = \frac{s+a}{(s+a)^2 - 1}$$

$$y = L^{-1}\left[\frac{s+a}{(s+a)^2 - 1}\right] = e^{-at}L^{-1}\left[\frac{s}{s^2 - 1}\right]$$

$$\boxed{y = e^{-at}\cosh t}$$

The solutions are,

$$\therefore \boxed{\begin{array}{l} x = e^{-at}\sinh t \\ \\ y = e^{-at}\cosh t \end{array}}$$

Check Your Progress

1. Find $L^{-1}\left(\frac{1}{s+5}\right)$ = ____________________

2. Find $L^{-1}\left(\frac{1}{s^5}\right)$ = ____________________

8.5 Recap

1. If $L[f(t)] = \bar{f}(s)$, then $f(t) = L^{-1}\left(\bar{f}(s)\right)$

2. $L^{-1}\left(\frac{1}{s^{n+1}}\right) = \frac{t^n}{n!}$

3. $L^{-1}\left(\frac{1}{s+a}\right) = e^{-at}$

4. $L^{-1}\left(\frac{s}{s^2+a^2}\right) = \cos at$

5. $L^{-1}\left(\frac{1}{s^2+a^2}\right) = \frac{\sin at}{a}$

6. $L^{-1}\left(\frac{1}{s^2-a^2}\right) = \frac{\sinh at}{a}$

7. Linearity Property : $L^{-1}[c_1 \bar{f}_1(s) + c_2 \bar{f}(s)] = c_1 f_1(t) + c_2 f_2(t)$

8. Shifting Property : $L^{-1}[\bar{f}(s-a)] = e^{at} f(t)$

9. Change of Scale Property : $L^{-1}[\bar{f}(as)] = \frac{1}{a}\bar{f}\left(\frac{t}{a}\right)$

10. Inverse L.T of Derivatives : $L^{-1}\left[\frac{d^n \bar{f}(s)}{ds^n}\right] = (-1)^n t^n f(t)$

11. I.L.T of Integrals : $L^{-1}\int_s^{\infty} \bar{f}(u)du = \frac{f(t)}{t}$

12. Multiplication by s : $L^{-1}[s\bar{f}(s)] = f'(t)$

13. Division by s : $L^{-1}\left[\frac{\bar{f}(s)}{s}\right] = \int_0^t f(u)du$

8.6 Exercise

1. Find the Laplace Transform of the following functions.

(i)	t^5-4t^3+3	(ii)	$\sin^4 t$
(iii)	$\cos 2t \sin 3t$	(iv)	$\sin 7t \cos 3t$

(v) $f(t) \begin{cases} 0, & 0 < t < 2 \\ t, & t > 2 \end{cases}$

2. Find the Laplace transform of the following functions

(i)	$t.e^{-3t}$	(ii)	$t\sin 3t$
(iii)	$t^2\cos 3t$	(iv)	$t\cosh 2t$
(v)	$e^{3t}\cos 6t-t^3+e^t$	(vi)	$\frac{1-\cos t}{t}$
(vii)	$\frac{\cos t-\cos 2t}{t}$	(viii)	$e^{2t}t^n$
(ix)	$te^{-2t}\sin 3t$	(x)	$(at+b)^2 e^{2t}$

3. Find the Laplace Transform of the following functions

(i)	$t^2+e^t\cos t$	(ii)	$e^{-t}\sin 3t+e^{6t}-1$
(iii)	$(1+e^{-t})^2$	(iv)	$e^{7t}\sin 2t$
(v)	$t\sin 2t$ sini5t	(vi)	$te^{2t}\cos 5t$
(vii)	$\sin^2 t \cos t$	(viii)	$\sin^3 t. \cos^2 t$
(ix)	$t^3\cos t$	(x)	$(\sin t-\cos t)^2$

4. Evaluate the following integrals using Laplase Transform

(i) $\int_0^\infty \frac{e^{-3t} - e^{-4t}}{t} dt$

(ii) $\int_0^\infty \frac{e^{-t} \sin t}{t} dt$

(iii) $\int_0^\infty t.e^{-2t} \cos t dt$

(iv) $\int_0^\infty t^3.e^{-t} \sin t dt$

(v) $\int_0^\infty te^{-3t} \sin t dt$

5. For a square wave function f(t) $\begin{cases} -1, & \text{for} \quad 0 < t < \frac{\pi}{2} \\ -1 & \text{for} \quad \frac{\pi}{2} < t < \pi \end{cases}$

show that the transform of this function is $\frac{1}{s} \tan \left(\frac{as}{\pi} \right)$

6. Find the inverse Laplase Transform aof the following functions.

(i) $\frac{6}{(s+2)^4}$ (ii) $\frac{3s-15}{2s^2-4s+16}$

(iii) $\frac{s-1}{(s^2-2s+5)}$ (iv) $\frac{3}{5^4}$

(v) $\frac{1}{s^2+4s+1}$ (vi) $\frac{1}{3s-5}$

(vii) $\frac{8}{s^2+4}$ (viii) $\frac{2s-8}{4s^2+25}$

7. Find the Inverse Laplace Transform of the following rational functions.

(i) $\frac{7s-1}{(s+1)(s+2)(s+3)}$ (ii) $\frac{3s+5}{s(s-2)(s+3)}$

(iii) $\frac{5s^2+34s+53}{(s+3)^2(s+4)}$ (iv) $\frac{5s+3}{(s-1)(s^2+2s+5)}$

(v) $\frac{s+2}{(s^2+1)(s^2+4)}$

8. Find

(i) $\log\left(\frac{s}{s^2+1}\right)$ (ii) $\log\left(\frac{s+2}{s-5}\right)$

(iii) $\log\left(\frac{s^2+9}{s^2+1}\right)$

9. Solve the following differential equations, using the method of Laplace Transforms.

(i) $\frac{d^2y}{dt^2} - \frac{dy}{dt} - 2y = 0$ given y (0) = -2, y'(0) = 5

(ii) $y''+4y'+5y = 4e^{3t}$, given y(0) = 2; y'(0) = 7

(iii) $\frac{d^2y}{dt^2} + y = t$ given y (0) = 1, y'(0) = -2.

(iv) $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = \sin t$, given that $y(0)$, $\frac{dy}{dt} = 0$, when $t = 0$

(v) $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 5y = e^{-t} \sin t$, given $y(0)$, $y'(0) = 1$.

(vi) $\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = \cos t$, given $y(0) = 0$, $y'(0) = -1$.

(vii) $\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 5y = e^{t}$, given $y(0) = 0$, $y'(0) = -1$.

10. Solve the following simultaneous differential equation using the Laplace transform.

(i) $\frac{dx}{dt} - y = e^{t}$; $\frac{dy}{dt} + x = \sin t$, Given that $x(0) = 1$ and $y(0) = 0$

(ii) $\frac{dx}{dt} - y = e^{t}$; $\frac{dy}{dt} - 3x + y = e^{2t}$, given $x(0) = 1$, $y(0) = 0$.

(iii) $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 2y = 0$, given $y(0) = 1$ and $y'(0) = -1$.

LESSON - 9
VECTOR DIFFERENTIATION

Learning Objectives

In this lesson our aim is to study about

- define of scalar function, vector function, Limit continuity and derivative of vector functions, velocity and acelaration.
- Define gradient of a scalar point function
- Determine unit normal to a surface
- Determine the tangent plane and normal to the surface.
- Properties of gradient

Structure

9.1 Introduction

In Calculas we deal with variables vectors (i.e.,) vectors which are varying in magnitutude or direction or both. Corresponding to each value of scalar variable t, if there exists a value of the vector $\vec{f}$, then $\vec{F}$ is called a vector function of the scalar variable t and is denoted as $\vec{F} = \vec{F}(t)$. For example the position of a particle that moves continiously on a curve in space varies with respect to time 't' (i.e.,) $\vec{r} = \vec{r}\ (t)$.

Also the position vector of the particle varies from point to point. Hence it can also be considered as a function of the point. If the points are specified by their rectangular cartesian co-ordinates (x,y,z) then $\vec{r}$ is a function of the scalar variables x,y,z (i.e.,) $\vec{r} = \vec{r}\ (x,y,z)$.

9.2 Vector Function

If f(f),g(t) and h(t) are the component function of $\vec{r}$ and we write $\vec{r}\,(t) = f(t)\,\vec{i} + g(t)\,\vec{j} + h(t)\,\vec{k}$. Where t denotes the inependent variable, because it represents time in most applications of vector function.

9.2.1 Limit of a vector function

The limit of a vector function $\vec{r}$ is defined by taking the limits of its component function as follows :

If $\vec{r}\ (t) = f(t)\ \vec{i} + g(t)\ \vec{j} + h(t)\ \vec{k}$

then

$$\lim_{t\to 0} \vec{r}\ (t) = \vec{i} \lim_{t\to 0} f(t) + \vec{j} \lim_{t\to 0} g(t) + \vec{k} \lim_{t\to 0} h(t)$$

Limit of vectors functions obeys the some rules as limits of real valued functions.

(or)

If for any small positive number ε, there corresponds a positive number. δ such that $\left|\vec{r}(t) - \vec{L}\right| < \varepsilon$ whenever $0 < \left|t - t_0\right| < \delta$.

9.2.2 Continuity of a vector function

A vector function $\vec{r}$ is said to be continuous at 'a'

if $\lim\limits_{t\to a} \vec{r}(t) = \vec{r}(a)$ exists and for any small positive number these corresponds a positive number S such that $|\vec{r}(t)-\vec{r}(a)| < \varepsilon$ function f,g and h are continous at 'a'.

9.2.3 Derivative of a vector function

The derivatives $\vec{r}^{\,1}$ of a vector function $\vec{r}$ is defined just as for real -valued functions.

i.e., $\dfrac{d\vec{r}}{dt} = \vec{r}^{\,1}(t) = \lim\limits_{h\to 0} \dfrac{\vec{r}(t+h)-\vec{r}(t)}{h}$

It can be shown that all the results of scalar differentiation can be extended to vector differentiation.

Some standard results

If $\vec{u},\vec{v},\vec{w}$ are differentiable, vector valued functions of the scalar variable 't'.

(i) $\dfrac{d}{dt}(\vec{u}+\vec{v}) = \dfrac{d\vec{u}}{dt}+\dfrac{d\vec{v}}{dt}$

(ii) $\dfrac{d}{dt}(\vec{u}\,.\,\vec{v}) = \vec{u}\,.\dfrac{d\vec{v}}{dt}+\dfrac{d\vec{u}}{dt}\,.\,\vec{v}$

(iii) $\dfrac{d}{dt}(\vec{u}\times\vec{v}) = \vec{u}\times\dfrac{d\vec{v}}{dt}+\dfrac{d\vec{u}}{dt}\times\vec{v}$

(iv) $\dfrac{d}{dt}\left(\dfrac{\vec{u}}{s}\right) = \dfrac{s\dfrac{d\vec{u}}{dt}-\vec{u}\dfrac{ds}{dt}}{s^2}$,where s is scalar

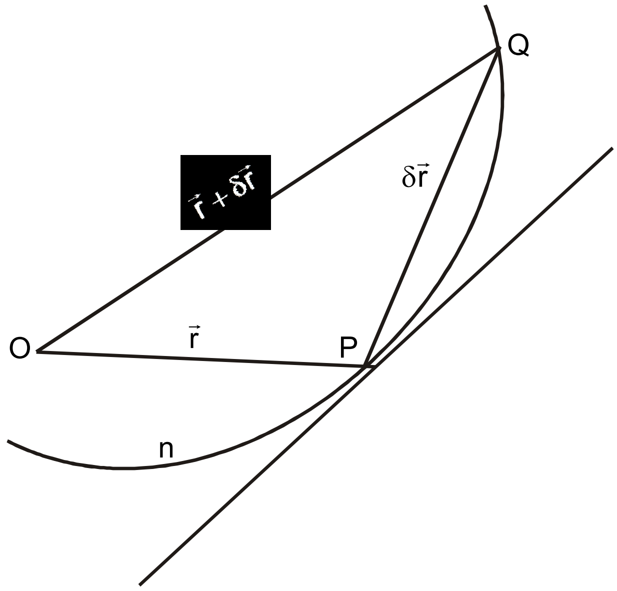
(v) $\dfrac{d}{dt}\left[\vec{u}\times(\vec{v}\times\vec{w})\right] = \dfrac{d\vec{u}}{dt}\times(\vec{v}\times\vec{w})+\vec{u}\times\left(\dfrac{d\vec{v}}{dt}\times\vec{w}\right)+\vec{u}\times\left(\vec{v}\dfrac{d\vec{w}}{dt}\right)$

9.2.4 Geometrical Significance of Vector Differentiation\

Let $\vec{r} = \vec{f}$ (t) be the vector equation of a curve in space.

Let P : Point on the curve

Q : Neighbouring point on the curve



Let

$\overrightarrow{OP}$ = $\vec{r} = \vec{f}$ (t)

OQ = $\vec{r} + \delta \vec{f}$

= $\vec{f}\,(\vec{r} + \delta\, \vec{r}\,)$

and

PQ = $\delta\, \vec{r}$

$\therefore$ $\dfrac{\delta\vec{r}}{\delta t}$ is a vector parallel to PQ

As Q $\to$ P (i.e., $\delta\, t \to 0$) chord PQ tends to tangent at P to the curve.

(i.e.,) $\lim\limits_{\delta t \to 0} \dfrac{\delta\vec{r}}{\delta t} = \dfrac{d\vec{r}}{dt}$ is a vector parallel to the tangent at P to the curve $\vec{r} = \vec{f}$ (t).

9.2.5 Physical Application of Derivatives of Vectors

Let t be time and $\vec{r}$ be the position vector of a moving point P of the particle.

Then, $\vec{v} = \dfrac{d\,\vec{r}}{dt}$ is the velocity at P and its direction is along the tangent at P.

If $\vec{v}$ is the velocity at time t of a moving point along a curve C.

Then, the acceleration of the particle at time t is

$$\therefore a = \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right) = \frac{d^2\vec{r}}{dt^2}$$

9.2.6 Partial Derivative of a vector function

Suppose $\vec{r}$ is a vector function of variable x,y and z.

Let $\vec{r} = \vec{f}(x, y, z)$

Then the partial derivative of $\vec{r}$ with respect to x is defined by

$$\frac{\partial \vec{r}}{\partial x} = \lim_{\Delta x \to 0} \frac{\vec{f}(x + \Delta x, y, z) - f(x, y, z)}{\Delta x}$$ if this limit exists.

$$\frac{\partial \vec{r}}{\partial y} = \lim_{\Delta y \to 0} \frac{\vec{f}(x, y + \Delta y, z) - f(x, y, z)}{\Delta y}$$

$$\frac{\partial \vec{r}}{\partial z} = \lim_{\Delta z \to 0} \frac{\vec{f}(x, y, z + \Delta z) - f(x, y, z)}{\Delta z}$$

Suppose $\vec{r}$ is a vector function of variable x,y and z.

Let $\vec{r} = \vec{f}(x, y, z)$

Then the partial derivative of $\vec{r}$ with respect to x is defined by

$$d\vec{r} = \frac{\partial \vec{r}}{\partial x}\, dx + \frac{\partial \vec{r}}{\partial y}\, dy + \frac{\partial \vec{r}}{\partial z}.dz$$

Where

$$\frac{\partial \vec{r}}{\partial x} = \lim_{\partial x \to 0} \frac{\vec{f}(x+\partial x, y, z) - f(x,y,z)}{\partial x} \text{ exists}$$

Similarly,

$$\frac{\partial \vec{r}}{\partial y} = \frac{\partial \vec{r}}{\partial z} \text{ are also exists.}$$

Example 1

A particle moves along the curve $x = t^3+1$, $y = t^2$, $z = 2t + 5$, where t is the time. Find the components of its velocity and acceleration at t =1 in the direction $\vec{i} + \vec{j} + 3\vec{k}$

Solution

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$= \left(t^3+1\right)\vec{i} + t^2\vec{j} + (2t+5)\vec{k}$$

Velocity $\vec{v} = \frac{d\vec{r}}{dt} = 3t^2\vec{i} + 2t\vec{j} + 2\vec{k}$

at t =1, is $\vec{v} = 3\vec{i} + 2\vec{j} + 2\vec{k}$

Acceleration $\vec{a} = \frac{d^2\vec{r}}{dt^2} = 6t\vec{i} + 2\vec{j}$

$$\left(\frac{d^2\vec{r}}{dt^2}\right)_{t=1} = 6\vec{i} + 2\vec{j}$$

Velocity of component in the direction of

$\vec{b} = \vec{i} + \vec{j} + 3\vec{k}$ at t=1 (Given)

$= \quad \vec{v} \, . \, \hat{b}$

$$= \quad \left(3\vec{\text{l}} + 2\vec{\text{j}} + 2\vec{\text{k}}\right) \ . \ \frac{\left(\vec{\text{i}} + \vec{\text{j}} + 3\text{k}\right)}{\sqrt{11}}$$

$$= \quad \frac{3+2+6}{\sqrt{11}}$$

$$= \quad \frac{11}{\sqrt{11}}$$

$$= \quad \sqrt{11}$$

Acceleration component in the direction of

$$\vec{\text{b}} \quad = \ \vec{\text{a}}.\hat{\text{b}}\left|\vec{\text{b}}\right| \ \text{at t = 1}$$

$$= \ \left(6\vec{\text{i}} + 2\vec{\text{j}}\right) \ . \ \frac{\left(\vec{\text{l}} + \vec{\text{j}} + 3\vec{\text{k}}\right)}{\sqrt{11}}$$

$$= \ \frac{6+2}{\sqrt{11}}$$

$$= \ \frac{8}{\sqrt{11}}$$

Example 2

A particle moves along a curve whose equations are $x = e^{-t}$, $y = 2\cos 3t$, $z = 2 \sin 3t$. Determine the velocity and accelaration at any time t and their magnitutdes at t = 0.

Solution

$$\vec{\text{r}} = x\vec{\text{i}} + y\vec{\text{j}} + z\vec{\text{k}}$$

$$\vec{\text{r}} = e^{-t} \ \vec{\text{i}} + 2\cos 3t\,\vec{\text{j}} + 2\sin 3t\vec{\text{k}}$$

$$\vec{\text{v}} = \frac{d\vec{\text{r}}}{dt} = -e^{-t} \ \vec{\text{i}} - 6\sin 3t\vec{\text{j}} + 6\cos 3t\vec{\text{k}}$$

Velocity :

$$\left(\frac{d\vec{r}}{dt}\right)_{t=0} = -\vec{i} + 6\vec{k}$$

Magnititude of the velocity = $|\vec{v}|$ = $\sqrt{1+36} = \sqrt{37}$

(i) Accelaration $\vec{a} = \left(\frac{d^2\vec{r}}{dt^2}\right)_{t=0} = e^{-t}\vec{i} - 18\cos 3t\vec{j} - 18\sin 3t\vec{k}$

$$= \vec{i} - 18\vec{j}$$

Magnitude of $\vec{a}$:

$$|\vec{a}| = \sqrt{1+324} = \sqrt{325} = 5\sqrt{13}$$

Example 3

If $\vec{r} = \vec{a}\cos nt + \vec{b}\sin nt$ where $\vec{a}, \vec{b}$ n are constant prove that $\frac{d^2\vec{r}}{dt^2} + n^2\vec{r} = 0$

Solution

Given $\vec{r} = \vec{a}\cos nt + \vec{b}\sin nt$

$$\frac{d\vec{r}}{dt} = -\vec{a}n\sin nt + \vec{b}n\cos nt$$

$$\frac{d^2\vec{r}}{dt^2} = -n^2\vec{a}\cos nt - n^2\vec{b}\sin nt$$

$$= -n^2(\vec{a}\cos nt + \vec{b}\sin nt)$$

$$= -n^2\vec{r}$$

$$\Rightarrow \frac{d^2\vec{r}}{dt^2} + n^2\vec{r} = 0$$

Hence proved.

Example 4

A particle moves along a curve whose equations are $x = e^{-t}$, $y = 2\cos 3t$, and $z = 2\sin 3t$ where t is the time, fixed.

(i) its velocity and acceleration

(ii) its magnitude of velocity and accelaration at t = 0

Solution

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$$\vec{r} = e^{-t}\,\vec{i} + 2\cos 3t\,\vec{j} + 2\sin 3t\vec{k}$$

Differentiating w.r.t. t

$$\text{Velocity} = \frac{d\vec{r}}{dt} = \frac{d}{dt}\left[e^{-t}\,\vec{i} + 2\cos 3t\,\vec{j} + 2\sin 3t\,\vec{k}\right]$$

$$= -e^{-t}\,\vec{i} - 6\sin 3t\vec{j} + 6\cos 3t\vec{k}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2} = \frac{d}{dt}\left(-e^{-t}\,\vec{i} - 6\sin 3t\vec{j} + 6\cos 3t\vec{k}\right)$$

$$= e^{-t}\,\vec{i} - 18\cos 3t\vec{j} - 18\sin 3t\vec{k}$$

(ii) At time t = 0, to find velocity and acceleration

$$\left(\frac{d\vec{r}}{dt}\right)_{t=0} = -e^{-0}\,\vec{i} - 6\sin 0\,\vec{j} + 6\cos 0\vec{k}$$

$$\vec{v} = -\vec{i} + 6\vec{k}$$

$$|\vec{v}| = \sqrt{1^2 + 6^2} = \sqrt{37}$$

$$\vec{a} = \left(\frac{d^2\vec{r}}{dt^2}\right)_{t=0} = e^{-0}\,\vec{i} - 18\cos 0\vec{j} - 18\sin 0\vec{k}$$

$$\vec{a} = \vec{i} - 18\vec{j}$$

$$|\vec{a}| = \sqrt{1^2 + 18^2} = \sqrt{325}$$

$\therefore$ magnitude of velocity = $\sqrt{37}$

Magnitude of acceleration = $\sqrt{325}$

Example 5

Find the unit tangent vector at any pt on the curive x =t^2+1, y = 4t -3, z = $2t^2$ -6t and Determine the unit tangent at the point where t = 2

Solution

A tangent to the curve at any point is defined as $\frac{d\vec{r}}{dt}$, where $\vec{r} = x\vec{i} + y\vec{i} + 2\vec{k}$

$$\therefore \ \vec{r} = (t^2 + 1)\vec{i} + (4t - 3)\vec{j}(2t^2 - 6t)\vec{k}$$

$$\frac{d\vec{r}}{dt} = 2t\vec{i} + 4\vec{j} + (4t - 6)\vec{k}$$

The magnitude of the vector is $\left|\frac{d\vec{r}}{dt}\right|$ = $\sqrt{(2t)^2 + 4^2 + (4t - 6)^2}$

The required unit tangent vector is

$$\hat{T} = \frac{\frac{d\vec{r}}{dt}}{\left|\frac{d\vec{r}}{dt}\right|}$$

$$= \frac{2t\vec{i} + 4\vec{j} + (4t - 6)\vec{k}}{\sqrt{(2t)^2 + 4^2 + (4t - 6)^2}}$$

(ii) At t = 2, the unit tangent vector is

$$\hat{T} = \frac{2(2)\vec{i} + 4\vec{j} + [4(2) - 6]\vec{k}}{\sqrt{4^2 + 4^2 + (8 - 6)^2}}$$

$$= \frac{4\vec{i} + 4\vec{j} + 2\vec{k}}{\sqrt{16+16+4}}$$

$$= \frac{4\vec{i} + 4\vec{j} + 2\vec{k}}{6}$$

$$= \frac{2}{3}\vec{i} + \frac{2}{3}\vec{j} + \frac{1}{3}\vec{k}$$

Check Your Progress - I

1. Find the velocity and accelaration at time t=1 of a particle which moves along the curve $x = 3t^2$, $y=t^2$, $z=t^3$.

2. For the curve $x = e^t \cos t$, $y = e^t \sin t$, $z=e^t$. Find the velocity and acceleration of the particle moving on the curve at t = 0.

3. Find the acceleration of the particle which moves that the curve $x = 2x\sin 3t$, $y = 2\cos 3t$, $z=3t$ at $t = \frac{\pi}{2}$.

9.3 Scalar and Vector point function

Definition of Scalar Point Function

If to each point (x,y,z) of a region R in space, there correspondence a number or a scalar ϕ (x,y,z), then ϕ is called a scalar point function and we say that a scalar field has been defined in R.

Examples

1. The temperature at any point within or on the surface of earth at a given time defines a scalar field.

2. The pressure P in a field varies according to its depth and p(x,y,z) is a scalar point function.\

3. $Q(x,y,z) = x^2+y^2+z^2$ defines a scalar field density, potential are also examples of scalar point functions and for their scalar fields.

Definition of Vector Point Function

If to each point (x,y,z) of a region R in space, there correspondence a vector $\vec{F}$ (x,y,z),then $\vec{F}$ is called a vector point function and we say that a vector field has been defined in R.

Examples

1. The velocity at any point (x,y,z) within a moving field at time t is a vector point functionand hence a vector field in the region of field.

2. $\vec{F}$ (x,y,z) = $2x\,\vec{i} + xy\,\vec{j} + xyz\,\vec{k}$ defined a vector field. For electric intensity are also examples of vector point function.

Level Surfaces

Let ϕ (x,y,z) be a scalar filed defined in region R. Then the set of all points satisfying the equation $\phi(x, y, z)$ = c (a constant), determines a surface called level surface. If c is an arbitrary constant, then for different values of c we get different level surfaces.

Examples

1. ϕ (x,y,z) = $x^2+y^2+z^2-16$ is a Level Surface

2. Equipotential or isothermal surfaces are also examples of level Surfaces.

9.3.1 Gradient of a Scalar point function

Let ϕ(x,y,z) be a scalar point function defined in a region of space. Then the vector point function is given by

$$\nabla\phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$$

$$\text{grad}\,\phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)\phi$$

is defined as the gradient of ϕ and denoted by grad ϕ.

Note 1 :

1. $\nabla = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}$ is a vector differential operator.

2. If f defines o scalar fieldm then $\nabla \phi$ defines a vector field.

Note 2 :

1. $\nabla \phi$ should not be written as $\phi \Delta$.

2. It ϕ is a constant $\nabla \phi = 0$.

3. $\nabla (\phi_1, \phi_2) = \phi_1 \; \Delta\phi_2 + \phi_3 \; \nabla \phi_1$

4. $\nabla , \left(\frac{\phi_1}{\phi_2}\right) = \frac{\phi_2 \nabla \phi_1 - \phi_1 \nabla \phi_2}{\phi_2^2}$ if $\phi_2 \neq 0$.

5. If v = f(u), then ∇ v = f^1(u). ∇ u.

9.3.2 Directional derivative of a scalar point function

Let ϕ (x,y,z) be scalar point functions in a region R. Let P be any point in this region and Q be a neighbouring point to P in that region in the given direction $\vec{a}$.

Then

$\lim_{Q \to P} \frac{\phi(Q) - \phi(P)}{PQ}$ if it exists in called the divectional derivative of P in the direction of $\vec{a}$.

But the direction derivative of ϕ at P in the direction of

$$\lim_{Q+P} \frac{\phi(Q) - \phi(P)}{PQ} = \lim_{\delta 3 \to 0} \frac{\delta\varphi}{\delta s} = \frac{d\phi}{ds}$$

$\therefore \frac{d\phi}{ds}$ represents the rate of change of ϕ with respect to the distance of P in the direction $\vec{a}$,

Theorem 1

The directional derivative of a scalar point function ϕ at a point (x,y,z) in the direction of unit vector $\vec{a}$ is given by $\nabla\phi.\hat{a}$

Proof

Let $\phi(x, y, z)$ be a scalar point function defined in a given region R.

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ be the position vector of the point p (x,y,z) in this region.

If s denotes the distance of P from some fixed point in the direction of $\vec{a}$.

Then δs. dentoes a small element at P in the direction $\vec{a}$.

Therefore $\frac{d\vec{r}}{ds}$ is a unit vector at p in that direction.

$$\therefore \frac{d\vec{r}}{ds} = \hat{a}$$

But $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$\frac{d\vec{r}}{ds} = \frac{dx}{ds}\vec{i} + \frac{dy}{ds}\vec{j} + \frac{dz}{ds}\vec{k} = \vec{a}$$

$$\therefore \nabla\phi.\vec{a} = \left(\vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}\right).\left(\vec{i}\frac{\partial x}{\partial s} + \vec{j}\frac{\partial y}{\partial s} + \vec{k}\frac{\partial z}{\partial s}\right)$$

$$= \frac{\partial\phi}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial\phi}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial\phi}{\partial z}\frac{\partial z}{\partial s}$$

$$= \frac{\partial\phi}{\partial s}$$

$\therefore \nabla\phi.\vec{a}$ = the directional derivative of ϕ in the direction $\hat{a}$

Theorem 2

Let $\vec{n}$ be a unit vector normal to the level surface $\phi(x,y,z) = c$ at the point $p(x,y,z)$.

Let n be the distance of P from some fixed point in the direction of $\vec{n}$ so that δn represents an element of P in the direction of the normal at P (i.e., $\hat{n}$)

Then $\nabla\phi = \frac{d\phi}{dn}.\hat{n}$.

Theorem

The maximum value of the directional derivative of ϕ at P is $|\nabla\phi|$

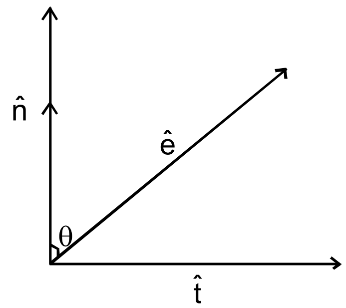
Proof

Let $\hat{t}$ be the unit tangent at P and $\hat{n}$ be the unit normal at p. Since $\nabla\phi$ is a vector normal to the surface we ahve

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \text{unit normal vector}$$

$$\therefore \nabla\phi = |\nabla\phi|.\hat{n}$$

The directional derivative of ϕ at p along any direction specified by the unit vector $\hat{e}$ is given by



$$\nabla\phi.\hat{e} = |\nabla\phi|\hat{n}.\hat{e}$$

$$= |\nabla\phi|\cos\theta$$

Where θ is the angle between $\hat{n}$ and $\hat{e}$. This value is a maximum when $\theta = 0$. That is the maximum value of the directional derivative of θ is $|\nabla\phi|$ and this value corresponds to the direction normal to the surface at p.

Theorem 3

The directional derivatives at a point maximum in the direction of the normal to the level surface at P and its magnitutde is $\nabla\phi$.

The maximum directional derivative at

P = $\frac{d\phi}{dn}$ = $|\nabla\phi|$

9.3.3 Equation of Tangent plane and Normal line to a level surface

Let the equation of the level surface be $\phi(x, y, z) = c$ let P(x, y, z) be a point on the level surface with position vector $\vec{r_1}$ so that $\vec{r_1} = x_1\vec{i} + y_1\vec{j} + z_1\vec{k}$

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ be the position vector of any point on the tangent plane at P.

If $\vec{r} - \vec{r_1}$ lies on the tanget plane at P and hence it is perpendicular to the $\nabla\phi$

$$\because \left(\vec{r} - \vec{r_1}\right).\nabla\phi = 0$$

$$\left[(x - x_1)\vec{i} + (y - y_1)\vec{j} + (z - z_1)\vec{k}\right].\left[\vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}\right] = 0$$

Therefore, $(x - x_1)\frac{\partial\phi}{\partial x} + (y - y_1)\frac{\partial\phi}{\partial y} + (z - z_1)\frac{\partial\phi}{\partial z} = 0$ .... (1)

This is the equation to the tangent plane at p (x_1, y_1, z_1)

Normal at P :

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ be the position vector of any point Q on the normal at P. Hence

$\overrightarrow{PQ} = \left(r - \vec{r_1}\right)$ lies along the normal at P to the surface. Hence it is parallel to $\nabla\phi$

$\therefore \left(r - \vec{r_1}\right) x \, \nabla\phi = 0$ .... (2)

This is the vector equation of the normal at P (x_1, y_1, z_1)

Cartesian equation of the normal at (x_1, y_1, z_1)

$$\left(\vec{r}-\vec{r_1}\right)=(x-x_1)\vec{i}+(y-y_1)\vec{j}+(z-z_1)\vec{k}$$

$$\nabla\phi=\vec{i}\,\frac{\partial\phi}{\partial x}+\vec{j}\,\frac{\partial\phi}{\partial y}+\vec{k}\,\frac{\partial\phi}{\partial z}$$

$\left(\vec{r}-\vec{r_1}\right)$ is parallel to $\nabla\phi$ if $\vec{r}-\vec{r_1}=p\nabla\phi$

where p is some scalar.

$$(x-x_1)\vec{i}+(y-y_1)\vec{j}+(z-z_1)\vec{k}=P\left[\vec{i}\,\frac{\partial\phi}{\partial x}+\vec{j}\,\frac{\partial\phi}{\partial y}+\frac{\partial\phi}{\partial z}\right]$$

Hence we get

$$\frac{x-x_1}{\frac{\partial\phi}{\partial x}}=\frac{y-y_1}{\frac{\partial\phi}{\partial y}}=\frac{z-z_1}{\frac{\partial\phi}{\partial z}} \qquad \text{................ (3)}$$

Consider the equation of the level surface be $\phi(x,y,z)=c$

Let $\vec{r}=x\vec{i}+y\vec{j}+z\vec{k}$ be any point on the surface

and $\nabla\phi$ = $\frac{\partial\phi}{\partial x}+\vec{i}+\frac{\partial\phi}{\partial y}\vec{j}+\frac{\partial\phi}{\partial z}\vec{k}$ is normal to the surface.

$\therefore$ $\frac{\partial\phi}{\partial x},\frac{\partial\phi}{\partial y},\frac{\partial\phi}{\partial z}$ are the direction ratio of the normal at P.

The normal at P is perpendicular to the tangent plane at P

If, (x_1,y_1,z_1) is a point on the tangent plane, then the equation of the tangent plane is

$A(x-x_1) + B\,(y-y_1) + C(z-z_1) = 0$

(i.e.,) $(x-x_1)\dfrac{\partial\phi}{\partial x} + (y-y_1)\dfrac{\partial\phi}{\partial y} + (z-z_1)\dfrac{\partial\phi}{\partial z} = 0$

The direction ratio are the values of $\dfrac{\partial\phi}{\partial x}, \dfrac{\partial\phi}{\partial y}, \dfrac{\partial\phi}{\partial z}$ evaluated at the point (x_1,y_1,z_1)

∴ The equation of the normal at (x_1,y_1,z_1) is

$$\frac{x-x_1}{\frac{\partial\phi}{\partial x}} = \frac{y-y_1}{\frac{\partial\phi}{\partial y}} = \frac{z-z_1}{\frac{\partial\phi}{\partial z}}$$

9.4 Properties of Gradient

Property 1

If f and g are any two scalar point functions then

$\nabla(f \pm g) = \nabla f \pm \nabla g$

Proof

$$\nabla(f+g) = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)(f+g)$$

$$= \vec{i}\frac{\partial}{\partial x}(f+g) + \vec{j}\frac{\partial}{\partial y}(f+g) + \vec{k}\frac{\partial}{\partial z}(f+g)$$

$$= \vec{i}\left(\frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}\right) + \vec{j}\left(\frac{\partial f}{\partial y} + \frac{\partial g}{\partial y}\right) + \overrightarrow{kj}\left(\frac{\partial f}{\partial z} + \frac{\partial g}{\partial z}\right)$$

$$= \left(\vec{i}\frac{\partial f}{\partial x} + \vec{j}\frac{\partial f}{\partial y} + \vec{k}\frac{\partial f}{\partial z}\right) + \left(\vec{i}\frac{\partial g}{\partial x} + \vec{j}\frac{\partial g}{\partial y} + \vec{k}\frac{\partial g}{\partial z}\right)$$

$$\nabla(f+g) = \nabla f + \nabla g$$

Similarly

$$\nabla(f - g) = \nabla f - \nabla g$$

$\therefore$ we get $\nabla(f \pm g) = \nabla f \pm \nabla g$

Property 2

If f and g are two scalar point function then

∇ (fg) = f (∇ g) + g (∇ f)

i.e., grad (fg) = f (gradg) + g (grad f)

Proof

$$\nabla(fg) = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)(fg)$$

$$= \vec{i}\frac{\partial}{\partial x}(fg) + \vec{j}\frac{\partial}{\partial y}(fg) + \vec{k}\frac{\partial}{\partial z}(fg)$$

$$= \vec{i}\left(f\frac{\partial g}{\partial x} + g\frac{\partial f}{\partial x}\right) + \vec{j}\left(f\frac{\partial g}{\partial y} + g\frac{\partial f}{\partial y}\right) + \vec{k}\left(f\frac{\partial g}{\partial z} + g\frac{\partial f}{\partial z}\right)$$

$$= f\left(\vec{i}\frac{\partial g}{\partial x} + \vec{j}\frac{\partial g}{\partial y} + \vec{k}\frac{\partial g}{\partial z}\right) + g\left(\vec{i}\frac{\partial f}{\partial x} + \vec{j}\frac{\partial f}{\partial y} + \vec{k}\frac{\partial f}{\partial z}\right)$$

∇ (fg) = f (∇ g) + g (∇ f)

i.e., grad (fg) = f (grad g) + g (grad f)

Property 3

Gradient of a constant is zero. (that is, $\nabla\,\phi = 0$, Here ϕ = C)

Let $\phi(x, y, z)$ be a constant

Then $\frac{\partial \phi}{\partial x} = 0, \frac{\partial \phi}{\partial y} = 0, \frac{\partial \phi}{\partial z} = 0$

$$\because \nabla\phi = \vec{i}\frac{\partial \phi}{\partial x} + \vec{j}\frac{\partial \phi}{\partial y} + \vec{k}\frac{\partial \phi}{\partial z} = 0\vec{i} + 0\vec{j} + 0\vec{k} = 0$$

Proerty 4

If f and g are two scalar point functions then

$$\nabla\left(\frac{f}{g}\right)=\frac{g(\nabla f)-f(\nabla g)}{g^2}$$

Proof

We have

$$\nabla\left(\frac{f}{g}\right)=\left(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z}\right)\left(\frac{f}{g}\right)$$

$$=\vec{i}\frac{\partial}{\partial x}\left(\frac{f}{g}\right)+\vec{j}\frac{\partial}{\partial y}\left(\frac{f}{g}\right)+\vec{k}\frac{\partial}{\partial z}\left(\frac{f}{g}\right)$$

$$=\vec{i}\left[\frac{g\frac{\partial f}{\partial x}-f\frac{\partial g}{\partial x}}{g^2}\right]+\vec{j}\left[\frac{g\frac{\partial f}{\partial y}-f\frac{\partial g}{\partial y}}{g^2}\right]+\vec{k}\left[\frac{g\frac{\partial f}{\partial z}-f\frac{\partial g}{\partial z}}{g^2}\right]$$

$$=\frac{1}{g^2}\left[g\left(\vec{i}\frac{\partial f}{\partial x}+\vec{j}\frac{\partial f}{\partial y}+\vec{k}\frac{\partial f}{\partial z}\right)\right]-f\left(\vec{i}\frac{\partial g}{\partial x}+\vec{j}\frac{\partial g}{\partial y}+\vec{k}\frac{\partial g}{\partial z}\right)$$

$$=\frac{1}{g^2}\left[g(\nabla f)-f(\nabla g)\right]$$

$$\nabla\left(\frac{f}{g}\right)=\frac{g(\nabla g)-f(\nabla g)}{g^2}$$

9.5 Simple Problems

Example 1

If $\phi(x,y,z) = x^2y+y^2x+z^2$. Find $\nabla\phi$ at the point (1,1,1)

Solution

$\phi(x,y,z) = x^2y+y^2x+z^2$

$$\frac{\partial\phi}{\partial x} = 2xy + y^2$$

$$\frac{\partial\phi}{\partial y} = x^2 + 2xy$$

$$\frac{\partial\phi}{\partial z} = 2z$$

$$\therefore \nabla\phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$$

$$= (2xy+y^2)\,\vec{i} + (x^2+2xy)\,\vec{j} + 2z\,\vec{k}$$

At point (1,1,1)

$$(\nabla\phi)_{(1,1,1)} = 3\vec{i} + 3\vec{j} + 2\vec{k}$$

Example 2

If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = \left|\vec{r}\right|$ Prove that

(i) $\nabla r = \frac{1}{r}\vec{r}$

(ii) $\nabla\left(\frac{1}{r}\right) = \frac{-\vec{r}}{r^3}$

(iii) $\nabla r^n = nr^{n-2}\vec{r}$

(iv) $\nabla f(r) = f'(r)\frac{\vec{r}}{r} = f'(r)\nabla r$

(v) ∇ (log r) = $\frac{\vec{r}}{r^2}$

(vi) ∇ f(r) x $\vec{r}$ = 0

Solution

(i) $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$ = r

=> $r^2 = x^2 + y^2 + z^2$ .... (1)

Diffy (1) 'p' w.r. to x we get

$$2r\frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$\frac{\partial r}{\partial x} = \frac{y}{r} \text{ and } \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\therefore \nabla r = \vec{i}\frac{\partial r}{\partial x} + \vec{j}\frac{\partial r}{\partial y} + \vec{k}\frac{\partial r}{\partial z}$$

$$= \frac{x\vec{i} + y\vec{j} + z\vec{k}}{r}$$

$$\therefore \nabla r = \frac{\vec{r}}{r}$$

(ii) $\nabla\left(\frac{1}{r}\right) = \hat{i}\frac{\partial}{\partial x}\left(\frac{1}{r}\right) + \hat{j}\frac{\partial}{\partial y}\left(\frac{1}{r}\right) + \hat{k}\frac{\partial}{\partial z}\left(\frac{1}{r}\right)$

$$= \frac{-1}{r^2}\left[\hat{i}\frac{\partial r}{\partial x}+\hat{j}\frac{\partial r}{\partial y}+\hat{k}\frac{\partial r}{\partial z}\right]$$

$$= \frac{-1}{r^2}\left[\hat{i}\frac{x}{r}+\hat{j}\frac{y}{r}+\hat{k}\frac{z}{r}\right]$$

$$= \frac{-1}{r^2}\left(\frac{\vec{r}}{r}\right) \qquad \text{by (1)}$$

$$\therefore \nabla\left(\frac{1}{r}\right) = \frac{\vec{r}}{r^3}$$

(iii) $\nabla r^n \quad = \hat{i}\frac{\partial}{\partial x}(r^n)+\hat{j}\frac{\partial}{\partial y}(r^n)+\hat{k}\frac{\partial}{\partial z}(r^n)$

$$= nr^{n-1}\left[\hat{i}\frac{\partial r}{\partial x}+\hat{j}\frac{\partial r}{\partial y}+\hat{k}\frac{\partial r}{\partial z}\right]$$

$$= nr^{n-1}\left[\frac{x\,\vec{i}+y\,\vec{j}+z\,\vec{k}}{r}\right]$$

$$\therefore \nabla r^n = nr^{n-2}\,.\,\vec{r}$$

(iv) $\nabla f(r) \quad = \vec{i}\,\frac{\partial}{\partial x}f(r)+\vec{j}\,\frac{\partial}{\partial y}f(r)+\vec{k}\,\frac{\partial}{\partial z}f(r)$

$$= f'(r)\left[\hat{i}\frac{\partial r}{\partial x}+\hat{j}\frac{\partial r}{\partial y}+\hat{k}\frac{\partial r}{\partial z}\right]$$

$$= f'(r)\ \frac{\left(x\vec{i} + y\vec{j} + z\vec{k}\right)}{r}$$

$$= f'(r)\ \frac{\vec{r}}{r}$$

$$\nabla f(r) = f'(r).\ \nabla r$$

(v) $$\nabla(\log r) = \vec{i}\frac{\partial}{\partial x}(\log r) + \vec{j}\frac{\partial}{\partial y}(\log r) + \vec{k}\frac{\partial}{\partial z}(\log r)$$

$$= \frac{1}{r}\left[\vec{i}\frac{\partial r}{\partial x} + \vec{j}\frac{\partial r}{\partial y} + \vec{k}\frac{\partial r}{\partial z}\right]$$

$$= \frac{1}{r}\ \frac{\left(x\vec{i} + y\vec{j} + z\vec{k}\right)}{r}$$

$$\nabla(\log r) = \frac{\vec{r}}{r^2}$$

(vi) $$\nabla f(r) = \frac{f'(r)}{r}\vec{r}$$

$$\nabla f(r) \times \vec{r} = \frac{f'(r)}{r}\vec{r}\,.\,\vec{r} = 0$$ since $\vec{r} \times \vec{r} = 0$

Problem 1

Find grad ϕ if ϕ = xyz at (1,1,1)

Solution

Let ϕ = xyz

$$\frac{\partial\phi}{\partial x} = yz, \frac{\partial\phi}{\partial y} = xz, \frac{\partial\phi}{\partial z} = xy$$

$$\nabla\phi = \vec{i}\,\frac{\partial\phi}{\partial x} + \vec{j}\,\frac{\partial\phi}{\partial y} + \vec{k}\,\frac{\partial\phi}{\partial z}$$

$$\nabla\phi = (yz)\,\vec{i} + (xz)\,\vec{j} + (xy)\vec{k}$$

At (1,1,1),

$$\nabla\phi = \vec{i} + \vec{j} + \vec{k}$$

Problem 2

If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ prove the following

(i) $\nabla r = \frac{1}{r}\vec{r}$

(ii) $\nabla r^n = nr^{n-2}\vec{r}$

(iii) $\nabla f(r) = f'(r)\frac{\vec{r}}{r}$

Solution

Given $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$$\therefore\ r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

$$r^2 = x^2 + y^2 + z^2 \qquad \text{...........} (1)$$

Partially differentiating w.r.t x,y,z, we get

$$2r\frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

$$2r\frac{\partial r}{\partial y} = 2y \Rightarrow \frac{\partial r}{\partial y} = \frac{y}{r} \qquad \text{..........} (2)$$

$$2r\frac{\partial r}{\partial z} = 2z \Rightarrow \frac{\partial r}{\partial z} = \frac{z}{r}$$

(i) $\nabla r = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)r$

$= \vec{i}\frac{\partial r}{\partial x} + \vec{j}\frac{\partial r}{\partial y} + \vec{k}\frac{\partial r}{\partial z}$

$= \frac{x}{r}\vec{i} + \frac{y}{r}\vec{j} + \frac{z}{r}\vec{k} = \frac{x\vec{i} + y\vec{j} + z\vec{k}}{r}$, Using (2)

$\nabla r = \frac{\vec{r}}{r}$

(iii) $\nabla r^n = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right)r^n$

$= \vec{i}\frac{\partial}{\partial x} + \left(r^n\right) + \vec{j}\frac{\partial}{\partial y} x \left(r^n\right) + \vec{k}\frac{\partial}{\partial z}\left(r^n\right)$

$= \vec{i}\, nr^{n-1}\frac{\partial r}{\partial x} + \vec{j}\, nr^{n-1}\frac{\partial r}{\partial y} + \vec{k} nr^{n-1}\frac{\partial r}{\partial z}$

$= nr^{n-1}\left[\vec{i}\frac{\partial r}{\partial x} + \vec{j}\frac{\partial r}{\partial y} + \bar{k}\frac{\partial r}{\partial z}\right]$

$= nr^{n-1}\left[\frac{x}{r}\vec{i} + \frac{y}{r}\vec{j} + \frac{z}{r}\vec{k}\right]$ using 2,

$= nr^{n-1}\left[\frac{x\vec{i} + y\vec{j} + z\vec{k}}{r}\right]$

$\nabla r^n = nr^{n-2}\vec{r}$

(iii) $\nabla [f(r)] = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z} \right) f(r)$

$$= \vec{i}\frac{\partial}{\partial x} f(r) + \vec{j}\frac{\partial}{\partial y} f(r) + \vec{k}\frac{\partial}{\partial z} f(r)$$

$$= \vec{i} f'(r)\frac{\partial r}{\partial x} + \vec{j} f'(r)\frac{\partial r}{\partial y} + \vec{k} f'(r)\frac{\partial r}{\partial y}$$

$$= f'(r)\left[\vec{i}\frac{\partial r}{\partial x} + \vec{j}\frac{\partial r}{\partial y} + \vec{k}\frac{\partial r}{\partial z} \right]$$

$$= f'(r)\left[\frac{x}{r}\vec{i} + \frac{y}{r}\vec{j} + \frac{z}{r}\vec{k} \right], \text{ using (2)}$$

$$= \frac{f'(r)}{r}\left[x\vec{i} + y\vec{j} + z\vec{k} \right]$$

$$= f'(r)\frac{\vec{r}}{r}$$

Problem 3

Find the directional derviative of xyz - $xy^2 z^3$ at the point (1,2,-1) in the directions of the vector $\vec{i} - \vec{j} - 3\vec{k}$

Solution

Let $\phi = xyz - xy^2z^3$

$$\frac{\partial \phi}{\partial x} = yz - y^2z^3$$

$$\frac{\partial \phi}{\partial y} = xz - 2xyz^3$$

$$\frac{\partial \phi}{\partial z} = xy - 3xy^2z^2$$

$$\nabla\phi = \left(yz - y^2z^3\right)\vec{i} + \left(xz - 2xyz^3\right)\vec{j} + \left(xy - 3xy^2z^2\right)\vec{k}$$

At (1,2,-1), we have

$$\nabla\phi \quad = \left(-2+4\right)\vec{i} + \left(-1+4\right)\vec{j} + \left(2-12\right)\vec{k}$$

$$= 2\vec{i} + 3\vec{j} - 10\vec{k}$$

Let $\vec{a} = \vec{i} - \vec{j} - 3\vec{k}$

$$\left|\vec{a}\right| = \sqrt{1^2 + 1^2 + 3^2} = \sqrt{11}$$

$$\vec{a} = \frac{\vec{i} - \vec{j} - 3\vec{k}}{\sqrt{11}}$$

Directional derivative in the directions of the vector $\vec{i} - \vec{j} - 3\vec{k}$

$$\nabla\phi.\hat{a} = \left(2\vec{i} + 3\vec{j} - 10\vec{k}\right) . \frac{\left(\vec{i} - \vec{j} - 3\vec{k}\right)}{\sqrt{11}}$$

$$= \frac{2-3+30}{\sqrt{11}} = \frac{29}{\sqrt{11}}$$

Problem 4

Find the maximum value of the dirctional derivative of the function $\phi = 2x^2 + 3y^2 + 5z^2$ at the point (1,1,-4)

Solution

Let $\phi = 2x^2 + 3y^2 + 5z^2$

$$\frac{\partial\phi}{\partial x} = 4x, \frac{\partial\phi}{\partial y} = 6y, \ \frac{\partial\phi}{\partial z} = 10z$$

$$\nabla\phi \quad = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$$

$$= 4x\vec{i} + 6y\vec{j} + 10z\vec{k}$$

At (1,1,-4)

$\nabla\phi = 4\vec{i} + 6\vec{j} - 40\vec{k}$

$|\nabla\phi| = \sqrt{4^2 + 6^2 + 40^2} = \sqrt{1652}$

$\therefore$ Maximum value of the directional derivative at the point (1,1,-4) is $\sqrt{1652}$

Problem 5

Find the unit vector normal to the surface $x^3 - xyz + z^3 - 1$ at the point (1,1,1)

Solution

$\phi = x^3 - xyz + z^3 - 1$

$$\nabla\phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$$

$$= \vec{i}\left(3x^2 - yz\right) + \vec{j}(-xz) + \vec{k}\left(-xy + 3z^2\right)$$

At the point (1,1,1)

$\nabla\phi = 2\vec{i} - \vec{j} + 2\vec{k}$

$|\nabla\phi| = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$

$\therefore$ Unit normal vector = $\dfrac{\nabla\phi}{|\nabla\phi|} = \dfrac{2\vec{i} - \vec{j} + 2\vec{k}}{3}$

Problem 6

Find the angle of intersection at the point (2,-1,2) of the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$

Solution

Let $\phi_1 = x^2 + y^2 + z^2 - 9$

$\phi_2 = x^2 + y^2 - z - 3$

$$\nabla\phi_1 = \vec{i}\frac{\partial\phi_1}{\partial x} + \vec{j}\frac{\partial\phi_1}{\partial y} + \vec{k}\frac{\partial\phi_1}{\partial z} = 2x\vec{i} + 2y\vec{j} + 2z\vec{k}$$

At (2,-1,2) $\nabla\phi_1 = 4\vec{i} - 2\vec{j} + 4\vec{k}$ .... (1)

$$|\nabla\phi_1| = \sqrt{16+4+16} = \sqrt{36} = 6$$

$$\nabla\phi_2 = \vec{i}\frac{\partial\phi_2}{\partial x} + \vec{j}\frac{\partial\phi_2}{\partial y} + \vec{k}\frac{\partial\phi_2}{\partial z}$$

$$= 2x\vec{i} + 2y\vec{j} - \vec{k}$$

At the point (2,-1,2), we have

$\nabla\phi_2 = 4\vec{i} - 2\vec{j} - \vec{k}$ .... (2)

$$|\nabla\phi_2| = \sqrt{16-14+1} = \sqrt{21}$$

Angle between the surfaces ϕ_1 and ϕ_2 is given by

$$\text{Cos } \theta = \frac{\nabla\phi_1.\nabla\phi_2}{|\nabla\phi_1||\nabla\phi_2|}$$

$$= \frac{\left(4\vec{i} - 2\vec{j} + 4\vec{k}\right) . \left(4\vec{i} - 2\vec{j} - \vec{k}\right)}{(6)\left(\sqrt{21}\right)}$$

$$= \frac{16+4-4}{6\sqrt{21}}$$

$$= \frac{8}{3\sqrt{21}}$$

$$\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)$$

Problem 7

Find ϕ, if $\nabla\phi = \left(6xy + z^3\right)\vec{i} + \left(3x^2 - z\right)\vec{j} + \left(3xz^2 - y\right)\vec{k}$

Solution

$$\nabla\phi = \vec{i}\,\frac{\partial\phi}{\partial x} + \vec{j}\,\frac{\partial\phi}{\partial y} + \vec{k}\,\frac{\partial\phi}{\partial z} \qquad \text{................ (1)}$$

Also,

$$\nabla\phi = \left(6xy + z^3\right)\vec{i} + \left(3x^2 - z\right)\vec{j} + \left(3xz^2 - y\right)\vec{k} \qquad \text{................ (2)}$$

Comparing (1) and (2) we get

$$\frac{\partial\phi}{\partial x} = 6xy + z^3 \qquad \text{................ (3)}$$

$$\frac{\partial\phi}{\partial y} = 3x^2 - z \qquad \text{................ (4)}$$

$$\frac{\partial\phi}{\partial z} = 3xz^2 - y \qquad \text{................ (5)}$$

Integrating (3) w.r.t. x we get

$$\int d\phi = \int \left(6xy + z^3\right)dx$$

$$= 3x^2y + xz^3 + f_1\,(y,z) \qquad \text{................ (6)}$$

Integrating (4) w.r.t "y", we get,

$$\int d\phi = \int \left(3x^2 - z\right)dy$$

$$\phi = 3x^2y - yz + f_2\,(x,z) \qquad \text{................ (7)}$$

Integrating (5) w.r.t z we get

$$\int d\phi = \int \left(3xz^2 - y\right)dz$$

$$= \frac{3xz^3}{3} - yz + f_3(x,y)$$

$$\phi = xz^3 - yz + f_3(x,y) \quad \ldots\ldots\ldots\ldots\ldots (8)$$

From (6), (7) and (8)

$\phi = 3x^2y + xz^3 - yz + c$. where c is the arbitrary constant.

Problem 8

Find the equation of the tangent plane and normal to the surface $2xz^2 - 3xy - 4x = 7$ at the point (1,-1,2)

Solution

Given $\phi = 2xz^2 - 3xy - 4x - 7$

$$\frac{\partial \phi}{\partial x} = 2z^2 - 3y - 4$$

$$\left(\frac{\partial \phi}{\partial x}\right)_{(1,-1,2)} = 2(2^2) - 3(-1) - 4 = 7$$

Now,

$$\frac{\partial \phi}{\partial y} = -3x$$

$$\left(\frac{\partial \phi}{\partial y}\right)_{(1,-1,2)} = -3(1) = -3$$

$$\frac{\partial \phi}{\partial z} = 4xz$$

$$\left(\frac{\partial \phi}{\partial z}\right)_{(1,-1,2)} = 4(1)(2) = 8$$

The equation of the tangent plane at (x_1,y_1,z_1) is

$$(x-x_1)\frac{\partial\phi}{\partial x}+(y-y_1)\frac{\partial\phi}{\partial y}+(z-z_1)\frac{\partial\phi}{\partial z}=0$$

At the point (1,-1,2) the equation is

$$(x-1)(7)+(y+1)(-3)+(z-2)(8)=0$$

7x - 3y + 8z -26 = 0

7x - 3y + 8z = 26

The equation of the normal at (x_1, y_1, z_1) is

$$\frac{x-x_1}{\frac{\partial\phi}{\partial x}}=\frac{y-y_1}{\frac{\partial\phi}{\partial y}}=\frac{z-z_1}{\frac{\partial\phi}{\partial z}}$$

$$\frac{x-1}{7}=\frac{y+1}{-3}=\frac{z-2}{8}$$

Problem 9

Find ϕ if $\nabla\phi=(y+\sin z)\vec{i}+x\vec{j}+x\cos z\vec{k}$

Solution

Let $\nabla\phi=\vec{i}\,\frac{\partial\phi}{\partial x}+\vec{j}\,\frac{\partial\phi}{\partial y}+\vec{k}\frac{\partial\phi}{\partial z}$ .... (1)

$\nabla\phi=(y+\sin z)\vec{i}+x\vec{j}+x\cos z\vec{k}$ .... (2)

Comparing (1) and (2) we get

$\frac{\partial\phi}{\partial x}=y+\sin z$ .... (3)

$\frac{\partial\phi}{\partial y}=x$ .... (4)

$\frac{\partial\phi}{\partial z}=x\cos z$ .... (5)

Integrating (3), (4) and (5) w.r.t x,y,z respect we get,

$$\phi = xy + x\,\sin z + f_1\,(y, z) \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ (6)$$

$$\phi = xy + f_2\,(x, z) \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ (7)$$

$$\phi = x\sin z + f_3(x, y) \qquad \ldots\ldots\ldots\ldots\ldots\ldots\ (8)$$

From (6), (7) and (8) we get

$\phi = xy + x\sin z + c$ where c is a constant.

Problem 10

If u = x+y+z, v = $x^2+y^2+z^2$, w = yz +zx + xy prove that grad u. grad v x grad w = 0

Solution

Given u = x+y+z, v = $x^2+y^2+z^2$, w = yz +zx + xy

$$\text{grad } u = \nabla u = \vec{i}\frac{\partial u}{\partial x} + \vec{j}\frac{\partial u}{\partial y} + \vec{k}\frac{\partial u}{\partial z}$$

$$= \vec{i} + \vec{j} + \vec{k}$$

$$\text{grad } v = \nabla v = \vec{i}\frac{\partial v}{\partial x} + \vec{j}\frac{\partial v}{\partial y} + \vec{k}\frac{\partial v}{\partial z}$$

$$= 2(x\vec{i} + y\vec{j} + z\vec{k})$$

$$\text{grad } w = \nabla w = \vec{i}\frac{\partial w}{\partial x} + \vec{j}\frac{\partial w}{\partial y} + \vec{k}\frac{\partial w}{\partial z}$$

$$= (y+z)\vec{i} + (z+x)\vec{j} + (x+y)\vec{k}$$

$$(\text{grade u}).(\text{grad v x gradw}) = \begin{vmatrix} 1 & 1 & 1 \\ 2x & 2y & 2z \\ y+z & z+x & x+y \end{vmatrix}$$

$$= 2\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ y+z & z+x & x+y \end{vmatrix}$$

$$= 2\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x+y+z & x+y+z & x+y+z \end{vmatrix}$$

$$= 2\,(x+y+z)\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix}$$

= 0, since two rows are identical

Problem 11

Find the directional derivative of $xyz = xy^2z^3$ a the point (1,2,-1) in the direction of the vector $\vec{i} - \vec{j} - 3\vec{k}$

Solution

Given $\phi = xyz = zy^2\, z^3$

$$\nabla\,\phi = \vec{i}\,\frac{\partial\phi}{\partial x} + \vec{j}\,\frac{\partial\phi}{\partial y} + \vec{k}\,\frac{\partial\phi}{\partial z}$$

$$= (yz-y^2z^3)\,\vec{i} + (\text{`}xz-2xyz^3)\,\vec{j} + (xy-3xy^2z^2)\,\vec{k}$$

$$\vec{n} = \frac{\vec{i} - \vec{j} - 3\vec{k}}{\sqrt{11}}$$

The directional derivative of ϕ in the directions of the vector $\vec{i} - \vec{j} - 3\vec{k} = \nabla \vec{\phi}.\hat{n}$

$$= \frac{\left[(yz - y^2z^3) - (x^2 - 2xyz^3) - 3(xy - 3xy^2z^2)\right]}{\sqrt{11}}$$

$$\nabla \phi.\ \hat{n}\ (1,2,-1) = \frac{(-2+4) - (-1+4) - 3(2-12)}{\sqrt{11}}$$

$$= \frac{29}{\sqrt{11}}$$

Problem 11

Find the unit normal to the surface $x^2+3y^2+2z^2=6$ at the point (2,0,1)

Solution

Let $\phi = x^2 + 3y^2 + 2z^2$

$$\nabla \phi = 2x\vec{i} + 6y\vec{j} + z\vec{k}$$

$$(\nabla \phi)_{(2,0,1)} = 4\vec{i} + 4\vec{k}$$

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{4\vec{i} + 4\vec{k}}{4\sqrt{2}} = \frac{\vec{i} + \vec{k}}{\sqrt{2}}$$

the unit normal vector at the point (2,0,1) to the given surface

$$= \frac{1}{\sqrt{2}}\left(\vec{i} + \vec{k}\right)$$

Problem 12

Find the maximum value of the directional derivative of the function $\phi = 2x^2+3y^2 +5z^2$ at the point (1,1,-4)

Solution

$\phi = 2x^2 + 3y^2 +5z^2$

$$\nabla \phi = \vec{i}\frac{\partial \phi}{\partial x} + \vec{j}\frac{\partial \phi}{\partial y} + \vec{k}\frac{\partial \phi}{\partial z}$$

$$= 4x\vec{i} + 6y\vec{j} + 10z\vec{k})$$

$$(\nabla \phi)\,(1,1,-4) = 4\vec{i} + 6\vec{j} - 40\vec{k}$$

Maximum value of the directional derivative at the point (1, 1, -4)

$$= \sqrt{16 + 36 + 1600}$$

$$= \sqrt{1652}$$

Problem 13

Find the angle between the normal to the surface $xy-z^2=0$ at the points (1,4,-2) and (-3, -3, 3)

Solution

$\phi = xy - z^2$

$$\nabla \phi = \vec{i}\frac{\partial \phi}{\partial x} + \vec{j}\frac{\partial \phi}{\partial y} + \vec{k}\frac{\partial \phi}{\partial z}$$

$$= y\vec{i} + x\vec{j} - 2z\vec{k}$$

$$(\nabla \phi)\ (1,4,-2) = 4\vec{i} + \vec{j} + 4\vec{k}$$

$$(\nabla \phi)\ (-3, -3, -3) = -3\vec{i} - 3\vec{j} - 6\vec{k}$$

Unit normal vector to the surface at the point (1, 4, -2) is

$$\hat{n}_1 = \frac{\nabla\phi}{|\nabla\phi|} = \frac{4\vec{i} + \vec{j} + 4\vec{k}}{\sqrt{33}}$$

Unit normal vector at the point (-3, -3, 3) is

$$\hat{n}_2 = \frac{-3\vec{i} - 3\vec{j} - 6\vec{k}}{\sqrt{54}}$$

It θ is the angle between the normal the $\cos\theta = \hat{n}_1.\hat{n}_2$

$$= \frac{-12 - 3 - 24}{\sqrt{33}.\sqrt{54}} = \frac{-39}{9\sqrt{22}} = \frac{-13}{3\sqrt{22}}$$

$$\therefore \theta = \cos^{-1}\left(\frac{-13}{3\sqrt{22}}\right)$$

Problem 14

If ϕ, $\nabla \phi = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$

Solution

$$\nabla \phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z} \qquad (1)$$

$$= (6xy + z^3)\vec{i} + (3x^2-z)\vec{j} + (3xz^2-y)\vec{k}$$

Comparing (1) and (2) we get

$$\frac{\partial\phi}{\partial x} = 6xy + z^3 \qquad (3)$$

$$\frac{\partial\phi}{\partial y} = 3x^2 - z \qquad (4)$$

$$\frac{\partial\phi}{\partial z} = 3zx^2-y \qquad (5)$$

Integrating, (3), (4) and (5) with respect to x,y,z we get

$(3) => \phi = 3x^2y+xz^3+f_1(y.z)$ (6)

$(4) => \phi = 3x^2y-yz+f_2(x,z)$ (7)

$(5) => \phi = xz^3-yz+f_3(x,y)$ (8)

From (6), (7) and (8) we get,

$\therefore\ \phi = 3x^2y+xz^3-yz+c$ where c is any arbitrary constant.

Problem 15

Find ϕ, if $\nabla\phi = (y+\sin z)\vec{i} + x\vec{j} + x\cos z\,\vec{k}$

Solution

we knowthat $\nabla\phi = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$ (1)

also $\nabla \phi = (y+\sin z)\vec{i} + x\vec{j} + x\cos z\vec{k}$ (2)

Comparing (1) and (2) we get

$$\frac{\partial \phi}{\partial x} = y + \sin z$$

$$\frac{\partial \phi}{\partial y} = x$$

$$\frac{\partial \phi}{\partial z} = x \cos z$$

Integrating above equation with respect to x,y,z we get,

$\phi = xy + x\sin z + f_1(y,z)$

$\phi = xy + f_2(x,z)$

$\phi = x\sin z + f_3(y,z)$

From the above equations,

$\therefore \phi = xy + x \sin z + c$, where c is a constant

Problem 16

Show that the surface $5x^2 - 2yz - 9x = 0$ and $4x^2y + z^3 - 4 = 0$ are orthogonal at (1, -1, -2)

Solution

Let $\phi_1 = 5x^2 - 2yz - 9x = 0$

$\phi_2 = 4x^2y + z^3 - 4$

$\nabla \phi_1 \quad = (10x-9)\vec{i} - 2z\vec{j} - 2y\vec{k}$

$(\nabla \phi_1)_{(1,-1,2)} \quad = \vec{i} - 4\vec{j} + 2\vec{k}$

$\nabla \phi_2 \quad = 8xy\vec{i} + 4x^2\vec{j} + 3z^2\vec{k}$

$\nabla \phi_{2\ (1,-1,2)} \quad = -8\vec{i} + 4\vec{j} + 12\vec{k}$

It two curves are orthogonal, $\nabla \phi_1, \nabla \phi_2 = 1$

∴ $\nabla \phi_1 \cdot \nabla \phi_2 = -8-16+24 = 0$

∴ The surface are orthogonal at the point (1, -1, 2)

Problem 17

Find the equation of the tangent plane and normal to the surface xyz = 4 at the point (1,2,2)

Solution

The equation of the surface is $\phi(x,y,z) = xyz - 4 = 0$

=> $\nabla \phi = yz\vec{i} + zx\vec{j} + xy\vec{k}$

$\nabla \phi_{(1,2,2)} = 4\vec{i} + 2\vec{j} + 2\vec{k}$

$\nabla \phi_{(1,2,2)}$ is a vector along the normal to the surface $\phi = 0$

The equation of the tangent plane is $\left(\vec{r} \cdot \vec{\alpha}\right) \cdot \nabla \phi = 0$ (1)

where $\vec{\alpha}$ is the position vector of a point in the plane and $\vec{r}$ is any vector in the plane.

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\alpha = \vec{i} + 2\vec{j} + 2\vec{k}$$

The equation of the tangent plane is

$$(1) => \left[(x\vec{i} + y\vec{j} + z\vec{k}) - (\vec{i} + 2\vec{j} + 2\vec{k})\right] . (4\vec{i} + 2\vec{j} + 2\vec{k}) = 0$$

$$4(x-1) + 2(y-2) + 2(z-2) = 0$$

$$4x+2y+2z - 12 = 0$$

$$2x+y+z - 6 = 0$$

The equation of the normal to the surface at the point (1,2,2) is

$$\frac{x - x_1}{\frac{\partial \varphi}{\partial x}} = \frac{y - y_1}{\frac{\partial \varphi}{\partial y}} = \frac{z - z_1}{\frac{\partial \varphi}{\partial x}}$$

$$\frac{x-1}{4} = \frac{y-2}{2} = \frac{z-2}{2}$$

$$\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-2}{1}$$

Check Your Progress - I

1. Find grad ϕ when $\phi = x^2 + y^2 + z - 2$ at (1,-1,2)

   ....

2. Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at (2,-1,2) in the direction of $2\vec{i} - 3\vec{j} + 6\vec{k}$

   ....

3. Find the unit normal vector $z = x^2 + y^2$ at the point (1,2,5)

   ....

4. For the angle intersection at the point (2,-1,2) of the surfaces $x^2 + y^2 + z^2 = 9$ and $x^2 + y^2 - 3 = 0$

....

9.6 Recap

1. If the position vector $\vec{r} = x\,\vec{i} + y\,\vec{j} + z\,\vec{k}$

Then

Velocity $\vec{v} = \dfrac{d\vec{r}}{dt}$ and Acceleration $\dfrac{d^2\vec{r}}{dt^2}$ at time t.

2. Gradient of a Scalar point function is

$$\nabla \phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right).\phi$$

where $\nabla = \vec{i}\dfrac{\partial}{\partial x} + \vec{j}\dfrac{\partial}{\partial y} + \vec{k}\dfrac{\partial}{\partial z}$ is a vector differential operator.

3. Directional derivatives of a scalar point function is

$$\frac{d\phi}{ds} = \nabla \phi . \vec{a} \qquad \left(\hat{a} = \frac{d\vec{r}}{ds}\right)$$

4. The equation of the normal at (x_1,y_1,z_1) is

$$\frac{x - x_1}{\frac{\partial \varphi}{\partial x}} = \frac{y - y_1}{\frac{\partial \varphi}{\partial y}} = \frac{z - z_1}{\frac{\partial \phi}{\partial z}}$$

5. The equation of the tangent plane is

$$(x-x_1)\frac{\partial \phi}{dx} + (y-y_1)\frac{\partial \phi}{dy} + (z-z_1)\frac{\partial \phi}{dz} = 0$$

6. The unit normal to the surface is

$$\hat{n} = \frac{\nabla\phi}{|\nabla\phi|}$$

7. Angle between the normal to the surface is

$$\cos\theta = \hat{n}_1 . \hat{n}_2$$

9.7 Exercise

1. If $\phi(x,y,z) = x^2-2y^2z^3$ Find $\nabla\phi$ at the point (1, -1, 2)

2. If $\phi = x^2yz^3$. Find $\nabla\phi$ at the point (1, 1, 1)

3. If $\phi = (x+xy^2+yz^2)$, find $\nabla\phi$

4. Find the directional derivative of $x^3+y^3+z^3$ at the point (1,-1,2) ini the direction of the vector $\vec{i}+2\vec{j}+\vec{k}$.

5. Find the directional derivative at $\phi(x,y,z) = x^2yz + 4xz^2$ at the point (1, -2, -1) in the direction of th evector $2\vec{i}-\vec{j}-2\vec{k}$.

6. Find the directional derivative of $\phi(x,y,z) = x^2-2y^2+4z^2$ at the point (1,1,-1) in the direction $2\vec{i}-\vec{j}-2\vec{k}$.

7. Find the directional derivative of the functino $\phi = xy+yz+zx$ in the direction of the vector $(2\vec{i}-3\vec{j})+6\vec{k}$ at the point (3,1,2)

8. Find the maximum value of the directional derivative of $\phi = x^3y\,z$ at the point (1,4,1)

9. Find the unit normal to the surface $z = x^2+y^2$ at the point (-1,-2,-5)

10. Find a unit vector normal to the surface $x^2y + 2xz^2 = 8$ at (1,0,2)

11. Find the unit normal to $xy^2z^3 = 1$ at (1,1,1)

12. If $\nabla \phi = 2xyz\vec{i} + x^2z\vec{j} + x^2y\vec{k}$, determine ϕ.

13. Find ϕ if $\nabla \phi = x(2yz+1)\vec{i} + x^2z\vec{j} + x^2y\vec{k}$.

14. Find the scalar point function whose gradient is $yz^2\vec{i} + (xz^2-1)\vec{j} + 2(xyz-1)\vec{k}$.

15. Find the angle between at inter section at (4, -3, 2) of the spheres $x^2+y^2+z^2 = 29$ and $x^2+y^2+z^2+4x-6y-8z-47 = 0$

16. Find the equation of the tangent to the surface $2xz^2 - 3xy - 4x = 7$ at the point (1, -1, 2). Also write down the equatio of the normal to the given surface at the point (1, -1, 2)

17. Find the equation of the tangent plane and normal to the surface $x^2+y^2+z^2 = 25$ at (4,0,3)

LESSON - 10

DIVERGENCE AND CURL OF A VECTOR POINT FUNCTION

Learning Objectives

In this lesson our aim is to study about

- Define Divergent, Curl of a vector point function
- Define solenodial and irrotational vector
- Important formula involving ∇ .

Structure

10.1 Introduction

The function that we have used in differential and Integral calculas have been real valued scalar functions. In this chapter we study functions whose values are vector such functions are used to study curves such functions are used to study curves in space are the motion of a particle in space.

10.2 Divergence of a vector point function

Let $\vec{F}(x, y, z)$ be a vector point function differentiatiable in a region R. Then the divergence of $\vec{F}$ is defined by $\nabla . \vec{F}$.

$$\nabla . \vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) \cdot \vec{F}$$

(i.e.,) The divergence of the vector point function $\vec{F}$ is

$$\nabla . \vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) \cdot \vec{F}$$

If $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$

$$\vec{F} = \sum \vec{i} \frac{\partial}{\partial x} \vec{F}$$

$$= \sum \vec{i} \frac{\partial}{\partial x} (F_1\vec{i} + F_2\vec{j} + F_3\vec{k}) = \sum \left(\vec{i}, \vec{j}\right) \frac{\partial F}{\partial x}.$$

$$= \sum \frac{\partial F}{\partial x}$$

$$= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

10.2.1 Solenoidal vector

If $\vec{F}$ is a vector point function such that $\nabla . \vec{F} = 0$. Then $\vec{F}$ is called a solenoidal vector.

10.3 Curl of vector point function

If $\vec{F}(x, y, z)$ be vector point function differentiable in a region R, the curl of $\vec{F}$ denoted as curl $\vec{F}$ is defined as

curl $\vec{F}$ = $\nabla x\vec{F}$.

(i.e.,)

$$\nabla x\vec{F} = \left(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z}\right)x\vec{F}$$

$$= \vec{i}\,x\frac{\partial F}{\partial x}+\vec{j}\frac{\partial \vec{F}}{\partial y}+\vec{k}\frac{\partial \vec{F}}{\partial z}$$

$$= \sum \vec{i}\,x\,\frac{\partial \vec{F}}{\partial x}$$

If $\vec{F} = F_1\vec{i}+F_2\vec{j}+F_3\vec{k}$

then

$$\therefore \nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \text{curl } \vec{F}.$$

10.3.1 Irrotational vector

if $\vec{F}$ is a vector point function such that $\nabla\, x\, \vec{F} = 0$. Then $\vec{F}$ is called a irrotaional vector.

$$\text{(i.e.,) } \nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = 0$$

Example 1

Find $\vec{F} = xy^2\vec{i} + 2x^2\,yz\vec{j} - 3yz^2\vec{k}$, find div $\vec{F}$ and curl $\vec{F}$. What are these values at the point (1, -1, 1).

Solution

$$\vec{F} = xy^2\vec{i} + 2x^2yz\vec{j} - 3yx^2\vec{k}$$

$$\text{div } \vec{F} = \nabla.\vec{F}$$

$$= \frac{\partial}{\partial x}\left(xy^2\right) + \frac{\partial}{\partial y}\left(2x^2yz\right) + \frac{\partial}{\partial z}\left(-3yz^2\right)$$

$$= y^2 + 2x^2z - 6yz$$

At the point (1, -1, 1)

div $\vec{F}$ = 1+2+6 = 9

$\boxed{\text{div } \vec{F} = 9}$

$$\text{curl } \vec{F} = \nabla x\vec{F}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & 2x^2yz & -3yz^2 \end{vmatrix}$$

$$= \vec{i}\left(-3z^2 - 2x^2y\right) - \vec{j}(0-0) + \vec{k}\left(4xyz - 2xy\right)$$

$$\text{Curl } \vec{F} = -\left(3z^2 + 2x^2y\right)\vec{i} + \left(4xyz - 2yz\right)\vec{k}$$

At the point (1, -1, 1),

$$= -(3-2)\vec{i} + (-4+2)\vec{k}$$

∴ Curl $\vec{F} = -\vec{i} - 2\vec{k}$

Example 2

If $\vec{F} = xz^3\vec{i} - 2xyz\vec{j} + xz\vec{k}$. Find div $\vec{F}$ at (1,2,0)

Solution

$\vec{F} \quad = xz^3\vec{i} - 2xyz\vec{j} + xz\vec{k}$

div $\vec{F} \quad = \nabla.\vec{F}$

$$= \frac{\partial}{\partial x}(xy^3) + \frac{\partial}{\partial y}(-2xyz) + \frac{\partial}{\partial z}(xz)$$

$$= z^3 - 2xz + x$$

At (1,2, 0)

div $\vec{F} \quad = 0+0+1 = 1$

curl $\vec{F} \quad = \nabla x\vec{F}$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz^3 & -2xyz & xz \end{vmatrix}$$

$$= \vec{i}(0+2xy) - \vec{j}(z-3xz^2) + \vec{k}(-2yz-0)$$

$$= 2xy\vec{i} - (z-3xz^2)\vec{j} - 2yz\vec{k}$$

At point (1, 2, 0),

Curl $\vec{F} = 4\vec{i}$

Example 3

Show that the vector $\vec{F} = 3y^4z^2\vec{i} + 4x^3z^2\vec{j} - 3x^2y^2\vec{k}$ is solenoidal.

Solution

Let $\vec{F}$ to be solenoidal, we have $\nabla.\vec{F} = 0$

Given

$$\vec{F} = 3y^4z^2\vec{i} + 4x^3z^2\vec{j} - 3x^2y\vec{k}$$

$$\nabla.\vec{F} = \frac{\partial}{\partial x}\left(3y^4z^2\right) + \frac{\partial}{\partial y}\left(4x^3z^3\right) + \frac{\partial}{\partial z}\left(-3x^2y\right)$$

$$= 0 + 0 + 0$$

$$\nabla.\vec{F} = 0$$

Here $\vec{F}$ is solenoidial.

Example 4

Show that the vector $3x^2y\,\vec{i} - 4xy^2\,\vec{j} + 2xyz\,\vec{k}$ is solenoidal.

Solution

$$\vec{F} = 3x^2y\,\vec{i} - 4y^2\,\vec{j} + 2xyz\,\vec{k}$$

$$\nabla.\vec{F} = \frac{\partial}{\partial x}\left(3x^2y\right) + \frac{\partial}{\partial y}\left(-4xy^2\right) + \frac{\partial}{\partial z}\left(2xyz\right)$$

$$= 6xy - 8xy + 2xy$$

$$\nabla.\vec{F} = 0$$

Here $\vec{F}$ is solenoidial.

Example 5

Find the value of 'a' so that the vector $\vec{F} = (z+3y)\vec{i} + (x-2z)\vec{j} + (x+az)\vec{k}$ solenoidal.

Solution

Given $\vec{F} = (z+3y)\vec{i} + (x-2z)\vec{j} + (x+az)\vec{k}$

$$\nabla.\vec{F} = \frac{\partial}{\partial x}(z+3y) + \frac{\partial}{\partial y}(x-2z) + \frac{\partial}{\partial z}(x+az)$$

$$= 0+0+a$$

$$\nabla.\vec{F} = a$$

For $\vec{F}$ to be solenoidial, $\nabla.\vec{F} = 0$

$$a = 0$$

Example 6

Prove that div $\vec{r}$ = 3 and curl $\vec{r}$ =0, where $\vec{r}$ is the position vector of the point (x,y,z).

Solution

Given $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$$\nabla.\vec{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)$$

$$= 1+1+1$$

$$\nabla.\vec{r} = 3$$

$$\text{curl } \vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$$

$$= \vec{i}\,(0) - \vec{j}\,(0) + \vec{k}\,(0)$$

$$= 0$$

since $\nabla x\vec{F} = 0,$

$\therefore$ $\vec{F}$ is irrotational

Example 7

Find the value of 'a' such that $\vec{F} = (axy-z^2)\vec{i} + (x^2+2yz)\vec{j} + (y^2-axz)\vec{k}$ is irrotational.

Solution

Given $\vec{F} = (axy-z^2)\vec{i} + (x^2+2yz)\vec{j} + (y^2-axz)\vec{k}$

Curl $\vec{F} = \nabla x\vec{F}$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ axy - z^2 & x^2 + 2yz & y^2 - axz \end{vmatrix}$$

$$= \vec{i}\,(2y-2y) - \vec{j}\,(-az+2z) + \vec{k}\,(2x-ax)$$

$$\nabla x\vec{F} = (a-2)\,z\,\vec{j} + (2-a)x\,\vec{k}$$

If $\vec{F}$ is irronational, then, $\nabla x\vec{F} = 0$

Hence co-efficients of $\vec{j}$ = 0 and co-efficients of $\vec{k}$ = 0

therefore, $\boxed{a = 2}$

Example 8

A Field $\vec{F}$ is of the form $\vec{F} = (6xy+z^3)\vec{i} + (3x^2-z)\vec{j} + (3xz^2-y)\vec{k}$. Show that $\vec{F}$ is conservative field and find a function ϕ such that $\vec{F} = \nabla \phi$.

Solution

Given $\vec{F} = (6xy+z^3)\vec{i} + (3x^2-z)\vec{j} + (3xz^2-y)\vec{k}$

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy + z^3 & 3x^2 - z^2 & 3xz^2 - y \end{vmatrix}$$

$$= \vec{i}\,(-1+1) - \vec{j}\,(-3z^2-3z^2) + \vec{k}\,(6x-6x)$$

$$= 0$$

∴ $\vec{F}$ is conservative field.

Since $\vec{F}$ is a conservative field, then exists a scalar function, ϕ such that $\vec{F} = \nabla \phi$.

∴ $(6xy+z^3)\vec{i} + (3z^2-z)\vec{j} + (3z^2-y)\vec{k} = \vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}$

$$\frac{\partial \phi}{\partial x} = 6xy + z^3$$

$$\frac{\partial \phi}{\partial y} = 3x^2 - z$$

$$\frac{\partial \phi}{\partial z} = 3xz^2 - y$$

Integrating we get,

$\phi = 3x^2y + xz^3 + f_1 (y,z)$

$\phi = 3x^2y - yz + f_2 (x,z)$

$\phi = xz^3 - yz + f_3 (x,y)$

from the above values of ϕ, we see that

$\phi = 3x^2y + xz^3 - yz + c$

Example 9

if $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ Show that $\nabla \times [f(r).\vec{r}] = 0$

Solution

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, $r^2 = x^2+y^2+z^2$

$f(r).\vec{r} = xf(r)\vec{i} + yf(r)\vec{j} + zf(r)\vec{k}$

$$\frac{\partial r}{\partial x} = \frac{x}{r};\ \frac{\partial r}{\partial y} = \frac{y}{r};\ \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\nabla \times \left[f(r).\vec{r}\right] = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xf(r) & yf(r) & zf(r) \end{vmatrix}$$

$$= \sum \vec{i} \left[\frac{\partial}{\partial y} zf(r) - \frac{\partial}{\partial z} yf(r)\right]$$

$$= \sum \vec{i} \left[zf'(r)\frac{\partial r}{\partial y} - yf'(r)\frac{\partial r}{\partial z}\right]$$

$$= \sum \vec{i} \left[\frac{yz}{r} - \frac{yz}{r}\right].f'(r)$$

$$= \frac{f'(r)}{r} \sum \vec{i} \left[yz - yz\right]$$

$$\therefore \nabla \times [f(r).\vec{r}] = 0$$

Hence Proved.

Example 10

Prove that $\operatorname{div}\left(\frac{\vec{r}}{r}\right) = \frac{2}{r}$

Solution

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, $r^2 = x^2+y^2+z^2$

$$\operatorname{div}\left(\frac{\vec{r}}{r}\right) = \nabla . \frac{\vec{r}}{r}$$

$$= \nabla \left(\frac{x \vec{i} + y \vec{j} + z \vec{k}}{r} \right)$$

$$= \frac{\partial}{\partial x} = \left(\frac{x}{r} \right) + \frac{\partial}{\partial y} = \left(\frac{y}{r} \right) + \frac{\partial}{\partial z} = \left(\frac{z}{r} \right)$$

$$= \sum \frac{r.1 - x \frac{\partial r}{\partial x}}{r^2}$$

$$= \sum \frac{r - \frac{x^2}{r}}{r^2}$$

$$= \sum \frac{r^2 - x^2}{r^2}$$

$$= \frac{r^2 - x^2 + r^2 - y^2 + r^2 - z^2}{r^3}$$

$$= \frac{3r^2 - r^2}{r^3}$$

$$\therefore \text{div} \left(\frac{\vec{r}}{r} \right) = \frac{2}{r}$$

Check Your Progress - I

1. If $\vec{F}$ is a solenoidal, then $\nabla . \vec{F}$ = ____________________

2. If $\vec{F}$ is a solenoidal, then $\nabla \times \vec{F}$ = ____________________

10.4 Vector Identities

Let ϕ be a scalar point function and $\vec{u}$ and $\vec{v}$ be vector point functions.

Then

(1) $\nabla.(\vec{u}+\vec{v}) = \nabla.\vec{u}+\nabla.\vec{v}$

(2) $\nabla x(\vec{u}+\vec{v}) = \nabla \text{ x } \vec{u}+\nabla \text{ x } \vec{v}$

(3) $\nabla.(\varphi\vec{u}) = \nabla\varphi.\vec{u}+\varphi\nabla.\vec{u}$

(4) $\nabla x(\varphi\vec{u}) = \nabla\varphi x\vec{u}+\varphi\nabla x\vec{u}$

(5) $\nabla.(\vec{u}x\vec{v}) = \vec{v}.\nabla x\vec{u}-\vec{u}.\nabla x\vec{u}$

$= \vec{v}.\text{curl }.\vec{u}-\vec{u}.\text{curl}\,\vec{v}$

(6) $\nabla x(\vec{u}\,x\,\vec{v}) = (\vec{v}.\nabla)\vec{u}-(\nabla.\vec{u})\vec{v}$ - $\left[(\vec{u}.\nabla)\vec{v}-(\nabla.\vec{v})\vec{u}\right]$

(7) $\nabla(\vec{u}.\vec{v}) = (\vec{v}.\nabla)\vec{u}+(\vec{u}.\nabla)\vec{v}+\vec{v}\,x(\nabla x\,\vec{u}.)+\vec{u}\,x(\nabla x\,\vec{v})$

(8) $\nabla.\nabla\phi = \nabla^2\phi$

(9) $\nabla x\nabla\phi = 0$

(10) $\nabla.(\nabla x\vec{F}) = 0$

(11) $\nabla.(\nabla x\vec{F}) = \nabla.(\nabla x\vec{F})-\nabla^2\vec{F}$, where $\vec{F}$ is a vector point function

Proof

(i) $\nabla.(\vec{u}+\vec{v}) = \left(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z}\right).(\vec{u}+\vec{v})$

$= \sum\vec{i}\frac{\partial}{\partial x}.(\vec{u}+\vec{v})$

$$= \Sigma\left[\vec{i}\,.\frac{\partial\vec{u}}{\partial x} + \vec{i}\,.\frac{\partial\vec{v}}{\partial x}\right]$$

$$= \sum\vec{i}.\frac{\partial\vec{u}}{\partial x} + \sum\vec{i}.\frac{\partial\vec{v}}{\partial x}$$

$$= \nabla.\vec{u} + \nabla.\vec{v}$$

(ii) $\nabla \times (\vec{u} + \vec{v}) = \sum\vec{i}\frac{\partial}{\partial x}\times(\vec{u} + \vec{v})$

$$= \sum\left[\vec{i}\times\frac{\partial\vec{u}}{\partial x} + \vec{i}\times\frac{\partial\vec{v}}{\partial x}\right]$$

$$= \sum\vec{i}\times\left(\frac{\partial\vec{u}}{\partial x}\right) + \sum\left(\vec{i}\times\frac{\partial\vec{v}}{\partial x}\right)$$

$$= (\nabla\times\vec{u}) + (\nabla\times\vec{v})$$

(iii) $\nabla.(\phi\vec{u}) = \sum\vec{i}.\frac{\partial}{\partial x}(\phi\vec{u})$

$$= \Sigma\,\vec{i}.\left[\frac{\partial\phi}{\partial x}\vec{u} + \phi\frac{\partial\vec{u}}{\partial x}\right]$$

$$= \sum\left(\vec{i}.\frac{\partial\phi}{\partial x}\vec{u} + \phi\,\vec{i}.\frac{\partial\vec{u}}{\partial x}\right)$$

$$= \Sigma\frac{\partial\phi}{\partial x}\,\vec{i}.\vec{u} + \phi\Sigma\,\vec{i}\,.\frac{\partial\vec{u}}{\partial x}$$

$$= \left(\Sigma\frac{\partial\phi}{\partial x}\vec{i}\right).\vec{u} + \phi\Sigma\,\vec{i}.\,\frac{\partial\vec{u}}{\partial x}$$

$$= \nabla\phi.\vec{u} + \phi(\nabla.\vec{u})$$

(iv) $\nabla x(\phi\vec{u}) = \sum \vec{i} x \frac{\partial}{\partial x}(\phi\vec{u})$

$$= \sum \vec{i} x\left(\frac{\partial\phi}{\partial x}\vec{u} + \phi\frac{\partial\vec{u}}{\partial x}\right)$$

$$= \Sigma\left(\vec{i} \text{ x } \frac{\partial\phi}{\partial x}\vec{u}\right) + \Sigma\left(\vec{i} \text{ x } \phi\frac{\partial\vec{u}}{\partial x}\right)$$

$$= \Sigma\frac{\partial\phi}{\partial x}\vec{i} x\vec{u} + \phi\Sigma\, \vec{i} x\frac{\partial\vec{u}}{\partial x}$$

$$= \nabla\phi \text{ x } \vec{u} + \phi\nabla x\vec{u}$$

(v) $\nabla.(\vec{u}x\vec{v}) = \Sigma\vec{i}\,.\frac{\partial}{\partial x}(\vec{u} \text{ x } \vec{v})$

$$= \Sigma\vec{i}\,.\left(\frac{\partial\vec{u}}{\partial x}\text{ x }\vec{v} + \vec{u}\text{ x }\frac{\partial\vec{v}}{\partial x}\right)$$

$$= \sum\vec{i}\frac{\partial u}{\partial x}x\vec{v} + \sum\vec{i}.\vec{u}x\frac{\partial\vec{v}}{\partial x}$$

$$= \Sigma\vec{i}\,.\left(\frac{\partial\vec{u}}{\partial x}\text{ x }\vec{v}\right) + \Sigma\vec{i}\,.\left(\vec{u}\text{ x }\frac{\partial\vec{v}}{\partial x}\right)$$

(Since dot and cross can be interchanged)

$$= (\nabla \text{ x } \vec{u}).\vec{v} - (\nabla \text{ x } \vec{v}).\vec{u}$$

$$= \vec{v}\,.\,\nabla \text{ x } \vec{u} - \vec{u}\,.\nabla \text{ x } \vec{v}$$

$$= \vec{v}.\text{curl}\,\vec{u} - \vec{u}\,\text{curl}\,\vec{v}$$

(vi) $\nabla x(\vec{u}x\vec{v}) = \sum \vec{i} x \frac{\partial}{\partial x}(\vec{u}x\vec{v})$

$$= \sum \vec{i}\, x \left(\frac{\partial \vec{u}}{\partial x} x\, \vec{v} + \vec{u}\, x \frac{\partial \vec{v}}{\partial x} \right)$$

$$= \sum \vec{i}\, x \left(\frac{\partial \vec{u}}{\partial x} x\vec{v} \right) + \sum \vec{i}\, x \left(\vec{u} x \frac{\partial \vec{v}}{\partial x} \right)$$

$$= \sum \vec{i}\, x \left(\frac{\partial \vec{u}}{\partial x} x\vec{v} \right) - \sum \vec{i}\, x \left(\frac{\partial \vec{v}}{\partial x} x\vec{u} \right)$$

$$= \sum \left[(\vec{i}.\vec{v}) \frac{\partial \vec{u}}{\partial x} - \left(\vec{i}. \frac{\partial \vec{u}}{\partial x} \right) \vec{v} \right] - \sum \left[(\vec{i}.\vec{u}) \frac{\partial \vec{v}}{\partial x} - \left(\vec{i}. \frac{\partial \vec{v}}{\partial x} \right) \vec{u} \right]$$

$$= \sum (\vec{i}.\vec{v}) \frac{\partial \vec{u}}{\partial x} - \sum \left(\vec{i}. \frac{\partial \vec{u}}{\partial x} \right) \vec{v} - \sum (\vec{i}.\vec{u}) \frac{\partial \vec{v}}{\partial x} + \sum \left(\vec{i}. \frac{\partial \vec{v}}{\partial x} \right) \vec{u}$$

$$= \sum (\vec{v}.\vec{i}) \frac{\partial \vec{u}}{\partial x} - \sum \left(\vec{i}. \frac{\partial \vec{u}}{\partial x} \right) \vec{v} - \sum (i.\vec{u}) \frac{\partial \vec{v}}{\partial x} + \sum \left(\vec{i}. \frac{\partial \vec{v}}{\partial x} \right) \vec{u}$$

$$= (\vec{v}.\nabla)\vec{u} - (\nabla.\vec{u})\vec{v} - (\vec{u}.\nabla)\vec{v} + (\nabla.\vec{v})\vec{u}$$

$$= [(\vec{v}.\nabla)\vec{u} - (\nabla.\vec{u}.)\vec{v}] - [(\vec{u}.\nabla)\vec{v} - (\nabla.\vec{v})\vec{u}]$$

(vii) $\nabla x(\vec{u}.\vec{v}) = \sum \vec{i} x \frac{\partial}{\partial x}(\vec{u}.\vec{v})$

$$= \sum \vec{i}\ x\ \left(\frac{\partial \vec{u}}{\partial x}\ .\ \vec{v} + \vec{u}\ .\ \frac{\partial \vec{v}}{\partial x} \right) \qquad (1)$$

Now

$$(\nabla x\vec{v}) = \vec{u}\, x \sum\left(\vec{i}\, x \frac{\partial \vec{v}}{\partial x}\right)$$

$$= \sum\left(\vec{u}.\frac{\partial v}{\partial x}\right)\vec{i} - \sum(\vec{u}.\vec{i})\frac{\partial \vec{v}}{\partial x}$$

$$\therefore \sum\left(\vec{u}.\frac{\partial v}{\partial x}\right)\vec{i} = \vec{u}\, x(\nabla x\, \vec{v}) + \vec{u}.\left(\sum \vec{i}\frac{\partial}{\partial x}\right)\vec{v}$$

$$= \vec{u}\, x(\nabla x\, \vec{v}) + (\vec{u}.\nabla)\vec{v} \qquad (2)$$

Similarly

$$\sum\left(\vec{v}.\frac{\partial y}{\partial x}\right)\vec{i} = \vec{v}\, x(\nabla x\, \vec{u}) + (\vec{v}.\nabla)\vec{u} \qquad (3)$$

using (2) & (3) in (1) we get

$$\nabla(\vec{u}.\vec{v}) = \vec{u}x(\nabla x\vec{v}) + (\vec{u}.\nabla)\vec{v} + \vec{v}x(\nabla x\vec{u}) + (\vec{v}.\nabla)\vec{u}$$

(viii) $\nabla.\nabla\phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right).\left(\vec{i}\frac{\partial\phi}{\partial x} + \vec{j}\frac{\partial\phi}{\partial y} + \vec{k}\frac{\partial\phi}{\partial z}\right)$

$$= \frac{\partial}{\partial x}\left(\frac{\partial\phi}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{\partial\phi}{\partial y}\right) + \frac{\partial}{\partial z}\left(\frac{\partial\phi}{\partial z}\right)$$

$$= \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

$$= \nabla^2\phi$$

This is called Laplacian of ϕ and is denoted by $\nabla\phi$

(ix) $\nabla x \nabla \phi$ $= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$

$$= \vec{i}\left[\frac{\partial^2\phi}{\partial y\partial z} - \frac{\partial^2\phi}{\partial z\partial y}\right] - \vec{j}\left[\frac{\partial^2\phi}{\partial x\partial z} - \frac{\partial^2\phi}{\partial z\partial x}\right] + \vec{k}\left[\frac{\partial^2\phi}{\partial x\partial y} - \frac{\partial^2\phi}{\partial y\partial x}\right]$$

$\because \nabla \text{ x } \nabla\phi = 0$ $\left(\because \frac{\partial^2\varphi}{\partial x\partial y} = \frac{\partial^2\varphi}{\partial y\partial x} \text{ if } \phi \text{ is continuous}\right)$

(x) $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

$$= \vec{i}\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right) - \vec{j}\left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}\right) + \vec{k}\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)$$

$\nabla.\left(\nabla x\vec{F}\right) = O$ $\left[\text{Since f is continous } \frac{\partial^2 f}{\partial x\partial y} = \frac{\partial^2 f}{\partial y\partial x}\right]$

(xi) $\nabla x\left(\nabla x\vec{F}\right)$

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

$$= \vec{i}\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right) - \vec{j}\left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right) + \vec{k}\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)$$

$$\nabla x\left(\nabla x\vec{F}\right) = \begin{vmatrix} \vec{i} & \vec{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right) & \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right) & \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) \end{vmatrix}$$

$$= \sum \vec{i}\left[\frac{\partial}{\partial y}\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right) - \frac{\partial}{\partial z}\left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\right]$$

$$= \Sigma\,\vec{i}\left[\frac{\partial}{\partial x}\left(\frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\right) - \left(\frac{\partial^2 F_1}{\partial y^2} + \frac{\partial^2 F_1}{\partial z^2}\right)\right]$$

$$= \Sigma\,\vec{i}\left[\frac{\partial}{\partial x}\left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\right) - \left(\frac{\partial^2 F_1}{\partial x^2} + \frac{\partial^2 F_1}{\partial y^2} + \frac{\partial^2 F_1}{\partial z^2}\right)\right]$$

$$= \Sigma\,\vec{i}\left[\frac{\partial}{\partial x}\left(\nabla.\vec{F}\right) - \nabla^2 F_1\right]$$

$$= \left(\Sigma\,\vec{i}\,\frac{\partial}{\partial x}\right)\nabla.\vec{F} - \Sigma\nabla^2 F_1\,\vec{i}$$

$$= \nabla\left(\nabla.\vec{F}\right) - \nabla^2\vec{F}$$

Simple Problems

Problem 1

Find $\nabla.\vec{F}$ and $\nabla x\vec{F}$, the vector point function $xy^2\vec{i} + 2x^2\,yz\vec{j} - 3yz^2\vec{k}$

Solution

Let $\vec{F} = xy^2\vec{i} + 2x^2yz\vec{j} - 3yx^2\vec{k}$

$$\nabla.\vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right).\ \left(xy^2\vec{i} + 2x^2yz\vec{j} - 3yz^2\vec{k}\right)$$

$$= \frac{\partial}{\partial x}\left(xy^2\right) + \frac{\partial}{\partial y}\left(2x^2yz\right) + \frac{\partial}{\partial z}\left(-3yz^2\right)$$

$$= y^2 + 2x^2z - 6yz$$

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & 2x^2yz & -3yz^2 \end{vmatrix}$$

$$= \vec{i}\left[\frac{\partial}{\partial y}\left(-3yz^2\right) - \frac{\partial}{\partial z}\left(2x^2yz\right)\right] - \vec{j}\left[\frac{\partial}{\partial x}\left(-3yz^2\right) - \frac{\partial}{\partial z}\left(xy^2\right)\right] + \vec{k}\left[\frac{\partial}{\partial x}\left(2x^2yz\right) - \frac{\partial}{\partial y}\left(xy^2\right)\right]$$

$$= \left(-3z^2 - 2x^2y\right)\vec{i} - \vec{oj} + \left(4xyz - 2xy\right)\vec{k}$$

Problem 2

If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ prove that $\nabla.\vec{r} = 3$ and $\nabla x\vec{r} = 0$

Solution

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$$\nabla.\vec{r} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right).\left(x\vec{i} + y\vec{j} + z\vec{k}\right)$$

$$= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1$$

$$\nabla\, x\, \vec{r} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$$

$$= \vec{i}(0-0) - \vec{j}(0-0) + \vec{k}(0-0)$$

$$= 0$$

Hence $\nabla.\vec{r} = 3$ and $\nabla x\vec{r} = 0$

Problem 3

Show that the vector $\vec{F} = 3y^4z^2\vec{i} + 4x^3z^2\vec{j} - 3x^2y^2\vec{k}$ is solenoidal.

Solution

$$\text{div}\vec{F} = \nabla.\vec{F}$$

$$= \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right).\left(3y^4\ z^2\ \vec{i} + 4x^3z^2\vec{j} - 3x^2y^2\vec{k}\right)$$

$$= \frac{\partial}{\partial x}\left(3y^4z^2\right) + \frac{\partial}{\partial y}\left(4x^3z^2\right) + \frac{\partial}{\partial z}\left(-3x^2z^2\right)$$

$$= 0 + 0 + 0 = 0$$

Here $\vec{F}$ is solenoidial.

Problem 4

Show that the vector $2xy\vec{i} + \left(x^2 + 2yz\right)\vec{j} + \left(y^2 + 1\right)\vec{k}$ is irrotational.

Solution

Let $\vec{F} = 2xy\vec{i} + \left(x^2 + 2yz\right)\vec{j} + \left(y^2 + 1\right)\vec{k}$

A vector $\vec{F}$ is said to be irrotational if $\nabla x\vec{F} = 0$

Now,

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy & x^2 + 2yz & y^2 + 1 \end{vmatrix}$$

= $\vec{i}\left[\frac{\partial}{\partial y}\left(y^2+1\right)-\frac{\partial}{\partial z}\left(x^2+2yz\right)\right]-\vec{j}\left[\frac{\partial}{\partial x}\left(y^2+1\right)-\frac{\partial}{\partial z}(2xy)\right]+\vec{k}\left[\frac{\partial}{\partial x}\left(x^2+2yz\right)-\frac{\partial}{\partial y}(2xy)\right]$

= $\vec{i}(2y-2y)-\vec{j}(0-0)+\vec{k}(2x-2x)$

= $0\vec{i}-0\vec{j}+0\vec{k}$

= 0

$\therefore$ $\vec{F}$ is irrotational

Problem 5

Find the value of a so that the vector $\vec{F}=(x+3y)\,\vec{i}+(y-2z)\vec{j}+(x+az)\vec{k}$ is solenoidal.

Solution

Given $\vec{F}$ is solenoidal

$\therefore$ $\nabla.\vec{F}=0$

$$\left(\vec{i}\frac{\partial}{\partial x}+\vec{j}\frac{\partial}{\partial y}+\vec{k}\frac{\partial}{\partial z}\right).\left[(x+3y)\vec{i}+(y-2z)\vec{j}+(x+az)\vec{k}\right]=0$$

$$\frac{\partial}{\partial x}(x+3y)+\frac{\partial}{\partial y}(y-2z)+\frac{\partial}{\partial z}(x+az)=0$$

$$1 + 1 + a = 0$$

$$a = -2$$

Problem 6

Find the value of the constant a,b,c so that the vector

$\vec{F}=(x+2y+az)\vec{i}+(bx-3y-z)\vec{j}+(4x+cy+2z)\vec{k}$ irrotational

Solution

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+2y+az & bx-3y-z & 4x+cy+2z \end{vmatrix}$$

$$= \vec{i}\left[\frac{\partial}{\partial y}(4x + cy + 2z) - \frac{\partial}{\partial z}(bx - 3y - z)\right] - \vec{j}\left[\frac{\partial}{\partial x}(4x + cy + 2z) - \frac{\partial}{\partial z}(x + 2y + az)\right]$$

$$+\vec{k}\left[\frac{\partial}{\partial x}(6x - 3y - z) - \frac{\partial}{\partial y}(x + 2y + az)\right]$$

$$= (c+1)\vec{i} - (4-a)\vec{j} + (b-2)\vec{k}$$

Given $\vec{F}$ is irrotational

i.e., $\nabla x\vec{F} = 0$

$$(c+1)\vec{i} + (a-4)\vec{j} + (b-2)\vec{k} = 0$$

$$(c+1)\vec{i} + (a-4)\vec{j} + (b-2)\vec{k} = 0\vec{i} + 0\,\vec{j} + 0\vec{k}$$

Equating on bothsides,

c + 1 = 0; a - 4 = 0; b - 2 = 0

i.e., c = -1, a =4, b = 2

Problem 7

If $\vec{A}$ and $\vec{B}$ are irrotational, prove that $\vec{A}x\vec{B}$ is solenoidal.

Solution

$\vec{A}$ and $\vec{B}$ are irrotatioal

$\therefore$ $\nabla x\vec{A} = 0$ and $\nabla x\vec{B} = 0$

By property,

$$\nabla.(\vec{A}x\vec{B}) = (\nabla \times \vec{A}).\vec{B} - (\nabla \times \vec{B}).\vec{A}$$

$$= 0 - 0$$

$$= 0$$

Hence $\vec{A}X\vec{B}$ is solenoidal.

Problem 8

If $\vec{F} = x^2y\vec{i} + y^2z\vec{j} + z^2x\vec{k}$ find curl curl $\vec{F}$

Solution

Let $\vec{F} = x^2y\vec{i} + y^2z\vec{j} + z^2x\vec{k}$

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y & y^2z & z^2x \end{vmatrix}$$

$$= \vec{i}\left(0 - y^2\right) - \vec{j}\left(z^2 - 0\right) + \vec{k}\left(0 - x^2\right)$$

$$\nabla x\vec{F} = -y^2\vec{i} - z^2 j - x^2\vec{k}$$

$$\text{Curl curl } \vec{F} = \nabla X\left(\nabla X\vec{F}\right) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & -z^2 & -x^2 \end{vmatrix}$$

$$= \vec{i}[0 + 2z] - \vec{j}[-2x - 0] + \vec{k}[0 + 2y]$$

$$= 2z\vec{i} + 2x\vec{j} + 2y\vec{k}$$

$$= 2\left[z\vec{i} + x\vec{j} + y\vec{k}\right]$$

Problem 9

Show that $r^n\,\vec{r}$ is an irrotational vector for any value of n but is solenoidal only if n = -3

Solution

Let $\vec{F} = r^n\vec{r}$

$$= r^n\left[x\vec{i} + y\vec{j} + z\vec{k}\right]$$

$= xr^n\vec{i} + r^n y\vec{j} + r^n z\vec{k}$

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ r^n x & r^n y & r^n z \end{vmatrix}$$

$= \sum \vec{i}\left[znr^{n-1}\frac{\partial r}{\partial y} - ynr^{n-1}\frac{\partial r}{\partial z}\right]$

$= \sum \vec{i}\left[znr^{n-1}\frac{y}{r} - ynr^{n-1}\frac{z}{r}\right]$

$= \sum \vec{i}\left[yznr^{n-2} - yznr^{n-2}\right] = \sum \vec{i}(0)$

$= 0\vec{i} + 0\vec{j} + 0\vec{k}$

$= 0$

$\therefore$ For all value of n, $\vec{F}$ is irrotational

Now, $\nabla.\vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right).\left(r^n x\vec{i} + r^n y\vec{j} + r^n z\vec{k}\right)$

$= \sum \frac{\partial}{\partial x}\left(r^n x\right) = \sum\left[r^n + xnr^{n-1}\frac{\partial r}{\partial x}\right]$

$= \sum\left[x^n + nr^{n-1}x^2\right]$

$= \sum\left[r^n + nr^{n-1}x^2\right]$

$= 3r^n + nr^{n-2}[x^2 + y^2 + z^2]$

$= 3r^n + nr^{n-2}(r^2)$

$= 3r^n + nr^n = (3+n)r^n$

If n = -3 we get $\nabla.\vec{F} = 0$

$\therefore$ $r^n\vec{r}$ is sonenoidal only if n = -3

Note : In this chapter, we assume ∇ as differential operator then $\nabla r = \frac{1}{r}\vec{r}$.

Problem 10

If $\vec{u} = \frac{1}{r}\vec{r}$ find div $\vec{u}$

Let $\vec{u} = \frac{1}{r}\vec{r}$

By property, $\nabla.(\phi\vec{u}) = \nabla\phi.\vec{u} + \phi\nabla.\vec{u}$

$$\nabla.\left(\frac{1}{r}\vec{r}\right) = \nabla\left(\frac{1}{r}\right).\vec{r} + \frac{1}{r}(\nabla.\vec{r})$$

$= \frac{-1}{r^2}\nabla r.\vec{r} + \frac{1}{r}(\nabla.\vec{r})$, Since $\nabla r = \frac{\vec{r}}{\vec{r}}, \nabla . \vec{r} = 3$

$= \frac{-1}{r^2}\ \frac{\vec{r}}{r} . \vec{r} + \frac{1}{r}(3)$

$= \frac{-1}{r^3}\vec{r}. \vec{r} + \frac{3}{r}$

$= \frac{-1}{r^3}r^2 + \frac{3}{r}$

$= -\frac{1}{r} + \frac{3}{r}$

$= \frac{2}{r}$

Problem 11

If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $|\vec{r}| = r$ prove the following

(i) $\nabla^2 f(r) = f''(r) + \frac{2}{r}f'(r)$

(ii) $\nabla^2(r^n) = n(n+1)r^{n-2}$

(iii) $\nabla^2\left(\frac{1}{r}\right) = 0$

Solution

$$\nabla^2 f(r) = \nabla \cdot [\nabla f(r)]$$

$$= \nabla . [f'(r) \nabla r]$$

$$= \nabla . \left[f'(r) \frac{\vec{r}}{r} \right], \qquad \because \quad \nabla r = \frac{\vec{r}}{r}$$

$$= \nabla . \left(\frac{f'(r)}{r} \vec{r} \right) \qquad \because \nabla . \left(\phi \vec{A} \right) = (\nabla \phi) . \vec{A} + \phi (\nabla . \vec{A})$$

$$= \nabla \left(\frac{f'(r)}{r} \right) . \vec{r} + \frac{f'(r)}{r} (\nabla . \vec{r}) \qquad \text{................ (1)}$$

Now,

$$\nabla \left(\frac{f'(r)}{r} \right) = \frac{r \nabla (f'(r)) - f'(r) \nabla r}{r^2}$$

$$= \frac{r f''(r) \nabla r - f'(r) \nabla r}{r^2}$$

$$= \frac{r f''(r) \frac{\vec{r}}{r} - f'(r) \frac{\vec{r}}{r}}{r^2}$$

$$= \frac{r f''(r) \frac{\vec{r}}{r} - f'(r) \frac{\vec{r}}{r}}{r^2}$$

$$= \left[\frac{f''(r)}{r^2} - \frac{f'(r)}{r^3} \right] \vec{r} \qquad \text{................ (2)}$$

and

$$\nabla . \vec{r} = 3$$

Substitute (2) in (1)

$$= \left[\frac{f''(r)}{r^2} - \frac{f'(r)}{r^2}\right]\vec{r}.\vec{r} + \frac{f'(r)}{r}(3)$$

$$= \left[\frac{f''(r)}{r^2} - \frac{f'(r)}{r^2}\right](r^2) + 3\frac{f'(r)}{r}$$

$$= f''(r) - \frac{f'(r)}{r} + 3\frac{f'(r)}{r}$$

$$= f''(r) + \frac{2}{r}f'(r)$$

(ii) $\nabla^2(r^n) = \nabla.(\nabla r^n)$

$$= \nabla.(nr^{n-1}\nabla r)$$

$$= \nabla.\left(nr^{n-1}\frac{\vec{r}}{r}\right)$$

$$= \nabla.(nr^{n-2}\vec{r})$$

$$= \nabla(nr^{n-2}).\vec{r} + nr^{n-2}(\nabla.\vec{r}) \qquad \because \nabla.(\phi\vec{A}) = (\nabla\phi).\vec{A} + \phi(\nabla.\vec{A})$$

$$= [n(n-2)^{n-3}\nabla r].\vec{r} + nr^{n-2}(3)$$

$$= \left[n(n-2)^{n-3}\frac{\vec{r}}{r}\right].\vec{r} + 3nr^{n-2} \qquad \because \nabla r = \frac{\vec{r}}{r}$$

$$= n(n-2)r^{n-4}\vec{r}.\vec{r} + 3nr^{n-2}$$

$$= n(n-2)r^{n-4}r^2 + 3nr^{n-2}$$

$$= n(n-2)r^{n-2} + 3nr^{n-2}$$

$$= nr^{n-2}[n-2+3] = n(n+1)r^{n-2}$$

(iii) $\nabla^2\left(\frac{1}{r}\right) = \nabla.\nabla\left(\frac{1}{r}\right)$

$$= \nabla . \left[\nabla \left(\frac{1}{r} \right) \right]$$

$$= \nabla . \left[\frac{-1}{r^2} \nabla r \right]$$

$$= \nabla . \left[\frac{-1}{r^2} \frac{\vec{r}}{r} \right] \qquad \because \nabla r = \frac{\vec{r}}{r}$$

$$= \nabla . \left[\frac{-1}{r^3} \vec{r} \right]$$

$$= -\nabla \left[\frac{1}{r^3} \right] . \vec{r} + \left(-\frac{1}{r^3} \right) (\nabla . \vec{r})$$

$$= -\left[\frac{-3}{r^4} \nabla r \right] . \vec{r} + \left(\frac{-1}{r^3} \right) (3)$$

$$= \frac{3}{r^4} \frac{\vec{r}}{r} . \vec{r} - \frac{3}{r^3}$$

$$= \frac{3}{r^5} \vec{r} . \vec{r} - \frac{3}{r^3}$$

$$= \frac{3}{r^5} r^2 - \frac{3}{r^3}$$

$$= \frac{3}{r^3} - \frac{3}{r^3}$$

$$= 0$$

Problem 12

If $\vec{A}$ and $\vec{B}$ are irrotational, show that $\vec{A} x \vec{B}$ is solenoidal.

Solution

Since $\vec{A}$ and $\vec{B}$ are irrotatioal $\nabla x\vec{A} = 0$ and $\nabla x\vec{B} = 0$

$\nabla.(\vec{A}x\vec{B})$ = $\vec{B}.(\nabla \times \vec{A}) - (\nabla \times \vec{B})$ = 0

$\therefore$ $\vec{A}x\vec{B}$ is solenoidal

Problem 13

Prove that if $\vec{F}$ is solenoidal curl. curl. curl (curl $\vec{F}$) - $\nabla^4\vec{F}$.

Solution

Since $\vec{F}$ is solenoidal $\nabla \times \vec{F} = 0$

curl (curl $\vec{F}$) $= \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}$

$= \nabla(\nabla \cdot \vec{F}) - \nabla^2 \vec{F}$

$= -\nabla^2 \vec{F}$ (since $\vec{F}$ is solenoidal)

curl. curl. curl (curl $\vec{F}$) $= \nabla \times \nabla(-\nabla^2 . \vec{F})$

$= \nabla^2(-\nabla^2\vec{F})$

$= \nabla^4 \vec{F}$

Problem 14

A vector function $\vec{F}$ is the product of a scalar function and the gradient of a scalar function. Show that $\vec{F}$ curl $\vec{F}$ = 0.

Solution

Let $\vec{F}$ = $u\nabla v$, where u and v are scalar point function

Now,

$$\text{Curl } \vec{F}) = \nabla \times \vec{F}$$

$$= \nabla \times (u \nabla v)$$

$$= \nabla u \times \nabla v + u (\nabla \times \nabla u)$$

$$= \nabla u \times \nabla v + 0 \qquad (\nabla \times \nabla v = 0)$$

$$\vec{F} \text{ curl } \vec{F} = u \nabla v. (\nabla u \times \nabla v) = 0$$

(Since two of the vectors of the scalar triple product are equal)

Problem 15

Prove that div $(\vec{A} \times \vec{r}) = \vec{A}$ cur $\vec{r}$, where $\vec{A}$ is a constant vector.

Solution

$$\text{div } (\vec{A} \times \vec{r}) = \nabla . (\vec{A} \times \vec{r})$$

$$= \vec{A} \text{ curl } \vec{r} - \vec{r} \text{ curl } A \quad (\text{curl } \vec{A} = 0)$$

$$= \vec{A} \text{ curl } \vec{r} - 0$$

$$= \vec{A} \text{ curl } \vec{r}$$

Problem 16

Prove that div $\left[r^n (\vec{a} \times \vec{r}) \right] = 0$

Solution

$$\text{div } (\phi \vec{A}) = \phi \nabla . \vec{A} \; . \nabla \phi$$

$$\text{div}\left[r^n(\vec{a} \times \vec{r})\right] = \nabla\left[r^n\left(\vec{a} \times \vec{r}\right)\right] = 0$$

$$= r^n \nabla \left(\vec{a} \times \vec{r}\right) + \left(\vec{a} \times \vec{r}\right) . \ \nabla r^n$$

$$= r^n\left[r.\nabla \times \vec{a} - \vec{a}.\nabla \times \vec{r}\right] + \left(\vec{a} \times \vec{r}\right)\left[nr^{n-1}\frac{\vec{r}}{r}\right]$$

$$= r^n (0\text{-}0) + nr^{n-2}\left(\vec{a} \times \vec{r}\right)\vec{r}$$

$$= 0$$

Problem 17

Prove that $\nabla \times (\nabla r^n) = 0$

Solution

$$\nabla \times (\nabla r^n) = \nabla \times (nr^{n-2}\vec{r})$$

$$= \nabla (nr^{n-2}) \vec{r} + nr^{n-2} \nabla \times \vec{r}$$

$$= n(n-2)\, r^{n-4}\, \vec{r} \times \vec{r} + nr^{n-2} \nabla \times \vec{r}$$

$$= 0+0$$

$$= 0$$

Problem 18

Prove that Curl $\left(\vec{a} \times \vec{r}\right) r^n$= n+2) $r^n \vec{a}$ -$nr^{n-2}\left(\vec{r} \times \vec{a}\right)\vec{r}$

Solution

$$\text{Curl}\left(\vec{a}\times\vec{r}\right) r^n = \nabla\times\left[\left(\vec{a}\times\vec{r}\right)r^n\right]\backslash$$

$$= (\nabla r^n)\times\left(\vec{a}\times\vec{r}\right) + r^n\,\nabla\times\left(\vec{a}\times\vec{r}\right)$$

$$= nr^{n-2}\,\vec{r}\times\left(\vec{a}\times\vec{r}\right) + r^n\left(2\vec{a}\right)$$

$$(\because\ \nabla r^n = nr^{n-2}.\vec{r}\ :\ \nabla\times\left(\vec{a}\times\vec{r}\right) = 2\vec{a})$$

$$= nr^{n-2}\left[\vec{r}.\vec{r}\,\vec{a}-\left(\vec{r}.\vec{a}\right)\vec{r}\right]+2\vec{a}r^n$$

$$= nr^n\,\vec{a}-nr^{n-2}\left(\vec{r}.\vec{a}\right).\vec{r}+2\vec{a}r^n$$

$$= (n+2)\,r^n.\,\vec{a} - nr^{n-2}\left(\vec{r}.\vec{a}\right).\vec{r}$$

Problem 19

Prove that $\nabla\times\left[\left(\vec{r}\times\vec{a}\right)\times\vec{b}\right]=\vec{b}\times\vec{a}$

Solution

$$\nabla\times\left[\left(\vec{r}\times\vec{a}\right)\times\vec{b}\right] = \nabla\times\left[\left(\vec{b}.\vec{r}\right)\vec{a}-\left(\vec{b}.\vec{a}\right)\vec{r}\right]$$

$$= \nabla\times\left[\left(\vec{b}.\vec{r}\right)\vec{a}\right] - \nabla\times\left[\left(\vec{b}.\vec{a}\right)\vec{r}\right]$$

$$= \nabla\left(\vec{b}\,.\,\vec{a}\right) x\,\vec{a} + \left(\vec{b}\,.\,\vec{r}\right)\nabla \times \vec{a} - \nabla\left(\vec{b}\,.\,\vec{a}\right) \times \vec{r} - \left(\vec{b}\,.\,\vec{a}\right)\nabla \times \vec{r}$$

$$= \vec{b} \times \vec{a} + \left(\vec{b}\,.\,\vec{r}\right) \times 0 - 0 - \left(\vec{b}\,.\,\vec{a}\right).\,0$$

[Since $\nabla \times \vec{a} = 0$, $\nabla\left(\vec{b}\,.\,\vec{a}\right) = \vec{b}$], $\nabla\left(\vec{b}\,.\,\vec{a}\right) = 0$, $\nabla \times \vec{r} = 0$

$$= \vec{b} \times \vec{a}$$

Problem 20

Prove that $\nabla \times \left[r\nabla\left(\frac{1}{r^3}\right)\right] = \frac{3}{r^4}$

Solution

$$\nabla \times \left(\frac{1}{r^3}\right) = \sum \vec{i}\,\frac{\partial}{\partial x}\left(\frac{1}{r^3}\right)$$

$$= \frac{-3}{r^4}\sum \vec{i}\,\frac{x}{r}$$

$$= \frac{-3}{r^5}.\vec{r} \qquad \left(\therefore \vec{r} = x\vec{i} + y\vec{j} + z\vec{k}\right)$$

$$\nabla\left[r\,\nabla\left(\frac{1}{r^3}\right)\right] = \nabla\left[r\left(\frac{-3}{r^4}\vec{r}\right)\right]$$

$$= \nabla\left[\frac{-3}{r^4}\right]\vec{r} - \frac{3}{r^4}\nabla.\vec{r}$$

$$= \frac{(-3)(-4)}{r^5}\,.\,\frac{\vec{r}}{r}.\vec{r} - \frac{3}{r^4}.3$$

$$= \frac{12}{r^6}\,.\,\vec{r}^{\,2} - \frac{9}{r^4}$$

$$= \frac{3}{r^4}$$

problem 21

If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, show that $\nabla \cdot \vec{r} = 3$

Solution

$$\nabla \cdot \vec{r} = \left(\frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}\right) \qquad \left(x\vec{i} + y\vec{j} + z\vec{k}\right)$$

$$= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z}$$

$$= 1+1+1$$

$$\therefore \boxed{\nabla \cdot \vec{r} = 3}$$

Check Your Progress - II

1. Find the value of $\nabla \times \vec{r}$

2. Find the equaiton of the tangent plane and normal line to the surface

   (i) $x^2 + y^2 - z = 0$ at $(2,-1,5)$

   (ii) $x^2 + y^2 + z^2 = 25$ at $(4,0,3)$

   ....

3. Find ϕ if $\nabla\phi = 2xyz^3\vec{i} + x^2z^3\vec{j} + 3x^2yz^2\vec{k}$

   ....

4. If $\nabla\phi = yz\vec{i} + zx\vec{j} + xy\vec{k}$ find ϕ

   ....

5. If $\vec{F} = x^2z\vec{i} - 2y^3z^2\vec{j} + xy^2z\vec{k}$. Find $\nabla.\vec{F}$ the point $(1,-1,1)$

   ....

10.5 Recap

1. Divergence :

$$\nabla \cdot \vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right).\vec{F}$$

2. Curl

$$\nabla \cdot \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

3. If $\vec{F}$ is a solenoidal, then $\nabla . \vec{F}$ = 0.

4. If $\vec{F}$ is a irrotational, then $\nabla . \vec{F}$ = 0.

5. Vector Identities :

   1. $\nabla\left(\vec{u} + \vec{v}\right) = \nabla.\vec{u} + \nabla\vec{v}$

   2. $\nabla x\left(\vec{u} + \vec{v}\right) = \nabla \text{ x } \vec{u} + \nabla \text{ x } \vec{v}$

   3. $\nabla\left(\phi\vec{u}\right) = \nabla\phi.\ \vec{u} + \nabla\phi.\ \vec{v}$

   4. $\nabla x\left(\phi\vec{u}\right) = \nabla\phi x\vec{u} + \phi\nabla x\ \vec{u}$

   5. $\nabla\left(\vec{u} \text{ x } \vec{v}\right) = \vec{v}.\nabla x\ \vec{u} - \vec{u}.\nabla x\ \vec{v}$

   6. $\nabla \text{ x } \nabla\phi = 0$

7. $\nabla \times \nabla \phi = \nabla^2\phi$

8. $\nabla . (\nabla \times \vec{F}) = 0$

9. $\nabla . (\nabla \times \vec{F}) = \nabla(\nabla . \vec{F}) - \nabla^2 \vec{F}$

10.6 Exercise

1. Show that $\nabla . (\phi \nabla \psi - \psi \nabla \phi) = \phi \nabla^2 \psi - \psi \nabla^2$

2. Show that $\nabla \phi \times \nabla \psi$ is solenoidal.

3. Prove that div. $\vec{r}$ = 3 and curl $\vec{r}$ =0,where $\vec{r}$ is the position vector of me point (x,y,z).

4. Compute divergence and curl of the following vector point functions

   (i) $\vec{F} = x^2z\vec{i} - 2y^3z^2\vec{j} + xy^2\vec{k}$ at the point (1,-1,1)

   (ii) $\vec{F} = xz^3\vec{i} - 2x^2y = \vec{j} + 2y = 4\vec{k}$ at the point (1,-1,1)

5. Show that the vector $3x^2\vec{i} - 4xy^2\vec{j} + 2xy = \vec{k}$ isi solenoidal.

6. Find theconstant 'a' so that the vector

   $\vec{F} = (x+3y)\vec{i} + (y-2z)\vec{j} + (x+az)\vec{k}$ is solenoidal.

7. Show that the vector $\vec{F} = (y2+2xz-1)\vec{i} + (2xy)\vec{j} + 2x^2z\vec{k}$ is irrotational.

8. If $\vec{a}$ is a constant vector prove that

   (i) div $\left(\vec{r} \times \vec{a}\right) = 0$

   (ii) curl $\left(\vec{r} \times \vec{a}\right) = -2\vec{a}$

   (iii) grand $\left(\vec{r} \times \vec{a}\right) = 0$

9. If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ and $|\vec{r}|$ show that

$\nabla \left[f(r)\vec{r} \right]$ = r f'(r) + 3f(r)

10. If $\vec{F}$ = (x+2y+az) $\vec{i}$ + (bx-3y-z) $\vec{j}$ + (4x+cy+2z) $\vec{k}$ is irrotational, Find a,b,c determine ϕ such that $\vec{F} = \nabla \phi$.

11. Prove that Curl $(\phi . \nabla \phi) = 0$

12. Prove that

(i) $\nabla . (\phi \nabla \psi) = \nabla \phi . \nabla \psi + \phi \nabla^2 \psi$

(ii) $\nabla \times (\phi \nabla \psi) = \nabla \phi \times \nabla \psi$

13. Given $\vec{u}$ = xyz $\vec{i}$ + $(2x^2z - y^2x)$ $\vec{j}$ + $x^2 3 \vec{k}$,

$\vec{u}$ = $x^2 \vec{i}$ + 2yz $\vec{j}$ + (1+2z) $\vec{k}$; $\phi = xy + yz + z^2$

14. Prove that $\nabla . (\phi \vec{F}) = \nabla \phi \vec{F} + \phi (\nabla . \vec{F})$

15. Prove that $r^n \vec{r}$ is solenoidal any, when n=-3

LESSON - 11
VECTOR INTEGRATION

Learning Objectives

In this lesson our aim is to study about

- Vector Integration
- Line integral
- Work done and Conservative field
- Surface Integral
- Volume of Integral

Structure

11.1 Introduction

11.2 Line Integral

11.3 Surface integral

11.4 Volume Integral

11.5 Recap

11.6 Exercises

11.1 Introduction

In integral calculas, we have defined indefinite integral, definite integral, line integral, surface integtral and volume integral for a real valued funciton. In this chapter we introduce these integrals for a vector valued function.

If $\vec{F}(t)$ be two vector valued functions. Such that $\frac{d}{dt}\vec{F}(t) = \vec{f}(t)$

Then $\vec{F}(t)$ is called integral of $\vec{f}(t)$ with respect in 't' and this is denoted in symbol by $\int \vec{f}(t)dt = \vec{F}(t)$.

In general if $\frac{d}{dt}\left[\vec{F}(t)+\vec{c}\right] = \vec{f}(t)$,where $\vec{c}$ is an arbitrary vector independent of t, then $\int \vec{f}(t)dt = \vec{F}(t)$+C. Here $\vec{F}(t)$ is called the infinite integral of $\vec{f}(t)$ and C is an arbitrary constant of integration.

As seen in the case of real valued function the definite integral for vector valued function is given by

$$\int_a^b \vec{f}(t)dt = \left[\vec{F}(t)+\vec{c}\right]_a^b = \vec{F}(b)-\vec{F}(a)$$

Also if, $\vec{f}(t) = f_1(t)\,\vec{i} + f_2(t)\,\vec{j} + f_3(t)\vec{k}$

then

$$\int_a^b \vec{f}(t)dt = \int_a^b [f_1(t)\,\vec{i} + f_2(t)\,\vec{j} + f_3(t)\vec{k}]\,dt$$

$$= \vec{i}\int_a^b f_1(t)dt + \vec{j}\int_a^b f_2(t)dt + \vec{k}\int_a^b f_3(t)dt$$

Example 1

If $\vec{f}(t) = (3t^2-1)\vec{i} + (2-6t)\vec{j} - 4t\vec{k}$ Find $\int_2^3 \vec{f}(t)\,dt$

Solution

Given $\vec{f}(t) = (3t^2-1)\vec{i} + (2-6t)\vec{j} - 4t\vec{k}$

$$\int_2^3 \vec{f}(t)\,dt = \int_2^3 \left[(3t^2-1)\vec{i} + (2-6t)\vec{j} - 4t\vec{k}\right]dt$$

$$= \left[(t^3-t)\vec{i} + (2t-3t^2)\vec{j} - 2t^2\vec{k}\right]_2^3$$

$$= \left[(27-3)\vec{i} + (6-27)\vec{j} - 18\vec{k}\right] - \left[(8-2)\vec{i} + (4-12)\vec{j} - 8\vec{k}\right]$$

$$= \left[24\vec{i} - 21\vec{j} - 18\vec{k}\right] - \left[6\vec{i} - 8\vec{j} - 8\vec{k}\right]$$

$$= 18\vec{i} - 13\vec{j} - 10\vec{k}$$

Example 2

The accelelaration of a particle at any time is $\vec{a} = 12\cos 2t\,\vec{i} - 8\sin 2t\,\vec{j} + 16t\,\vec{k}$. If the velocity $\vec{v}$ and the displacement are zero at time t=0. Find the velocity and displacement at any time t.

Solution

$$\vec{a} = (12\cos 2t)\vec{i} - (8\sin 2t)\vec{j} + 16t\vec{k}$$

$$\vec{a} = \frac{dv}{dt}$$

$$\vec{v} = \int\left[(12\cos 2t)\vec{i} - (8\sin 2t)\vec{j} + 16t\vec{k}\right] \, d \times s$$

$$= \vec{i}\,[6\sin 2t] + \vec{j}\,(4\cos 2t) + 8t^2\vec{k} + c$$

when t = 0, $\vec{v} = 0 \Rightarrow 0 = 4\vec{j} + \vec{c}$

$$\boxed{\vec{c} = -4\vec{j}}$$

$$\therefore \vec{v} = (6\sin 2t)\vec{i} + (4\cos 2t - 1)\vec{j} + 8t^2\vec{k}$$

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\therefore \vec{r} = \int\left[(6\sin 2t)\vec{i} + (4\cos 2t - 1)\vec{j} + 8t^2\right] dt$$

$$= (-3\cos 2t)\vec{i} + 4\left(\frac{\sin 2t}{2} - t\right)\vec{j} + \frac{8t^3}{3}\vec{k} + \vec{c}$$

Given thbat when t=0, $\vec{r}$ =0,we have

$$0 = -3\vec{i} + \vec{c}_1$$

$$\boxed{c_1 = 3\vec{i}}$$

$$\vec{r} = (3\text{-}3\cos 2t)\vec{i} + 4\left(\frac{\sin 2t}{2} - t\right)\vec{j} + \frac{8t^3}{3}\vec{k}$$

$$\vec{r} = (3\text{-}3\cos 2t)\vec{i} + (2\sin 2t\text{-}4t)\vec{j} + \frac{8t^3}{3}\vec{k}.$$

Check Your Progress - I

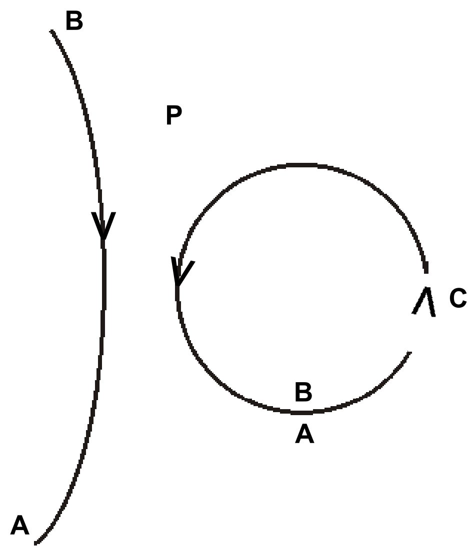
1. Evaluate $\int_2^4 [(3t^2-1)\vec{i} + (2-6t)\vec{j} - 4t\vec{k}]dt.$

2. Evaluate $\int_0^{\pi/2} (3\sin t\,\vec{i} + 2\cos t\,\vec{j})dt.$

3. The acceleeration of a particle at time t is given by $\vec{a}$ =18 cos3t $\vec{i}$ -8sin2t $\vec{j}$ +6t $\vec{k}$. Find the velocity and displacement at time t given that velocity and displacement are zero at time t =0.

11.3 Line Integral

Let C be a curve in space.The orientation of the curve C is defined by a direction C.There are two possible directions along C namely A to B and B to A.If the direction from A to B is defined as the positive direction,the other one is taken as the negative direction.



In case of a closed curve C the initial point and the finial point B coincide and the direction Ato B measured anticlockiwise is taken as positive and the other direction is taken as negative.,

Let $\vec{r}$ (t) = x(t) $\vec{i}$ +xy(t) $\vec{j}$ + z(t)$\vec{k}$ be the parrametric representation of a curve. Let for any two points A and B on the curve the parametric values be $t=t_1$ and $t=t_2$ respectively.

Then, we know that, $\frac{d}{dt}(\vec{r})$ is the tangent vector to this curve.

If $\vec{r}$ (t) is continious and has continous derivatives then the curve C possesses a unique tangent at each part of the curve. Such a curve is called a smooth curve. Suppose a curve is formed by join of a finite number of smooth curve it is said to be piecewise continious. For example, a circle is a smooth curve. A Square is a piecewise smooth curve.

Let $\vec{F}(t) = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ be a vector point function defined along a curve C.

Let $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ be the position vector of any point on this curve.

Let the are length along this curve be measured from a fixed point A. If s denotes thte arc length from A to any point P(x,y,z)

we know that

$\frac{d\vec{r}}{ds} = \vec{t}$ is a unit vector along the tangent to the curve at P.The component of $\vec{F}$ along the tangent is given by $\vec{F} \cdot \frac{d\vec{r}}{ds}$.

The integral of this component along C measured from the point A to the point B is given by $\int_A^B \vec{F}.\frac{d\vec{r}}{ds}.ds$.

This integral is called the line integral of $\vec{F}$ along C. This integral is also called the tangential line integral of $\vec{F}$ alon C.

We note that, $\int_c \left(\vec{F}.\frac{d\vec{r}}{ds}\right).ds = \int_c \vec{F}.d\,\vec{r}$

The line integral is a scalar function.

Note (1)

When,

$\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$

$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$

$\vec{F}.d\vec{r} = F_1dx + F_2dy\vec{j} + F_3dz$

$$\therefore \int_A \vec{f}.d\vec{r} = \int_C (F_1dx + F_2dy + F_3dz)$$

Note (2)

If the equation of the curve is given in paramatric form say x = x(t), y=y(t),z = z(t) and the parametric values of A and B are $t = t_1$ and $t = t_2$.

Then

$$\int_c \vec{F}.d\vec{r} = \int_{t_1}^{t_2} \left(F_2\frac{dx}{dt} + F_2\frac{dy}{dt} + F_3\frac{dz}{dt}\right)dt$$

Note (3)

If C is a simple closed curve the line integral of $\vec{F}$ around C is called the circulation of $\vec{F}$ around C and is called the circulation of $\vec{F}$ about C and

$$\oint \vec{F}.d\vec{r} = \oint (F_1dx + F_2dy + F_3dz)$$

Application line Integral

Suppose $\vec{F}$ is a force acting upon a particle, which moves along a curve C in space and $\vec{r}$ be the position vector of the particle at a point on C. Then work done by the particle at P along C is $\vec{F}.d\vec{r}$ and the total work done by $\vec{F}$ in this displacement along a curve C is given by the line integral $\int_c \vec{F}.d\vec{r}$.

11.3.2 Theorem on Line Integral

The necessary and sufficient condhitions for the integral $\int_A^B \vec{F}.d\vec{r}$ be independent of the path joining A and B is that $\vec{F}$ is the gradient of some scalar point function (i.e., $\vec{F} = \nabla \phi$. if and only if $\int_A^B \vec{F}.d\vec{r} = \phi_B - \phi_A$.

Note 1

If $\vec{F} = \nabla \phi$, then use know that $\nabla \times \phi = 0$ Hence the necessary and sufficient conditions for $\int_C \vec{F}.d\vec{r}$ to the independent is that curl $\vec{F}$ =0.

Note 2

In a closed curve the initial point and finial point coincide. Therefore, if $\int \vec{F}.d\vec{r}$ is independent of the path of integration, then $\int_C \vec{F}.d\vec{r}$ = 0 for any closed curve.

Note 3

If $\int_A^B \vec{F}.d\vec{r}$ = 0 is independent of path then $\vec{F}$ is called a conservative force. Therefore, the necessary and condition for $\vec{F}$ to be conservative force is $\vec{F} = \nabla \phi$.

Example 1

If $\vec{F} = (3x^2 + +6y)\vec{i} - 14yz\,\vec{j} + 20xz^2\vec{k}$, evaluate $\int_C \vec{F}.d\vec{r}$, from the straight line joining (0,0,0) and (1,1,1)

Solution

$\vec{F}$ = $(3x^2 + +6y)\vec{i}$ - $14yz\ \vec{j}$ + $20xz^2\vec{k}$

$\vec{r}$ = $x\vec{i}$ + $y\vec{j}$ + $z\vec{k}$

$d\vec{r}$ = $dx\vec{i} + dy\vec{j} + dz\vec{k}$

$\vec{F}\cdot d\vec{r}$ = $(3x^2 +6y)dx - 14yz\ dy + 20xz^2dz$

When the points moves from (0,0,0) to (1,1,1) is given in parametric form is x=y=z=t.

$x = t$; $y = t^2$; $z = t^3$

$dx = dt$; $dy = 2dt$; $dz = 3t^2dt$

$$\int_C \vec{F}.d\vec{r} = \int_{t=0}^{1} (3t^2 + 6t)dt - 14(t)(t)dt + 20t(t^2)dt$$

$$= \int_0^1 (3t^2 + 6t - 14t^2 + 20t^3)dt$$

$$= \int_0^1 (6t - 11t^2 + 20t^3)dt$$

$$= \left[6\frac{t^2}{2} - 11\frac{t^3}{3} + 20\frac{t^4}{4}\right]_0^1$$

$$= \frac{6}{2} - \frac{11}{3} + \frac{20}{4}$$

$$= 3 - \frac{11}{3} + 5$$

$$= \frac{13}{3}$$

Note : The value of line integral is not the same along all these three paths but depends upon the path.

Example 2

If $\vec{F} = 3xy\,\vec{i} - y^2\,\vec{j}$, Evaluate $\int_C \vec{F}.d\vec{r}$, where C is a curve in the xy plane $y = 2x^2$, from (0,0) to (1,2)

Solution

$$\vec{F} = 3xy\,\vec{i} - y^2\,\vec{j}$$

$$d\vec{r} = dx\vec{i} + dy\vec{j}$$

$$\vec{F} \cdot d\vec{r} = 3xy\,dx - y^2\,dy$$

$$\int \vec{F}.d\vec{r} = \int [3xyzdx-y^2dy)$$

$$\int \vec{F}.d\vec{r} = \int \left[3x.6x^2dx - 4x^4.4xdx\right]$$

$$= \int_0^1 (6x^3-16x^5)dx$$

$$= \left[\frac{6x^4}{4} - \frac{16x^6}{6}\right]_0^1$$

$$= \frac{3}{2} - \frac{8}{3} = \frac{9-16}{6}$$

$$= \frac{-7}{6}$$

Problem 3

Find the workdone is moving a particle once round the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1, z=0$ under the field of force given by $\vec{F}=(2x-y+z)\vec{i}+\left(x+y-z^2\right)\vec{j}+(3x-2y+4z)\vec{k}$. Is the field conservative?

Solution

$$\vec{F}=(2x-y+z)\vec{i}+\left(x+y-z^2\right)\vec{j}+(3x-2y+4z)\vec{k}$$

$$d\vec{r}=dx\vec{i}+dy\vec{j}+dz\vec{k}$$

$$\vec{F}.d\vec{r}=(2x-y+z)\,dx+\left(x+y-z^2\right)dy+(3x-2y+4z)\,dz$$

C is the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1, z=0$

$$\Rightarrow dz=0$$

$$\int_c \vec{F}.d\vec{r}=\int_c (2x-y)dx+(x+y)dy$$

The parametric equaiton of the ellipse are

x = 5 cosθ, y = 4 sin θ

dx = -5 sinθ dθ, dy = 4 cos θ dθ

θ values from 0 to 2π.

$$\int_c \vec{F}.d\vec{r}=\int_c^{2\pi}\left[(10\cos\theta-4\sin\theta)(-5\sin\theta d\theta)+(5\cos\theta+4\sin\theta)4\cos\theta\right]d\theta$$

$$=\int_0^{2\pi}\left[20\left(\sin^2\theta+\cos^2\theta\right)-34\sin\theta\cos\theta\right]d\theta$$

$$= \int_{c}^{2\pi} (2\theta - 17\sin\theta) d\theta$$

$$= \left[20\theta + \frac{17\cos 2\theta}{2} \right]_0^{2\pi}$$

$$= 40\pi + \frac{17}{2} - 0 - \frac{17}{2}$$

$= 40\ \pi$ units.

∴ Since C is a closed curve and the work done round C is not equal to zero, the field is not conservative.

Problem 4

Find the total work done in moving a particle in a force field given by $\vec{F} = 3xy\,\vec{i} - 5z\,\vec{j} + 10x\,\vec{k}$ along the curve $x = t^2+1$, $y=2t^2$, $z=t^3$, from $t=1$ to $t=2$.

Solution

$\vec{F} \quad = 3xy\,\vec{i} - 5z\,\vec{j} + 10x\,\vec{k}$

$d\vec{r} \quad = dx\vec{i} + dy\vec{j} + dz\vec{k}$

$\vec{F}\cdot d\vec{r} \quad = 3xy\,dx - 5zdy + 10xdz$

$$\int_C \vec{F}.d\vec{r} = \int_C (3xy\,dx - 5zdy + 10xdz)$$

Work done by the force $= \int_C \vec{F}.d\vec{r}$

Given $\quad x = t^2+1 \quad ; \quad y = 2t^2 \quad ; \quad z = t^3$

$\quad dx = 2tdt \quad ; \quad dy = 4tdt \quad ; \quad dz = 3t^2dt$

$\vec{F}\cdot d\vec{r} \quad = 3(t^2+1).2t^2.2t\,dt - 5t^3 4t dt + 10(t^3+1)\,3t^2 dt$

$= 12t^5 + 10t^4 + 12t^3 + 30t^2\,dt$

$$\int_C \vec{F}.d\vec{r} = \int_1^3 (12t^5 + 10t^4 + 12t^3 + 30t^2)\,dt$$

$$\int_C \vec{F}.d\vec{r} = \left[\frac{12t^6}{6} + \frac{10t^5}{5} + \frac{12t^4}{4} + \frac{30t^3}{3}\right]_1^2$$

$= 2[64-1] + 2[32-1] + 3[16-1] + 10[8-1]$

$= 303$ Units

Problem 5

Find $\int_C \vec{F}.d\vec{r}$, where $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$ along the arc of the curve $\vec{r} = \cos t\,\vec{i} + \sin\vec{j} + t\vec{k}$ from from $t = 0$ to $t = 2\pi$.

Solution

The parametric equation of the curve are $x = \cos t$; $y = \sin t$; $z = t$

Let $\vec{F} = t\vec{i} + \cos t\,\vec{j} + \sin\vec{k}$

$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$

$= (-\sin t\,\vec{i} + \cos t\,\vec{j} + \vec{k})dt$

$$\vec{F}\cdot d\vec{r} = (-t\sin t + \cos^2 t + \sin t)dt$$

$$\int_C \vec{F}.d\vec{r} = \int_0^{2\pi} (-t\sin t + \cos^2 t + \sin t)dt$$

$$= \left[(t\cos t - \sin t) + \frac{1}{2}\left(t + \frac{\sin 2t}{2}\right) - \cos t\right]_0^{2\pi}$$

$$= 2\pi + \pi - 1 + 1$$

$$= 3\pi$$

Problem 6

If $\vec{F} = yz\vec{i} + zx\vec{j} - xy\vec{k}$, Find $\int_C \vec{F}.d\vec{r}$ where C is given by $x = t$, $y = t^2$, $z = t^3$ from p (0,0,0) to Q (2,4,8).

Solution

$$\vec{F} = yz\vec{i} + zx\vec{j} - xy\vec{k}$$

$$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$\vec{F}\cdot d\vec{r} = yzdx + zxdy - xydz$$

Given $x = t$; $y = t^2$; $z = t^3$

$dx = dt$; $dy = 2tdt$; $dz = 3t^2dt$

$$\int_C \vec{F}.d\vec{r} = \int_0^2 (t^2\, dt + t^4\, 2tdt - t^3 .\, 3t^2\, dt$$

$$= \int_0^2 (t^5 + 2t^5 - 3t^5)\, dt$$

$$= 0$$

Problem 7

Show that $\vec{F} = (2xy + z^3)\vec{i} + x^2\vec{j} + 3x^2z^2\vec{k}$ is a conservative vector field and find a function ϕ such that $\vec{F} = \nabla \phi$. Also Find the work done in moving an object in this field from (1,-2,1) to (3,1,4).

Solution

If $\vec{F}$ is a conservative field, we have to prove that $\nabla \times \vec{F} = 0$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix}$$

$$= \vec{i}\,(0-0) - \vec{j}\,(3z^2-3z^2) + \vec{k}\,(2x-2x)$$

$$= 0$$

$\therefore$ $\vec{F}$ is a conservative field.

$\therefore$ $\vec{F} = \nabla \phi$

(i.e.,) $(2xy+z^3)\vec{i} + x^2\vec{j} + 3xz^2\vec{k} = \frac{\partial \phi}{\partial x}\vec{i} + \frac{\partial \phi}{\partial y}\vec{j} + \frac{\partial \phi}{\partial z}\vec{k}$

Equating the co-effiocient of $\vec{i}$, $\vec{j}$, $\vec{k}$, we get

$$\frac{\partial\phi}{\partial x} = 2xy+z^3 \quad ; \quad \phi = x^2y+xz^3+f_1(x,z) \qquad (1)$$

$$\frac{\partial\phi}{\partial y} = x^2 \quad ; \quad \phi = x^2y+f_2(x,z) \qquad (2)$$

$$\frac{\partial\phi}{\partial z} = 3xz^2 \quad ; \quad \phi = xz^3 + f_3(x,y) \qquad (3)$$

From (1), (2) and (3) we get

$$f = x^2y + xz^3 + C$$

Work done in moving an object from A (1, -2, 1) to B (3,1,4)

$$\text{work done} = \phi_B - \phi_4$$

$$= \left[x^2y + xz^3 + c\right]_{(1,-2,1)}^{(3,1,4)}$$

$$= (9+192+C) - (-2+1+C)$$

$$= 202 \text{ Units}$$

Problem 9

Find $\int_C \vec{F}.d\vec{r}$, we have $\vec{F} = (x^2-y^2)\vec{i} + 2xy\vec{j}$ and C is the square bounded by the coordinate axis and the lines x = a and y =a

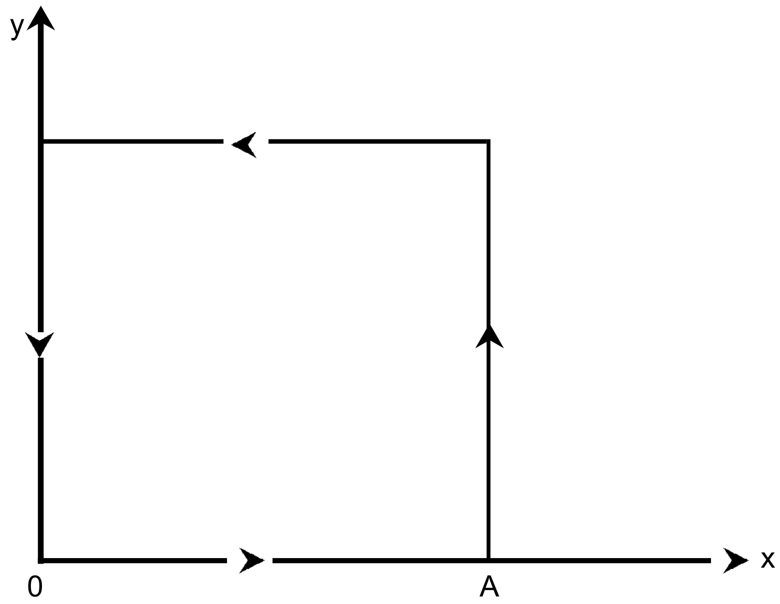
Solution

Given $\vec{F} = (x^2-y^2)\vec{i} + 2xy\vec{j}$

$$d\vec{r} = dx\vec{i} + dy\vec{j}$$

$$\vec{F}\cdot d\vec{r} \quad = \quad (x^2-y^2)dx + 2xydy$$

$$\int_C \vec{F}.d\vec{r} \quad = \quad \int_{OA} \vec{F}.d\vec{r} + \int_{AB} \vec{F}.d\vec{r} + \int_{BC} \vec{F}.d\vec{r} + \int_{CO} \vec{F}.d\vec{r}$$



Along OA, x varies from 0 to a and y = 0 and dy = 0

$$\therefore \int_{OA} \vec{F}.d\vec{r} = \int_0^a x^2dx = \frac{a^3}{3} \qquad (1)$$

Along AB, y varies from 0 to a and x = a and dx = 0

$$\therefore \int \vec{F}.d\vec{r} = \int_0^a 2aydy = \left[ay^2\right]_0^a = a^3 \qquad (2)$$

Along BC, x varies from a to 0 and y = a, $\therefore$ dy = 0

$$\therefore \int_{BC} \vec{F}.d\vec{r} = \int_a^0 (x^2-a^2) = \left[\frac{x^3}{3} - a^2x\right]_a^0 = \frac{2a^3}{3} \qquad (3)$$

Along CO, y varies from a to 0 and x = 0 , ∴ dx = 0

$$\therefore \int_{C0} \vec{F}.d\vec{r} = \int_a^0 0.dy = 0 \qquad (4)$$

$$\therefore \int_C \vec{F}.d\vec{r} = \frac{a^3}{3} + a^3 + \frac{2a^3}{3} + 0$$

$$= 2a^3$$

Problem 10

For the vector function $\vec{F} = 2xy\vec{i} + (x^2+2yz)\vec{j} + (y^2+1)\vec{k}$ determine the value of integral $\int_C \vec{F}.d\vec{r}$ around the unit circle with centre at the origin inthe xy-p lane.

Solution

$$\vec{F} = 2xy\vec{i} + (x^2+2yz)\vec{j} + (y^2+1)\vec{k}$$

C is the curve $x^2+y^2=1$, z = 0

$$\vec{F}\cdot d\vec{r} = 2xydx + x^2dy$$

$$\int \vec{F}.d\vec{r} = \int (2xydx + x^2dy)$$

Along C, $x = \cos\theta$, $y = \sin\theta$

$dx = -\text{sini}\theta d\theta$, $dy = \cos\theta\, d\theta$

$$\int_C \vec{F}.d\vec{r} = \int_0^{2\pi} [2\sin\theta.\cos\theta\,(-\sin\theta) + \cos^2\theta\,\cos\theta]\,d\theta$$

$$= \int_0^{2\pi} [-2\sin^2\theta\,\cos\theta + \cos^3\theta]\,d\theta$$

$$= \int_0^{2\pi} \cos\theta\ (-2\sin^2\theta + 1 - \sin^2)\ d\theta$$

$$= \int_0^{2\pi} (1 - 3\sin^2\theta)\ d\ (\sin\theta)$$

$$= \left[\sin\theta - \frac{3\sin^2\theta}{3} \right]_0^{2\pi}$$

$$= 0$$

Check Your Progress - I

1. Evaluate $\int_c \vec{F}.d\vec{r}$ where $\vec{F} = x^2y^2\vec{i} + y\vec{j}$ and c is $y^2 = 4x$ in the xy plane from (0,0) to (4,4)

....

2. Find the word done is moving a particle once around a circle $x^2 + y^2 = 4$ if

$\vec{F} = (2x - y + 2z)\vec{i} + (x + y - z^2)\vec{j} + (3x - 2y - 5z)\vec{k}$

....

3. If $\vec{F} = (2y + 3)\ \vec{i} + xz\vec{j} + (yz - x)\vec{k}$ find $\int_c \vec{F}.d\vec{r}$ along c where

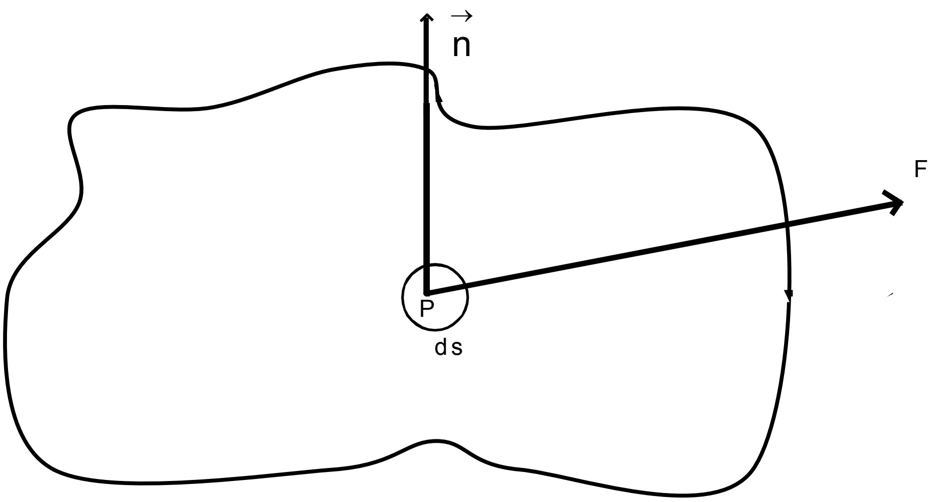
(i) the straight line joining (0,0,0) to (1,1,1)

(ii) x = t, y = t^2, z = t^3 from t = 0 to t = 1

....

11.4 Sufrace Integrals

Let S be a two sided surface. If one side of the surface is arbitrary chosen at the positive side, the other side of the surface is taken as the negative side. However in a closed surface the outer side is taken as the positive side and the inner side is taken as the negative side.



A unit normal vector $\hat{n}$ at any point of the positve side of S is called the outward drawn normal and its direction is considered to be positive.

Let 'ds' be an elementary surface area on the positive side of S.

Let $\hat{n}$ be the outward drawn normal at P.

Let $\vec{ds}$ be a vector in the direction of $\hat{n}$ and is of magnitutde $\vec{ds}$, then $\vec{ds} = \hat{n}\,ds$.

Let $\vec{f}$ (x,y,z) be a vector function defined at each point (x,y,z) of the surface S.

The integral $\iint_S \vec{F}.\vec{ds}$ or $\iint_S \vec{F}.\hat{n}\vec{ds}$ is called the Sufrace Integral of $\vec{F}$ over S.

Physically, a sufrace integral of a vector point function $\vec{F}$ represents the normal flux through the surface. If $\vec{F}$ represents the velocity of a fluid. Then the surface integral of $\vec{F}$ over a closed surface represents flow of fluid through the surface.

There are also Surface integrals of the type

$$\iint_S \vec{F} \times \hat{n}\, ds, \quad \iint_S \phi\, \hat{n}\, ds$$

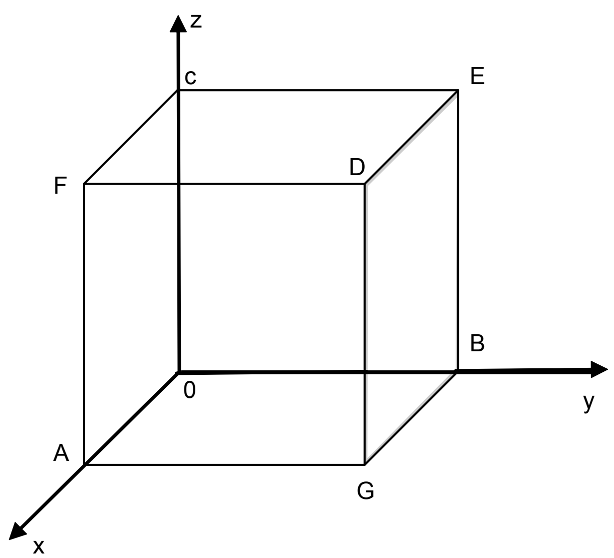
Note :

1) If S is a closed surface, the outer surface is usually chosen as the positive side.

2) $\int \phi d\vec{s}$ and $\int_S \vec{F} x d\vec{s}$, where ϕ is a scalar point function are also surface integral.

Example 1

If $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$, evaluate $\iint_S \vec{F}.\vec{n}\, ds$ where S is the Surface of the cube bouded by x = 0, x = 1, y = 0, z = 0, z = 1



Solution

Given : $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$

For the force AGDF, $x = 1$, $\vec{n} = \vec{i}$

$$\iint_{AGD} \vec{F}.\vec{n}ds = \int_0^1\int_0^1 4z.\vec{i}.\vec{i}\ dy\,dz$$

$$= \int_0^1\int_0^1 4z\ dy\,dz$$

$$= \int_0^1 4z.(y)_0^1\ dz$$

$$= 4\left[\frac{z^2}{2}\right]_0^1 = 2$$

For the face OBEC, $x = 0$, and $\vec{n} = -\vec{i}$

$$\iint_{OBEC} \vec{F}.\vec{n}ds = \int_0^1\int_0^1 -(-y^2\vec{j} + yz\vec{k}).\ \vec{i}\ dydz$$

For the face BEDG, $y = 1$ and $\vec{n} = \vec{j}$

$$\iint_{BEGD} \vec{F}.\vec{n}ds = \int_0^1\int_0^1 (4xz\vec{i} - \vec{j} + z\vec{k}).\ \vec{j}\ dx\,dz$$

$$= \int_0^1\int_0^1 -\ dx\,dz$$

For the face OAFC, $y = 0$ and $\vec{n} = -\vec{j}$

$$\therefore \iint_{OAFC} \vec{F}.\vec{n}\ ds = \int_0^1\int_0^1 4xz\vec{i}\ (-\vec{j})\ dx\,dz = 0$$

For the face CFDE, z = 1 and $\vec{n} = \vec{k}$

$$\therefore \iint_{CFDE} \vec{F}.\vec{n}\, ds = \int_0^1 \int_0^1 (4x\vec{i} - y^2\vec{j} + y\vec{k})\, dxdy = \int_0^1 \int_0^1 y dxdy = \frac{1}{2}$$

For the face OAGB, z = 0 and $\vec{n} = -\vec{k}$

$$\therefore \iint_{OAGB} \vec{F}.\vec{n}\, ds = \int_0^1 \int_0^1 (-y^2\vec{j})(-\vec{k})\, dxdy = 0$$

$$\therefore \iint_{S} \vec{F}.\vec{n}\, ds = 2+0-1+0+\frac{1}{2}+0$$

$$= \frac{3}{2}$$

Example 2

Evaluate $\iint_S \vec{F}.\vec{n}\, ds$, where $\vec{F} = z\vec{i} + x\vec{j} - 3y^2z\vec{k}$, and S is the surface of the cylinder $x^2+y^2=16$ included in the first Octant between z = 0 and z = 5.

Solution

$$\vec{F} = z\vec{i} + x\vec{j} - 3y^2z\vec{k}\ ; \quad \phi = x^2+y^2-16=0$$

$$\nabla\phi = \nabla(x^2+y^2-16)$$

$$= 2x\vec{i} + 2y\vec{j}$$

$\therefore \vec{n}$ = unit normal vector to the cylinder S.

$$= \frac{2x\vec{i} + 2y\vec{j}}{2\sqrt{x^2+y^2}} = \frac{x\vec{i} + y\vec{j}}{4} \qquad (\because x^2+y^2=16)$$

$$\iint_S \vec{F}.\vec{n}\, ds = \iint_R \vec{F}.\vec{n} \frac{dxdz}{|\vec{n}.\vec{j}|}$$ Where R is projection of surface S on xz - plane

$$= \iint_R (z\vec{i} + x\vec{j} - 3y^2z\vec{k}) . \left(\frac{x\vec{i} + y\vec{j}}{4}\right) \frac{dx\,dz}{\frac{y}{4}}$$

$$= \iint_R \left(\frac{xz + xy}{4}\right) \frac{dx\,dz}{\frac{y}{4}}$$

$$= \int_0^5 \int_0^4 \left(\frac{xz}{\sqrt{16 - x^2}} + x\right) dx\,dz$$

$$= \int_0^5 \left[\left(-\sqrt{16 - x^2}\right)z + \frac{x^2}{2}\right]_0^4$$

$$= \int_0^5 (4z+8)\, dz$$

$$= \left[2z^2 + 8z\right]_0^5$$

$$= 90$$

Example 3

Evaluate $\iint \vec{F}.\vec{n}\, ds$, where $\vec{F} = 18z\vec{i} - 12\vec{j} + 3y\vec{k}$, and S is the part of the plane 2x+3y+6z=12, which is located in the first quadrant.

Solution

$$\iint_S \vec{F}.\vec{n}\, ds = \iint_S \vec{F}.\vec{n} \frac{dxdy}{|\vec{n}.\vec{k}|}$$

Where R is the projection of S in the xy - plane

$$\vec{n} = \frac{2\vec{i}+3\vec{j}+6\vec{k}}{\sqrt{2^2+3^2+6^2}} = \frac{2\vec{i}+3\vec{j}+6\vec{k}}{7}$$

$$\vec{n}.\vec{k} = \left(\frac{2\vec{i}+3\vec{j}+6\vec{k}}{7}\right)\vec{k}$$

$$= \frac{6}{7}$$

$$\iint_S \vec{F}.\vec{n}\, ds = \iint_R (18z\vec{i} - 12\vec{j} + 3y\vec{k}) . \left(\frac{2\vec{i}+3\vec{j}+6\vec{k}}{7}\right) . \frac{dxdy}{\vec{n}.\vec{k}}$$

$$= \iint_R (18z\vec{i} - 12\vec{j} + 3y\vec{k}) . \frac{2\vec{i}+3\vec{j}+6\vec{k}}{7} . \frac{dxdy}{6/7}$$

$$= \iint_R \frac{36z-36-18y}{6} dxdy$$

$$= \iint_R \left[\frac{36(12-2x-3y)-36+18y}{6}\right]\frac{dxdy}{6}$$

$$= \iint_R \left[\frac{36-12x}{6}\right]dxdy$$

$$= \int_{x=0}^{6}\int_{y=0}^{\frac{12-2x}{3}} (6-2x)\, dy\, dx$$

$$= \int_0^t (6-2x)\left(\frac{12-2x}{3}\right)dx$$

$$= \left[24x - 6x^2 + \frac{4}{3}.\frac{x^3}{3} \right]_0^6$$

= 144-216+96

= 24.

Example 4

Evaluate $\iint \vec{F}.\vec{n}\, ds$, where $\vec{F} = yz\vec{i} + zx\vec{j} + xy\vec{k}$, and S is the part of the surface of the sphere $x^2+y^2+z^2=1$, which lies in the first octant.

Solution

$\vec{F} = yz\vec{i} + zx\vec{j} + xy\vec{k}$

$\nabla \phi = \nabla (x^2+y^2+z^2-1) = 2(x\vec{i} + y\vec{j} + z\vec{k})$

$$\vec{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2(x\vec{i} + y\vec{j} + z\vec{k})}{2\sqrt{x^2+y^2+z^2}} = (x\vec{i} + y\vec{j} + z\vec{k})$$

$$\vec{F}\cdot\vec{n} = (yz\vec{i} + zx\vec{j} + xy\vec{k}) . (x\vec{i} + y\vec{j} + z\vec{k})$$

$= 3xyz$

$$\iint_S \vec{F}.\vec{n}\, ds = \iint_R \vec{F}.\vec{n}.\frac{dx.dz}{|\vec{n}.\vec{k}|}$$

$$= \iint_R 3xyz.\frac{dxdz}{z}$$ where R is the projection of S on the ny - plane

$$= \iint_R 3xy\, dx\, dz$$

Le us convert this into polar coordinate,

x - rcosθ, y = rsinθ, dxdz = rdθdr

$$\iint_S \vec{F}.\vec{n}\, ds = \int_{\theta=0}^{\pi/2} \int_{r=0}^{1} 3r\cos\theta.r\sin\theta.\ rd\theta dr$$

$$= \int_{\theta=0}^{\pi/2} \int_{r=0}^{1} 3r^3\sin\theta.\cos\theta.\ d\theta dr$$

$$= 3\left[\frac{r^4}{4}\right]_0^1 \int_{\theta=0}^{\pi/2} \sin\theta.\cos\theta.d\theta$$

$$= \frac{3}{4}\left[\frac{\sin^2\theta}{2}\right]_0^{\pi/2}$$

$$= \frac{3}{8}$$

11.5 Volume Integral

Let S be a closed Surface enclosing a voluve V in a region R and $\vec{F}$ be a vector point function in the region R. Then $\iiint_V \vec{F}dv$ is called the volume integral of $\vec{F}$ over the volume V.

Similarly, if ϕ is a scalar point function defined in the region R enclosing a volume V. then

$\iiint_V \phi dv$ is also a volume integral of ϕ over V.

Example 1

Evaluate $\iiint\limits_V \nabla.\vec{F}dv$, if $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ and is V is the volume of the region enclosed by the cube $0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq z \leq 1$.

Solution

$$\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$$

$$\nabla.\vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) . (x^2\vec{i} + y^2\vec{j} + z^2\vec{k})$$

$$= \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(z^2)$$

$$= 2(x+y+z)$$

$$\iiint\limits_V \nabla.\vec{F}dv = 2\int_0^1\int_0^1\int_0^1 (x+y+z)\, dz\, dy\, dx$$

$$= 2\int_0^1\int_0^1 \left[xz + yz + \frac{z^2}{2}\right]_0^1 dy\, dx$$

$$= 2\int_0^1\int_0^1 \left(x + y + \frac{1}{2}\right) dy\, dx$$

$$= 2\int_0^1 \left[xy + \frac{y^2}{2} + \frac{y}{2}\right]_0^1 dx$$

$$= 2\int_0^1 \left(x + \frac{1}{2} + \frac{1}{2}\right) dx$$

$$= 2\int_0^1 (x+1)\, dx = 2\left[\frac{x^2}{2} + x\right]_0^1 = 2\left[\frac{1}{2} + 1\right] = 3$$

Example 2

Evaluate $\iiint_V \vec{F}.dv$, where $\vec{F} = 2xz\vec{i} - x\vec{j} + y^2\vec{k}$ and is V is the volume of the region enclosed by the cylinder $x^2+y^2=a^2$ between z = 0, z = 0.

Solution

$$\vec{F} = 2xz\vec{i} - x\vec{j} + y^2\vec{k}$$

$$\iiint_V \vec{F}.dv = \iiint_V (2xz\vec{i} - x\vec{j} + y^2\vec{k})dv$$

We can evaluate that integral, using cylindrical coordinates $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, and $dv = rdz\, d\theta\, dr$.

$$\iiint_V \vec{F}.dv = \int_0^a \int_0^{2a} \int_0^c \left[2r\cos\theta.z)\vec{i} - (r\cos\theta)\vec{j} + r^2 \sin^2\theta\vec{k}\right] (rdz\, d\theta dr$$

$$= \int_0^a \int_0^{2a} \left[2r\cos\theta.\frac{z^2}{2}\vec{i} - (r\cos\theta)z\vec{j} + (r^2 \sin^2\theta)^2\vec{k}\right]_0^C (d\theta\, rdr)$$

$$= \int_0^a \int_0^{2a} \left[C^2r\cos\theta\vec{i} - r\cos\theta\vec{j} + cr^2 \sin^2\vec{k}\right] (d\theta.rdr)$$

$$= \int_0^a \int_0^{2a} \left[c^2r(\cos\theta)\vec{i} - cr\cos\theta\vec{j} + cr^2\left(\frac{1-\cos 2\theta}{2}\right)\vec{k}\right](d\theta.rdr)$$

$$= \int_0^a \left\{c^2r\left(\sin\theta\right)_0^{2\pi}\vec{i} - \left(cr\sin\theta\right)_0^{2\pi}\vec{j} + \frac{cr^2}{2}\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi}\vec{k}\right\}rdr$$

$$= \int_0^a (0+cr^2\pi\vec{k}).\ rdr$$

$$= \int_0^a [cr^3\pi \vec{k}]dr$$

$$= \left[\frac{cr^4}{4}\pi \vec{k}\right]_0^a$$

$$= \frac{ca^4}{4}\pi \vec{k}$$

Check Your Progress - II

1. Evaluate $\iint_s \left(y^2z^2 i + z^2x^2\vec{j} + x^2y^2\vec{k}\right).\hat{n}ds$ where S is the part of the sphere $x^2 + y^2 + z^2 = 1$ above the xy plane

....

2. Evalute $\iint_s \vec{F}.\hat{n}\,ds$ where

$\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ and S is the surface of the cube bounded by x=0, x = 1, y = 0, y=1, z=0, z=1

....

3. Evaluate $\iiint_v \nabla.\vec{F}dv$, where $\vec{F} = 2x^2y\vec{i} - y^2\vec{j} + 4xz^2\vec{k}$ and V is the region in the first octant bounded by the cyclinder $y^2 + z^2 = 0$ and x =0, x = 2

....

4. Evaluate $\iiint_v \nabla.\vec{F}dv$ where $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ and v is the volume enclosed by the cube x =0, x =1, y =0, y =1, z =0, and z =1

....

11.6 Recap

1. The definite integral for vector valued function is given by

$$\int_a^b \vec{f}(t)dt = \vec{i}\int_a^b \vec{f_1}(t)dt + \vec{j}\int_a^b \vec{f_2}(t)dt + \vec{k}\int_a^b \vec{f_3}(t)dt$$

Where $\vec{f}(t) = f_1(t)\vec{i} + f_1(t)\vec{j} + f_1(t)\vec{k}$

2. Line Integral :

$$\int_a \vec{F}.d\vec{r} = \int_c F_1dx + F_2\,dy + F_3\,dz$$

3. Surface Integral :

If $\hat{n}$ is the unit normal at any point on the sufrace S.

then $\iint_S \vec{F}.\hat{n}$ is defined as the surface integral of A over S.

4. Volume Integral :

Let $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ be a unit vector function defined and continious on a closed surface S, enclosing a volume V.

Then, $\iiint_V \vec{F}.dv$ is defined as a volume integral of a over V.

11.7 Exercises

1. Evaluate $\int_c \vec{F}.d\vec{r}$, where $\vec{F} = xy\vec{i} + yz^2\vec{j} + y^2z\vec{k}$ from (0,0,0) to (1,1,1) along the curve $x = t^2$, $y = t^3$, $z = t^4$.

2. If $\vec{F} = (3x^2+6y)\vec{i} - 14yz\vec{j} + 20xz\vec{k}$, evaluate $\int_c \vec{F}.d\vec{r}$ along the curve C, $x = t$, $y = t^2$, $x = t^3$, $x = t^3$, from (0,0,0) to (1,1,1).

3. Find the work done in moving a particle in the field at force $\vec{F} = 3x^2\vec{i} + (2xz-y)\vec{j} + z\vec{k}$, along the curve C defined by $x^2=4y$, $3x^2=8z$ from x = 0, to x = 2.

4. Find the work done in moving a particle in the force field $\vec{F} = 3x^2\vec{i} + (2xz-y)\vec{j} + z\vec{k}$ along the line joining (0,0,0) to (2,1,3)

5. Evaluate $\iint\limits_S \vec{F}.\vec{n}\, ds$, where S is the region bounded by $x^2+y^2=4$, z = 0, z = 3 and $\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$.

6. Evaluate $\iint\limits_S \vec{F}.\hat{n}\, ds$, where $\vec{F} = (y-z+2)\vec{i} + (yz+4)\vec{j} - xz\vec{k}$, where S is the surface of the cube bounded by x = 0, x = 2 y = 0, y = 2, z = 0, z = 2.

7. Evaluate $\iint\limits_S \vec{F}.\hat{n}\, ds$, if $\vec{F} = 18z\vec{i} - 12\vec{j} + 3y\vec{k}$ and S is the surface 2x+3y+6z = 12 is the first octant.

8. Evaluate $\iint\limits_S \vec{F}.\hat{n}\, ds$, over the surface of the sphere $x^2+y^2+z^2=a^2$ along the xoy plane if $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$.

9. Evalute $\iiint\limits_C \nabla.\vec{F}dv$, if $\vec{F} = 2x^2y\vec{i} - y^2\vec{j} + 4xz^2\vec{k}$ and V is the region in the first octant bounded by the cylinder $y^2+z^2=9$ and bounded by x = 0 and x = 2.

10. Evalute $\iiint\limits_C \nabla.\vec{F}dv$, where $\vec{F} = 2xy\vec{i} + yz^2\vec{j} + xz\vec{k}$ bounded by x = 0, y = 0, z = 0, x = 2, y = 1, z = 3.

LESSON - 12
INTEGRAL THEOREMS

Learning Objectives

In this lesson, our aim is to study about,

- State Green's Theorem, Stoke's Theorem, and Gauss divergence Theorem
- To verify those theorems for some examples

Structure

12.1 Introduction

In this chapter, The following theorems in vector integrations are of importance from thereoretical and practical considerations :

1. Green's theorem in a plane
2. Stoke's theorem
3. Gauss Divergence theorem.

Green's theorem in a plane provided a relationsip between a line integral and a double integral.

Stoke theorem, which is a generalisation of Green's theorem, provies a relationship between a line integral and a surface integral.Infact, Green's theorem can be deduced as a particular case of stoke's theorem. Gauss divergence theorem provides a relatinoship between a surface integral and a volume integral.

We shall give the statements of the above theorems (without proof) below and apply them to solve problems.

12.3 Gauss Divergence Theorem

Statement :

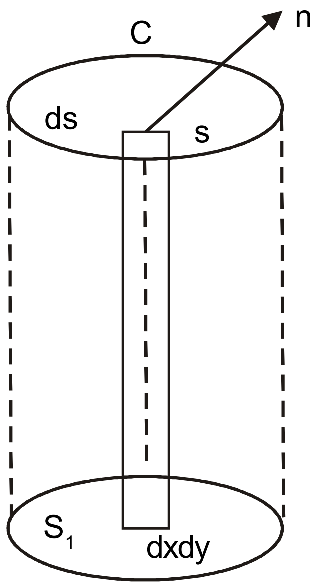
The surface integrals of the normal component of a vector point function $\vec{F}$ over a closed surface S is equal to volume integral of divergence of $\vec{F}$ taken over the volume V enclosed by a surface S.

(i.e.,) $\iint_S \vec{F} \, . \, \hat{n} ds = \iiint_V \nabla \, . \, \vec{F} dv$

Where $\vec{F}$ is the unit vector along the outward drawn normal at any point P on the surface S.

Proof

Let S be a closed Surface and any line drawn parallel to co-ordinate axes cut S in at most two points.



Let S_1 and S_2 be the Surfaced the top and bottom of S.

Let the equation of the top and bottom part of the Surfaces be represented by z = f(x,y) and z = ϕ(x,y) respectively. Let the projection of the Surface 'S' on the xy - plane by the region R.

If $\vec{F} = F_1\vec{i} + F_2\bar{j} + F_3\vec{k}$

Then

$$\iiint \frac{\partial F_3}{\partial z} dV = \iiint \frac{\partial F_3}{\partial z} \, dx\, dy\, dz$$

$$= \iint_R \left[\int_{z=\phi(x,y)}^{z=f(x,y)} \frac{\partial F_3}{\partial z} dz \right] dx\, dy$$

$$= \iint_R \left[F_3(x,y,z)\right]_{z=\phi(x,y)}^{z=f(X,y)} dx\, dy$$

$$= \iint_R [F_3\,(x,y,f) - F_3\,(x,y,\phi)\, dx\, dy$$

For the upper Surface S_2,

dx dy = cos γ_2, ds = $\vec{k}.\vec{n}_2$ ds since the normal vector $\vec{n}_2$ to S_2 makes an acute angle γ_2 with $\vec{k}$.

For the lower surface S_2, dx dy = -cosγ, ds_1 = -$\vec{k}.\vec{n}$ ds_1.

Since the normal vector $\vec{n}_1$ to S_1 makes obtuse angle γ, with $\vec{k}$.

Then

$$\iint_R \left[F_3(x,y,f)\right] dx\, dy = \iint_{S_2} F_3\, \vec{k}.\vec{n}_2\, ds_2$$

$$\iint_R F_3(x,y,\phi)\, dx\, dy = -\iint_{S_1} F_3\, \vec{k}.\vec{n}_1\, ds_1$$

$$\iint_R \left[F_3(x,y,f)\right] dx\, dy = \iint_R F_3\,(x,y,\phi)\, dx\, dy = \iint_{S_2} F_3\, \vec{k}.\vec{n}_2\, ds_2 +$$

$$\iint_{S_1} F_3\, \vec{k}.\vec{n}_1\, ds_1$$

$$= \iint_S F_3 \vec{k}.\vec{n} \; ds$$

$$\therefore \iiint_V \frac{\partial F_3}{\partial z}.dv = \iint_S F_3 \vec{k}.\vec{n} \; ds \qquad (1)$$

Similarly, projecting S on the other coordinates planes, we get,

$$\iiint_V \frac{\partial F_2}{\partial y}.dv = \iint_S F_2 \vec{j}.\vec{n} \; ds \qquad (2)$$

and

$$\iiint_V \frac{\partial F_1}{\partial x}.dv = \iint_S F_1 \vec{i}.\vec{n} \; ds \qquad (3)$$

Adding (1), (2) and (3) we get,

$$\iiint_V \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dv = \iint_S (F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}) \; \vec{n} \; ds$$

$$\therefore \iiint_V \nabla.\vec{F} dv = \iint_S \vec{F}.\vec{n} \; ds$$

Note

We have proved this theorem by considering a special situation that any line drawn parallel to the coordinates axes do not cut S in morethan 2 points. However the theorem can be proved to surface which are such that any line drawn parallel to co-ordinate planes cut 'S' in morethan 2 points.

Example 1

Verify Gauss's theorem for $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ taken over the region bounded by the planes $x = 0$, $x = a$, $y = 0$, $y = a$, $z = 0$, $z = a$.

Solution

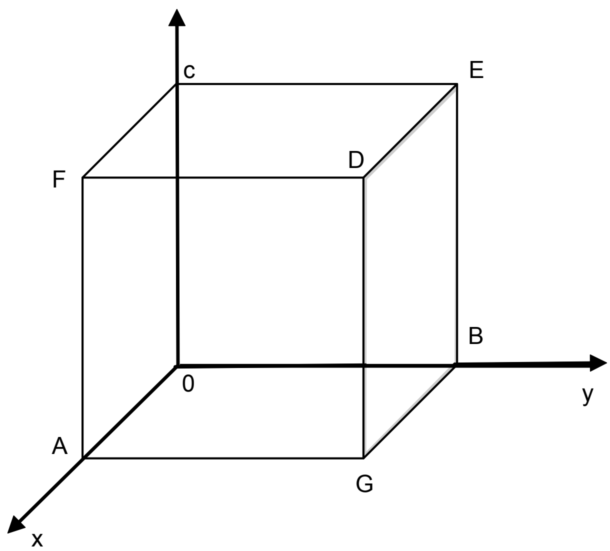
$\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$

Along the face AGDF, x = a, $\vec{n} = \vec{i}$

$$\therefore \iint_{x=0} \vec{F}.\vec{n}ds = \int_0^a \int_0^a (a\vec{i} + y\vec{j} + z\vec{k})\,\vec{i}\; dy\,dz$$

$$= \int_0^a \int_0^a a\; dy\,dz$$

$$= a^3$$



Along the face x = 0, $\vec{n} = -\vec{i}$

$$\iint_{x=0} \vec{F}.\vec{n}ds = \int_0^a \int_0^a (-0)\; dy\,dz = 0$$

Along the face y = a, $\vec{n} = \vec{j}$

$$\iint \vec{F}.\vec{n}ds = \int_0^a \int_0^a a\,dx\,dz = a^3$$

Along the face y = 0, $\vec{n} = -\vec{j}$

$$\iint \vec{F}.\vec{n}ds = 0$$

Along the face z = a, $\vec{n} = \vec{k}$

$$\iint \vec{F}.\vec{n}ds = \int_0^a \int_0^a a\,dx\,dy = a^3$$

Along the face z = 0, $\vec{n} = -\vec{k}$

$$\iint \vec{F}.\vec{n}ds = 0$$

$$\iint_S \vec{F}.\vec{n}ds = a^3+0+a^3+0+a^3+0 = 3a^3 \qquad (1)$$

$$\nabla.\vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{k}\frac{\partial}{\partial z}\right) . \left(x\,\vec{i} + y\,\vec{j} + z\,\vec{k}\right)$$

$$= 1+1+1$$

$$= 3$$

$$\therefore \iiint_V \nabla.\vec{F}.dv = \iiint_V 3.dv = 3v = 3a^3 \qquad (2)$$

From (1) and (2) we get,

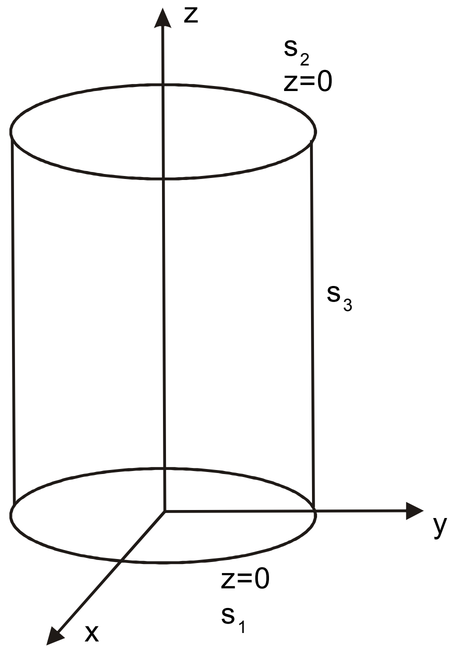
$$\iint_S \vec{F}.\vec{n}\, ds = \iiint \nabla.\vec{F} dv$$

Hence verified

Example 2

Verify divergence theorem for $\vec{F} = 4x\vec{i} - 2y^2\vec{j} + z^2\vec{k}$ taken over the region bounded by $x^2 + y^2 = 4$, $z = 0$ and $z = 3$.

Solution



$\vec{F} = 4x\vec{i} - 2y^2\vec{j} + Z^2\vec{k}$.

The total surface consists of the bottom surface S_1 (z=0), the top Surface S_2 (z=3) and curved Surface S_3 ($x^2+y^2=4$).

Along the Surface z = 0, $\vec{n} = -\vec{k}$

$$\iint_{S_1} \vec{F}.\vec{n}\, ds = \iint_{S_1} (4x\,\vec{i} - 2y^2\,\vec{j})(-\vec{k}) = 0$$

Along the Surface z = 3, $\vec{n} = \vec{k}$

$$\iint_{S_2} \vec{F}.\vec{n}\, ds = \iint_{S_1} (4x\,\vec{i} - 2y^2\,\vec{j} + 9\,\vec{k}).\vec{k}\, ds$$

$$= \iint_{S_2} 9ds = 9 \times 4\pi$$

$$= 36\pi$$

Along the Surface S_3,

$$\vec{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{\nabla(x^2 + y^2 - 4)}{|\nabla\phi|}$$

$$= \frac{2x\,\vec{i} + 2y\,\vec{j}}{\sqrt{4x^2 + 4y^2}}$$

$$= \frac{2(x\,\vec{i} + y\,\vec{j})}{4}$$

$$= \frac{x\,\vec{i} + y\,\vec{j}}{2}$$

$$\iint_{S_3} \vec{F}.\vec{n}\, ds = \iint_{S_3} (4x\,\vec{i} - 2y^2\,\vec{j} + z^2\,\vec{k}) . \left(\frac{x\,\vec{i} + y\,\vec{j}}{2}\right).dS$$

$$= \iint_{S_3} (2x^2 - y^3)\, dS$$

On S_3, x = 2cosθ ; y = 2 sinθ ; ds = 2 dθ dz

$$\iint_{S_3} \vec{F}.\vec{n}\, dS = \int_{\theta=0}^{2\pi} \int_{z=0}^{3} (8\cos^2\theta - 8\sin^3\theta)\, 2\, dz\, d\theta$$

$$= 16 \int_{0}^{2\pi} (\cos^2\theta - \sin^3\theta)\, [z]_0^3\, d\theta$$

$$= 48 \int_{0}^{2\pi} (\cos^2\theta - \sin^3\theta)\, d\theta$$

$$= 48\,\pi$$

$$\iint_{S_3} \vec{F}.\vec{n}\, dS = 0+36\pi + 48\pi = 84\pi \qquad (1)$$

$$\nabla.\vec{F} = \left(\vec{i}\,\frac{\partial}{\partial x} + \vec{j}\,\frac{\partial}{\partial y} + \vec{k}\,\frac{\partial}{\partial z}\right) . \left(4x\,\vec{i} - 2y^2\,\vec{j} + z^2\,\vec{k}\right)$$

$$= 4\text{-}4y\text{+}2z$$

$$\iiint_{v} \nabla.\vec{F} dv = \int_{x=-2}^{2} \int_{y=-\sqrt{4-x^2}}^{y=\sqrt{4\text{-}x^2}} \int_{z=0}^{3} (4 - 4y + 2z) dxdydx$$

$$= \int_{-2}^{2} \int_{-\sqrt{4}-x^2}^{\sqrt{4-x^2}} \left[4z - 4yz + z^2\right]_0^3 dx\, dy$$

$$= \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (21 - 12y) dydx$$

$$= \int_{-2}^{2} \left[21y - 6y^2\right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx$$

$$= \int_{-2}^{2} 42\sqrt{4-x^2}\,dx$$

$$= 42\left[\frac{x}{2}\sqrt{4-x^2} + \frac{4}{2}\sin^{-1}\left(\frac{x}{2}\right)\right]_{-2}^{2}$$

$$= 42\left[0 + \frac{2\pi}{2} - \left(0 - \frac{2\pi}{2}\right)\right]$$

$$= 42 \text{ x } 2\pi$$

$$\iiint_V \nabla.\vec{F}dv = 84\pi \qquad (2)$$

From (1) and (2) we get,

$$\iint_S \vec{F}.\vec{n}\,ds = \iiint_V \nabla.\vec{F}dv$$

Hence verified.

Example 3

Using divergence theorem, evaluate $\int_S \vec{F}.\vec{n}\,ds$ where $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ and S is the Surface of the cube bounded by the planes x=0, x =2, y =0, y =2, z = 0, z = 2.

Solution

By Gauss's divergence theorem

$$\iint_s \vec{F}.\hat{n}ds = \iiint_v \nabla.\vec{F}dv$$

$$\nabla.\vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{j}\frac{\partial}{\partial z}\right).\left(4xz\vec{i} - y^2\vec{j} + yz\vec{k}\right)$$

$$= 4z - 2y + y$$

$$= 4z - y$$

$$\iiint_V \nabla.\vec{F}dv = \int_0^2\int_0^2\int_0^2 (4z - y)\,dz\,dy\,dx$$

` $$= \int_0^2\int_0^2 \left[2z^2 - yz\right]_0^2 dy\,dz$$

` $$= \int_0^2\int_0^2 (8 - 2y)\,dy\,dz$$

$$= \int_0^2 \left(8y - y^2\right)_0^2\ dx$$

$$= \int_0^2 12dx = (12x)_0^2$$

$$\iint_S \vec{F}.\vec{n}\,ds = 24 \qquad \text{(by (1))}$$

Example 4

Find $\int_S \vec{F}.\vec{n}\,ds$ for the vector $\vec{F} = x\vec{i} - y^2\vec{j} + 2z\vec{k}$ over the sphere $x^2+y^2+(z-1)^2 = 1$.

Solution

By Gauss's divergence theorem

$$\iint_s \vec{F}.\hat{n}ds = \iiint_v \nabla.\vec{F}dv$$

$$\nabla . \vec{F} = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{j}\frac{\partial}{\partial z} \right) . (x\vec{i} - y^2\vec{j} + 2z\vec{k})$$

$$= 1 - 1 + 2$$

$$= 2$$

$$\iint_S \vec{F}.\vec{n}ds = \iiint_v 2.dv = 2v$$

$$= 2x\frac{4}{3}\pi.1^3 \quad \left(\therefore \text{volume } v = \frac{4}{3}\pi.r^3 \right)$$

$$= \frac{8\pi}{3}$$

Example 5

Evaluate $\iint \vec{F}.ds$; if F = $x^3.\vec{i} + y^3\vec{j} + z^3\vec{k}$ and S is the surface of sphere $x^2+y^2+z^2 = a^2$. Using Gauses divergence theorem.

Solution

By Gauss divergence theorem,

$$\iint_S \vec{F}.ds = \iiint_V \nabla\vec{F}dv$$

$$= \iiint_V (3x^2 + 3y^2 + 3z^2)dxdydz$$

Changing to spherical coordinates $x = r \sin\theta. \cos\phi$

$y = r \sin\theta \sin\phi$,

$z = r \cos\theta$

r various from 0 to a

θ various from 0 to π

ϕ various from 0 to 2π

$$\iint_s \vec{F}.\vec{n}ds = \int_{r=0}^{a}\int_{\theta=0}^{\pi}\int_{\phi=0}^{2\pi} (3r^2)(r^2)\sin\theta\, dr\, d\theta\, d\phi)$$

$$= 3\int_0^a\int_0^{\pi} r^4 \sin\theta\, [\phi]_0^{2\pi}\, d\theta\, .\, dr$$

$$= 3 \times 2\pi \int_0^a r^4 (-\cos\theta)_0^{\pi}\, dr$$

$$= 12\pi\left[\frac{r^5}{5}\right]_0^a$$

$$= \frac{12\pi a^5}{5}$$

12.4 Stoke's Theorem

Statement

Let S be on open surface bounded a closed curve C, and $\vec{F}$ be a vector point function defined on the surface S and $\vec{n}$ be unit outward drawn normal at any point P on S.

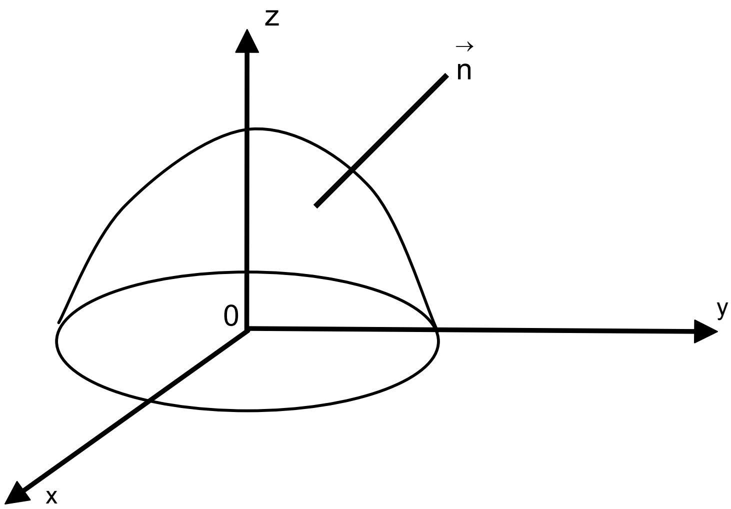
Then,

$$\int_c \vec{F}.d\vec{r} = \iint_S \nabla x \vec{F}.\vec{n}\, ds$$

Example 1

Verify Stoke's theorem for $\vec{F} = (2x-y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$ where S is the upper half surface of the sphere $x^2+y^2+z^2 = 1$ and C its boundary.

Solution



The boundary curve C is the circle $x^2+y^2=1$, $z = 0$. The parametric equations are $x = \cos\theta$, $y = \sin\theta$. θ various from 0 to 2π.

$$\vec{F} = (2x-y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$$

$$d\vec{r} = \vec{i}\,dx + \vec{j}\,dy + \vec{k}\,dz$$

$$\vec{F}\cdot d\vec{r} = (2x-y)dx - y\,z^2dy - y^2z\,dz$$

$$\int_c \vec{F}.d\vec{r} = \int_0^{2\pi} (2\cos\theta - \sin\theta)(-\sin\theta)\,d\theta$$

$$= \int_0^{2\pi} \left[(-\sin 2\theta) + \frac{1-\cos 2\theta}{2}\right] d\theta$$

$$= \left[\frac{\cos 2\theta}{2} + \frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_0^{2\pi}$$

$$= \left(\frac{1}{2} + \pi - 0\right) - \left(\frac{1}{2} + 0 - 0\right)$$

$$= \pi \qquad (1)$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x - y & -yz^2 & -y^2 z \end{vmatrix}$$

$$= \vec{i}\,(-2yz+2yz) - \vec{j}\,(0,0) + \vec{k}\,(1)$$

$$= \vec{k}$$

Now,

$$\iint_S \nabla \times \vec{F} . \vec{n} ds = \iint_S \vec{k} . \vec{n}\, ds$$

$$= \iint_S \vec{k} . \vec{n}\, ds \; \frac{dxdy}{\vec{k} . \vec{n}}$$

Where R is the projection of S in the xy - plane

$$= \iint_S dxdy$$

$$= \pi \qquad (2)$$

From (1) and (2) we get,

$$\int_c \vec{F} . d\vec{r} = \iint_S \nabla \times \vec{F} . \vec{n} ds$$

Hence Stoke's theorem is verified

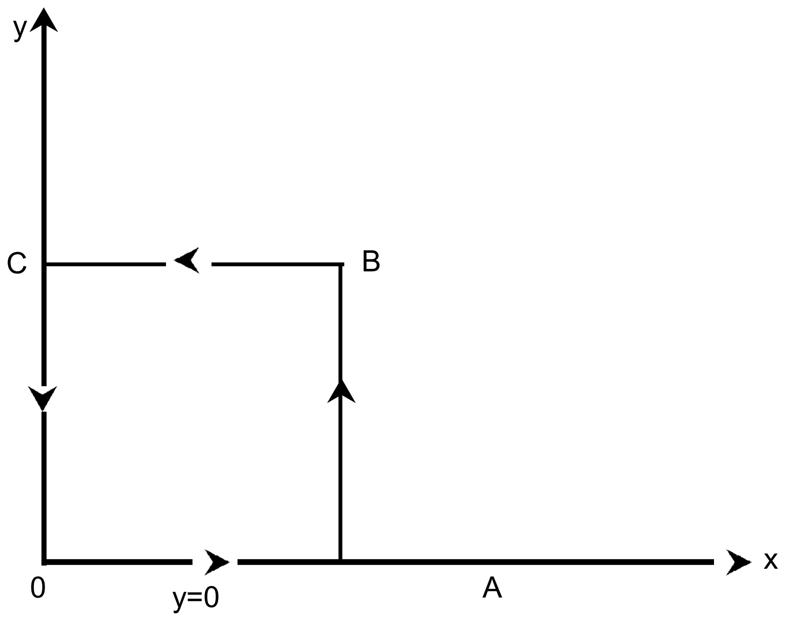
Example 2

Verify Stoke's Theorem for $\vec{F} = x^2\vec{i} + xy\vec{j}$, taken the square in the xy - plane whose sides are x = 0, x = a, y = 0, y = a.

Solution

$$\vec{F} = x^2\vec{i} + xy\vec{j}$$

$$\int_c \vec{F}\cdot d\vec{r} = \int_{OA}\vec{F}.d\vec{r} + \int_{AB}\vec{F}.d\vec{r} + \int_{BC}\vec{F}.d\vec{r} + \int_{CO}\vec{F}.d\vec{r}$$



Along OA, y = 0, dy = 0

$$= \int_{OA}\vec{F}.d\vec{r}$$

$$= \int_0^a x^2dx$$

$$= \frac{a^3}{3}$$

Along AB, x = a, dx = 0

$$\int_{AB} \vec{F}.d\vec{r} = \int_0^a aydy = \frac{a^3}{2}$$

Along BC, y = a, dy = 0

$$\int_{BC} \vec{F}.d\vec{r} = \int_a^0 (x^2 \vec{i} + ax).dx\ \vec{i}$$

$$= \int_a^0 x^2 dx$$

$$= -\frac{a^3}{3}$$

Along CO, x = 0, dx = 0

$$\int_{CO} \vec{F}.d\vec{r} = 0$$

$$\int_C \vec{F}.d\vec{r} = \frac{a^3}{3} + \frac{a^3}{2} - \frac{a^3}{3} + 0 = \frac{a^3}{2} \qquad (1)$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & xy & 0 \end{vmatrix}$$

$$= \vec{i}(0) - \vec{j}(0) + \vec{k}(y)$$

$$= \vec{k}y;\ \vec{n} = \vec{k}$$

$$\iint_S \nabla x \vec{F}.\vec{n} = \int_0^a \int_0^a y\,\vec{k}.\vec{k}\,dxdy$$

$$= \int_0^a \int_0^a ydxdy$$

$$= \int_0^a \left[\frac{y^2}{2}\right]_0^a dx$$

$$= \frac{a^2}{2}[x]_0^a = \frac{a^3}{2} \qquad (2)$$

From (1) and (2) we get

$$\int_C \vec{F}.d\vec{r} = \iint_S \nabla x \vec{F}.\vec{n}\,ds$$

Stoke's theorem is verified

Example 3

Verify Stoke's Theorem when $\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$ and Surface S is the part of the sphere $x^2 + y^2 + z^2 = 1$ along xy - plane

Solution

$$\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$$

$$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$F.d\vec{r} = ydx + zdy + xdz$$

C is the curve $x^2+y^2=1$, $z = 0$; $dz = 0$

$$\int_c \vec{F}.d\vec{r} = \int_c ydx \qquad x = \cos t \Rightarrow dx = -\sin t\, dt \,;\ y = \sin t$$

$$= \int_0^{2\pi} -\sin^2 dt$$

$$= -\int_0^{2\pi} \left(\frac{1-\cos 2t}{2}\right)dt$$

$$\int_c \vec{F}.d\vec{r} = \frac{-1}{2}\left[t - \frac{\sin 2t}{2}\right]_0^{2\pi} = -\pi \qquad (1)$$

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix}$$

$$= \vec{i}(-1) - \vec{j}(1) + \vec{k}(-1)$$

$$= -\left(\vec{i} + \vec{j} + \vec{k}\right)$$

$$\therefore \vec{n} = \frac{x\vec{i} + y\vec{j} + z\vec{k}}{\sqrt{x^2+y^2+z^2}} = x\vec{i} + y\vec{j} + z\vec{k}$$

$$\iint_S \nabla x \vec{F}.\vec{n}\,dx = -\iint_S (x+y+z)dx$$

Transforming to spherical polar coordinates

x = rsinθcosϕ ; y = rsinθ. sinϕ ; z = r sinθ and r = 1

ds = r^2 sinθ. dθ. dϕ

$$\iint_S \nabla x \vec{F}.\vec{n}\,dx = -\int_{\theta=0}^{\pi/2}\int_{\phi=0}^{2\pi} (\sin\theta.\cos\phi + \sin\theta.\sin\phi + \cos\theta).\sin\theta\,d\theta.\,d\phi$$

$$= -0+0-\int_0^{\pi/2} (\cos\theta.\sin\theta\,d\theta).\int_0^{2\pi} d\phi$$

$$= -\left[\frac{\sin^2\theta}{2}\right]_0^{\pi/2}.2\pi$$

$$= -\pi \qquad (2)$$

From (1) and (2)

$$\int_c \vec{F}.d\vec{r} = \iint_S \nabla x \vec{F}.\vec{n}\,ds$$

Hence verified

Example 4

Prove that $\int_c \vec{r}.d\vec{r} = 0$

Solution

By Stoke's theorem, $\int_c \vec{F}.d\vec{r} = \iint_S \nabla x \vec{F}.\vec{n}\, ds$

Take $\vec{F} = \vec{r}$, but r = $x\vec{i} + y\vec{j} + z\vec{k}$

Then,

$$\nabla x\vec{F} = \nabla x\vec{r}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & z & x \end{vmatrix} = 0$$

$$\therefore \int_C \vec{r}.d\vec{r} = 0$$

Example 5

Show that $\int_c \vec{A} \times \vec{r}.d\vec{r} = 2\iint_S \vec{n}.\vec{A}\, ds$, where A is a constant vector.

Solution

By Stoke's theorem

$$\int_c \vec{F}.d\vec{r} = \iint_S \nabla x \vec{F}.\vec{n}\, ds$$

Take $\vec{F} = \vec{A} \times \vec{r}$,

Then, RHS

$$= \iint_S \nabla x(A x \vec{r})\vec{n} dS$$

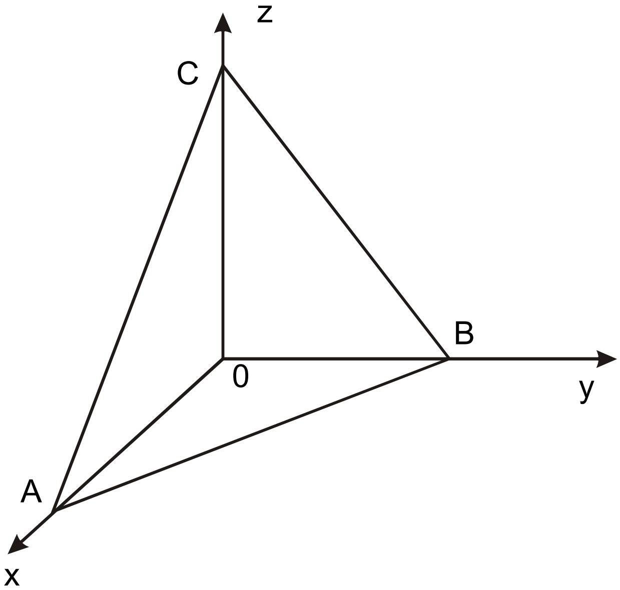
$$\left(\text{since } \nabla x(\vec{A} x \vec{r}) = 2\vec{A}\right)$$

$$= 2\iint_S \vec{n}.\vec{A}.dS$$

Hence Proved.

Example 6

Verified Stoke's Theorem for the fumnction $\vec{F} = (x+y)\vec{i} + (2x-z)\vec{j} + (y+z)\vec{k}$, taken over the triangle ABC cut from the plane 3x+2y+z=6 by the coordinate plane.



Solution

$\vec{F} = (x+y)\vec{i} + (2x-z)\vec{j} + (y+z)\vec{k}$

$$\int_c \vec{F}.d\vec{r} = \int_C \left[[(x+y)\, dx + (2x-z)\, dy + (y+z)\, dz\right]$$

Along AB, z = 0 => dz = 0

x varies from 0 to 2,

y varies from 0 to 3

$$\int_{AB} \vec{F}.d\vec{r} = \int_{C} [(x+y)\,dx + 2xdy]$$

$$= \int_{2}^{0} \left[\left(x + \frac{6-3x}{2}\right)dx + 2x\left(\frac{-3}{2}dx\right)\right]$$

$$= \int_{2}^{0} \left(\frac{6-7x}{2}\right) dx$$

$$= \frac{1}{2}\left[6x - \frac{7x^2}{2}\right]_2^0$$

$$= \frac{1}{2}\,[-12+14] = 1$$

Along BC,

$$\int_{BC} \vec{F}.d\vec{r} = \int_{BC} [-zdy + (y+z)\,dz]$$

$$= \int_{2}^{0} -(6-2y)\,dy + (y+6-2y)\,(-2dy)$$

$$= \int_{2}^{0} (-6+2y - 2y -12+4y)\,dy$$

$$= \int_{2}^{0} (-18+4y)\,dy$$

$$= \left[-18y + 2y^2\right]_3^0$$

$$\int_{BC} \vec{F}.d\vec{r} = 54-18 = 36$$

$$\int_{CA} \vec{F}.d\vec{r} = \int_{CA} (xdx + zdz)$$

$$= \int_{6}^{0} \left[\frac{6-z}{3}\left(-\frac{dz}{3}\right) + zdz\right]$$

$$= \int_{6}^{0} \left[\frac{-(6-z)}{9} + z\right] dz$$

$$= \int_{6}^{0} \frac{-6-10z}{9} dz$$

$$= \frac{1}{9} \left[-6z + 5z^2\right]_6^0$$

$$= \frac{1}{9} [36-180] = -16$$

$$\int_{CA} \vec{F}.d\vec{r} = 1+36-16=21 \qquad (1)$$

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y & 2x-z & y+z \end{vmatrix}$$

$$= \vec{i}\,(1+1) - \vec{j}\,(0-0) + \vec{k}\,(2-1)$$

$$= 2\vec{i} + k$$

$$\nabla \phi = \left(\vec{i}\frac{\partial}{\partial x} + \vec{j}\frac{\partial}{\partial y} + \vec{j}\frac{\partial}{\partial z} \right) (3x+2y+z-6)$$

$$= 3\vec{i} + 2\vec{j} + \vec{k}$$

$$\vec{n} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{3\vec{i}+2\vec{j}+\vec{k}}{\sqrt{14}}$$

$$\iint_S \nabla x \vec{F}.\vec{n}\, ds = \iint_S \left(2\vec{i}+\vec{k}\right) . \frac{\left(3\vec{i}+2\vec{j}+\vec{k}\right)}{\sqrt{14}} ds$$

$$= \iint_R \frac{7}{\sqrt{14}} . \frac{dxdy}{\vec{n}.\vec{k}},$$ Where R is the projective of S on xy - plane

$$= 7\iint_R dx\,dy \qquad \left(\text{Since } \vec{n}.\vec{k} = \frac{1}{\sqrt{14}}\right)$$

$$= 7\iint_R dx\,dy$$

$$= 7 \text{ x } \frac{1}{2} \text{ x } 2 \text{ x } 3$$

$$= 21 \qquad (2)$$

From (1) and (2) we get,

$$\int_C \vec{F}.d\vec{r} = \iint_S \nabla x \vec{F}.\vec{n}\, ds$$

∴ The Stoke's theorem is verified

Example 7

Use Stoke’s theorem to evaluate $\iint_S \nabla x \vec{F}.\vec{n}\, ds$ where S is the upper half of the hemi-sphere of radius a, centre at the origin and $\vec{F} = 2y\vec{i} - x\vec{j} + z\vec{k}$.

Solution

By Stoke's theorem

$$\iint_S \nabla x \vec{F} . \vec{n}\, ds = \int_C \vec{F}.d\vec{r}$$

$$\vec{F} = 2y\vec{i} - x\vec{j} + z\vec{k}$$

$$d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$\int_C \vec{F}.d\vec{r} = \int_6^0 (2ydx\text{-}xdy + zdz)$$

C is the circle, $x^2+y^2 = a^2$ in the xy plane.

∴ $x = a\cos\theta \Rightarrow dx = -a\sin\theta\, d\theta$

$y = a\sin\theta \Rightarrow dy = a\cos\theta\, d\theta$

$$\int_C \vec{F}.d\vec{r} = \int_6^{2\pi} [2a \sin\theta\, (-a\sin\theta) - a \cos\theta . a\cos\theta] d\theta$$

$$= -\int_0^{2\pi} a^2 (\cos^2\theta + 2\sin^2\theta)\, d\theta$$

$$= -\int_0^{2\pi} \left(1 + \frac{1-\cos 2\theta}{2}\right) d\theta$$

$$= -\frac{a^2}{2} \int_0^{2\pi} (3-\cos 2\theta)\, d\theta$$

$$= -\frac{a^2}{2} \left[3\theta - \frac{\sin\theta}{2}\right]_0^{2\pi}$$

$$= -\frac{a^2}{2}\ (6\pi)$$

$$\iint_S \nabla x \vec{F}.\vec{n}\, ds = -3\pi a^2$$

12.5 Green's theorem in plane

Ler R be a closed curve C. Let M and N be continuous functions of x and y having continuous partial derivatives in R. Then

$$\oint_c M dx + N dy = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\ dy$$

Where C is traversed in the positive direction (Anticlock wise direction)

Green's theorem in plane (vector form)

$$\text{Mdx + Ndy} = \left(M\vec{i} + N\vec{j}\right).\left(dx\vec{i} + dy\vec{j}\right)$$

Where

$$\vec{F} = M\vec{i} + N\vec{j}$$

$$\vec{r} = x\vec{i} + y\vec{j}$$

$$d\vec{r} = dx\vec{i} + dy\vec{j}$$

Also,

$$\nabla x\vec{F} = \begin{vmatrix} \vec{i} & \overline{j} & \vec{u} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & 0 \end{vmatrix}$$

$$= -\frac{\partial N}{\partial z}\vec{i} + \frac{\partial M}{\partial z}\vec{j} + \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)\vec{k}$$

$$\therefore \ \nabla \text{ x } \vec{F} \ . \ \vec{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

The Green's Theorem states.

$$\oint \left(Mdx + Ndy\right) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx \ dy$$

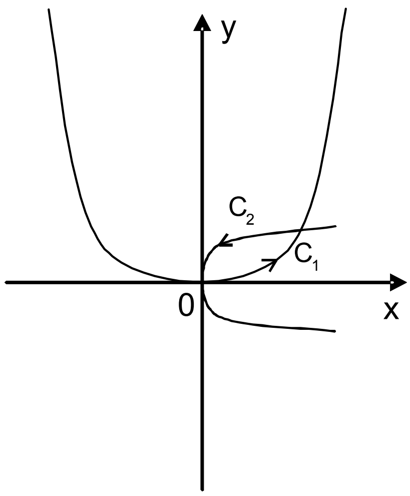
$$\therefore \ \oint \vec{F}.d\bar{r} = \iint_R \nabla x\vec{F}.\vec{k}dR$$

Where dR = dxdy

Example 1

Verify Green's theorem in the plane for $\int_c 3x^2 - 8y^2 + \left(4y - 6xy\right) dy$ where C is the boundary of the region defined by y = $\sqrt{x}$ and y = x^2.

Solution



Let y = $\sqrt{x}$ and y = x^2 are the parabola's whose points inntersection are (0,0) and (1,1).

$$\oint \left(Mdx + Ndy\right) = \int_{C_1} Mdx + Ndy + \int_{C_2} Mdx + Ndy \qquad (1)$$

Here $M = 3x^2 - 8y^2$; $N = 4y - 6xy$

Where C_1 is the curve $y = x^2$

$dy = 2x\,dx$ and x varies form 0 to 1 and C_2 is the curve $y^2 = x$

$2y\,dy = dx$ and y varies from 1 to 0

$$\int_{C_1} Mdx + Ndy = \int_0^1 (3x^2-8x^4)\,dx + (4x^2-6x^3)\,2x\,dx$$

$$= \int_0^1 (3x^2+8x^3 - 20x^4)\,dx$$

$$= 1+2-4$$

$$= -1$$

$$\int_{C_2} Mdx + Ndy = \int_1^0 (3y^4-8y^2)\,2y\,dy + (4y-6y^3)\,dy$$

$$= \int_1^0 (6y^5-22y^3 + 4y)\,dy$$

$$= \frac{5}{2}$$

$\therefore$ (1) => $\oint (Mdx + Ndy) = -1 + \frac{5}{2} = \frac{3}{2}$ (3)

Here $M = 3x^2-8y^2$; $N = 4y - 6xy$

$$\frac{\partial M}{\partial y} = -16y \quad ; \quad \frac{\partial N}{\partial x} = -6y$$

$$\therefore \quad \frac{\partial N}{\partial y} - \frac{\partial M}{\partial y} = 10y$$

$$\iint_R \left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial y}\right) dx\,dy = \int_{x=0}^{1} \int_{y=x^2}^{y=\sqrt{x}} 10y\,dy\,dx$$

$$= \int_0^1 \left(5y^2\right)_{x^2}^{\sqrt{x}} dx$$

$$= 5\int_0^1 (x - x^4)\,dx$$

$$= 5\left[\frac{x^2}{2} - \frac{x^5}{2}\right]_0^1$$

$$= \frac{3}{2} \qquad (3)$$

From (2) and (3) we get,

$$\therefore \oint (M dx + N dy) = \iint_R \left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial y}\right) dx\,dy$$

Hence Green's theorem is verified.

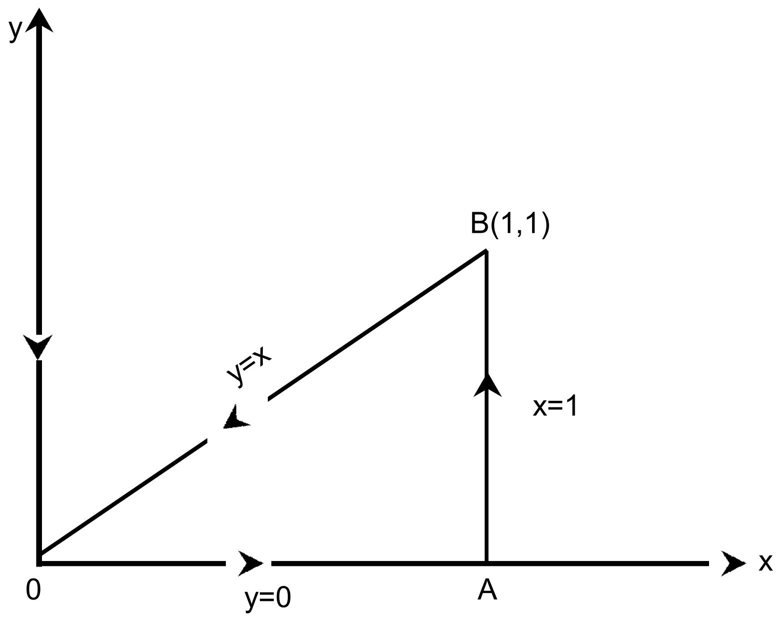
Example 2

Compute $\int_C (xy - x^2)\,dx + x^2 y\,dy$ over the triangle bounded by the lines $y = 0$, $x = 1$, $y = x$ and verify Green's theorem.

Solution

Here $M = xy - x^2$; $N = x^2y$

$$\frac{\partial M}{\partial y} = x \quad ; \quad \frac{\partial N}{\partial y} = 2xy$$



$$\therefore \oint (Mdx + Ndy) = \int_{OA} + \int_{AB} + \int_{BO} \qquad (1)$$

Along OA, y = 0, => dy = 0

$$\int_{OA} Mdx + Ndy = \int_0^1 -x^2\, dx = -\frac{1}{3}$$

Along AB, x = 1, => dx = 0

$$\int_{AB} Mdx + Ndy = \int_0^1 y\, dy = \frac{1}{2}$$

Along BO, y = x, => dy = dx

$$\int_{BO} Mdx + Ndy = \int_0^0 (0+x^3dx) = -\frac{1}{4}$$

$$\therefore \int_C Mdx + Ndy = -\frac{1}{3}+\frac{1}{2}-\frac{1}{4}=-\frac{1}{12} \qquad (2)$$

$$\iint_R \left(\frac{\partial N}{\partial y}-\frac{\partial M}{\partial y}\right) dx\, dy = \int_0^1 \int_0^x (2xy - x)\, dy\, dx$$

$$= \int_0^1 \left(xy^2 - xy\right)_0^x dx$$

$$= \int_0^1 (x^3 - x^2)\, dx$$

$$= \frac{1}{4}-\frac{1}{3}$$

$$= -\frac{1}{12}$$

From (1) and (2) we get,

$$\oint (Mdx + Ndy) = \iint_R \left(\frac{\partial N}{\partial y}-\frac{\partial M}{\partial y}\right) dx\, dy$$

$\therefore$ Hence Green's theorem is verified

Example 3

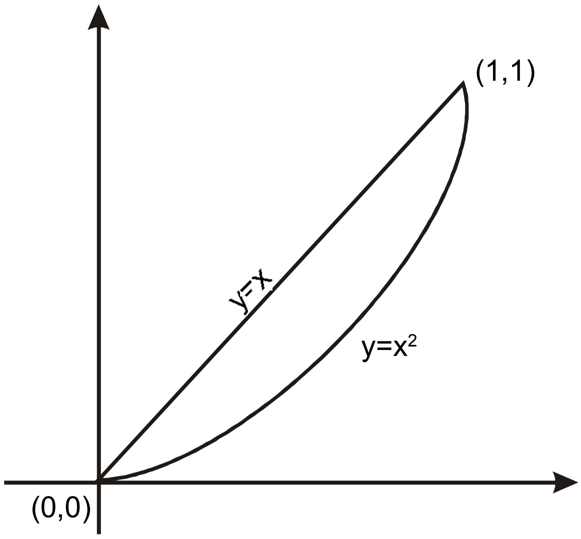
Verify Green's theorem in the plane for $\oint_C (xy+y^2)\, dx + x^2\, dy$, where C is the closed curve of the region bounded by y = xand $y = x^2$.

Solution

Green's theorem in a plane states that, $\oint_C (\mu dx+ndy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial x}\right) dxdy$

Here $M = xy+y^2$; $N = x^2$

$\frac{\partial M}{\partial y} = x+2y$; $\frac{\partial N}{\partial y} = 2x$



put $y = x$ and $y = x^2$ intersect at (0,0) and (1,1)

Along $y = x^2$, $dy = zxdx$

$$\oint_{C_1}(Mdx + Ndy) = \int_0^1 (x^3+x^4)\, dx + x^2.2xdy$$

$$= \int_0^1 (3x^3+x^4)\, dx$$

$$= \left[3.\frac{x^4}{4} + \frac{x^5}{5}\right]_0^1$$

$$= \frac{3}{4} + \frac{1}{5}$$

$$= \frac{19}{20}$$

Along y = x => dy = dx

$$\oint_{C_2} (Mdx + Ndy) = \int_0^1 3x^2 dx$$

$$= 3\left(\frac{x^3}{3}\right)_0^1$$

$$= -1$$

$$\therefore \int_C Mdx + Ndy = \frac{19}{20} - 1 = \frac{-1}{20}$$

$$\iint_R \left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial y}\right) dx\,dy = \iint_R (2x-x-2y)\,dx\,dy$$

$$= \int_{x=0}^{1} \int_{y=x^2}^{x} (x-2y)\,dx\,dy$$

$$= \int_0^1 \left[xy - y^2\right]_{x^2}^{x} dx$$

$$= \left[\frac{x^5}{5} - \frac{x^4}{4}\right]_0^1$$

$$= \frac{1}{5} - \frac{1}{4}$$

$$= \frac{-1}{20}$$

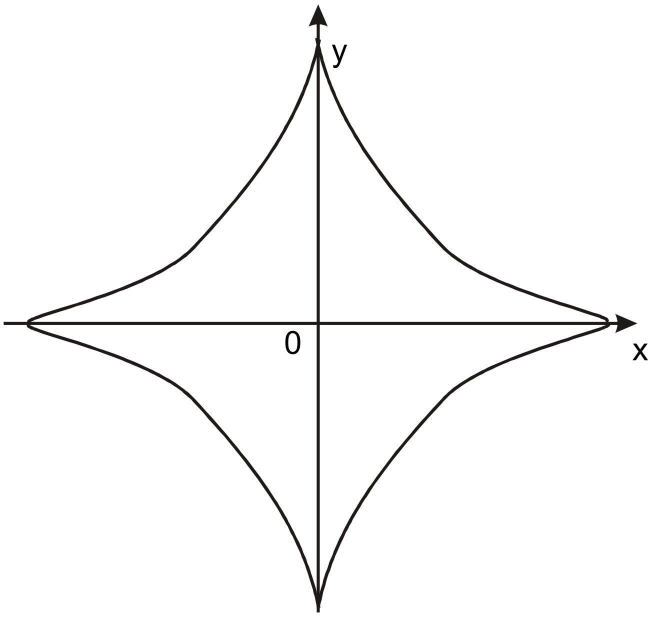
From (1) and (2) we get

So that the the theorem is verified

Example 5

Find the area of the curve $x^{2/3} + y^{2/3} = a^{2/3}$ using Green's theorem.

Solution



The Parametric equation of the curve are

$x = a\cos^3\theta$; $y = a\sin^3\theta$

$dx = -3a\cos^2\theta.\sin\theta\, d\theta$; $dy = 3a\sin^2\theta.\cos\theta\, d\theta$

By Green's theorem

$$\text{Area} = \frac{1}{2}\oint (xdy - ydx) = \frac{1}{2}\oint [3a\cos^3\theta(3a\sin^2\theta\cos\theta) - 3a\sin^3\theta(-3a\cos^2\theta.\sin\theta]\, d\theta$$

$$= \frac{1}{2} 9a^2.4 \int_0^{\pi/2} (\cos^4\theta\sin^2\theta + \sin^4\theta\cos^2\theta)\, d\theta$$

$$= 4 \times \frac{9a^2}{2} \int_0^{\pi/2} \cos^2\theta \sin^2\theta \, d\theta$$

$$= 18a^2 \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= \frac{9\pi a^2}{8}$$

Example 4

Show that the area bounded by a closed curve C is given by $\frac{1}{2}\oint$ (xdy-ydx).

Solution

Green's theorem states $\oint$ (Mdx + Ndy) = $\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right)$ dx dy.

Take m = -y ; N = x

Then,

$$\oint_C (-ydx + xdy) = \iint_R (1+1)\, dx\, dy$$

$$= \iint_R 2dx\, dy = 2A$$

$$A = \frac{1}{2}\oint (xdy - ydx)$$

Note :

Take $x = r\cos\theta$; $y = r\sin\theta$

$dx = -r\sin\theta \, d\theta$; $dy = r\cos\theta \, d\theta$

Area in polar coordinates = $\frac{1}{2}\oint_C r^2 (\cos^2\theta + \sin^2\theta)\, d\theta = \frac{1}{2} r^2\, d\theta$.

Example 6

Find the area of the ellipse $x = a\cos\theta$; $y = b\sin\theta$

Solution

$$\text{Area} = \frac{1}{2}\oint_C (xdy - ydx)$$

$$= \frac{1}{2}\int_0^{2\pi} [a\cos\theta\,(b\cos\theta) - b\sin\theta\,(-a\cos\theta)]\, d\theta$$

$$= \frac{1}{2}\, ab \int_0^{2\pi} (\cos^2 + \sin^2\theta)\, d\theta$$

$$= \frac{1}{2} ab\, .\, 2\pi$$

$$= \pi\, ab.$$

Example 7

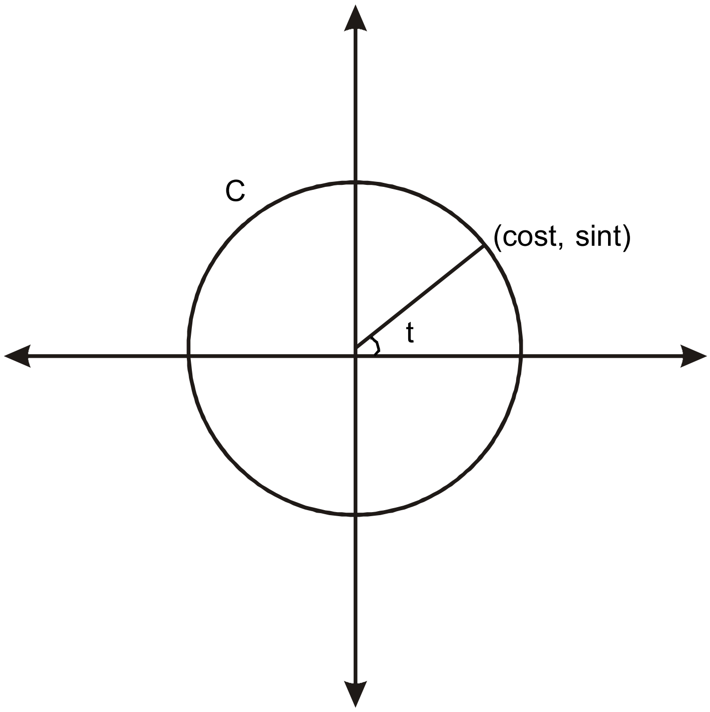
Verify Green's theorem for $\int_C (x-2y)dx + xdy$ where C is the circle $x^2+y^2 = 1$.

Solution

The parametric equation of the circle is $x = \cos t$, $y = \sin t$

Where t varies 0 to 2π.

$$\int_C Mdx + Ndy = \int_C (x-2y)dx + xdy$$



$$= \int_0^{2\pi} (\cos t - 2\sin t)(-\sin t)\, dt + \cos t (\cos t)\, dt$$

$$= \int_0^{2\pi} (-\sin t.\cos t + 2\sin^2 t + \cos^2 t) dt$$

$$= \int_0^{2\pi} \left(\frac{-1}{2}\sin 2t + \frac{3}{2} - \frac{1}{2}\cos 2t\right) dt$$

$$= \frac{3}{2}(2\pi)$$

$$= 3\pi \qquad (1)$$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy = \iint_R (1+2)\, dx\, dy$$

$$= \iint_R 3dx\, dy$$

$$= 3\iint_R dx\,dy$$

= 3 (area of the circle)

$$= 3\pi \qquad (2)$$

∴ From (1) & (2) the Green's theorem is verified

Check Your Progress

1. Verify Green's Theorem for the Integral $\int_c (x^2 - 2xy)dx + (x^2y + 3)dy$ along the boundary of the region defined by $y^2 = 8x$ and $x = 2$.

   ....

2. Verify Green's Theorem for the integral $\int_c (x - y)dx + (x + y)dy$ taken around the boundary of the area in the first quadrant the curves $y = x^2$ and $y^2 = x$

   ....

3. Evaluate $\int_c (x^2 + y^2)dx - 2xydy$ where C is the rectangle bounded by $x = 0$, $y = 0$ $x = a$, $y = b$ using Green's Theorem.

   ....

12.6 Recap

1. Gauss Divergence's Theorem.

$$\iint_S \vec{F}.\vec{n}\,ds = \iiint_V \nabla.\vec{F}\,dv$$

2. Stoke's theorem

$$\int_C \vec{F}.d\vec{r} = \iint_S \nabla x \vec{F}.\vec{n}\, ds$$

3. Green's Theorem

$$\oint_C Mdx + Ndy = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\, dy$$

12.7 Exercise

1. If A = $(3x^2+6y)\vec{i}$ - $14yz\ \vec{j}$ $+20xz\vec{k}$, evaluate $\int \vec{F}.d\vec{r}$ along the curve C, x=t, $y=t^2$, $x = t^3$ from (0,0,0) to (1,1,1).

2. Verify the Gauss divergence theorem for $\vec{F} = x^2\vec{i} + y^2\vec{j} + z^2\vec{k}$ taken over the region bounded by the planes x = 0, x = 1, y = 0, y =1, z = 0, z = 1.

3. Verify Gauss divergence theorem for $\vec{F}$ = $(x^3-yz)\vec{i} - zx^2y\vec{j} + z\vec{k}$ over the cube x = 0, x = a ; y = 0, y = a ; z = 0, z = a.

4. Evaluate $\iint_S (y^2z^2\vec{i} + z^2x^2\vec{j} + x^2y^2\vec{k}).\ d\vec{s}$ over the Surface $x^2+y^2+z^2 = 9$ in the xoy - plane.

5. Prove that $\iint_S \left[x(y-z)\vec{i} + y(z-x)\vec{j} + z(x-y)\vec{k}\right] ds = 0$ where S is any closed Surface.

6. Evaluate $\iint_S (4x\vec{i} + 3y\vec{j} - 2z\vec{k}).ds = 0$ where S is the closed region bounded by x = 0, y = 0; z = 0, 2x+2y+z=4.

7. Verify Stoke's theorem's for $\vec{F}$ = $(x^2+y^2)\vec{i} - 2xy\vec{j}$ + taken round the rectangle R bounded by the lines x = 0, x = a ; y = 0, y = b.

8. Verify Stoke's theorem's for $\vec{F} = x^2 \vec{i} + xy \vec{j}$ the Surface of a Square lemina bounded by $x = -1$, $x = 1$; $y = -1$, $y = 1$.

9. Verify Stoke's theorem, for $\vec{F} = y\vec{i} + 2x\vec{j} + z\vec{k}$ and C is the circle $x^2+y^2=1$ in the xy - plane and S is the plane area bounded by C.

10. We Stoke's theorem to evaluate $\iint_S \text{curl } \vec{F}.\vec{n}\, ds$

11. Evaluate $\int$ $(yzdx+2xdy + xy\, dz)$ where C is the curve $x^2+y^2=1$, $z = y^2$.

12. Show that $\int_C \left(\vec{a} \times \vec{r}\right).d\vec{r} = 2\vec{a} \iint_S ds$ a being a constant vector.

13. Evaluate $\iint_S \nabla \times \vec{F}.\vec{n}$ ds, where S is the Surface $x^2+y^2+z^2 = a^2$ above xoy - plane and $\vec{F} = y\vec{i} + (x-2xz)\vec{j} - xy\vec{k}$.

14. Use Stoke's theorem, to evaluate $\int_C (xydx+xy^2dy)$ where C is the square in xy - plane with the vertices (1,0), (-1,0), (0,1) and (0,-1).

15. Verify Green's theorem in the plane $\int_C \left[(x^2-2xy)\, dx + (x^3y+1)\, dy\right]$ where C is the boundary given by $y^2=8x$ and $x = 2$.

16. Verify Green's thorem, for $\int_C (3x^2-8y^3)\, dx + (4y - 6xy)\, dy$ where C is the bounary of the region defined by $x = 0$, $y = 0$, $x + y = 1$.

17. Evaluate by Green's theorem $\int_C (x^2+y^2)\,dx - 2x\,dx\,dy$, where C is the rectangle bounded by y =0, x = 0 ; x = b, x = 0.

18. Find the area of the Parabola $y^2=4x$ and $x^2=4y$.

19. Verify Green's theorem, in the plane for $\oint (xy + y^2)d + x^2dy$ where C is the chord curve of the region bounded by y = x and $y = x^2$.

20. Verify Green's theorem for $\int \left[(x^2+y)\,dx - xy^2\,dy\right]$ taken round the boundary of the square with verifies (0,0), (1,0), (1,1) and (0,1).

MODEL QUESTIONS

BCA - COMPUTER APPLICATIONS

FIRST YEAR - SECOND SEMESTER

Allied Paper - II

MATHEMATICS - II

Time : 3 Hrs. Max marks 75

PART A - (10 x 2 = 20 marks)

Answer any TEN questions

1. Integrate x^4e^x with respect to "x".

2. Evaluate : $\int_0^{\pi/2} \cos^8 x dx$.

3. Find the value of $\int_0^{\frac{\pi}{2}} \sin^6 x \cos^5 x dx$.

4. Solve : $(D^3-3D^2-6D+8)y = 0$

5. Form the PDE by eliminating a and b from $z = xy + y\sqrt{x^2+a^2+b^2}$

6. Form the PDE by eliminating f from $z = f(x,y)$.

7. Find the value of $L\left(e^{-at}\right)$.

8. What is the value of $L^{-1}[f(s+a)]$?

9. If $\phi(x,y,z) = x^2y-2y^2z^3$ and find $\nabla\phi$ at the point (1, -1, 2).

10. Compute curl $\vec{F}$ at the point (1,-1,1) for the vector function $\vec{F} = x^2z\vec{i} - 2y^3z^2\vec{j} + xy^2z\vec{k}$.

11. If $\vec{f}(t) = \left(3t^2-1\right)\vec{i} + (2-6t)\vec{j} - 4t\vec{k}$ then find $\int_0^1 \vec{f}(t)dt$.

12. State Green's theorem.

PART B - (5 x 5 = 25 marks)

Answer any FIVE questions

13. Find the Fourier's coefficient a_0 and b_n for the function f(x) = $\left(\frac{\pi - x}{2}\right)^2$ in $(0,2\pi)$.

14. Find the Fourier sine series for the function f(x) = x^2 in $(0,\pi)$.

15. Solve $p^2+q^2=1$

16. Solve : $(y+z)p + (z+x)q = x+y$.

17. Find the value of

    (a) $L\left[te^{-3t}\right]$, (b) L[sin3tcos2t]

18. Find ϕ if $\nabla \phi$ = $\left(6xy + z^3\right)\vec{i} + \left(3x^2 - z\right)\vec{j} + \left(3xz^2 - y\right)\vec{k}$.

19. If $\vec{F} = 3xy\vec{i} - y^2\vec{j}$, evaluate $\int_C \vec{F}.d\vec{r}$ where C is the curve on the xy plane $y = 2x^2$ from (0,0) to (1,2).

PART C - (3 x 10 = 30 marks)

Answer any THREE questions

20. Find the reduction formula for I_n = $\int_0^{\pi/2} \cos^n x dx$.

21. Solve : $(D^2+2D+4)y=e^{-2x}. \cos 3x$

22. Solve : $y''+4y'+5y=4e^{3t}$; and given that y(0) = 2; y'(0) = 7.

23. Prove that

    (a) $\text{div}\left(\frac{\vec{r}}{r}\right) = \frac{2}{r}$ (b) $\nabla.\left(r^n\vec{r}\right) = (n+3)r^n$.

24. Verify Green's theorem in the plane for $\int_C \left(xy + y^2\right)dx + x^2dy$ where C is the chord curve the region bounded by y = x and $y = x^2$.